



Supervisor's Training centre

NORTH CENTRAL RAILWAY, JHANSI



HAND BOOK FOR C&W SUPERVISORS

Hand Book for C&W Supervisors released

by

SHRI HARISH CHANDRA JOSHI

General Manager

North Central Railway

On Occasion of Mechanical Officers Meet

27/08/2011



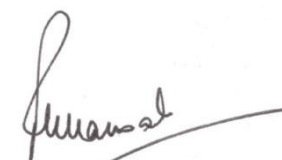
*Chief Mechanical Engineer
North Central Railway
Allahabad*

Carriage & wagon Supervisors play a vital role in train operations. It is our duty to regular update their knowledge. This will enable them to ensure safe running of trains and also assist in train operations.

It is matter of great pride that Supervisor Training Centre, Jhansi has made a hand book for C & W Supervisors. This contains the latest modifications, instructions for both Coaching & Wagon. Latest fittings in Coaching have also been included. This handbook will be very useful to the Supervisors. This should be distributed to all C & W supervisors during Training at STC.

I wish to congratulate the Principal & Instructors of STC for this effort.

Date: 29.06.2011


(A. K. KANSAL)



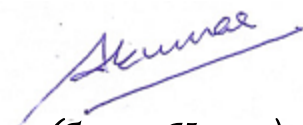
*Chief Workshop Engineer
North Central Railway
Allahabad*

Carriage & wagon supervisors are back bone of Mechanical Department & play important role in train operation. This can be done effectively if they are having sound technical knowledge in their field.

Supervisor Training Centre has issued Hand Book of Carriage & Wagon Supervisor, which is very useful for C & W supervisor latest development in the Rolling Stock have also been covered effectively.

I hope, after gone through this book supervisors can update/ upgrade their knowledge which will result efficient train operation & reduction of line failure & punctuality will improve.

Date: 06.05.2011


(Anoop Kumar)



*Chief Rolling Stock Engineer
North Central Railway
Allahabad.*

Carriage & wagon supervisors play a vital role in moving the wheels of the nation. It's not only duty of C&W Supervisor to update their knowledge and skill in their field but this is necessity of the time to perform their duties with thorough technical knowledge for effectiveness & efficiency.

It is a matter of great pleasure that Supervisors Training Centre has prepared a hand book for the C&W Supervisors. I appreciate the hard work done by STC in bringing the relevant information in the form of book,

The hand book will be useful, if it is referred to frequently.

Date: 13.06.2011



(A. K. Misra)



*Chief Workshop Manager
North Central Railway
Workshop Jhansi*

Supervisors Training Centre, Jhansi is playing a very important role in the training of supervisors of Mechanical Department of NCR. Besides this, it is also conducting training on Disaster Management for supervisors of all departments of NCR.

This Hand book is inspired by our CME, Shri A.K.Kansal, who gave instructions to STC that training material should be published in the form of a book and handed over to the trainees. I compliment Shri T.S.Bhati, Principal,STC, Shri N.K.Singh, Chief Instructor-STC and Shri Rahul Kumar, Trainee J.E.(Carriage and Wagon) who have taken great pains for compilation of the material contained in this training material.

I would also like to thank our CWE, Shri Anoop Kumar as well as CRSE, Shri A.K.Misra who have gone through the training material at draft stage and have given many valuable inputs before it was published in the final form.

We hope that the C&W Supervisors who are given this booklet will find it very useful in their day-to-day working.

Date: 14.06.2011

*Ashesh Agrawal
(Ashesh Agrawal)*



*Principal
Supervisor's Training Centre
Jhansi
Date: 04.05.2011*

Preface

Carriage & wagon rolling stock is back bone of Indian Railway. C & W Supervisors play a vital role to maintain C & W Rolling stock, in operative, maintain punctuality & reduce line failure, detachment of coach/wagon. It is duty of C&W supervisor to update/ upgrade his knowledge in his field.

Supervisor Training Center is imparting training to C & W supervisor it is basic moral duty of training center to upgrade and update the knowledge of C & W supervisor. This is a step towards this effort. A compendium in the form of book of carriage & wagon with latest development in the field of rolling stock is made. The purpose of this book is to upgrade the knowledge of C&W supervisor & punctuality will improve Line Failure, Detachments will reduce.

I express my sincere thanks to CME Shri A.K.Kansal for his inspiration & direction to issue a teaching material in form of book, I also express my sincere thank to CWE Shri Anoop Kumar, CRSE Shri A.K.Mishra and CWM Shri Ashesh Agrawal for their valuable guidance.

Content of this book has been collected from different references, this book should not be quoted for any technical/ legal references.

This is first attempt in this direction, advice / suggestions are hearty welcome. There may be some error in the printing of book or context of topic. I shall be personally obliged for giving your valuable advice / suggestions so that corrective actions may be taken on the matter.

Thanks

With Regards

(Tej Singh Bhati)

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INTRODUCTION OF CARRIAGE & WAGON

Role of C&W in Railways

A. Related with Open Line working :-

- i) To ensure and co-operate in safer running of rolling stock.
- ii) To attend required schedule maintenance & running repairs of rolling Stocks till the stocks are again due for P.O.H.
- iii) To assist in time running of trains to maintain the punctuality.

B. Related with Work -Shop working :-

- i) To attend Periodical Overhauling of Rolling Stocks.
- ii) To adopt required modifications.
- iii) To maintain proper records of all the rolling stocks running in Indian Railways.

CLASSIFICATION OF ROLLING STOCK

Rolling Stock: - It is a term used for the stocks of coaching, freight (Goods) & Locomotive.

Coaching Stock: - All coaching vehicles including self propelled units such as rail cars, electrical multiple unit, luggage & brake van fit to run with coaching stock are known as coaching stock.

Goods Stock: - All goods stock other than coaching stock whether attached to passenger or goods trains are known as goods stock.

Vehicle: - It is a term used only for coaching stock. There are two types of Coaching Stock.

- 1) **Passenger coaching vehicle (PCV):** - A vehicle in which whole or partial portion is being utilized for carrying passengers.
- 2) **Other coaching vehicle (OCV):** Saloons, inspection cars, medical cars, tourist cars, parcels & horse van, composite luggage Power Cars, Pantry Cars & brake van are known as OCV.

Types Of Coaches

- 1) **ICF:** Integral Coach Factory, Perambur (Chennai) & Rail Coach Factory, Kapurthala (Punjab)
- 2) **LHB COACHES:** LHB body with FIAT bogie (Linke Holfmann Busch – German) Maximum Permissible Speed 160 kmph
- 3) **Hybrid Coach:** LHB Body on Modified ICF Bogie – Running in same of the Duranto Rake.

DIMENSIONS OF COACH SHELL

	ICF	LHB/ HYBRID
Length Over Buffer	22296 (mm)	24000(mm)
Length Over Head Stock	21336 (mm)	23540(mm)
Width	3245 (mm)	3030(mm)
Height From Rail Level	3886 (old)4025 (new) (mm)	4025(mm)
Codal Life	25(Years)	30(Years)

MAIN COMPONENTS OF ICF BOGIE

UNDERFRAME:-

- | | |
|----------------|----------------------|
| i) Sole bar | iii) Transom |
| ii) Head stock | iv) Longitudinal bar |

PRIMARY SUSPENSION:-

- | | |
|-------------------------------|---------------------------------|
| i) Dash pot | v) Axle box safety bolt |
| ii) Dash pot spring | vi) Axle box wing & lug |
| iii) Dash pot protection tube | vii) Safety strap & safety loop |
| iv) Air vent screw | viii) Axle box & axle box plate |

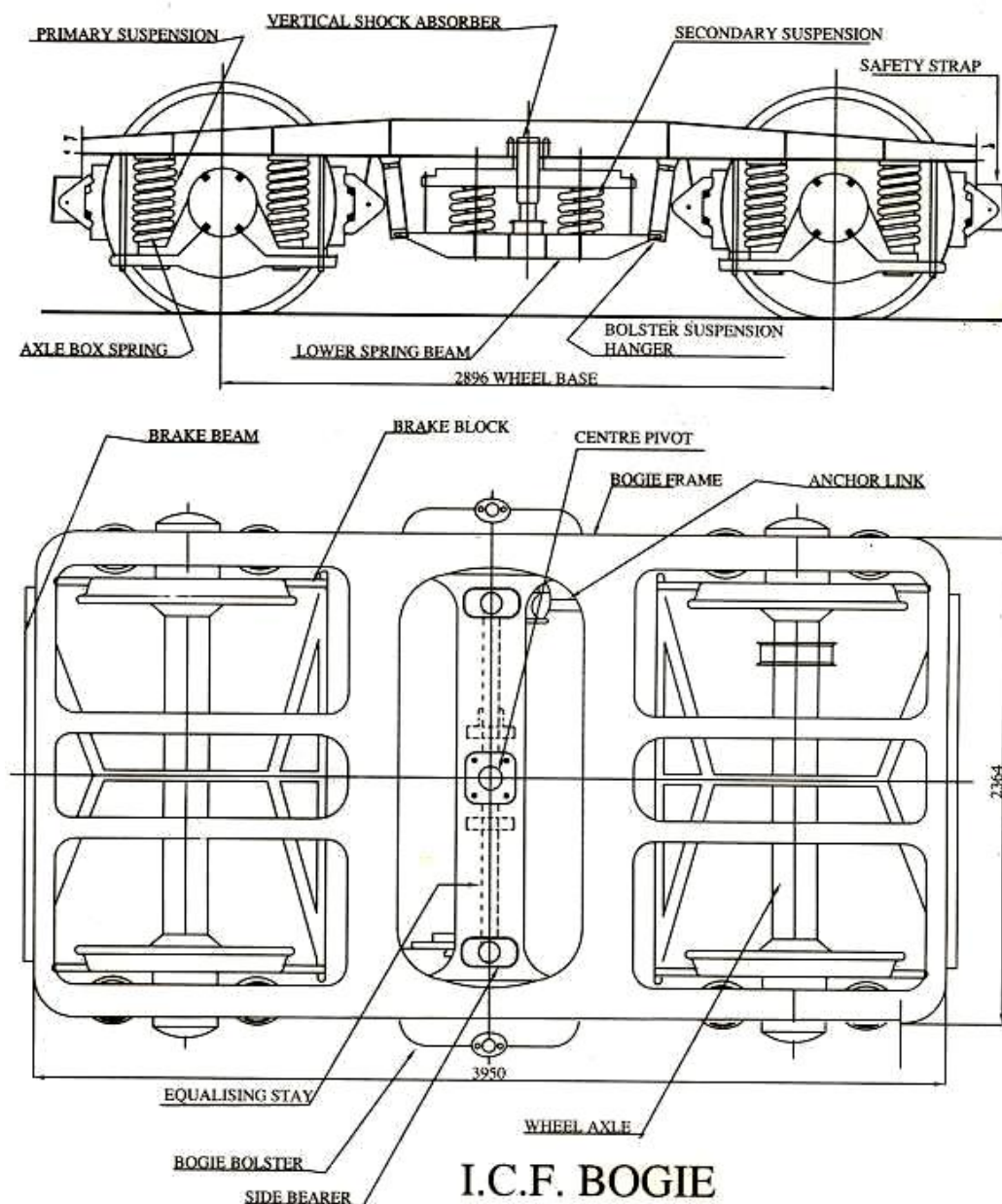
SECONDARY SUSPENSION:-

- | | |
|--|--|
| i) Bogie bolster | viii) Anchor link |
| ii) Bogie bolster lower plank | ix) Centre pivot cotter, split pin & cup |
| iii) Suspension link, link pin & stone | x) Silent block |
| iv) Bolster spring | xi) Side bearer housing |
| v) Vertical shock absorber | xii) Side bearer metal plate |
| vi) Safety strap & safety loop | xiii) Side bearer bronze wearing piece |
| vii) Equalizing stay rod | |

BRAKE GEAR:-

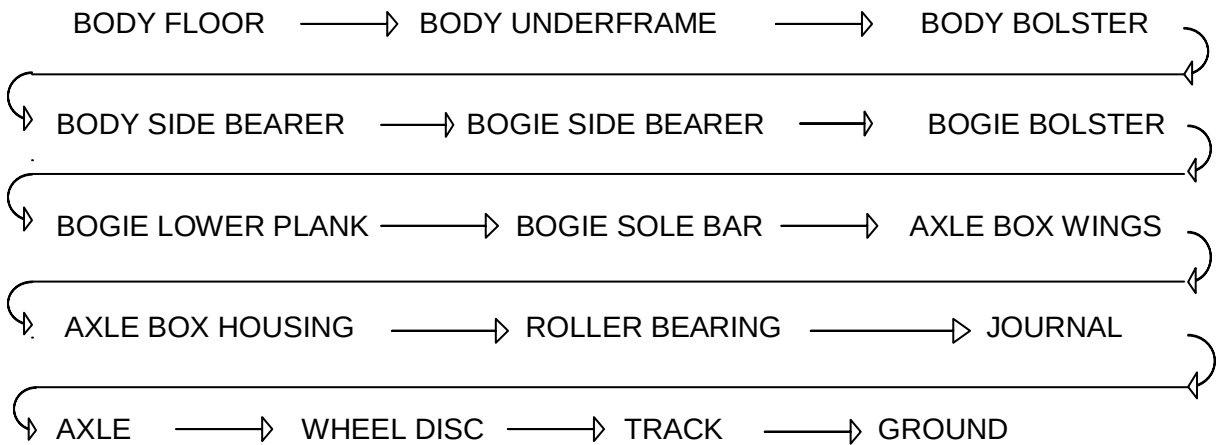
- | | |
|--|---------------------------|
| i) Brake beam | v) Floating lever |
| ii) Brake beam hanger & safety bracket | vi) Curved pull rod |
| iii) Brake safety wire rope | vii) Equalizing truss bar |
| iv) Brake shoe & key | viii) Palm end |

SALIENT FEATURES OF ICF A/C (ALL COIL) BOGIE

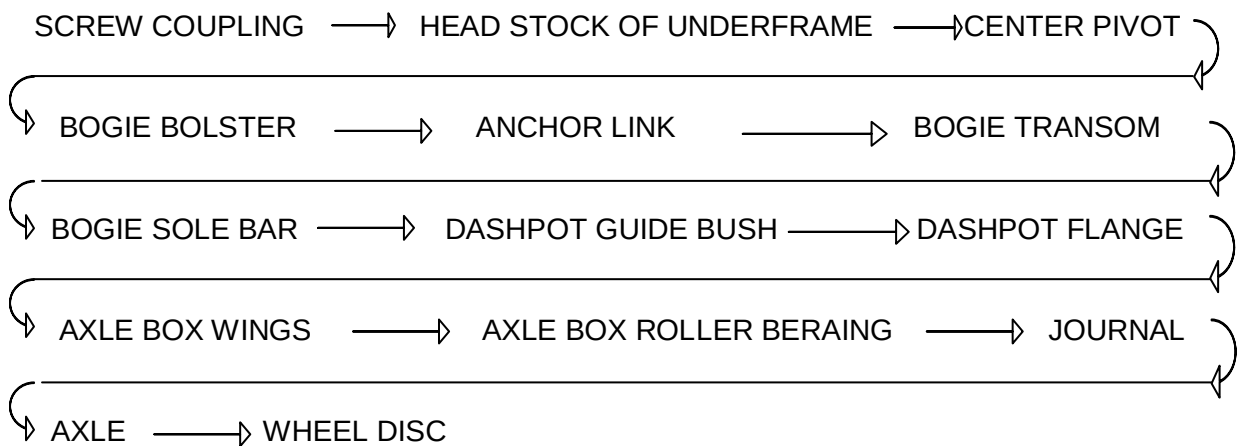


- 1) *Bogie is designed to run on Indian Broad Gauge Track (1676 mm).*
 - 2) *Provision of coil spring at primary & secondary suspension so that bogie is known as All Coil Bogie.*
 - 3) *Bogie Head Stock is provided with pressed T- section and sole bar is with pressed I- section, but at the location of link brackets it is in box section.*
 - 4) *Transom – Previously it was in C-section but now a days it is in Box section to be more robust.*
 - 5) *Wheel Base of bogie is 2896 mm.*
 - 6) *Weight Transmission - By 2 side bearer located at distance of 1600 mm.*
 - 7) *Guidance of bogie Lateral and Longitudinal both with the use of Centre Pivot pin located at the center of bolster.*
 - 8) *Wheel Guidance lateral and longitudinal both with the use of 2 nos. of Dash Pot guide per Axle Box Wings welded at sole bar.*
 - 9) *Axle Capacity - 13 T – For Non A/C coach
16 T – for A/C coach and WLRRM coach*
 - 10) *Roller Bearing – Double Roll Self Aligned Spherical Roller Bearing.*
 - 11) *Axle – Solid and Straight*
 - 12) *Wheel Diameter – 915 mm (New)
– 825 mm (Condemn)
– 837 mm (workshop release)*
- Ref: Rly. Board's Letter No. G2/ M(c)/151/2 vol.- V dated 25/01/2011*
- 13) *Shock Absorber – Provided on Secondary suspension between Bolster and lower plank (2 nos. in each Bogie).
2 nos. of lateral shock absorber are being provided in ICF Bogie to be utilized for Hybrid Coach.*
 - 14) *Vertical Hydraulic Dampers – 2 nos. per Axle Box Vertical telescopic hydraulic Dashpots are provided.*
 - 15) *Fitment of brake block - Clasp type brake block arrangement is provided with the use of brake shoe head and brake beam.*
 - 16) *2 nos. equalizing stay rods per bogie are utilized to maintain the distance between both the lower planks and to maintain lateral thrust occurring during run.*
 - 17) *Provision of Anchor link – 2 nos. per bogie with the provision of silent bushes are provided diagonally between bogie transom and bolster to work as a media to transmit the draw and braking force from trolley to body and body to trolley vice versa.*
 - 18) *Piston Stroke – In conventional type Air Brake system 90 ± 10 mm and in BMBC within 32mm should be maintained.*
 - 19) *Provision of Running Clearance:-*
 - a) *'A' Clearance: - 43^{+0}_{-3} mm (For 13 ton) and 27^{+0}_{-3} (For 16 ton)
It is a clearance to be provided between axle box crown & safety bolt.*
 - b) *'B' Clearance: - It is a clearance to be provided between bolster top & bottom of sole bar that should be 40 ± 5 mm to all type of bogie.*
 - 20) *Riding index: - ICF bogie – 3.25 to 3.50*
 - 21) *Truss bar Hanger: - Strength with double eye hole.
Length: 235 mm (New), 205 mm (Old)*
 - 22) *Journal Size: - Dia. – 120x113.5 mm (sleeve mounted), 120x130.5 mm (direct mounted)*
 - 23) *Journal Centre: - 2159.5 mm*
 - 24) *Speed: - Fit to run up to 110 kmph. (Trial has been conducted up to 140 kmph.)*

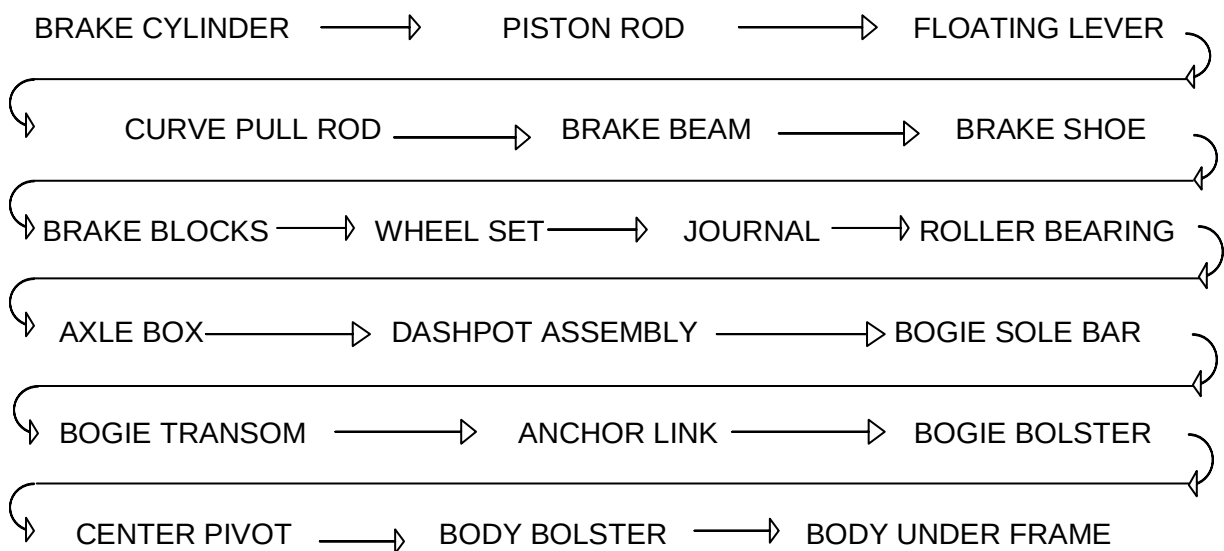
WEIGHT TRANSMISSION OF ICF COACH



DRAFT FORCE TRANSMISSION OF ICF COACH

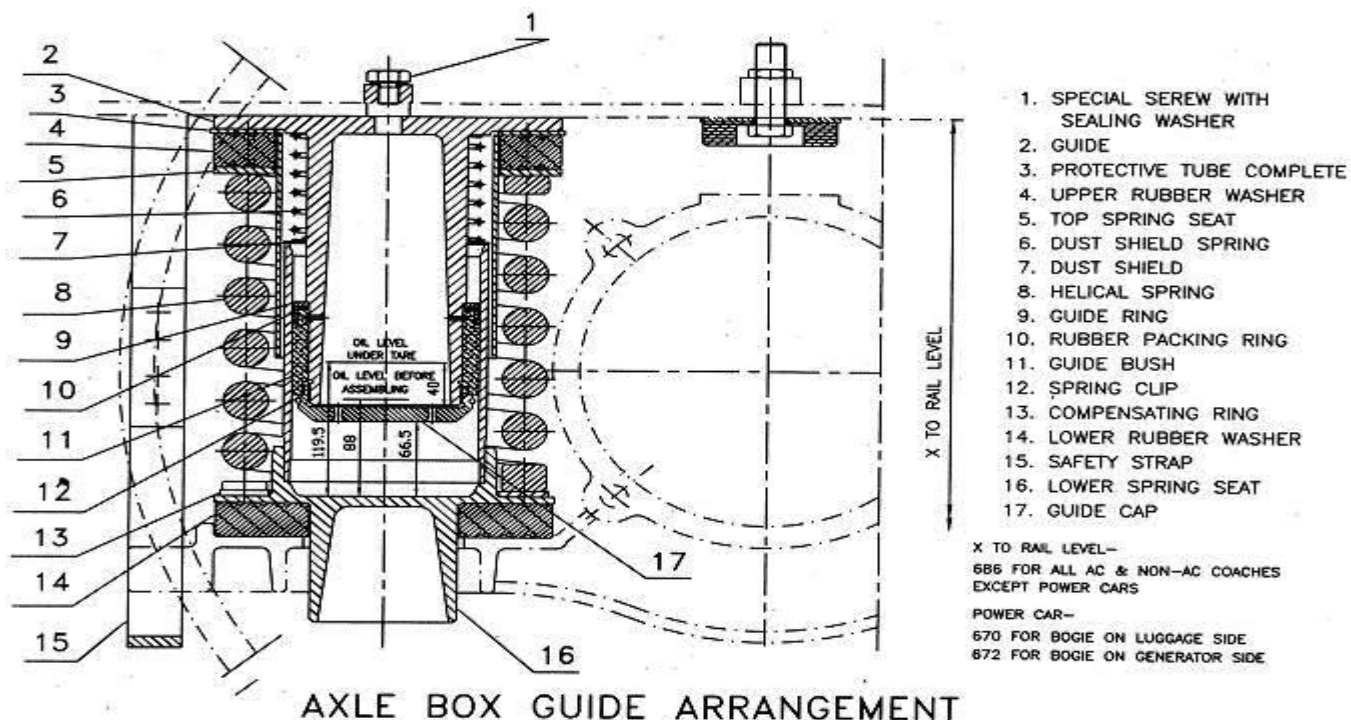


BRAKING FORCE TRANSMISSION OF ICF BOGIE TO BODY



1. AXLE BOX GUIDE WITH DASH POT ARRANGEMENT

Axle box guides are of cylindrical type welded to the bottom flanges of the bogie side frame with close dimensional accuracy. These guides together with lower spring seats located over the



axle box wings which house the axle box springs and also serve as shock absorbers. These guides are fitted with guide caps having nine holes of diameter 5 mm equidistant through which oil in the lower spring seat passes under pressure during dynamic oscillation of coach and provide necessary damping to primary suspension to enhance better riding quality of coach. This type of rigid axle box guide arrangement eliminates any longitudinal or transverse relative movement between the axles and the bogie frame. The quantity of oil required to achieve **40 mm** oil level above the guide cap in modified arrangement is approximately **1.6 liters** and in unmodified arrangement is approximately **1.4 liters**. As it is not possible in open line to distinguish between modified and unmodified arrangements, **40 mm** oil level is standardized for both.

	wire dia	Free Height Non A.C.	Free Height A.C.
Axle Box Spring	33.5 mm	360^{+2}_{-0} mm	375^{+2}_{-0} mm
Bolster Spring	42 mm	385^{+2}_{-0} mm	400^{+2}_{-0} mm

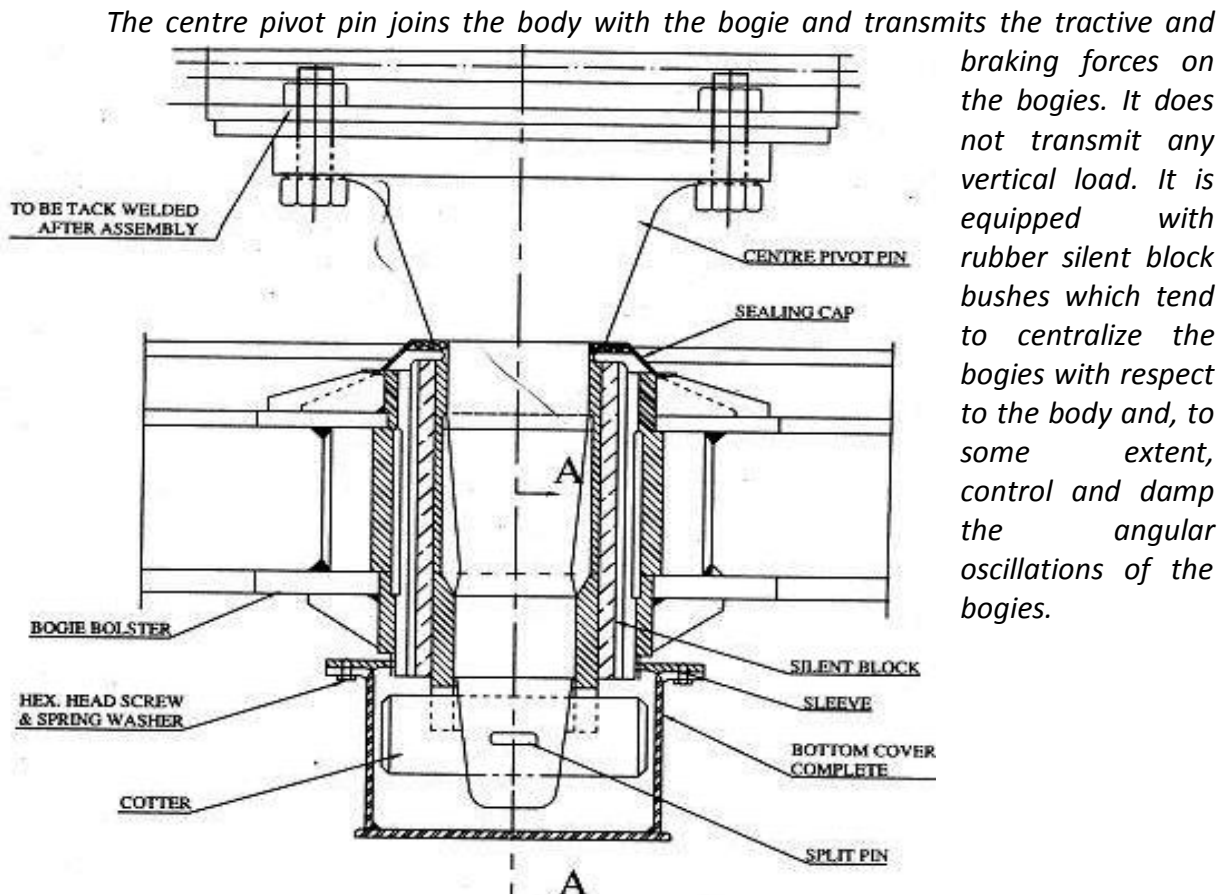
Common Defects Found In Axle Guide Assembly, Causes And Their Remedial Action :

Sr. No.	Defect	Reasons	Remedies
1.	Perished rubber packing ring.	Poor quality of rubber packing ring	1. Replace rubber packing ring at every examination. 2. Use only rubber packing rings conforming to IRS specifications.
2.	Axle guide found worn on one side	Initial difference in wheel diameters on same axle more	1. Maintain difference in wheel diameters on same axle within 0.5 mm. Use wheel diameter gauge with minimum 0.2 mm accuracy.

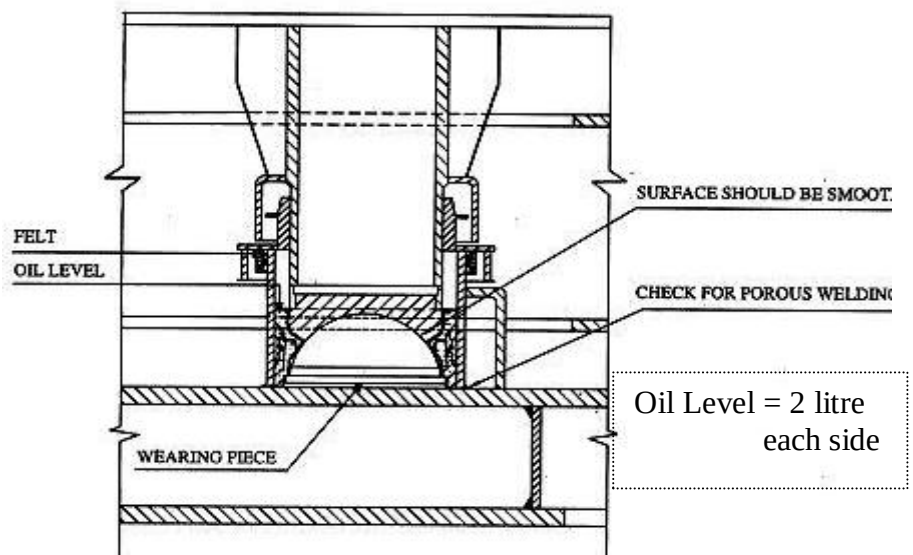
		than 0.5 mm. Coach is not leveled.	2. Level the coach. 3. The squareness and alignment of axle box guides should be checked with alignment gauges and corrected. 4. Vent holes should be sealed with gaskets screw tightened well after topping.
3.	Axle box springs rubber upper spring seat (protective tube)	-do-	-do-
4.	Guide bush worn.	-do-	-do-
5.	Lower spring seat surface worn.	-do-	-do-
6.	Guide ring broken	Axle guide is hitting lower spring seat. Weld joint of lower spring seat and tube is porous or cracked.	1. Guide securing bolt should not project out of guide cap. 2. Use good quality upper and lower rubber washers and correct number of compensating rings in the axle box guide assembly. 3. Adjust ABC clearance on leveled track. 4. Strip and re-weld lower spring seat correctly.
7.	Broken/distorted spring clip of guide cap.	-do-	-do-
8.	Guide cap securing assembly broken.	-do-	-do-
9.	Lower spring seat scored and dent mark on guide cap.	-do-	-do-
10.	Dust shield spring broken/distorted.	-do-	-do-
11.	Dust shield twisted or damaged.	-do-	-do-
12.	Guide threads damaged.	-do-	-do-
13.	Leakage from lower	-do-	-do-

2. **Air Vent Screws:** On the bogie side frames, directly above the dash-pots, tapped holes are provided for replenishing oil in the dash pots. Special screws with copper asbestos washers are screwed on the tapped hole to make it air tight.

3. **Bogie Bolster Suspension:** The bolster rests on the bolster coil springs - two at each end, located on the lower spring beam which is suspended from the bogie side frame by means of bolster-spring-suspension (BSS) hangers on either side. The two anchor links diagonally positioned are provided with silent block bushes. The links prevent any relative movement between the bogie frame and coach body.
4. **Springs:** In ICF bogie, helical springs are used in both primary and secondary suspension. The springs are manufactured from peeled and centre less ground bar of chrome vanadium/chrome molybdenum steel.
5. **Centre pivot arrangement:**



6. **Side Bearers:**
The side bearer arrangement consists of a machined steel wearing plate immersed in an oil bath and a floating bronze-wearing piece with a spherical top surface kept in it, on both sides of the bogie bolster. The coach body rests on



SIDE BEARER ARRANGEMENT

the top spherical surface of these bronze-wearing pieces through the corresponding attachments on the bottom of the body-bolster. The whole arrangement is provided with a cover to prevent entry of dust in the oil sump.

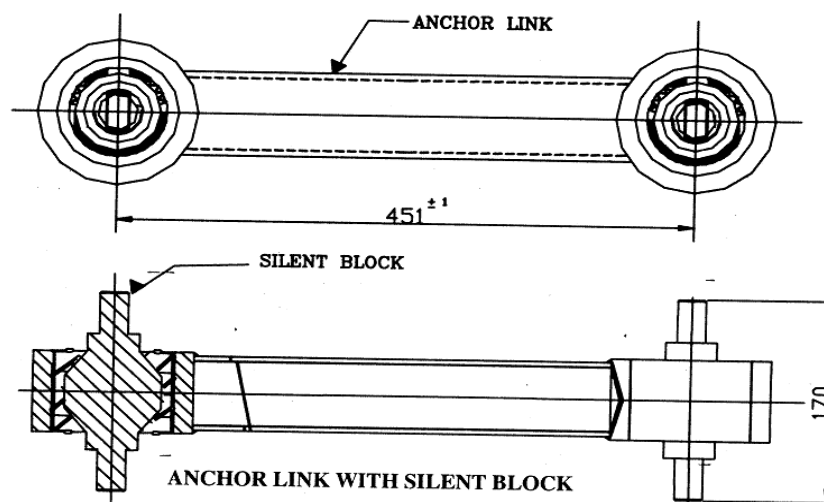
Wear limit for wearing plate:

New size : 10 mm Condemning size : 8.5 mm

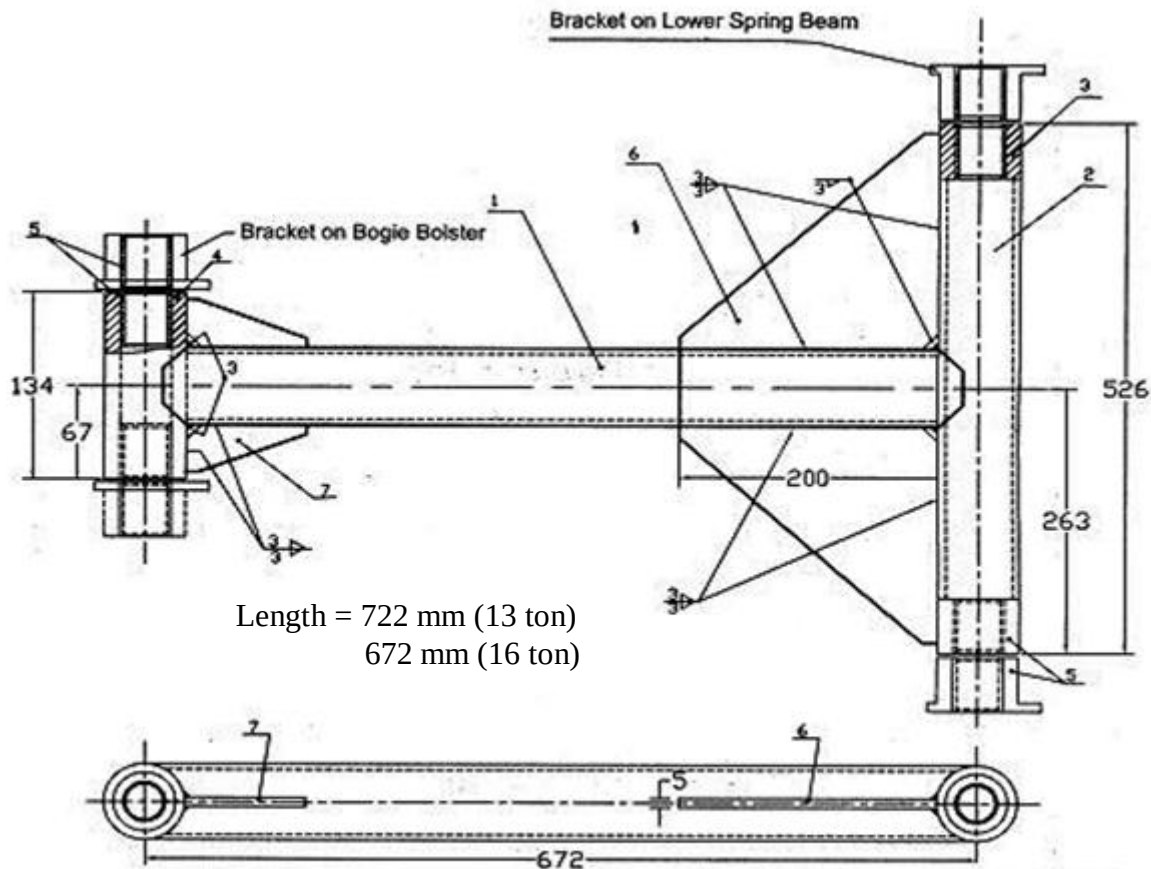
Wear limit for wearing piece:

New size : 45 mm Condemning size : 42 mm

7. **Anchor Link:** The floating bogie bolster which supports the coach body is held in position longitudinally by the anchor links which are pinned to the bolster sides and the bogie Transoms. One anchor link is provided on each side of the bolster diagonally across. The links can swivel universally to permit the bolster to rise and fall and sway side wards. They are designed to take the tractive and braking forces. The anchor links are fitted with silent block bushes. Now a day's modified anchor Links are being used in ICF bogies. (Similar as FIAT Bogie)



8. **Silent Block:** This is a synthetic rubber bush fitted in anchor link and center pivot of ICF bogies to transmit force without shock and reduce noise.
9. **Brake Rigging:** Brake rigging is provided to control the speed of the coach by transferring the braking force from the brake cylinder to the wheel tread. Brake rigging can be divided into two groups i.e. Bogie mounted brake rigging and coach under frame mounted brake rigging.
- A. **Coach Under Frame Mounted Brake Rigging:** In 16.25 t axle load bogie the four lever used in bogie brake rigging are each with lever ratio of **1:1.376** hence the total Mechanical Advantage in a bogie is **5.504**
In 13 t axle load bogie the four levers used in bogie brake rigging are each with lever ratio of **1:1** hence the total Mechanical Advantage in a bogie is **4**
- B. **Bogie Mounted Brake Rigging:** Bogie brake rigging has been modified to incorporate a total mechanical advantage of **7.644** per bogie for non-ac coaches and **8.40** per bogie for ac coaches
10. **Equalizing Stays:** This device has been provided on bogies between the lower spring plank and the bolster to prevent lateral thrust on the bolster springs which have not been designed to take the lateral forces. These links have pin connections at both ends and, therefore, can swivel freely.



**EQUALISING STAY FOR LOWER SPRING BEAM
(16.25 t Axle Load Bogie) (RDSO Sketch 88105)**

11. **Bolster Spring Suspension (BSS) Hangers:** In the secondary suspension, the bolster is supported on helical coil springs which are placed on the lower spring plank. The lower spring plank is suspended from the bogie side frame through BSS hangers on hanger blocks.
12. **Shock Absorbers:** Hydraulic shock absorbers with capacity of ± 600 kg at a speed of 10 cm/sec are fitted to work in parallel with the bolster springs to provide damping for vertical oscillations.

LIFTING OF THE BODY FROM THE BOGIE:-

A) Required Tools & Equipments:-

- a) With the use of 4 no. of Mechanical Jacks having capacity of 10 ton each OR
- b) With the use of 4 no. of Hydraulic jacks having capacity of 10 ton each OR
- c) With the use of 02 no. of Electrical operated Traveling Crane having capacity of 20/25 ton each.
- d) With the use of 4 no. of Whiting jacks having capacity of 20/25 ton each.

B) Other Tools:-

- a) Complete set of Spanners.
- b) Different types of hammers.
- c) Wooden Wedges & Packing.
- d) Trestles.
- e) Complete set of Gas cutting & welding equipment.
- f) Complete Tool kit

C) Items to be disconnected before lifting of a body:-

- 1) a) Removal of centre pivot cotter [If lifting is being done by E.O.T cranes or whiting jacks].
b) Removal or Unscrewing of centre pivot studs [If lifting is being done by mechanical/hydraulic jacks]
- 2) Unscrewing of air vent screw of dash pot.
- 3) Disconnection of Dynamo belt.
- 4) Disconnection of S.A.B pull rod.
- 5) Disconnection of lateral shock absorber if connected
- 6) Disconnection of axle box safety loops.
- 7) Disconnection of commode chute if infringing.
- 8) Disconnection of foot board.
- 9) Disconnection of dummy carrier if infringing.
- 10) Inserting required thickness of wooden packing between upper portion bolster & sole bar.

Buffer Height Adjustment: causes of low buffer height in ICF coaching stock

- a) excessive wear of wheel (circumference)
- b) Due to loss of proper stiffness of coil spring provided on primary and secondary suspension.
- c) Due to excessive wear on side bearer's metal & bronze piece.

Piece	New size	Condemning size
Steel	10 mm	8.5 mm
Bronze	45 mm	42 mm

- d) Due to excessive wear on link brackets, stone & pin provided on secondary suspension arrangement.
 - Maximum buffer height = 1105 mm [In empty condition]
 - Minimum buffer height = 1030 mm [In loaded condition]
 - Minimum buffer height of coaching stock should not be less than 1090 mm at the time of releasing of coach from POH Workshop.

Procedure to achieve buffer height for ICF coaches. (Newly Introduced- Hytrel Packing)

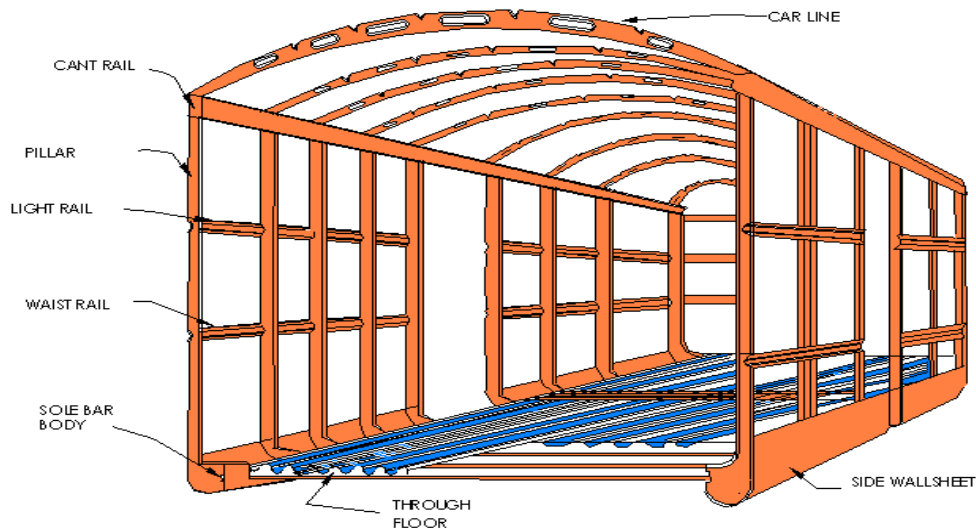
- To achieve buffer height a standard size of wooden packing pieces are used which are kept below the Dash Pot Flange on axle Box wings of primary suspension.
- New Wheel dia :- 915 mm
- Condemning :- Solid Wheel: 813 mm.

Sl. No.	Wheel Diameter (mm)	Thickness of Wooden Packing (mm)
1.	Below 889 up to 863	13
2.	Below 863 up to 839	26
3.	Below 839 up to 819	38
4.	819 & below	48

Note:-

- 1) If required 6 mm compensating metallic ring can also be used. If further required wooden packing in half's can be utilized in 8 mm & 12 mm thickness but, it should be ensured that total thickness of wooden half packing + compensating ring should not be more than 20 mm.
- 2) Gap between axle box lug & safety loop also should not to be less than 40 mm.
- 3) Running clearances i.e. 'A' & 'B' also should be ensured.

ICF COACH SHELL / BODY



Salient Features of ICF Coach Shell / Body

1. **All Metal:-** ICF coach shell is made up of steel channels frames of thin sections except the seats & luggage bunks which are made up of wooden members.
2. **Light Weight:-** The weight of coach shell is reduced due to less use of wooden members. Anti Corrosive Corten Steel (IRSM 41) is used for body fabrication purpose. As thickness of roof shell is 1.6 mm, corrugated floor side panel & end panel is only 2 mm. The use of gusset plate, knee & rivets are also avoided in under frame. Hence weight of ICF shell is reduced by 26% to 32% in respect to weight of IRS coach shell which was used in the past.
3. **Integral Construction:-** The shell of ICF coach is made with frame, Body side pillars, roof carlines, doors & windows, waist rail, light rail & cant rail are welded together with sole bar through side pillars. Corrugated flooring, side panels, end panels & roof are welded together by means of homogenous welding. End pillars, stanchions and side pillars are also connected with paneling work. This type of structure gives the integral construction of coach body which forms the big tubular hollow construction which is light in weight.
4. **Anti telescopic construction:-** The shell of ICF Coach is designed to bear 45 tones of vertical load and 200 tones of longitudinal impact on side buffers. The coach body so designed that it is more strengthen at end portion as well as in passenger seating portion and less at the gallery portion. Due to which maximum kinetic energy can be absorbed by the end portion as get damaged during accident and rest kinetic energy also can be shared by corrugated flooring and other members of body shell, resulted passenger accommodation area of middle portion of shell is safe in view of damaging.
With the result of these properties of shell entering of end portion of one coach shell is avoided to in adjacent coach body shell. For that shell construction is known as anti telescopic construction.
5. **Stress Skin Construction:-** The construction of the body shell in the end at side panel is multi point tag welded. As side panel is welded at waist rail, light rail, cant rail and car lines by means of CO₂ welding provided at perfect distance. 70 % of total developed stresses are absorbed by corrugated trough flooring. Thus this multi point welding

property of the end & side panel is enough to minimize developed stresses of panels during run.

6. **Aerodynamic shell:** - To minimize air resistance during running, the shell body is specially constructed, the roof and turn under are made in curved shape which minimize air resistance at high speeds.
7. **Anti corrosive:-**To achieve anti corrosive property Corten steel **IRSM - 41** (max at turn under and lavatory portions) is being utilized for paneling purpose. At the time of manufacturing sand blasting, grit blasting is also given on panel sheet which is helpful to prepare rough surface for painting resulted less chances of corrosion. Three coats of bituminous anti-corrosive paints are also given at welded portion and for other portion red- oxides paint is applied, for anti-corrosive treatment. Holes in trough flooring are also provided for proper drainage of water. Especially 200 x 135 mm size elliptical holes are given in turn under portion for proper drainage of rubbish water coming from window shell. With the help of these facilities and precaution corrosion probabilities can be minimized. 19 mm thick ply or 12 mm thick compregre ply and 2.00 mm thick PVC flooring is utilized.
8. **Heat resistance :-**
To develop heat resistance property in coach shell following precaution or facilities has been provided:-
 - a) Silver / aluminum paint coat is provided on outer side of roof which reflects the sun rays coming outside the coach.
 - b) In roof ceiling layers of insulating materials like asbestos / glass wool is provided which is bad conductor of heat resulted direct transmission of heat inside is minimized.
 - c) In carlines, elliptical holes are also provided for proper air circulation from one compartment to another.
 - d) No. of ventilators are also provided on top of roof to exit smokes, gases and to circulate fresh air.
 - e) Limpet sheet is also provided inside of the roof (2mm thick) which is bad conductor of heat.

WHEEL & AXLE

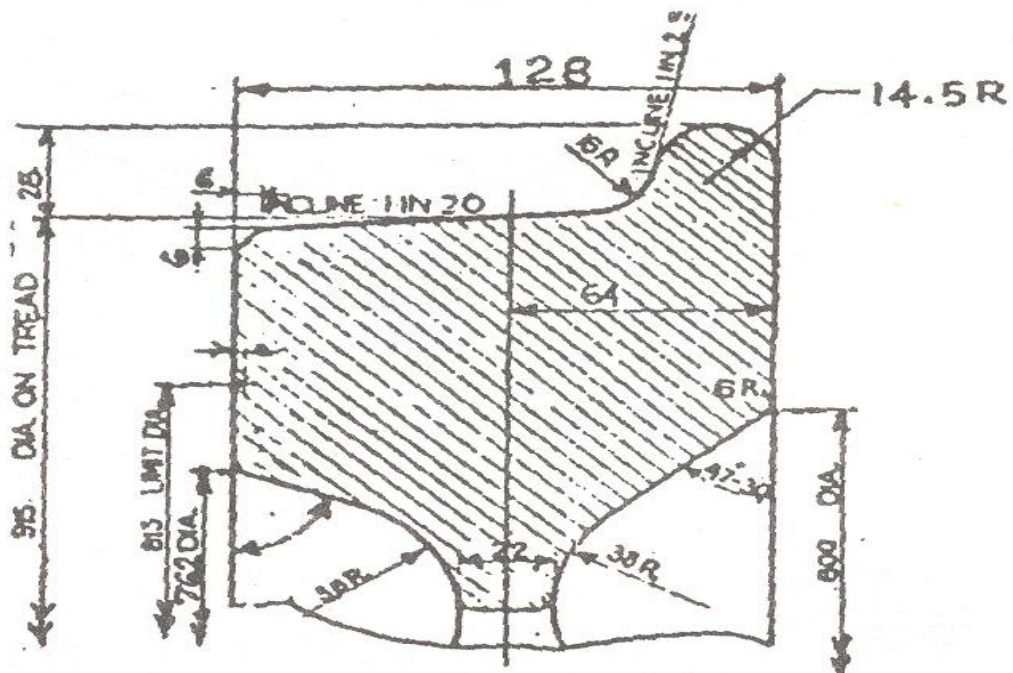
- 1) Width of wheel – 127 mm.
- 2) Wheel diameter – 915 mm (new), 825 mm (Condemn)
- 3) Axle capacity – 13 T(Non AC) -16 T(AC).
- 4) Journal – Direct mounted 120 x 130.5 mm
- 5) Journal centre – 2159.5 mm
- 6) Permissible diameter variation in wheel **(Coaching)**
 - a) On same axle – 0.5 mm (for machining purpose only)
 - b) On same bogie – 5.0 mm
 - c) On same coach – 13.0 mm

(Goods)

 - a) On same axle – 0.5 mm (for machining purpose only)
 - b) On same bogie – 13.0 mm
 - c) On same coach – 25.0 mm

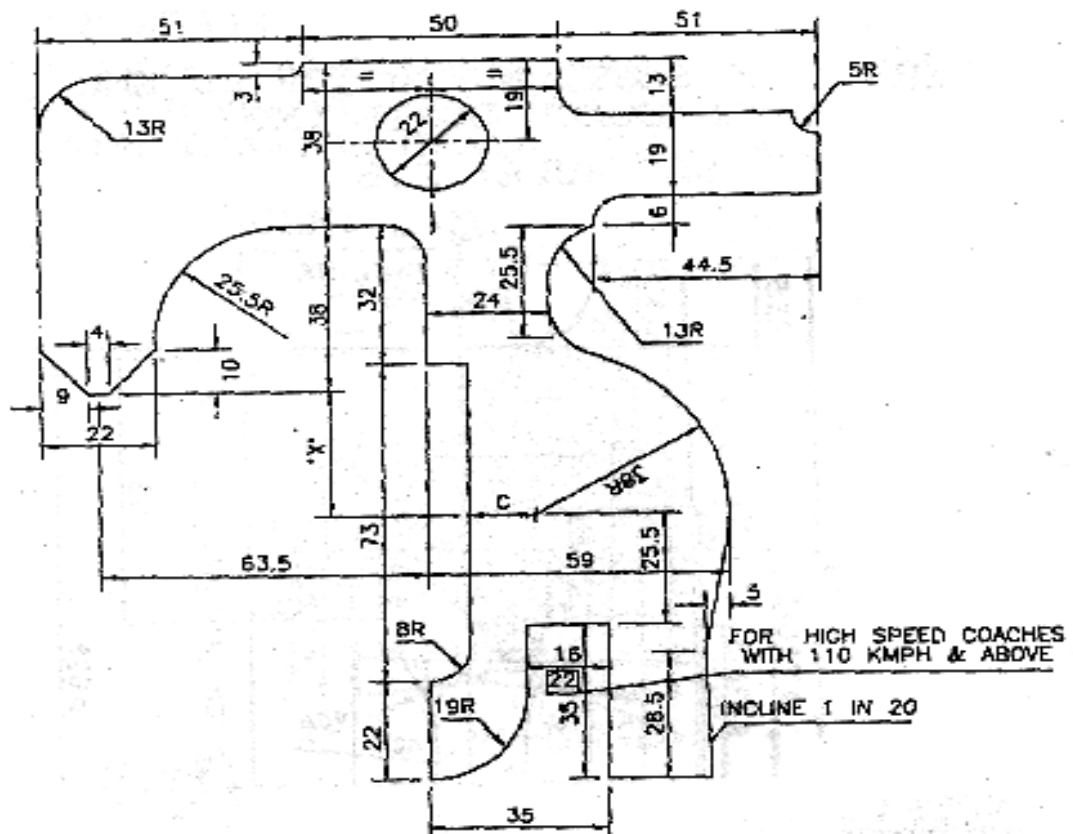
WHEEL PROFILE

Ref: IRCA Rule Book Part – IV (Plate 29)



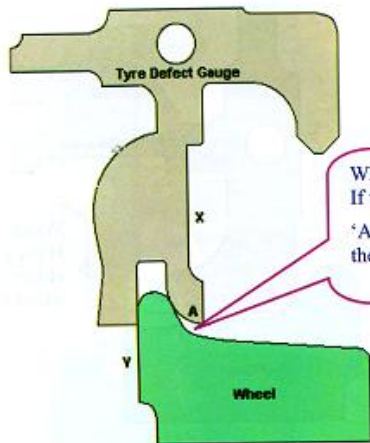
WHEEL DEFECT GAUGE

Ref: IRCA Rule Book Part – IV (Plate 39)

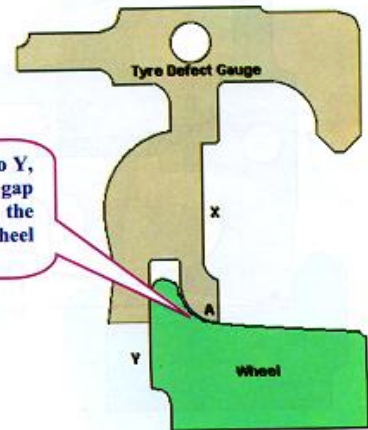


WHEEL DEFECTS

Thin Flange

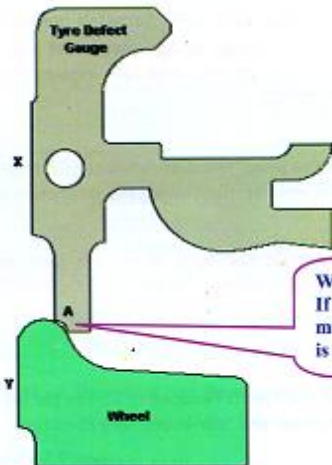


When X is parallel to Y,
If there is gap between
'A' and the root of flange,
the wheel is Serviceable

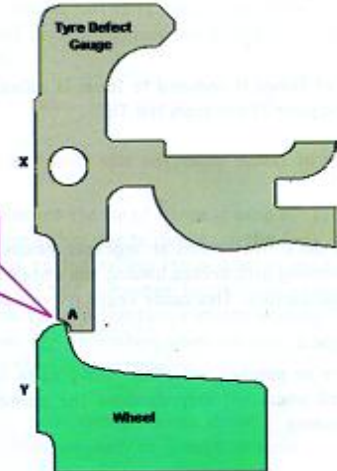


When X is Parallel to Y,
If there is no gap
between 'A' and the
root of flange, the wheel
is rejectable

Sharp Flange

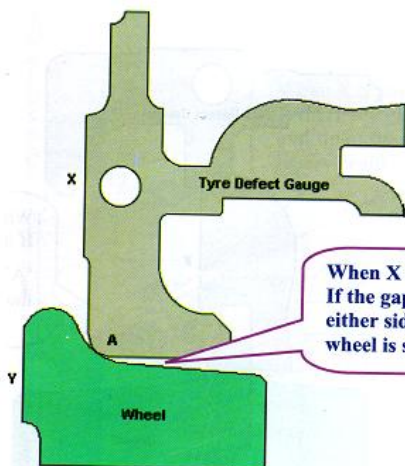


When X is parallel to Y,
If there is Gap in the
middle at A, the Wheel
is serviceable

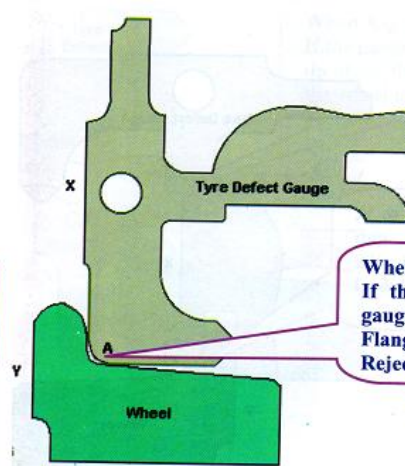


When X is parallel to Y,
If there is gap on either
side of A, the Wheel is
rejectable

Radius too small at the root of flange

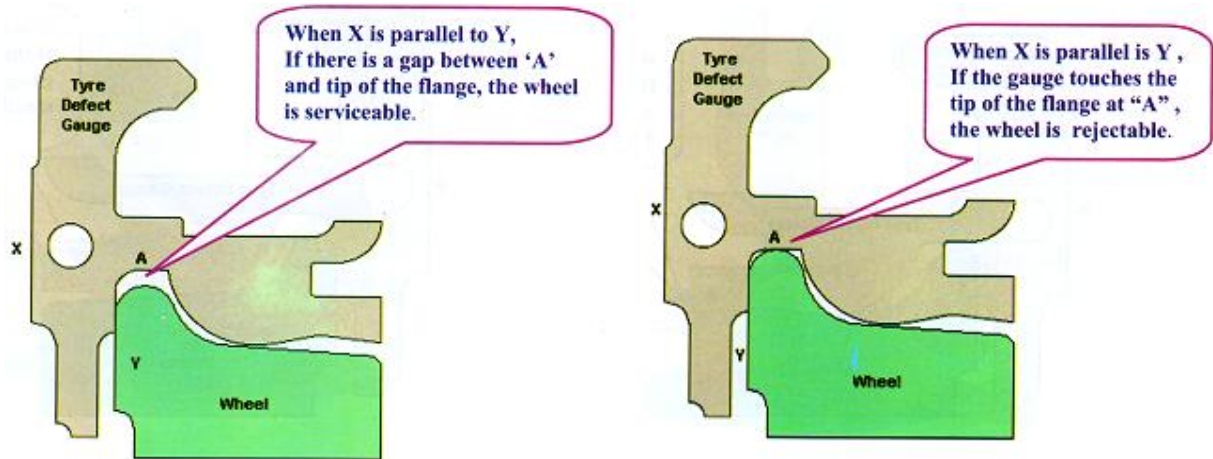


When X is parallel to Y,
If the gap is available at
either side of 'A', the
wheel is serviceable.

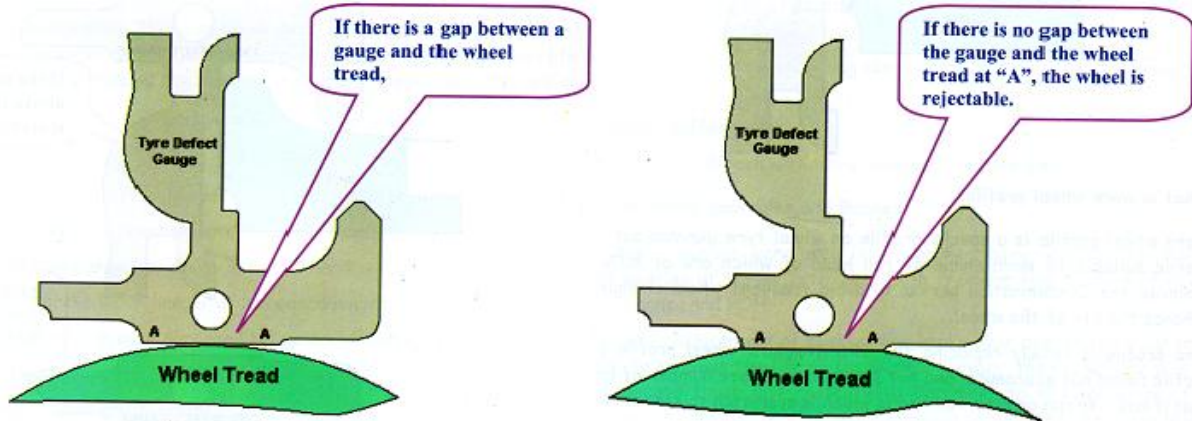


When X is parallel to Y ,
If there is a gap between
gauge and the Root of
Flange at A , the Wheel is
Rejectable

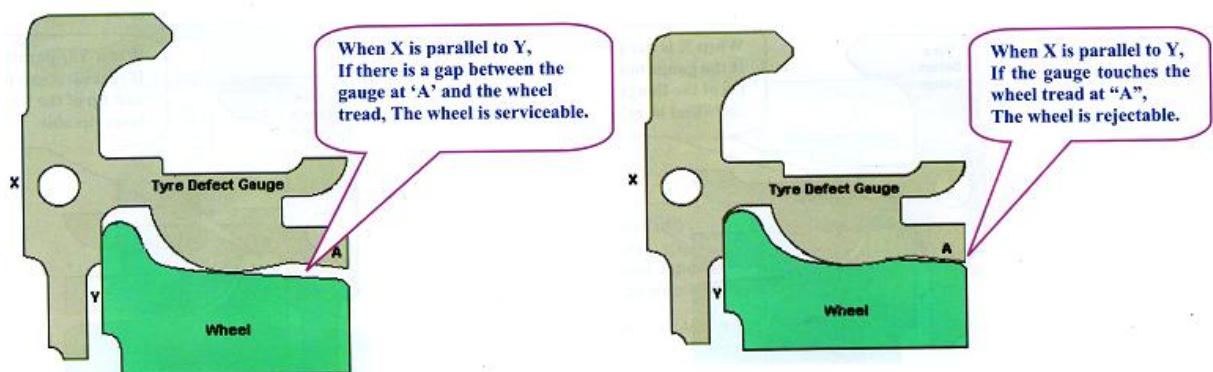
Deep Flange

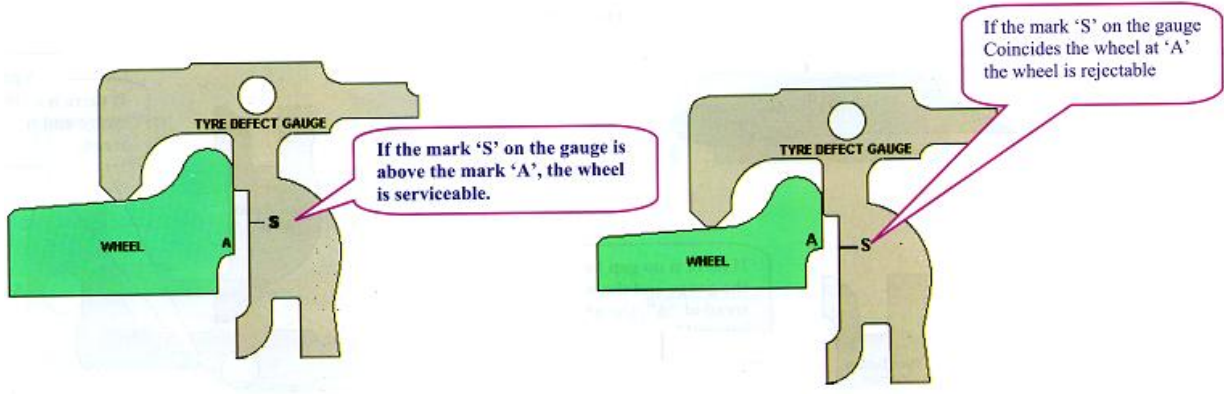


Flat faces on Tyre



Hollow Tyre



Thin Tyre	
 If the mark 'S' on the gauge is above the mark 'A', the wheel is serviceable. If the mark 'S' on the gauge Coincides the wheel at 'A' the wheel is rejectable	
Wheel Defects	Causes
Thin Flange	When the flange thickness reduces from 28.5mm (New) to 16 mm (Condemn) or less, then the flange is called thin flange. Flange thickness is measured at a depth of 13 mm from the tip of the flange. Repercussion: - Chances of bursting of point due to entering of flange between Tongue rail and Stock rail.
Sharp Flange	When the radius given at the tip of flange is worn out from 14.5mm (New) to 5 mm (Condemn) or less is called Sharp Flange. Repercussion: - Shearing of fish plate bolts at rail joints.
Radius too small at the root of flange	New Radius of flange at the root is 16R, when it is reduced to 13R or below, it is called Radius too small at the root of flange. Repercussion: - Excessive lateral play result in chances of mounting of flange over rail.
Deep Flange	The New height of the flange is 28.5mm, when it increased up to 35mm or more is called Deep Flange Repercussion: - Shearing of fish plate bolts at rail joints.
Flat faces on Tyre	Flatness on wheel circumference is called Flat faces on tyre. - For Coaching Stock it is allowed up to 50 mm - For Goods Stock it is allowed up to 60 mm Repercussion: - Chances of rail fracture due to hammering effect on rail.
Hollow Tyre	If the groove on the wheel tread is up to 5 mm or more, it is called Hollow tyre. Repercussion:- Chances of entanglement of tongue rail nose with wheel.
Thin Tyre	If the remaining thickness of tyre is less than 25 mm, it is called thin tyre. Repercussion: - probability of breakage of tyre.

INTERMEDIATE WORN WHEEL PROFILE FOR COACHING STOCK

Worn Wheel Profile:- Worn wheel profile is a special profile on wheel disc derived out of standard wheel profile suitable to worn shape of rail head (which are of 80% track).

This is to minimize condemnation of disc to avoid frequent wheel changing re-profiling & enhance the life of the wheel.

Average Wheel Wear: 2.5 to 3 mm per year for mail/Exp train.

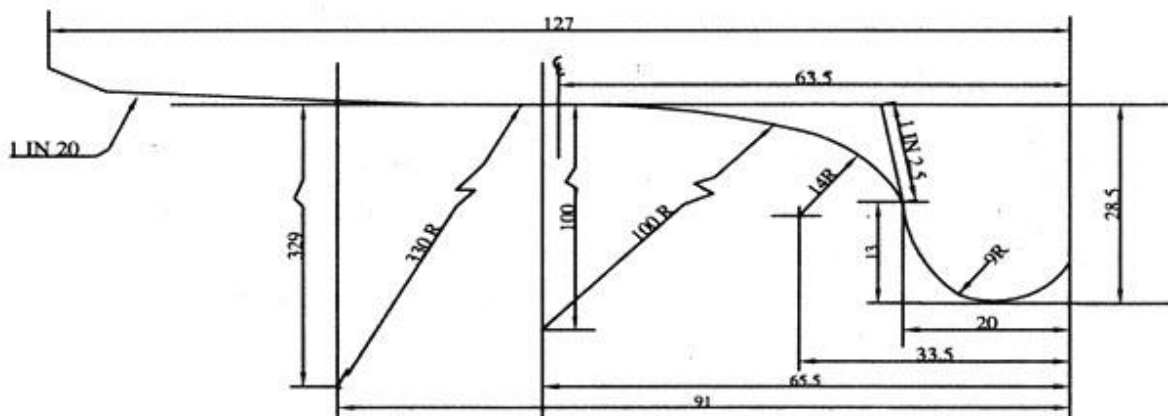
Three intermediate worn wheel profile are developed to increase the life of wheel

- 25 mm, 22 mm & 20 mm

TABLE – INTERMEDIATE WORN WHEEL PROFILE

Thickness of Flange (mm)	D1 (mm)	D2 (mm)	D3 (mm)	R1 (mm)	R2 (mm)	R3 (mm)	R4 (mm)
25	38.5	65.5	91	11.5	14	100	330
22	35.5	65.5	91	10	14	100	330
20	33.5	65.5	91	9	14	100	330

Worn Wheel Profile 20 mm flange Thickness



Benefits of Worn Wheel Profile:

- It increases the life of wheel.
- It decreases machining cost.
- Less fuel consumption of the engine.
- It increases the wheel lateral oscillation

Thermal Wheel Defects: The following guidelines are issued to make the visual inspection of wheels during maintenance more focused and effective. The following wheel conditions should be paid special attention during the visual inspections of solid and tired wheel discs used on coaches and EMU's.

In addition to normal checks exercised on wheel condition during primary / secondary maintenance of coaches, a detailed inspection of wheels should be done when the coaches are received in sick line for attention for either scheduled or out of course attention. The wheel sets shall be inspected for the following conditions and action taken as indicated for each condition:

1. **Shattered Rim:** a wheel with a fracture on the tread or flange must be withdrawn from service. This does not include wheels with localized pitting or flaking without presence of any other rejectable condition.



Shattered Rim

2. **Spread Rim** - If the rim widens out for a short distance on the front face, an internal defect may be present. Spreading of the rim is usually accompanied by a flattening of the tread, which may or may not have cracks or shelling on the tread. Such wheels must be withdrawn from service.



Spread Rim

This condition should not be confused with a uniform curling over of the outer edge of the rim around the entire wheel, which is called rim flow. Rim flow is not a defect.

3. **Shelled Tread** - Shelling can be identified by pieces of metal breaking out of the tread surface in several places more or less continuously around the rim. Shelling takes place when small pieces of metal break out between the fine thermal checks. These are generally associated with small skid marks or "chain sliding." Such wheels should be withdrawn from service.



Shelled Tread

4. **Thermal Cracks** - Thermal cracks appear on a wheel due to intense heating of the wheel arising out of severe brake binding. Such cracks occur on the tread and generally progress across the tread in a transverse & radial direction. Whenever such a crack becomes visible on the outer face of the rim or a tread crack has reached the outer edge (non-gauge face) of the rim; the wheel should be withdrawn from service. If a crack becomes visible on the outer flange face the wheel should be withdrawn from service.



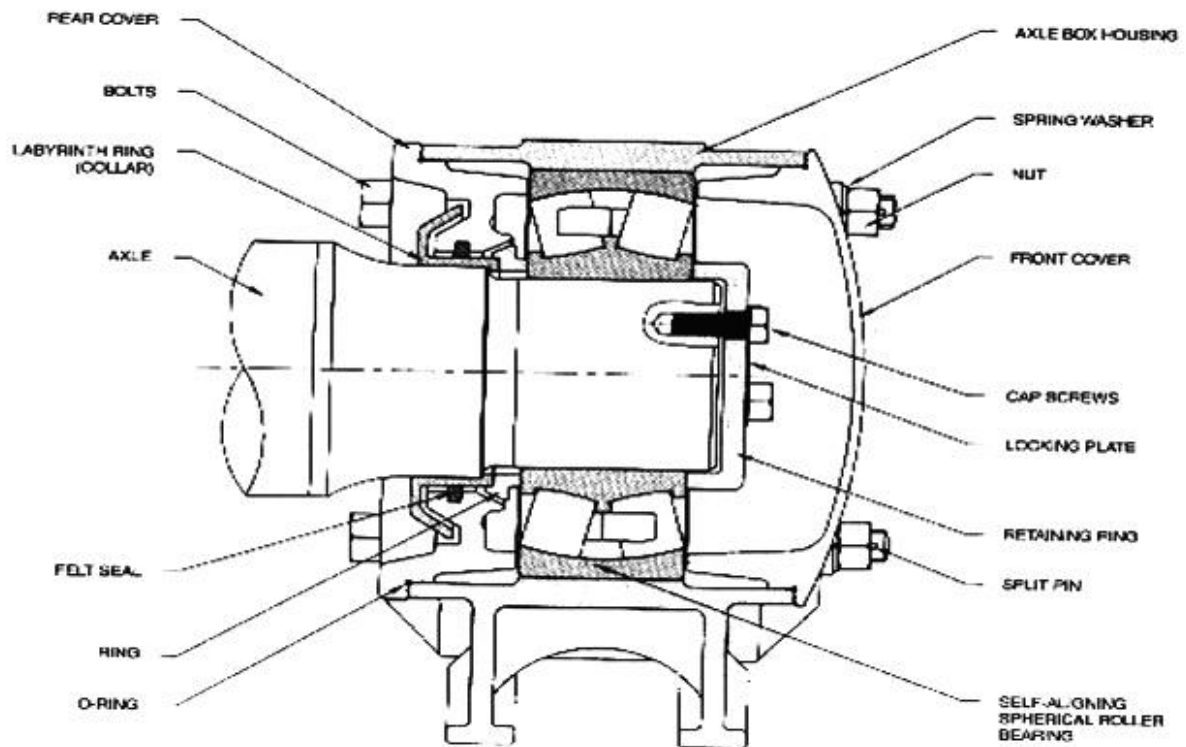
Thermal cracks

Wheels involved in severe brake binding should be examined carefully during the maintenance to rule out the possibility of rejectable thermal cracks. Such wheels may be identified by presence of flats (even within acceptable limits) and severe discoloration or blue/black heating marks on the tread.

5. **Heat checks** - Thermal cracks are deeper and need to be distinguished from fine superficial cracks visible on the tread on or adjacent to the braking surface. These are called heat checks, which are usually denser than the thermal cracks. Heat checks are caused on the tread due to heating and cooling cycles undergone by the wheel during normal braking. Such wheels do not need to be withdrawn but should be carefully distinguished from the rejectable thermal cracks



DOUBLE ROW SELF ALIGNED SPHERICAL ROLLER BEARING



DIRECT MOUNED ROLLER BEARING ARRANGEMENT

<i>Bearing make</i>	<i>Radial clearance in mm</i>
SKF	0.105 to 0.296 mm
FAG/NORMA	0.080 to 0.185 mm
NEI/NBC	0.080 to 0.190 mm

INSPECTION OF OTHER ROLLER BEARING COMPONENTS

The following components other than roller bearing should be inspected during roller bearing maintenance in the workshop.

▪ Axle end holes ▪ End locking plates ▪ End locking bolts ▪ Retaining Ring	▪ Collar ▪ Felt ring ▪ Rear and Front Cover ▪ Axle box housing
---	---

DEFECTS

- | | |
|---|--|
| <ol style="list-style-type: none"> 1) rollers- <ol style="list-style-type: none"> (a) Flaked (b) Pitted (c) Burn (d) Dented (e) Excessive worn (f) Smearing 2) Cage broken 3) Corrosion | <ol style="list-style-type: none"> 4) Locking stud loose or broken 5) Roller seized 6) Over greasing 7) Less grease 8) Contaminated grease 9) Felt ring broken 10) Retainer ring broken 11) Excessive lateral play |
|---|--|

DRAW & BUFFING GEAR

Draw Gear: It is a vital component of rolling stock, which is utilized to connect the one rolling stock to another to form a train & also to transmit draft forces from engine to last vehicle. It is located at both the ends in the centre of the body under frame head stock. Mainly two types of draft gear are being utilized in Indian Railways.

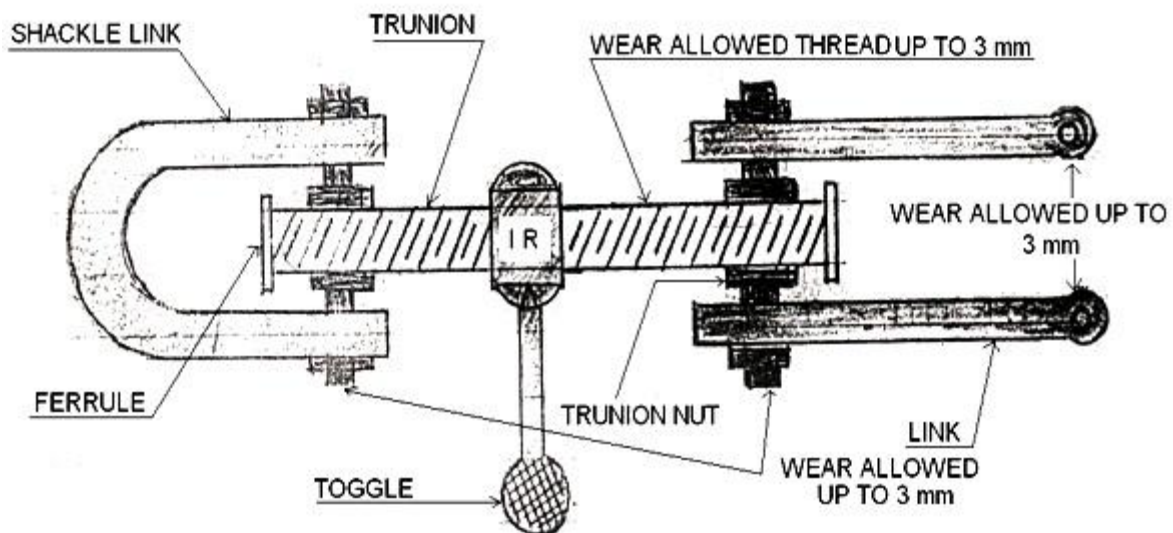
- i. Conventional Draft Gear
- ii. Centre Buffer Coupler

Main components of conventional draw gear are as under:-

- | | |
|------------------------------|-----------------------|
| I. Draft Hook. | VI. Washer |
| II. Draft Links | VII. Bent Pin(U-Pin) |
| III. Draft Key | VIII. Hexagonal Nut. |
| IV. Draft Spring/ Draft Pad. | IX. Screw Coupling |
| V. Cotter | |

Parts of Screw Coupling.

- | | |
|--------------------|---------------------|
| I. Shackle. | V. Screw Rod. |
| II. Link. | VI. Washer. |
| III. Trunnion Nut. | VII. Gravity Lever. |
| IV. Ferrule. | |



In 1984 use of Enhanced Screw Coupling was started, which is again modified in 1998. To identify this coupling a Dumbal mark is stenciled at both the side of coach end body.

Length of coupling when fully opened – 997 mm

Length of coupling when fully Tight – 751 mm

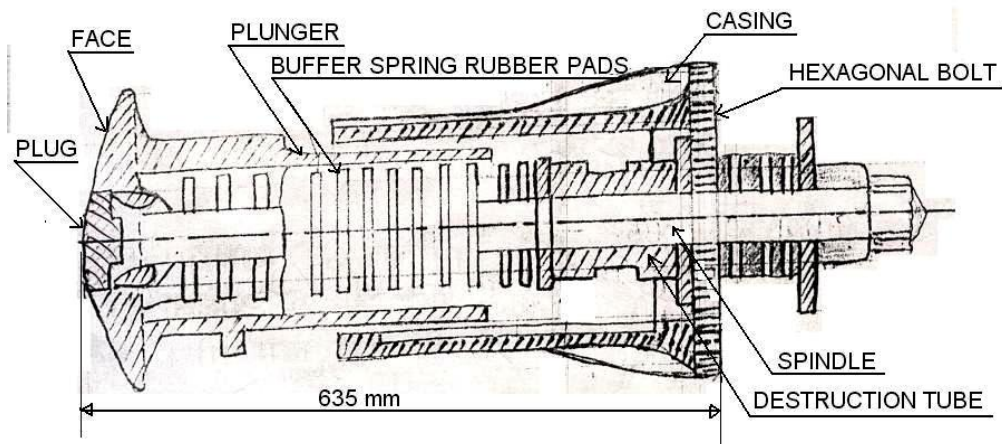
Modification:

Sl. No.	Description	Non Modified	Modified	Remark
1	Working Capacity	36 Tonnes	36 Tonnes	
2	Proof Load Capacity	60 Tonnes	70 Tonnes	
3	Breakage Capacity	Draw Bar – 108 T S/ Coupling – 112 T	130 T for both	
4	Stamping Mark	C – 60.61	IS – 5517	

Note: Proof Load Capacity of Enhanced Screw Coupling is increased from 70 T to 75 T. This must be used in all coaches including 24 coach trains.

BUFFING GEAR: - Two numbers of buffers are provided on body head stock to absorb the longitudinal impacts occurred during run on both ends, at a distance of 1956 mm. The role of buffers is also to transmit push impact to its trailing end stock. The main components of Buffing Gear are as under:-

- | | |
|-------------------------------------|-------------------------|
| 1. Buffer Plunger | 6. Recoil rubber Washer |
| 2. Buffer Socket with securing bolt | 7. Washer |
| 3. Buffer Spindle & Plug | 8. Nut |
| 4. Buffing Pad | 9. Cotter |
| 5. Destruction Tube | |



BUFFER ASSEMBLY IN ICF COACHES

Mainly Buffers are of two types:-

Long Case Buffer – Length from head stock – 635 mm

Short Case Buffer – Length from head stock – 458 mm (4 WH)

Other data:-

Max. Height in Empty condition	– 1105 mm
Min. Height in Loaded condition	– 1030 mm
Allowed variation in height at same end	– 64 mm
Allowed variation with adjacent vehicle	– 75 mm
Max. Plunger Travel	– 127 mm
Min. Plunger Travel	– 51 mm
Number of Buffing Pads in each Buffer	– 14 to 16 Nos.
Capacity of Buffing Pads	– 1030 kg m (New Type)

CLASSIFICATION OF COACHING MAINTENANCE DEPOTS.

According to number of based coaches (holding Capacity), depot is classified into three categories.

Sl. No	Depot	Number of based coaches
1	Minor	50 to 100
2	Medium	100 to 250
3	Major	Above 250

Standard Facilities:-

1. Covered accommodation
2. Flooring & pit for repair & examination
3. Office & store facilities
4. Sick line yard
5. Machinery & plant

Covered Accommodation:

1. Track length of under covered accommodation for any type of sick line must be at least 4% of the holding capacity(based coaches)
2. Total track length should not be less than 4 coaches in view of repair purpose whenever working space required for each coach is 35m
OR
Track length should not be less than 140m for any type of sick line.
3. It is essential to provide 50% track length under a covered area with pit examination facilities.
4. In pit there should be proper light arrangement.
5. It should also be ensured that it is provided with drainage facilities with 1% inclination & required number of man holes.
6. At least hoist crane in capacity 3 to 5 tones also should be provided across the track.
7. The width of the covered accommodation should be normally 15 meters and should normally cover two tracks under it. The gap between two adjacent tracks, where overhead crane is provided should not be less than 7.5 meters. If overhead gantry is not provided the can be maintained to 6 meters.
8. It should be ensured that proper space is provided beside the track for free and easy movement of transport vehicles like fork lift, lister, trolleys and truck etc.
9. Entire covered accommodation must have adequate lighting arrangement for workers without eye strain.

Machinery And Plant: To avoid heavy manual labour, wastage of manpower and to provide efficient working of depot, suitable adequate machinery and plant is required which are as under.

Sl. No. Machine & Plant

1. Self propelling whiting jacks
2. Coach shunter-portable wrench type.
3. Welding plant 200 amp capacity with double load
4. Gas cutting & welding equipment
5. Vacuum exhauster
6. Air compressor(150cfm)
7. 2 tones hoist with tram beam.
8. Sawing machine.

9. Diesel jeep with 2 trolleys.
10. Wood cutting saw machine.
11. Fork lift truck.
12. Hand shearing machine.
13. Portable furnace
14. Centre lathe 230mm (9") chuck.
15. Wheel lathe.
16. Manipulator/fixture for bogie.
17. Ultrasonic testing apparatus.
18. Tool post grinder.

TOOLS:

1. Pneumatic hand tools
(a) Grinder (b) Drill (C) Chipper/buster (d) Riveter
2. Electric power tools.
(a) Pop riveting tool gun (b) Drill (c) Bolt tighter/torque wrench.
3. Hand tools including torque wrenches as required.

Sl.No. Test benches and miscellaneous items.

1. D.V. testing bench.
2. Air Brake cylinder overhauling testing bench.

Primary Depot and Secondary Depot

Sl.No	Primary Depot	Secondary Depot
1.	Maintenance works attended by based depot is called primary depot Maintenance.	Maintenance works attended by terminating depot other than based depot is called secondary depot Maintenance.
2.	Preparation of DRS card is done by primary depot.	Only cross checking of items as per DRS card or only shortage, missing should be provided by secondary depot.
3.	Primary maintenance depot is responsible to prepare history card of coach.	Intimation to primary depot is essential whenever any major repair/maintenance is attended.
4.	It is duty of primary depot to ensure proper supply of brake van equipment for all originating trains.	Secondary maintenance depot is responsible to ensure only if there is any shortfall.
5.	Primary maintenance depot is responsible for all types of schedules of coaches.	Secondary maintenance does not have responsibility other than trip schedule.
6.	It is duty of primary maintenance depot to send the coaches for POH or NPOH if due or required.	It is not duty of secondary depot but it assist in sending the coaches for POH or NPOH through primary depot.

MAINTENANCE SCHEDULES TO BE FOLLOWED IN COACHING DEPOTS

To maintain coaching stock in good condition, the following preventive maintenance schedules are prescribed to be carried out in carriage depots on divisions where rake has been based for primary maintenance.

1. **Trip schedule**- After every trip by primary maintenance depot.
2. **Schedule A** - Monthly (**1 month ± 7 days**)
3. **Schedule B** - Quarterly (**3 months ± 15 days**)
4. **IOH** - **9 months $\pm \begin{smallmatrix} 30 \\ 0 \end{smallmatrix}$ days**
5. **POH** - **18 months $\pm \begin{smallmatrix} 30 \\ 0 \end{smallmatrix}$ days**

Primary maintenance schedules are required to be carried out by the base depots to which coaches are allotted. In emergency, when due to any reason coaches cannot reach their base depots and primary maintenance schedules become due, A & B schedules should be undertaken by the carriage depots where the coaches are available. All schedules should be carried out by primary maintenance depot.

1. TRIP SCHEDULE :-

Trip schedule is attended per trip of the rake. The trip is being attended by Primary depots. No need to detach the coach from the rake during trip schedule. Total distance traveled by passenger rake in a trip up and down is more than 3500 km. Following procedure is adopted during the trip schedule:-

1. All under gear parts are thoroughly examined.
2. All moving parts are lubricated.
3. Complete examination of draw & buffing gear for its proper functioning. Lubrication is essential.
4. Coupling should be free in its screw i.e. ensure easy movement of coupling.
5. Proper examination of primary suspension arrangement.
6. Ensure the leakage of dash pot and oil level of dash pot.
7. Proper securing of safety strap and safety loop.
8. Proper examination of secondary suspension. Ensure the working of spring, shock absorber, safety strap & safety loop.
9. Proper examination of wear in suspension link bracket, pin & shackle stone.
10. Examine the proper function of shock absorber & securing bolt.
11. Examination of equalizing stay rod for its proper securing.
12. Examination of proper securing of bolts & cotters & silent bushes of centre pivot.
13. Ensure the proper function of side bearer or its oil level.
14. Changing of worn brake blocks & pin & adjustment of brake power.
15. Proper cleaning of coach from inside & outside & disinfections.
16. Spraying of pesticides elements.
17. Checking of all joints & pipe joints & other fittings & filling of water tank.
18. Proper opening & closing of vestibule doors.
19. Checking of amenity & safety items.
20. All falls plate examination of vestibule.
21. Testing of alarm signal, guard van valve & its gauge.
22. Preparation of DRS card & brake power certificate.

New Policy (Recommendations) for enhancements of POH/IOH schedules of Coaching Stock.

1. The revised POH periodicity from 12 to 18 months is applicable to all Mail/Express coaches for which Railway shall arrange transportation of bogies from and to work shop.
2. A marking on the coach below return date shall be specified to distinguish 18 months periodicity.
3. The general sequence of coach will remain as per existing coaching maintenance manual.
4. The items of trip schedules; 'A' and 'B' schedules will remain same.
The coach will be given 2 quarterly 'B' schedules before IOH. The work specified for IOH schedule to mechanical & electrical work in appendix C & D respectively as specified by CAMTECH Pamphlet No CAMTECH 2008 coach POH/1.0 in Jan-2008.
5. CMIs, SMIs and technical circulars/pamphlets issued time to time by RDSO schedules followed for necessary modification and replacements.
6. As per requirement of bogie as unit exchange, the bogies should be collected from workshop considering transportation time plus two days before spare.
7. **The periodicity of overhauling of DV is changed from 24 months to 18 months (during POH)**
8. Work shop to switch over PU painting at workshop as advised by RDSO.

2. **SCHEDULE 'A'** : Schedule 'A' is required to be given **every month $\pm$ 7 days** at the nominated primary maintenance depot within the normal primary maintenance time on a washing/pit line. A coach need not to be detached from the rake for Schedule 'A' examination unless it requires such repairs which cannot be attended to on the washing line or within the prescribed maintenance time on the washing line.

'A' schedule maintenance:-

- (i). All items of primary/secondary maintenance schedule.
- (ii). Test the working of brake cylinders for proper application and release.
- (iii). Thorough inspection of brake pipe, feed pipe and their connecting pipes to brake cylinder, distributor valve, Auxiliary reservoir and hose coupling for leakage and attention.
- (iv). Carry out manual brake release test on every coach to ensure proper functioning of release lever of distributor valve.
- (v). Micro switch of ACP (Alarm Chain Pulling) should be tested by electrical staff for proper functioning.
- (vi). Clean Dirt collector filter with kerosene and refit.
- (vii). Test the working of slack adjuster in under frame mounted air brake system.
Repair/Replace the defective slack adjuster.
- (viii). Examine loops/ brackets and their securing devices and rectify.
- (ix). Examine for wear and replace if required brake hanger pins, brake blocks and brake heads.

The following items of work should be attended during Schedule 'A' examination, i.e., monthly examination:-

- (i). All items of primary/secondary maintenance schedule.
- (ii). Intensive cleaning of coaches.
- (iii). Intensive cleaning of lavatory pans and commode with specified cleaning agent.
- (iv). Thorough flushing of tanks.

- (v). Checking of water pipes, flush pipe, flushing cocks, push cocks, etc., for ease of operation and free flow of water.
- (vi). Thorough disinfection of all compartments.
- (vii). Thorough inspection and repairs of draw gear.
- (viii). Thorough inspection and repairs of buffers.
- (ix). Oil in hydraulic dash pots should be checked to detect oil leakage from them through defective seals or through vent screws. Add/replenish with specified grade of oil if oil level is below 40 mm in tare condition to ensure better riding comfort. Similarly oil in side bearer baths should be checked when the oil is below the plug and replenished with specified grade of oil so that wear plate is fully covered by oil.
- (x). Inspection and repairs of commode chute.
- (xi). Thorough check and repairs of sliding doors and vestibule doors for easy and smooth operation and correct alignment and all wearing parts, loose screws, etc.

3. SCHEDULE 'B' : Schedule 'B' is required to be given every **three months $\pm$ 15 days** at the nominated primary maintenance depot within the normal time allowed for primary maintenance on a washing line in rake. Coach need not be detached from the rake for purpose of this examination unless it requires such repairs which cannot be attended to on the washing line or within the prescribed maintenance time on the washing line.

The following items of work should be attended.

Air brake system

- (i). Same as 'A' schedule

Other assembly maintenance

- (i). Besides brake system other items should be attended as given below:
- (ii). All items of Schedule 'A'
- (iii). Painting of lavatories from inside.
- (iv). Thorough inspection and repairs of brake gear components.
- (v). Thorough checking of trough floor, turn under, etc., from underneath for corrosion.
- (vi). Touching up of painted portion, if faded or soiled.
- (vii). Overhauling & testing of alarm chain apparatus.
- (viii). Testing of guard van valve.
- (ix). Greasing of equalizing stay rod.

4. IOH:

- (i). IOH is required to be given **every nine months $\pm$ 30 days** at the nominated primary depot.
- (ii). Coaches are required to be detached from the rake and taken to the sick line for examination and repairs.
- (iii). For maintenance of major break-down/ mal-functioning of any subassembly etc. the decision whether the coach is to be detached from the formation for attending to maintenance/replacement of major subassembly is dependent on maintenance requirements, operational convenience, time availability etc. The decision is taken by the Engineer (C&W). Coach failure report should be made.
- (iv). At depot, the coach that is detached for IOH is taken over to the washing line for cleaning, lubrication and minor maintenance. The coach that are detached due to a major defect in the distributor valve, brake cylinder, Auxiliary reservoir etc, is taken to the pit line for the replacement of such sub-assemblies, on unit exchange basis. The detachment of coach is carried out so as to make the maintenance or testing

activities convenient and faster so that the coach is made ready for use without delay.

PROCEDURE: The activities performed to detach a coach with Air Brake system are as under-

- i) Safety precautions shall be taken to prevent injury while detaching/attaching a coach.
- ii) Remove the clamps on the cut-off angle cocks. Close the cut-off angle cock of both feed pipe and brake pipe on both sides of the coach that has to be detached.
- iii) Close the cut-off angle cocks of the feed and brake pipe of adjacent coaches. This is to ensure that the air pressure locked up in the air hose coupling gets vented to atmosphere through the vent hole of the cut-off angle cock.
- iv) Observe above mentioned safety measures to close all the four cut-off angle cocks on either side of the coach to be detached so that while opening air hose coupling, it may not cause injury due to air pressure inside.
- v) Release the brake of the coach to be detached by pulling the manual release lever of the distributor valve.
- vi) Open the Feed Pipe and Brake Pipe hose coupling from both sides of the coach.
- vii) If the air pressure of brake cylinder does not vent by pulling the manual release valve of distributor valve, open the brake cylinder vent plug to drain the air pressure.
- viii) Uncouple Screw coupling and detach the coach.
- ix) Observe all other safety measures as prescribed.

The following items of work should be attended during IOH.

Air brake system maintenance:

- (i). Check brake cylinder for loose rocker arm plate and change on Bogie Mounted system.
- (ii). Brake cylinder should be checked for smooth functioning and prescribed stroke. Defective brake cylinders shall be sent for repairs.
- (iii). Guard's van valve should be tested.
- (iv). Test BP & FP air pressure measuring gauges with master gauge and replace if found defective. A set of two master gauges should be kept for this purpose at every Primary Maintenance Depot and each master gauge should be sent one after the other to the base workshops for testing, repairs and calibration.
- (v). Thoroughly clean Dirt collector filter in kerosene or replace on condition basis.
- (vi). Check working of PEASD & PEAV by hearing the hissing sound of exhaust air. After resetting with the help of key the exhaust of air should stop. Replace the defective PEASD/PEAV.

Other assemblies maintenance:

- (i). All items of Schedule 'B'
- (ii). Thorough repairs of running gear including running out of bogies where considered necessary. Bogies which are working on rake links earning more than **nine months** must be run out and unit exchanged with overhauled bogie received from workshop.
- (iii). Touching up damaged paint of coaches on outside as well as inside.
- (iv). Thorough cleaning and removal of dust, rust, dirt, etc., accumulated at the pillars through the turn under holes, with coir brush and compressed air.
- (v). Thorough examination and repairs of upholstery, cushions, curtains, etc.

- (vi). *Thorough checking and full repairs of all window shutters, safety catches, safety latches, staples and hasps of compartment, lavatory, body side and vestibule doors for ease of operation.*
- (vii). *Thorough checking and repairs of UIC vestibules, their rubber flanges metal frames, doors, fall plate, locking gear, etc., for ease of operation and safety.*
- (viii). *Thorough checking and repairs of all cracks and worn out portions of flooring of the compartments.*

Engineer (C&W) of Primary Coaching Maintenance Depots should be fully familiar with the vulnerable areas of ICF coaches for corrosion, viz., sole bar at doorways, lavatories and adjoining areas, corridor sides - more so in case of those SLRs which are used for Fish, Salt, etc. For facilitating inspection of sole bars even spaced elongated holes of (215x127 mm) are already provided in the turn under.

Special attention should be taken for the following:-

- (i). *Pocket between sole bars and turn under should be thoroughly cleaned through the inspection opening of the sole bars and inspected with the help of torch light or inspection lamps.*
- (ii). *Drain holes provided in the trough floors should be kept clean and unclogged. If during the cleaning of these drain holes any accumulation of water is observed, the affected area should be very carefully inspected for possible corrosion.*
- (iii). *A register should be maintained of the primary maintenance coaches on the subject.*
- (iv). *During this lifting schedule, bogies/under frame members and body including trough floors of integral type coaches should be thoroughly examined and all parts of running gears are repaired/ replaced as necessary. The bogie frames should be particularly checked to detect damage, cracks or deformation and necessary repairs carried out. Where it is not possible for the maintenance depot to do these repairs or are prohibited to be done in the maintenance depots, the bogies should be sent to the shops for carrying out these repairs.*
- (v). *The detailed table of maintenance activities to be carried out during IOH schedule is enclosed as appendix-G.*
- (vi). *The date of intermediate lifting should then be stenciled at the appropriate place in schedule chart on the end panel*

Note:

- 1) Intermediate Overhauling of Shatabdi/Rajdhanni Exp. Coaches are attended in nominated workshop only.***
- 2) Intermediate overhauling of newly built coaches are to be attended after 12 months only wheels are to be replaced***

RPC-4 Revision-Jan 2007

No. 95/M/C/141/1

Sub:- Revised maintenance pattern of coaching train- running up to 3500 km in round trip with terminal attention at the other ends.

<i>Sl. N.</i>	<i>Category of train</i>	<i>Preventive maintenance at pit line</i>	<i>Under gear/ Brake System maintenance at pit</i>	<i>Internal cleaning, passenger amenity fittings and watering</i>	<i>External cleaning nominated line with proper facilities</i>	<i>En route/ terminating</i>	<i>Brake system check prior to start at plate form/at other end</i>
1	Mail/Exp. One way run > 3500 km	At primary end	At both the ends	At both the ends	At both the ends	En route after every 250-350 km location to be decided by Rly for each train. Terminating examination at terminating station.	Complete air brake testing with fresh BPC
2	Mail Exp one way run <3500 km but round trip run >3500 km	---do---	---do---	---do---	---do---	---do---	---do---
3(a)	Mail Exp round trip run up to 3500 km	---do---	Only at primary end	---do---	At primary end	---do---	Only continuity check if stabled at platform for 2 hrs, otherwise brake power check with endorsement on BPC engine

3(b)	Inter connected Mail Exp round trip run up to 3500 km	---do---	To be done after 3500 km or 96 hrs after issue of original BPC whichever is earlier at primary end	At primary end & each terminal	---do---	---do---	---do---
4	Passenger trains with toilet including interconnected passenger shuttles	---do---	---do---	At primary end	---do---	---do---	---do---
5	Passenger trains without toilet	---do---	To be done after 3500 km or 7 days, after issue of original BPC whichever is earlier only at primary end	---do---	---do---	Once a day at primary end or at nominated terminal	---do---

Note:- Internal cleaning, passenger amenity attention and watering may be done at platform line or nominated stabling line provide stipulated facilities are available at such line, in case the rake is stabled in yard for more than 6 hrs, positive safety arrangements should be made for the rake and in case the security is considered inadequate, the rake should be taken to pit line for attention to under gear as given under column(4) above.

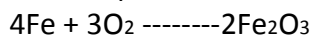
Difference Between IOH, POH & NPOH

Sl.No	IOH	POH	NPOH
1.	<i>It stands for intermediate overhauling.</i>	<i>It stands for periodic overhauling.</i>	<i>It stands for non periodic overhauling.</i>
2.	<i>IOH of coach is attended after 9 months of POH.</i>	<i>POH of Passenger carrying vehicle M/Exp -18 months,</i>	<i>Time Period is not fixed for any vehicle for NPOH</i>
3.	<i>IOH of coach is allowed in sick line where facility to lift the coach is available.</i>	<i>POH is allowed in nominated workshops only.</i>	<i>NPOH is done only in workshop or in nominated major sick line.</i>
4.	<i>During IOH all the parts of under gear thoroughly examined and replaced if necessary.</i>	<i>At the time of POH all parts of under gear are dismantled and should be replaced if there is any wear and tear.</i>	<i>Only indicated defects & safety items are attended.</i>
5.	<i>At the time of IOH painting of coach is not necessary, only required places are touched up.</i>	<i>Painting of whole coach is necessary.</i>	<i>Complete painting of coach is not necessary, only required places are touched up.</i>
6.	<i>Profiling of wheel is necessary.</i>	<i>Profiling of wheel is necessary.</i>	<i>Profiling of wheel is on need based.</i>

CORROSION AND ITS PREVENTION

Introduction: When metals are put into use in various forms they are exposed to environment containing liquids, gases etc. As a result of this the surface of metal starts deteriorating. This type of deterioration or destruction may be direct chemical attack or electrochemical attack.

Definition: corrosion is a chemical process of oxidation with metal to its surroundings, covering it into metal oxide, carbonates, hydroxide and sulfides. Oxidation takes place only when steel surface is exposed to atmosphere or moisture. Chemical reaction is as follows:



Example: rusting of iron. When iron is exposed to moist atmospheric conditions rusting of iron takes place. During this exposure a layer of reddish scale and powder of oxide is formed and iron becomes weak.

Effect of corrosion: Corrosion of materials is liable to performance of the product; lose their strength, ductility and certain other mechanical and physical properties.

With the introduction of all steel coaches corrosion has become a major problem. Once starts it is very difficult to control it. This requires replacement of the component. This is much costlier than to save the existing part by proper and timely attention.

Corrosion in ICF coaches: Corrosion in ICF coaches is very common. Corrosion repairs to coaches are mainly carried out during POH in workshops. Corrosion repairs are also done during midlife rehabilitation of coaches that are 12 to 13 years old especially at CRWS, Bhopal.

During POH all the under frame members are thoroughly inspected to locate corroded members. Corrosion is indicated by flaking of paint, flaking of metal, pitting and scale formation. Components those are not visible from both sides such as sole bar and trough floor should be examined by tapping with a spiked hammer.

Particular attention should be paid to the more vulnerable members and locations given below.

- 1) Sole bars, body pillars, turn under and trough floor below lavatories in all types of coaches and luggage compartments of SLRS.
- 2) Sole bars, body pillars, turn under and pillars above lifting pads.
- 3) Sole bars, body pillars behind the sliding doors of SLRS
- 4) Sole bars, body pillars, turn under at the door corners & near coach body bolster.
- 5) Headstock

Inspection of under frame member for corrosion attention should be done as per technical pamphlet no 7602(rev: 1)

Reason of corrosion in ICF Coach:

- 1) Accumulation of water, dust and salty discharge under luggage compartment in coaches.
- 2) Incorrect fitness of side panels.
- 3) Galvanic cell formation between steel and aluminium near window area.
- 4) Seepage of water at corners and ends due to water accumulation on floor.
- 5) In sufficient surface preparation before welding.
- 6) Frequent use of concentrated acids for the cleaning of toilets.
- 7) Leaky push cocks, flusher valves.

- 8) *Missing/defective commode chutes resulting in splashing of toilet discharge leads to corrosion of under frame members.*
- 9) *Carrying of perishables items like fish in SLRS and Parcel vans and insufficient cleaning after unloading.*
- 10) *Entry of water through gaps in window sills.*
- 11) *Cracks in body panels and roof left unattended.*
- 12) *Painting defects left unattended.*
- 13) *Damage to under frame and trough floor due to flying ballast in dynamic condition.*
- 14) *Acid spillage from batteries.*

Need of Corrosion prevention:

- 1) *To avoid premature detachment of coaches and wagons in service.*
- 2) *Corrosion makes wagons unfit for loading. This has a bad effect on earning capacity of railways.*
- 3) *There will be a shortage of wagons fit for loading if wagons are detached for corrosion repairs.*
- 4) *Detachment of coaches for corrosion repairs has an adverse effect on the normal composition of trains.*
- 5) *In monsoon season seepage of water through corroded panels spoils the consignment and railway is forced to pay compensation for the damage.*
- 6) *Manpower material and time involved in corrosion repairs can be controlled through proper anticorrosion measures.*
- 7) *Losses of railway revenue i.e. losses to the nation.*
Always remember 'A STICH IN TIME SAVES NINE' while dealing with corrosion.

Inspection during POH

- 1) **Inspection of sole bars, body pillars and turn under:** *Examine visually and with the help of a spiked hammer from below the coach and the inspection holes in the turn under. If corrosion is suspected at places without inspection holes 100mm dia hole should be cut at the bottom of turn under for examination. If corrosion is noticed in the bottom half of the sole bar the trough floor to be cut to a width of 300mm for inspection. In case of heavy corrosion the side wall to be cut to a width of 500mm.*
- 2) **Inspection of headstock:** *Examine visually inner and outer headstock, stiffening behind buffers and the junction of sole bar and the headstock for corrosion. Examine the base buffer assembly carefully.*
- 3) **Trough floor:** *Examine trough floor adjoining the lavatories and under the luggage compartment of SLRS and Parcel vans for corrosion with the hammer.*

Corrosion Repairs During POH

- 1) **Repairs to under frame members:** *Repairs to under frame members should be carried out as per RDSO pamphlet no C7602 for ICF coaches. Corrosion resistant steel sheet for trough floor, pillars, sidewalls and roof should conform to IRS M-41-97. Electrode IRS class B2 of approved brands. Paint red oxide zinc chromate primer is-2074-62. Bituminous anti corrosive solution to IRS-P30-96.*
- 2) **Repairs to Headstock:** *Only 8mm thick sheet is to be used headstock repairs.*
- 3) **Repairs to Sole bar:** *The new sole bar section to be welded from both inside and outside.*

- 4) Repairs to Side Wall Members: For repairs to side and end wall members interior fittings interior panels & window frames are to be stripped. Repairs to be done as per RDSO sketch No. 76019.
- 5) Repairs to Trough Floor: For trough floor repairs plywood flooring to be stripped. Repairs to be done as per RDSO instructions.
- 6) Repairs to Roof: Special attention to be paid at locations where gutter moldings are welded and where ventilators are fitted. RDSO instructions to be followed

HOW TO MINIMIZE CORROSION

Corrosion in rolling stock can not be eliminated altogether. Hot and humid conditions in our country are helpful for corrosion. A change in climate also has an adverse effect. However timely action during repairs and maintenance will minimize corrosion.

A) DURING POH

- 1) Thorough inspection giving extra attention to areas prone to corrosion.
- 2) Turn under repairs to be carried out with 5mm thick plates.
- 3) Only 8mm thick SS sheets to be used for head stock repairs.
- 4) Use stainless steel trough floor and inlays for toilets.
- 5) Use of 13mm compreg floor board instead of plywood.
- 6) Use PVC sheets for toilets and compartment floor.
- 7) Use stainless steel plates with drain holes in doorways.
- 8) Provision of tubular structure below lavatory area.
- 9) Corten steel is used for panel repairs.
- 10) Apply two coats of primer and three coats bituminous solution on all under gear members.

B) IN OPEN LINE

- 1) During pit line examination check thoroughly all under gear and under frame components, trough floor and headstock etc. for corrosion. If corrosion is noticed take proper anticorrosive measures.
- 2) Drain holes and drain pipes should be clear so that water stagnation is eliminated.
- 3) All water leakage to be arrested at the earliest.
- 4) Proper repairs to damaged PVC floor.
- 5) Gaps in window sills to be filled up.
- 6) Deficient/defective commode chutes to be made good.
- 7) Hosing of coach interior is to be avoided.
- 8) Avoid strong acids for toilet cleaning.
- 9) Body patches to be painted, carry out paint touchup where paint is peeled off.

During IOH all vulnerable areas are to be properly inspected after Cleaning of turn under holes.

How to apply anti corrosive paint in coaching stock.

- I. 1st coat ----- zinc chromate
- II. 2nd coat ----- zinc chromate, red oxide
- III. 3rd coat ----- bituminous thin black solution
- IV. 4th coat ----- Bituminous red brown solution
- V. 5th coat ----- Bituminous primer thick black
- VI. 6th coat ----- bituminous primer silver gray

Exterior paint schedule for coaches

At every 5th POH of a coach or if the condition of paint is not good adopt 9 days painting schedule. (A—schedule) otherwise choose 5days paint schedule.

A—schedule (9—days)

- 1st day – Remove old paint
- 2nd day – One coat of red oxide zinc chromate primer
- 3rd day – One coat of brush filler followed by spot putty
- 4th day – Filler 2nd coat (spot putty if necessary)
- 5th day – Rub down with silicon carbide paper
- 6th day – One coat of under coat
- 7th day – Flat with silicon carbide paper. One coat of enamel finishing.
- 8th day – Flat with silicon carbide paper. 2nd coat of enamel finish
- 9th day – Lettering and miscellaneous work

C--- Schedule(5—days)

- 1st day – Washing with soap solution touchup damages with primer
- 2nd day – Spot putty if necessary and one coat of under coat
- 3rd day – Flat with silicon carbide paper apply one coat of finishing Enamel
- 4th day – Flat with silicon carbide paper apply second coat of Finishing enamel.
- 5th day – Lettering and miscellaneous work

Suggestions to Prevent Corrosion: These small steps will go a long way in minimizing corrosion in rolling stock.

- 1) Supervisors involved in maintenance of rolling stock should be familiar with areas prone to corrosion.
- 2) Supervisors should educate their technicians about areas prone to corrosion.
- 3) Identify corrosion prone areas and inspect them thoroughly during pit line examination, sick line attention, ROH/IOH.
- 4) Suitable preventive measures to be adopted to save the affected component. In case of heavy corrosion replace the component.
- 5) Ensure painting of wagons during ROH. painting/ paint touchup during IOH and sick line attention.
- 6) Supervisors should educate their cleaning staff so that they follow proper cleaning technique.
- 7) Ensure water tightness of covered wagons.
- 8) Educate Shunting staff so that they perform smooth shunting without damaging the rolling stock.
- 9) Ensure proper cleaning of wagons by the contract staff after Unloading.
- 10) Electrical staff to be counseled about the corrosive effect of acids from batteries.

FACILITIES REQUIRED FOR MAINTENANCE OF 24 COACH TRAINS

(Railway Bd.'s letter no. 98/M(C)/137/19 Pt. I dt. 28.7.99 & dt. 05.05.2000)

a) *Infra structural Requirements*

- (i)** 24 coach length fully equipped pit line.
- (ii)** High pressure jet cleaning pipeline with plant for cleaning at primary pit line. Mechanized external cleaning is preferable.
- (iii)** Water hydrants for 24 coach length at en route watering stations with 20 minutes stoppage at nominated stations
- (iv)** Availability of the prescribed air brake maintenance and testing equipment.

b) *Coach Design related Requirements*

- (i)** Air brake with twin pipe graduated release system
- (ii)** Only enhanced capacity draw gear and screw coupling to RDSO sketch No. 79061 and 79067 are to be provided on the rake

c) *Maintenance Practices and system related requirements*

- (i).** The integrity of the rakes to be maintained.
- (ii).** Primary maintenance of the rake should be done in one hook without splitting
- (iii).** Minimum maintenance time of 6 hours on the pit during primary maintenance
- (iv).** Trains leakage rate to be maintained within prescribed limits by using rake test rig.
- (v).** Provision of proportionate brake system on the locomotive in good working order
- (vi).** Provision of audio visual alarm system on the locomotive
- (vii).** In case of double-headed diesel locos maximum traction motor current will be restricted to **650 Amperes** and in case of double headed WAP1/WAP3 electric locos, the traction motor current limit will be **750 Amperes** as prescribed in RDSO 's instructions for operation of main line air brake trains - C-9408.

d) *Operational requirements*

- (i).** Communication between driver and guard should be provided through suitable means.
- (ii).** Special care to ensure no gap between coach buffers after tightening the coupler.
- (iii).** No additional coach attachment beyond **24 coaches** will be permissible.

Note : As per Railway Board Instruction now one occupied saloon & one parcel van can be attached with 24 coaches rake.

NMG COACHES

The coaches with age group of 20 – 21 years or later being converted to air brake coaches during POH for the purpose to carry Automobile (Express) Auto Carrier are known as NMG coaches and the rakes formed with these coaches are called NMG rakes.

Following are the features:-

- 1. Load carrying capacity is 12 ton minimum.*
- 2. Speed 75 kmph Min. & 100 kmph Max.*
- 3. The life of such converted coach is 30 years from the date of original manufacture.*
- 4. Wider opening of 2800 mm × 2200 mm.*
- 5. Improved adjustable central locking arrangement to avoid damage to the vehicle during transit.*
- 6. The periodicity of POH is 24 months.*

Maintenance Pattern for NMG rakes

(As per Rly. Bd's Letter No. 91/M (C)/650/1 dated 29.5.2000)

In order to optimize utilization of NMG rakes it has been decided to introduce the following maintenance pattern:

- i) NMG rakes may be run on goods pattern with intensive examination at both the ends, following other conditions for reroute detention in case at stabling at road side stations. In case of close circuit runs up to 2500 km, the rake may be run on round trip basis.*
- ii) Close circuit rake must be clearly identified and should have a nominated base depot where adequate trained staff and spares should be available. Each NMG coach should be marked with the nominated POH workshop and return date.*
- iii) The maintenance schedules of the NMG coaches will continue to be on the coaching pattern to be carried out by the base depot.*
- iv) Using these coaches as parcel vans for running on piecemeal basis on passenger carrying trains is strictly prohibited.*
- v) Each coach should be stenciled at a suitable place on its end panel, the code name of the base depot and a schedule chart. The date and station code of the depot where a particular schedule is carried out should be stenciled at the appropriate place in the schedule chart immediately when the schedule is completed.*

EXAMINATION OF TRAINS

Ref: IRCA Rule Book Part – IV (Ch. 3)

Rolling In Examination: There are certain types of defects in rolling stock which can only be detected during motion of train. To detect such type of defects rolling in examination is adopted.

This type of examination is carried out on all through passing trains and terminating trains.

To carry out such type of examination C & W staff and supervisor will take position on both sides of platform / line in which train is being received.

During examination following defects are detected.

1. Unusual sound of flat faces on tyre of wheel of any vehicle of train.
2. Whistling of hot axle boxes.
3. Any hanging part or loose fitting of vehicle.
4. Any Spring broken.
5. Brake binding of any vehicle.
6. Any component or spring suspension bracket loose/broken.
7. Abnormal behavior of vehicle.
8. Any other defects by which safety infringement.

Rolling Out: Such type of examination is carried out to minimized detachment of vehicle particularly due to flat faces due to brake binding on tyre. It is carried out on all through passing trains and originating trains.

The procedure of conducting such type of examination is similar to rolling in examination. Only staff and supervisor will take position for conducting examination ahead of engine instead of last vehicle. And will ensure that the brakes of all vehicles running with train are in released condition and there is no any leftover defects during halt of examination by which safety infringement.

1. Examination of originating trains

- i) Before dispatching all trains must be examined by the mechanical train examining staff to ensure that all coaches on the train are in fit condition and without reject able defects (for reject able defects, please refer to IRCA Conference Rules, Part IV). On formation of a rake and after its placement for Examination, washing, cleaning and watering, the station master (SM) shall pass necessary memo to the Engineer (C&W). After carrying out all necessary work, the Engineer (C&W) shall communicate fitness of the train to Station Master. Normally, Railways have standard forms for the use of Station Masters and Engineers for this purpose. Railways, where such forms are not used, should also start using these forms as uniform practice for the guidance of both Engineer (C&W) and Station Master. The Station Master shall not dispatch the train unless the fitness certificate, in the prescribed form, is received from the Engineer (C&W).
- ii) The level of the air pressure/vacuum on the train engine and the brake van gauges as well as the percentage of operative cylinders should be recorded on a prescribed certificate and signatures of the driver and the guard of the train should be obtained by the Engineer (C&W) as per the procedure laid down by each Railway. No train should be allowed to leave with an inoperative/defective Brake cylinder on any coach after pit attention. Trains which have been attended on pit line should have **100% brake power**. Trains which are attended on platform or where secondary examination has been dispensed with or en route should have minimum **90% brake power**.

2. En route/Terminating Examination of Passenger Trains

- i. *Sr.DME/ DME in charge shall nominate the site for carrying out Rolling IN/Rolling OUT examination after personal inspection of site. While nominating the site following should be kept in view:*
 - a) *Site shall provide unobstructed view of under gear from both sides.*
 - b) *Speed of the train shall not be more than **30 kmph**,*
 - c) *It should cover the entire length of train,*
 - d) *Should have adequate space for fixing the lighting arrangement and for staff.*
- ii. *For rolling in examination of train it has to be ensured that proper lighting arrangement is provided on both the sides of the track at nominated spots for examination of under gear parts during night. Focusing of lights shall be done by keeping a coach on the line and adjusting the angle of light to illuminate under gear and bogie. C&W staff should take position at nominated rolling in place on both the sides of the track before the arrival of train.*
- iii. *As the train passes the nominated point, C&W staff should watch out vigilantly for loose/hanging/broken under gear parts of the coaches, any unusual sound coming from the coaches or any other abnormality in the coaches.*
- iv. *After train comes to halt, it should be ensured that the train is protected from both the sides (with the stop board/red flag during day time and red lamp during night time) before commencing the examination of the train. It should be ensured that a suitable indication board is placed at conspicuous location visible to the driver indicating that C&W staff is at work.*
- v. *Temperature of the axle boxes should be measured preferably with the help of the electronic temperature measuring device.*
- vi. *Brake release shall be checked by physically moving the brake beam. However, in case where train locomotive has to be detached, brakes of all coaches shall first be manually released. For checking the release of brakes the hook may be used. Other under gear parts should be examined visually to ensure that the train is safe to run further. During night the lamps/search light shall be used for illumination.*
- vii. *Repairs if required should be carried out promptly to avoid detention to train to the extent possible.*
- viii. *Lavatories of the coaches should be properly cleaned using High pressure water jet machine provided at nominated stations during halt of the train. Any complaint from passengers should be attended promptly to the satisfaction of the passenger*
- ix. *After attending to any required repairs stop board/red flag should be removed.*
- x. *Carriage controller (CCR) should be informed about any out of course work done.*
- xi. *CCR shall repeat the out of course work done to the Primary Maintenance (PM) depot after corrective action.*
- xii. *At the train examination stations where locomotives are changed on through trains, the level of air pressure/vacuum created on the locomotive and brake van gauges should be recorded on the certificate to be issued to the guard and driver on prescribed form. The inoperative/blanked cylinders, if any, should also be written in the certificate for their information. This certification should be an endorsement on the original brake power certificate; no fresh brake power certificate needs to be issued.*

3. Pit Examination of Passenger Trains

a. Washing and cleaning of coaches: Use recommended solutions for cleaning as per RDSO specification No. M&C/PCN/101/2001 or use cleaning agents approved by CME of the Railway.

b. Platform cleaning and washing: Wherever washable aprons are available on the platforms, the time available before the terminating trains are pulled out into the yard, should be utilized for inside sweeping and toilet cleaning.

c. External Cleaning / Washing

- I) Place the rake/coaches on the washing pit provided with equipments required for washing and cleaning. It should be ensured that the rake/coach is protected with proper board/signal for safety of the staff working on washing/cleaning job to prevent movement/disturbance in the activity. Scotch blocks with locking arrangement should protect lines and keys should be kept with Engineer (C&W) till the time rake is under maintenance.
- II) Before starting washing and cleaning of side wall, ensure that the glass shutters and louver shutters of that side are lowered. Remove dirt/dust accumulated on shutters by compressed air or duster.
- III) Remove old reservation charts/labels on the body panels. Splash water on old charts so that they are wet for easy separation. Care should be taken to avoid any damage to the paint.
- IV) The cleaning solution should be spread/rubbed with nylon brush or sponge brushes and then rubbed thoroughly to clean the panels. Extra attention should be given to oily and badly stained surfaces.
- V) Destination boards may be removed and cleaned with brush/duster.
- VI) Clean the external surface by high pressure jet where facilities are available.
- VII) All exterior panels including end panels should be hosed with water and brushed with diluted soft soap (detergent solution) The strength of the solution may be increased or decreased according to RDSO specification M&C/PCN/101/ 2001.

d. Cleaning of Toilet

- I) Before starting cleaning of toilets ensure that all repairs in the toilets have been carried out and after cleaning no employee should enter in the toilet.
- II) Doors and walls should be cleaned with water sprayed by high pressure jet up to waist level. Apply specified solution and rub thoroughly with sponge brush/ nylon bristle brush.
- III) Indian style lavatory pans have to be cleaned by thorough rubbing with concentrated solution of recommended cleaning agent.
- IV) Western style commode shall be cleaned as (iii) however due care should be taken that recommended solution should not fall on commode lid which may damage/spoil it.
- V) The flooring should be rubbed with nylon bristles/sponge brush and cleaned with recommended cleaning agent. The drain holes should be cleaned thoroughly for easy discharge of water.
- VI) The mirrors in toilet should be cleaned with light wet cloth. Recommended solution should be used for cleaning the dirty portion of glasses.
- VII) After all the washing and cleaning in the toilets mentioned above, the toilets should be thoroughly cleaned with water jets and water should be flushed out. All fittings and floor should be wiped dry with a cloth.
- VIII) After cleaning, spray deodorant in the toilet to remove the bad odor.

e. Internal cleaning of upper class AC and sleeper coaches

- i) Collect the cigarette ends from all Ash trays, news paper from magazine bag and waste from dust bin. Sweep the whole coach with broom in sleeper coaches. Clean the floor of AC coaches with vacuum cleaner.
- ii) Remove dust from floor, berths/seat, magazine nylon wire mesh bag fitted on panels and fan guards with duster. Use of vacuum cleaner is excellent in such areas.
- iii) Also remove dust/dirt from under the berths, window sill, sliding door, railing corner and all corner & crevices of coach interior with vacuum cleaner if provided.
- iv) Ceiling panels, wall panels, cushion berths, fittings, table top, etc. should be cleaned with duster and stain marks on these should be removed by use of recommended soft detergent.
- v) Aluminum frames, strips, and other metal fittings, etc. should be cleaned with recommended cleaning agent.
- vi) FRP window frames, louvers, etc. should be cleaned with recommended solution and rubbed out by nylon brush or sponge /duster to remove stain marks.
- vii) Alarm chain handle and its holding bracket should be washed and cleaned.
- viii) The coach flooring should be rubbed with hard coir brush and PVC flooring should be rubbed with nylon bristles/sponge brush and cleaned with recommended cleaning agent.
- ix) In AC coaches, the amenity fittings and toilet fittings such as coat hanger, stools, arm rest, foot rest, towel hanger, etc. should be cleaned with duster. Stains on these items should be removed with recommended detergent solution.
- x) The compartment carpet should be cleaned with vacuum cleaner. Every month, the carpet should be cleaned thoroughly by taking it out from compartment and if necessary they should be dry cleaned in every three to four months. Before re-laying the carpet, the compartment floor should be thoroughly cleaned.
- xi) Spray recommended air freshener in the coach. No employee should be allow to enter the coach for any purpose/work after complete cleaning
- xii) Curtains in the AC Coaches and Tourist Cars should be removed for periodical washing and cleaning. Faded and damaged curtains should be replaced on condition basis.
- xiii) Precaution should be taken to prevent nuisance of cockroaches in AC coaches and pantry cars by periodical spray of recommended insecticides
- xiv) No repair works on Electrical train light/fan/AC) or Mechanical account should be left to be carried out after washing and cleaning of the coach..

f. Internal Cleaning of GS, SLR

- I) Cleaning of GS, guard and passenger compartments of SLR should be done as mentioned above wherever applicable.
- II) If necessary clean the wooden seat and their frames with recommended detergent solution and water.
- III) Interior surfaces of parcel and luggage vans should be cleaned with recommended detergent and water.

g. Cleaning of buffers and screw couplings

- i) Buffer plungers should be scrubbed with a scraper to remove dirt and muck. Thereafter, they should be wiped clean with cleaning oil and rubbed with coir rope.
- ii) Screw coupling threads should be cleaned with wire brush to remove all dirt and dust. Thereafter, it should be cleaned and given a light coat of oil. Oiling should be done on slack adjuster also.

SAFETY & AMENITY FITTINGS

Amenity Fittings:- The fittings which are provided inside the coach for Luxurious & Comfortable & also for non strenuous journey are called as “Amenity Fittings”.

Safety Fittings:- The fittings which are fitted in the coach for safety of passengers & their luggage are called as “Safety Fittings”.

Amenity Fittings	Compartment			
	AC Coach	1st Class Coach	2nd Class Coach	Sleeper Coach
Folding or Fixed Table	Y	Y	N	Y
Tumbler Holder	Y	Y	N	N
Waste Paper Basket	Y	N	N	N
Mirror with Shelf below	Y	Y	N	Y
Coat Hook	Y	Y	Y	Y
Foot rest for upper Berth riding	Y	Y	N	Y
Fans	Y	Y	Y	Y
Upper Berth	Y	Y	N	Y
Luggage Racks	Y	Y	Y	Y
Light in Compartment	Y	Y	Y	Y

Furnishing Fittings:-

Shower Bath	Y	Y	N	N
Wash Basin	Y	Y	Y	Y
Towel Rail	Y	Y	N	N
Push Cock & Lota Shelf	Y	Y	Y	Y
Commode Rail	Y	Y	Y	Y
Mirror & Shelf	Y	Y	Y	Y
Bottle Opener	Y	N	N	N
Liquid Soap Container	Y	N	N	N

Safety Fittings: -	AC Coach	1st Class Coach	2nd Class Coach	Sleeper Coach
Luggage locking wire	Y	Y	N	Y
Alarm Chain	Y	Y	Y	Y
Upper Birth Safety Rail	Y	Y	N	N
Doors latch & catch	Y	Y	Y	Y
Doors	Y	Y	Y	Y
Window Shutters	Y	Y	Y	Y
Fire Extinguisher	Y	N	N	N
Commode Rail	Y	Y	Y	Y
Vestibule Safety brackets	Y	Y	N	Y
Window Safety Bars	N	Y	Y	Y

BRAKE POWER CERTIFICATE (B.P.C)

This is certificate jointly signed by guard, driver and C & W supervisor prepared in triplicate by C & W supervisor after ensuring vehicle attached in train is fit to run and required amount of vacuum/pressure is maintain in engine and brake van/last vehicle. It contains train no., engine no., load, break up of load, brake power of the train, amount of vacuum/pressure in engine and brake van and first and last two vehicles number respectively.

Revised format of BPC for coaching trains was advised vide board's letter No.98/M(C)/137/19 Pt. dated 25.08.06 S 121/2. As detailed below, certain practical situations/limitations where in the original BPC become invalid, need to look into:-

- (i) Two trains starting from different stations and amalgamating into one train at an en route station.*
- (ii) Train originating from one station and disintegrating into two /more trains at an en route station*
- (iii) Trains requiring revalidation of BPC as per provisions of RPC-4 category 3 (a) , 3(b) and 4 at terminal station or wherever train engine is changed at en route station, If no TXR staff is posted at such station.*

To take care situation as above, it has been decided that.

- (a) For trains as in (i) above The BPCs for individual trains shall be clubbed at the intermediate amalgamating point, revalidated and the train worked up to the destination.*
- (b) For trains under the category (ii) above, the originating station shall issue separate BPCs for all the trains which are to be formed after disintegration at any intermediate station. At the intermediate station the respective BPCs shall move along with the train after revalidation.*
- (c) Whenever revalidation of BPC is to be carried out at NON TXR station, It is proposed that the exercise be carried out jointly by Guard and Driver as is prevalent for GDR checks of freight*

For the purpose of checking brakes continuity and revalidation of BPC , whenever required , it must b ensured that the value of BP,FP in the locomotive and the rear most brake van are recorded a fresh each time whenever the train engine is changed or the rake composition is altered.

There are four types of B.P.

Description	Colour	Brake Power %	Validity	Remarks
Coaching Train	White	100%	Max. 3500 Km	
Vacuum Bk. Safe to run exam. (Goods)	Pink	85%	800 to 1000 Km	
Ordinary Intensive exam	Green	90%	One Trip/ Up to destination	
Premium Rake exam	Green	95%	12+3 days	
CC Rake Exam	Yellow	100% -ordinary For – spl. class For–A-spl. class	i) 4500 Km or 20 Days ii) 6000 „ „ 30 Days iii) 7500 „ „ 35 Days	

11. Driver remarks & en route endorsements by Engineer(C&W)

Station/ Rly. Where examined	Engine No.	No. of vehicles without brake power			Percentage of brake Power	Signature of Engineer
		Isolated D.V.	Malfunctioning SAB	Brake Rigging		

IMPORTANT

9. The incoming Driver shall handover the certificate to relieving Driver. If he is leaving the train without relief, it shall be deposited with the authority nominated to receive it, who will give it to the outgoing Driver.
10. The outgoing Driver & Guard will satisfy themselves from the coach nos. given in item 10 that the certificate pertains to their train.
11. It is responsibility of the Driver of train to satisfy himself that the brake power certificate is proper and valid, before working the train shall handover the certificate to relieving driver. If he is leaving the train without relief, it shall be deposited with the authority nominated to receive it, who will give it to the outgoing Driver.
12. For the all types of coaches fitted with air springs in secondary suspension, all air springs should be in inflated condition. In case one or more air springs are in deflated condition, they should be isolated by isolating cock for air springs. Speed restriction of 60 kmph shall be imposed until the condition of air spring is rectified.

Brake Van Equipment: Similarly, other brake van equipment for which Mechanical Train Examining staff is responsible to supply, should be provided according to the instructions of each Railway. As per RDSO's letter no. MC/CB/28 dt 19.5.2000, racks have to be provided in the SLRS for provision of portable control telephones, portable train lighting equipments, portable fire extinguisher, wooden wedges/skids and stretcher. Railways can modify existing emergency equipments rooms in the guard's compartment to provide racks for keeping the above mentioned items except fire extinguisher

In view of emergency use, all originating trains should be provided following items in front & rear SLRs :-

- i) Fire Extinguishers DCP type -in engine (to be supplied by Loco Shed) Fire Extinguishers DCP type in AC Coach, SLR, Pantry car (to be supplied by C&W)
- ii) Wooden Wedges (To be supplied by C & W)
- iii) Wooden or Steel Ladder (To be supplied by Operating dept.)
- iv) Stretcher (To be supplied by Medical dept.)
- v) First Aid Box (To be supplied by Medical dept.)
- vi) Electrical Box (To be supplied by ETL dept.)
- vii) Field Telephone Set (To be supplied by S & T dept.)

Formation Of Block Rake: For the purpose of maintaining the coaches & rakes in good condition & to avoid public complaints the Chief Mechanical Engineer in consultation with the Chief Operating Superintendent Of The Railway shall form BLOCK RAKES for each of the long distance trains & the inter railway trains & also nominate spare of coaches adequate no. of the block rakes to replace sick block rake coaches.

Destination Board: Each coach on originated rake should be provided with destination board of approved size by the Primary Maintenance Depot.

Fire Extinguisher: Approved type of Fire Extinguishers should be provided on all originating trains according to number prescribed by railways in Brake Van , Postal Van , Dining Cars AC Coaches etc. These fire extinguishers should be periodically checked after **every 3 months & completely refilled after 1 year.**

These Fire Extinguishers should not be over due for testing & refilling.

Deficiency Rolling Stock (DRS) For Coaching Stock: Railway should devise system for detecting deficiencies. Reports of deficiencies/defects in Rolling Stock (DRS) reports in the Performa should be prepared for each mail/express/passenger originating train in duplicate by the Engineer (C & W) Electrical (TL) and should be signed jointly with the RPF representatives. Reports for mechanical deficiencies should be prepared on specified Performa and may be altered by each Railway on the basis of the items most prone to theft on their system. This should be done soon after the maintenance of the rake is complete in the sick/washing lines. In case the train starts from the platform itself, these reports should be prepared by jointly by C & W and electrical department. DRS Cards should be prepared in duplicate in which original copy is kept as a record copy and carbon copy is handed over to train guard. After the coaches have been jointly checked and DRS reports have been made, the coaches should be pad locked/key locked and the key and report should be sent to the platform Engineer *(C & W)

Coach Maintenance History Card: Every coaching depot shall have computers for maintaining the coach maintenance history in a software programmed which should be compatible with the programmed of the coaching workshop.

The “Coach Maintenance History Card” (MHC) for each of its coaches. The card will contain records of maintenance schedules including POH and special repairs in shops. It will also show the history of the coach from the time the coach is placed in service will its condemnation and will give details of all major repairs wheel changing bogie changing etc.

The complete history book of each coach, consisting of maintenance history cards, date card, trial card etc. will however, be maintained by the base workshop. When a coach is sent for POH or special repairs, a copy of its maintenance history card should be sent by its base depot to the workshop for record in its complete history books.

INDO – GERMAN MODIFICATIONS

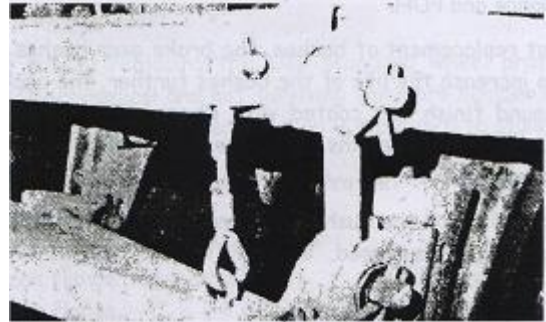
Description Of Modification: These modification were adopted with the technical guidance of German team so that known as Indo- German modification are as under :-

1. Use of nylon 66 bushes with case hardened N5 ground finish Class-ii pins:- previously resin bushes with N7 ground finish class -2 pin were used.

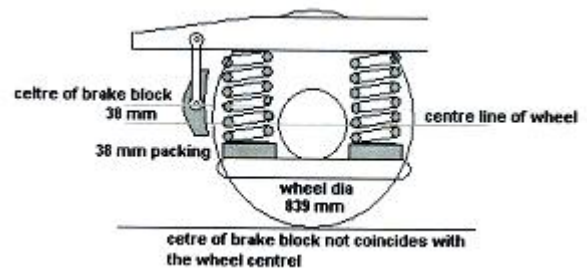
2. Modified design of brake shoe key and brake shoe head, previously width of the slot of brake shoe head was 60mm, it is reduced to 50mm .In new design of brake block key ,the camber is increased and length is decreased .



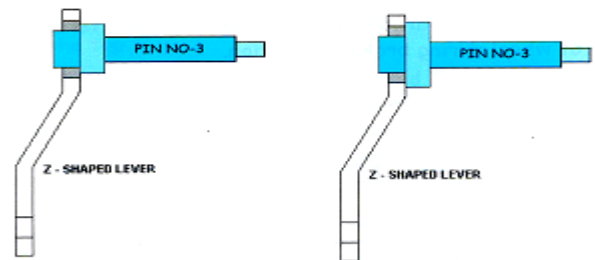
3. Provision of safety wire rope arrangement for brake beam, previously G shape safety brackets were used with brake beam but now wire rope safety brackets are used which are safe with less probability of breakage.



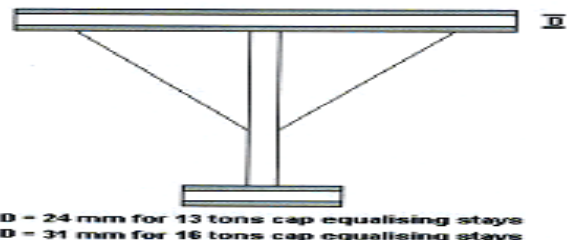
4. Fitment of modified brake block hanger in increasing length 235 mm from 205 mm. Length of the hanger is increased resulted no climbing cases of brake beam over to wheel.



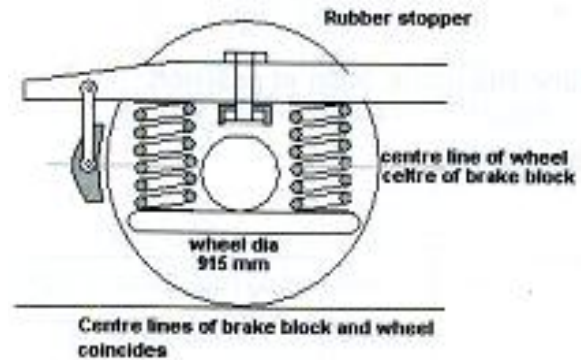
5. Provision of modified lever hanger pin. The width of the hexagonal head bolt is increased from 46mm to 51 mm to avoid inserting probability of hexagonal head bolt in z arm bushing dia.



6. Fitment of equalizing stay rod of 16 ton axle load also in 13 ton axle load bogies. With the fitment of 16 tone capacity equalizing stay rod breakage chances are reduced.



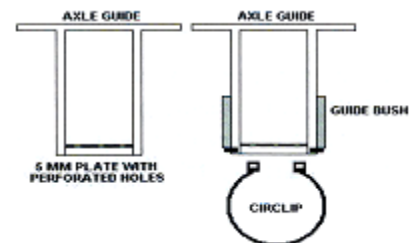
7. Provision of rubber stopper for hexagonal head bolt. Rubber pads of 10, 20, 35, 45mm with 4mm thick compensating rings are being utilized to maintain the A clearances in ICF bogie.



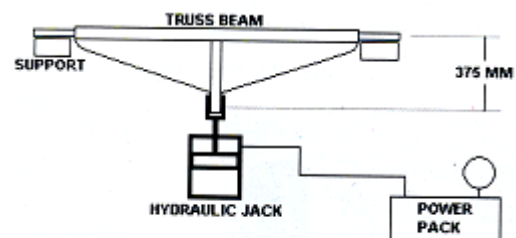
8. Slack adjuster articulation arrangement. It was the modification for vacuum brake coaches to avoid failure of SAB due to bending effects during negotiation on curve.



9. Provision of locking arrangement of axle box guide. To prevent the dropping of guide cap in pot. Now a days integral type guide with cap are being utilized.



10. Use of integral type of buffer false plate for reclamation. In this modification the process of flat buffer reclamation is given.
11. Use of high capacity buffer pads. In this modification high capacity buffer pad of 1030 kg m is utilized, no use of 515 kg m capacity pads is recommended.
12. Dimensional check report for bogie covering the aspects of avoidance of welding joint under axle guide. Under this modification the squareness of bogie should be checked by work shop staff on arrival of coaches for P.O.H.
13. Modified brake beam for air brake coaches and testing procedure. Under this modification only 16 T capacity brake beams are to be used in 16 & 13 T bogies.



LATEST MODIFICATION ON COACHES

- 1) Change in securing arrangement of anchor link, similar as in Fiat Bogie.
- 2) Use of Hopper type Shutters in place of froster Shutters.
- 3) Provision of ventilator with Arc shape fins in Lavatory for proper circulation of Air.
- 4) Provision of Resetting Handle on End panel for the resetting of ACP.
- 5) Increase in hole dia of PEAV from 4 to 8 mm.
- 6) Provision of APD for Guard van valve and its handle.
- 7) Reduction in width of brake shoe head by 5 mm from inside.

RIDING INDEX

Ride Index: It has been found that human sensations are dependent on acceleration, rate of change of acceleration (impulse) and displacement. In other words, the product of these values could be used as measure of comfort/discomfort.

Riding is a quality of comfort experienced by passenger, depends not only upon acceleration, rate of change of acceleration & vibration but also some other factors as like noise, moisture temperature etc.

Calculation Formula given by Dr. Sperling,

$$\text{Ride Index} = 0.896 \sqrt[10]{\frac{\sum b^3}{\sum f}} F(f)$$

where

b : Amplitude of acceleration

f : Frequency of acceleration

$F(f)$: a correction factor dependant on the frequency

Note : The accelerations referred to above are vehicle body accelerations, vertical or lateral, measured on the floor level just above the center pivot location.

The ride index is just a number with no units and its value gives us an indication of the riding comfort of a vehicle. The index is easily calculable during field trials by measuring the vertical/lateral accelerations using standard accelerometers.

Ride Index gradations are as follows:

<u>RI</u>	<u>Appreciation</u>	<u>Fatigue Limit</u>
1.0	Very Good	>24 Hrs
1.5	Almost Very Good	-Do-
2.0	Good	-Do-
2.5	Nearly Good	13 Hrs
3.0	Passable	5.6 Hrs
3.5	Still Passable	2.8 Hrs
4.0	Able To Run	1.5 Hrs
4.5	Not Able To Run	45 Mts.
5.0	Dangerous	15 Mts.

RI criteria applicable on Indian Railways:

	<u>Preferred Limit</u>	<u>Max.</u>
Coaches	3.25	3.5
Wagons	4.25	4.5
Loco	3.75	4.0
EMU/DMU	-	4.0

The ride index as described above gives the average riding quality of a vehicle over the chosen length of track (generally one kilometer). However, individual acceleration peaks also have an effect on the comfort of the passengers. Accordingly, limits for maximum acceleration values have also been laid down for coaches and locomotives. For details, the Third Criteria Committee Report of RDSO may be referred.

The Following Measures should be given to maintain Ride Index in ICF coach:-

- 1.** *By proper checking of primary suspension arrangement.*
 - *Checking free height & height variation.*
 - *Telescopic hydraulic dash pot & oil level in it (1.6 liters in modified & 1.4 liters in non-modified).*
 - *Gap between safety loop & axle box lug should be within limit i.e. 40 mm (old 20 mm).*
- 2.** *Proper pairing of springs on secondary suspensions.*
 - *Free height of spring should be within limit.*
 - *Ensuring proper working of shock absorber.*
- 3.** *Proper checking of side bearer, oil & bearing piece should be within limit.*
 - *Oil quantity should be 2.0 liters.*
 - *Thickness of bronze piece should not be less than 42 mm (45 mm – new size)*
 - *Thickness of metal piece should not be less than 8.5 mm (10 mm – new size)*

[6 mm dia hole is provided at the centre of bronze piece for better lubrication / working]
- 4.** *Proper checking of silent bushes fitted in bolster for proper matching of centre pivot.*
- 5.** *Proper checking of buffing gears.*
 - *Proper contact of buffer plunger.*
 - *Plunger stroke should not be more than 127 mm & less than 51 mm.*
 - *Wear on rubber pads should be within limit.*
 - *All securing bolts & nuts should be properly fitted.*
- 6.** *Proper checking of draft gear.*
 - *Wear on rubber pads / coiled spring should be within limit.*
 - *Coupling should be in proper tight position.*
 - *Other securing nuts, washers, cotters also should be in proper position.*
- 7.** *There should be proper bushing of anchor links. Checking of silent bushes therein.*
- 8.** *By checking silent bushes working / clearance of self alignment double row spherical roller bearing.*
- 9.** *All brake gear pins should be provided with proper bushing.*
- 10.** *SAB curdle roller should be lubricated.*
- 11.** *Piston should be proper working condition, means piston travel should be uniformly, and it should not be sticky.*
- 12.** *Berth should be provided with Dunlop cushioning.*
- 13.** *Dynamo pulley & belt should not be loose.*
- 14.** *It should also be ensured that there should not be any wheel defect as like flat faces (not more than 50 mm) deep flange, skidded wheel, sharp flange, thin flange.*
- 15.** *In excess of this the Ride Index also can be improved by ensuring p/way maintenance, signal aspects & engine man ship of the driver.*

LHB – COACHES

Linke Holfmann Busch GMBH – German



Need for new technology- : Indian railway has been manufacturing passenger coaches of “Schlirien” design for more than last 50 years.

Although continuous efforts were being put to upgrade these coaches, but a need was always felt to imbibe technology in-use developed countries so as to affect a quantum jump in quality of coaches.

“M/s Alstom LHB” are one of the leading manufacturers in transport sector having presence in most of the European countries.

Coaches manufactured by them are running in many countries across the world.

For this contract the bogies were designed and made by M/S FIAT, Switzerland which is now a part of Alstom group.

LHB coach body is designed and manufactured by leading German company **Linke Holfmann Busch GMBH** and bogies are designed by **FIAT Switzerland**. Combination of Coach of LHB and Bogies is known as **Alstom LHB Coach**.

HOW IT BENEFITS RAILWAYS

A longer coach: LHB coaches are approximately 2-meters longer than the conventional ICF type coaches. This means “more travel space” “increased seating capacity”, “wider bays and doorways” etc.

A lighter coach: Per meter length, weight of LHB coach is approximately “10%” lesser than the conventional coach. This not only means lower haulage costs but also less wear and tears of the coaches and track.

A higher speed coach: LHB coaches are designed to run at a maximum speed of 180 kmph. Even for speeds of 200 kmph, no major changes are required.

Lesser maintenance:

- Use of superior materials with longer life.
- Superior braking with Wheel slide protection (WSP) system based on microprocessor if the variation of speed of wheel of a Coach is more than 10 kmph. Brakes of the particular coach are released automatically by microprocessor unit till it is again reset. It protects the brake binding of the Coach.
- Bogie with less moving parts.
- Items of wear & tear shall not require replacement/renewal before 10 Lakh km.
- Use of stainless steel and less bogie moving parts shall reduce maintenance requirements.
- Entrance doors flush with side wall allowing automatic car washing.

HOW IT BENEFITS THE PASSENGER

Better ride quality

- Improved ride comfort - ride index reduced from over 3.0 to 2.5 at a speed of 160 kmph.
- Plush interiors of international standards.
- Improved air-conditioning through better duct designing & humidity control.
- Bigger size sealed windows filled with "argon" gas for a panoramic view & heat insulation.
- Modular "oriental" & "western" style toilets with Controlled discharge toilet system (CDTS) This system works on electro- pneumatic principle where in, the waste generated from the coach lavatories during run is collected in a retention tank and is disposed off far away from the city limits on meeting certain predefined conditions to avoid soiling of station premises.
- Well equipped pantry with hot cases, deep freezer, bottle coolers etc.
- Flush type swiveling berth reading light.
- Polycarbonate transparent centre tables.



European Style Toilet

Indian Style Toilet

More safe: Safety of passengers is of paramount importance, so a number of precautionary measures have been adopted in LHB design coaches, like :-

- Four emergency exit windows for faster passenger evacuation during emergencies.
- Wider vestibule design for smooth inter coach movement.
- Convenient to operate emergency alarm pull operation and fire- retardant furnishing.
- Tight lock centre buffer coupler makes coaches anti-climbing
- Crashworthiness.

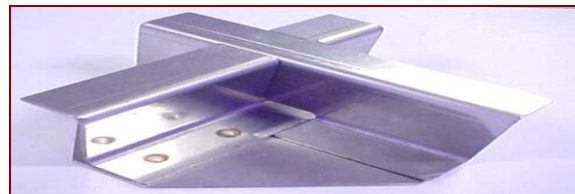
DETAILED DESIGN FEATURES OF LHB COACHES

- Length over body - 23540 mm
- Max. Distance between Inner wheels - 12345 mm
- Max. Buffer drop - 75 mm
- Max. Tare weight - 40.29 t
- No of passengers - 78 (Chair car)
-56 (Ex. Chair car)
- Test speed potential - 180 kmph
- Ride index of coach - 2.5 to 2.75
- Maintenance criteria - Items of wear & tear not require replacement / renewal before 1 million km
- Use of stainless steel and less bogie moving parts reduce maintenance requirements

	LHB	CONVENTIONAL ICF
➤ Test speed	- 180 kmph	140 kmph
➤ Fit to run at	- 160 kmph	130 kmph
➤ Ride index of coach	-2.5 approx	3 approx
➤ No major changes required for 200 kmph		

COACH SHELL

- Stainless steel material
 - Sidewall & roof structural members in ferritic steel
 - Sidewall and end wall sheets of ferritic steel
 - Roof sheet & trough floor in austenitic steel
 - IRSM-41 steel for all other structural members.
- Structural joints
 - Inter-locking joints of vertical & horizontal members



USE OF COMPOSITE MATERIALS

- **“Alucobond” Lavatory Ceiling Panel & Aluminium Honey Comb Partition Panels**

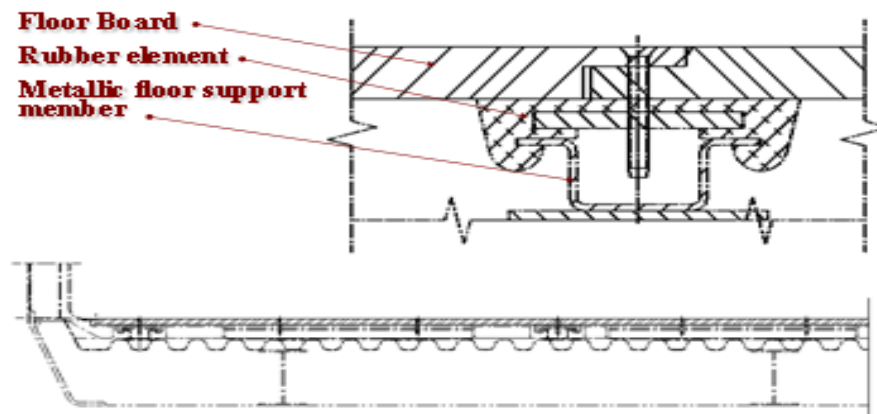
- Improved aesthetics
- Very good surface finish
- Corrosion resistance
- Better rigidity
- Higher strength to weight ratio
- Can undergo processes such as



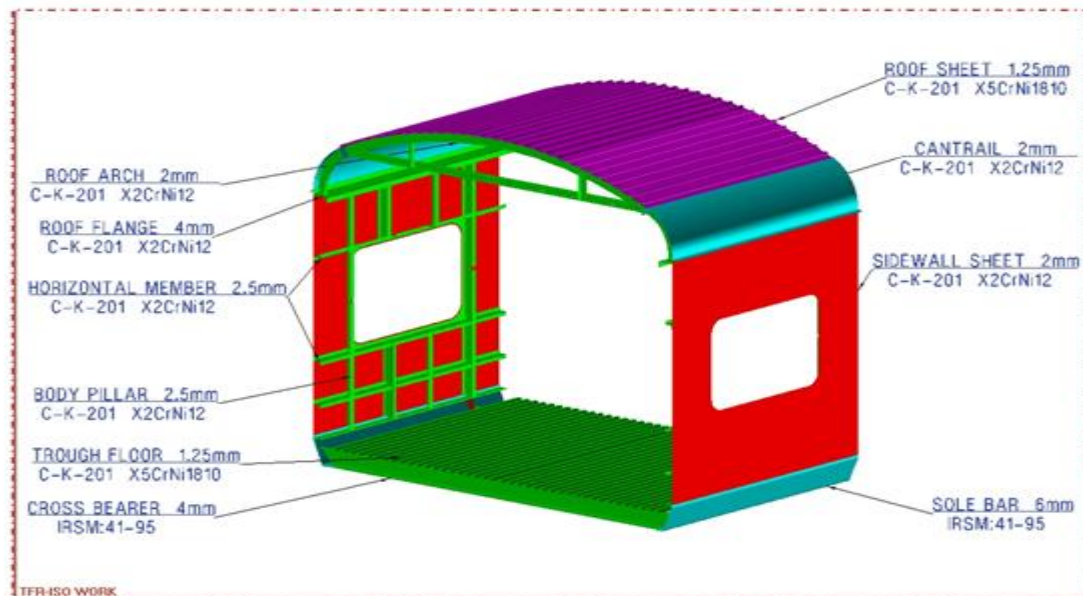
rolling, bending, cutting, welding etc. Without loosing out on it's properties

- Flooring Boards – Cork Sandwiched Between Compreg To Absorb Noise

RUBBER DE-COUPLING ELEMENTS IN FLOORING RESULT IN SUPERIOR NOISE AND VIBRATION ABSORPTION

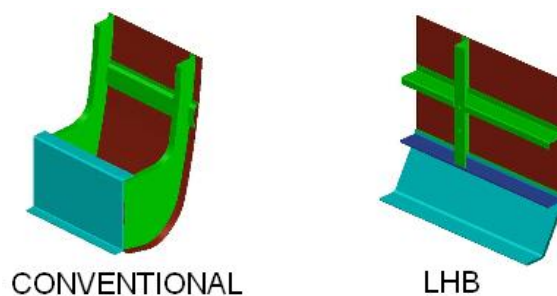


Shell structure



SIDEWALL TO UNDER FRAME JOINT

- Turn under has been eliminated
- Clear approach for sand blasting and painting
- No accumulation of water and muck
- Pillar rests on sole bar as compared to load transfer through a vertical welded joint in conventional. Coaches



SHELL CORROSION PROTECTION

- Weld able aluminium paint to avoid bi-metallic corrosion
- Sand blasting and immediate painting
- Noise & corrosion protection coat "BARYSKIN V60DB" sprayed on shell interior
- "RESONAFLEX-ALU" insulation avoids water collection

SUPERIOR INSULATION

- BARYSKIN V60DB" or eq. Sound insulation PU paints on full coach shell interior, provide anti drumming sound insulation as well as corrosion resistance
- Sound insulation of 31 dB
- Coating of 2-3 mm thickness in the coach interior, 6-8 mm in body bolster area
- Extremely good fire retardant of class Sr-4 to DIN 5510



RESONAFLEX THERMAL INSULATION

BARYSKIN V60DB. Sound insulation PU paints

BOGIE DESIGN FEATURES

- Adaptation of latest design for improved ride comfort.
- Designed for operating speed of 160 kmph, tested at 180 kmph and has a potential for operation up to 200 kmph.
- Structurally superior "y-frame" bogie construction making it lighter and yet stronger.
- Improved primary and secondary suspension for smooth and comfortable riding, obtained by use of flexi coil spring.
- Difficult to maintain dash-pot in ICF coach has been eliminated.
- Extensive use of electrometric rubber-metal components
- Equipped with hydraulic shock absorbers conforming to UIC standard with higher life cycle rating.
- Axle guidance is provided by an articulated control arm through a resilient bush which will give better negotiation on curves.

WHEELS, AXLE AND BRAKES

- Forged wheels IRS R-19 of 915 mm dia. Max usable worn dia: 845 mm.
- Forged axles.
- Cartridge taper roller bearing for axles of "TIMKEN make".
- Graduated twin pipe release axle mounted disc brake with "EP" feature.
- Passenger alarm systems available with 19 mm choke.

COACH INTERIORS

- Ergonomically designed seats as per Indian anthropometric data.
- Large windows with good visibility.
- Aluminium luggage racks with inbuilt reading lamps.

- *Insulation against noise.*
- *Use of fire- retardant FRP for paneling.*
- *Hand-safe feature in all automatic sliding doors.*
- *Anti-skid PVC flooring.*
- *UIC vestibules and auto closing vestibule door.*

COACH INTERIORS

- *Seats have powder coated aluminum frame. Seat cushion & back-rest of moulded “PU Foam”. Gas spring used for reclining mechanism.*
- *16 mm sandwich of “macore wood & cork” has been used as floor panel. This is light weight, water & impact resistant.*
- *Controlled discharge toilet system (CDTS) has a micro – processor controlled intermediate waste tank which discharges waste only at speeds above 30 kmph.*

DOORS:

- *LHB coach has :*
 - *4 body side doors.*
 - *2 compartment doors.*
 - *3 lavatory doors.*
 - *2 vestibule doors.*
 - *1 pantry door.*
 - *2 roller shutters for pantry storage area.*
- *Hinged type swing sandwich doors.*
- *Stainless steel body filled with Phenolic resin for better heat insulation.*
- *Inside panel of door made up of “FRP”.*
- *Provided with sealed window glass unit.*

NEW DESIGN OF LAVATORY DOORS

- *No interface with door frame.*
- *In built micro switch for WC indicator.*
- *Breakage free sunk-in handle for pulling the door.*
- *Forked latch for keeping the door closed when not in use.*



SEATS

- *78 seats in Second AC Chair car and 56 seats in Executive class chair car*
- *Fitted with table in the middle of the coach resulting in all seats not facing in one direction.*
- *Seats have powder coated aluminum frame. Seat cushion & backrest of moulded PU foam. Gas spring used for reclining mechanism.*
- *Ergonomically designed seats with adjustable back rest.*
- *Fitted with better quality of upholstery and woven roving for better fire characteristics.*
- *Provided with 3 step foot rest, snack table, magazine bag and bottle holder.*
- *Provided with coach attendant seat in entrance area.*

LUGGAGE RACK:

- Made from Aluminum extrusions lengths and tempered safety glass
- Can with stand distributed load of 1000 N (100 kg) per meter length
- Halogen reading lights, for individual seats, fitted into outer extrusion and wiring hidden by polycarbonate cover
- Provided with movable coat hooks



PASSENGER EMERGENCY ALARM: 5 passenger emergency alarms per coach have been provided at following locations(2 in passenger compartment & 3 in lavatories)

WINDOW UNITS: Three types of windows have been used in LHB coaches:

- Fixed window – 12 Nos.
- Emergency open-able window- 4 Nos.
- Hopper type window for lavatory – 3 per coach



EMERGENCY WINDOW



HOPPER WINDOW

SEALED WINDOW GLASS UNITS: The window glass unit characteristics are:

- K value not less than 1.6 W/m²K. Where 'K' value gives the heat insulation power of the glass, i.e. the quantity of energy which passes through it under well defined conditions.
- Transparency > 39 %
- Reflection > 40 %
- Total energy absorption < 21%
- The sealed window units consist of 8.4 mm outer laminated and 4 mm tempered inner glass with 6 mm Krypton/Argon gas filled.

EMERGENCY OPEN ABLE WINDOW:

- It is similar to the fixed unit.
- Four units are provided each coach to allow emergency evacuation of passengers
- A handle connected to the rubber profile opens the glass unit of the emergency window

WATER TANKS: 3 types of water tanks are provided in LHB coach:

- **685 liters:** 2 stainless steel water tanks of 685 liters capacity each. These tanks are installed in under frame are fixed with frames and are secured by safety belts. These tanks constitute the fresh water reserves in the passenger coaches. Water level indicator has been provided in these tanks.

- **450 liters:** One stainless steel water tanks of 450 liters capacity. These tanks are installed in under frame, fixed with frames and is secured by safety belts. This tank installed in the under frame constitute the fresh water reserves for the generator car.
- **30 liters :** 3-stainless steel water tanks of 30 liters capacity installed one in each lavatory in the roof and are continuously fresh-water-fed by means of pumps. They maintain the good running of water supply of the barometrical capacitor at each station. These tanks are installed alone and are fixed with 2 supports which are equipped with belts.

COACH LIGHTING:

- Fluorescent tubes 18 watts
- Incandescent lamps 10 watts (for night light and pantry)
- Halogen lamps 10 watts (reading lights)

Fluorescent tubes used as general lighting, vestibule, toilet, pantry and emergency working at 110 volts AC/DC with inverter and with polycarbonate diffusers

Technical data of the inverter for fl. Lamp	(size – 48x36x252)
Nominal input voltage	110 V AC/ 50 Hz (88 -138 v range)
Technical data of the inverter for reading lights	(size – 36x31x296)
Nominal input /output voltage	110 V DC / 12 V DC
Power	60 W / 12 V

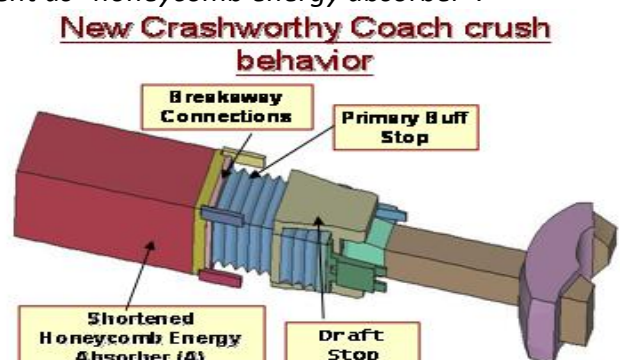


CRASH-WORTHINESS:

- In order to prevent injuries and casualties due to second collisions of passengers with objects and fittings inside the passenger coaches, various features & Modifications have been incorporated inside coaches like all fitting's hooks and corners eliminated and coated with Poly-Urethane Foam
- Re-designed for energy absorbing coach ends, which collapse in a planned manner under any impact, keeping the passenger area intact.
- "FRP doors" to aid collapse at door-way level.
- Buffer mounting area with a box structure, which will ensure end to end contact between the colliding coaches & improve chances of energy absorption in the coach ends.
- Modified "CBC draft gear" with crush element as "honeycomb energy absorber".

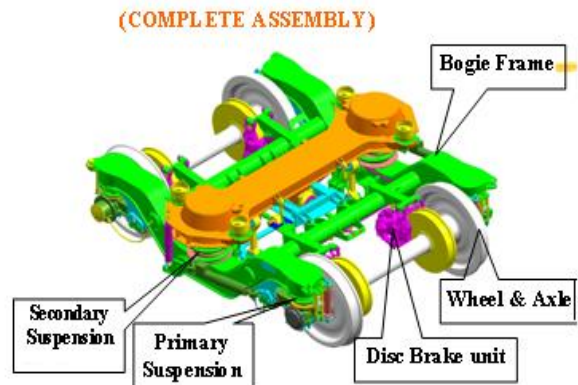


Honeycomb energy Absorber



FIAT BOGIE

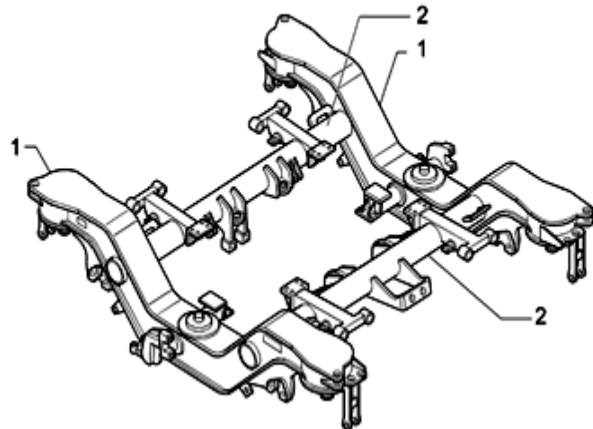
- Design obtained from M/S Alstom-LHB, Germany as part of LHB-TOT contract.
- Adapted at RCF for various coach variants.
- Fit to run up to 180 kmph.
- Superior ride quality
- Bump stops in primary and secondary suspension.
- Miner pads in secondary suspension.
- Yaw, lateral and vertical dampers.
- Nested spring sets with 2 helical and one rubber spring in secondary.
- Less wear and tear



BOGIE FRAME

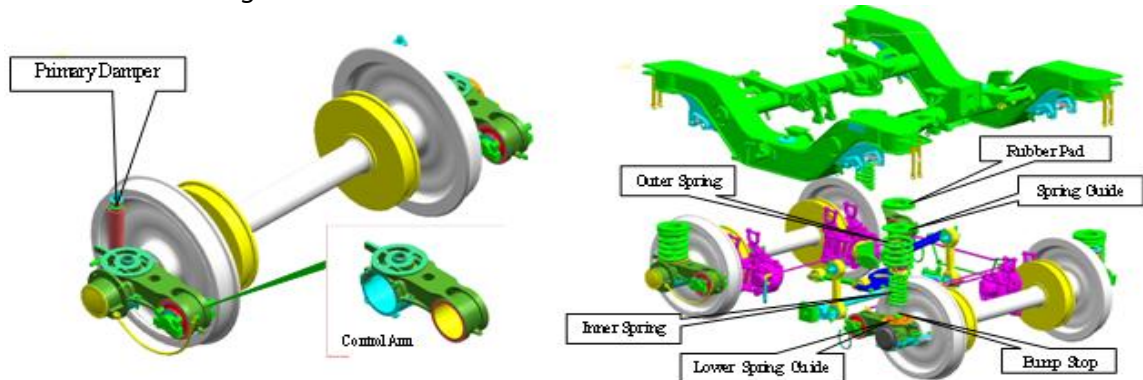
- Solid welded frame –steel sheets(ST-52) and forged, steel cast parts to material GS20MN5V; DIN 17182 (weld able).
- Two side frames connected by two cross beams –support brake units. Various brackets on frame.
- The bogie frame rests on the primary suspension spring units and supports the vehicle body by means of a bolster beam. The bolster beam is connected to the bogie frame by secondary suspension.

- 1.-SIDE FRAME
- 2.-CROSS BEAM(Ø140X Ø168X2506 SEAMLESS PIPE)



Primary suspension

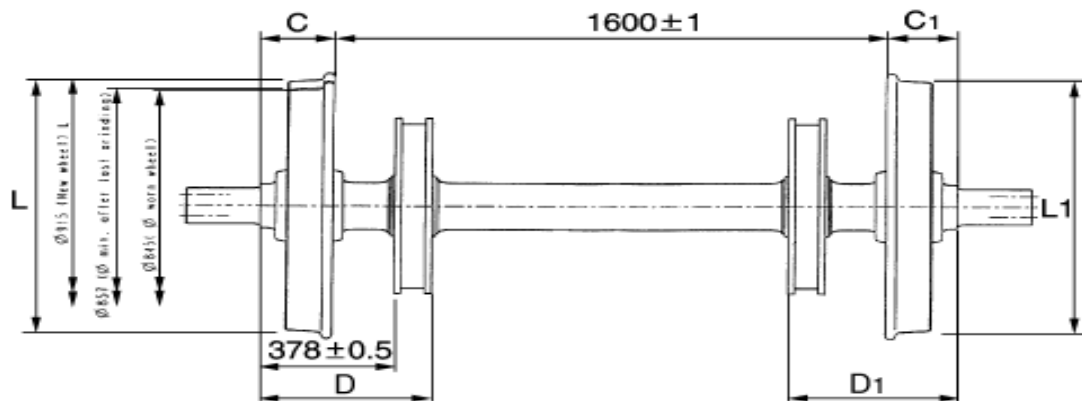
- Two coil springs, one vertical damper, articulated control arm, elastic joints connecting the axle bearing to the bogie frame.
- Better curve negotiation.



Wheel and Axles

- Two brake disks (4), diameter 640 mm and width 110 mm.
- Two wheel disc of tread dia 915 (New), 845 (worn).

Dimensions of wheel set:



The following differences must be respected:

$$|L - L_1| \leq 0.5 \text{ mm}$$

$$|C - C_1| \leq 0.5 \text{ mm}$$

$$|D - D_1| \leq 0.5 \text{ mm}$$

Wheel diameter: New wheel dia 915mm, worn wheel dia 845mm

Wheel Re-profiling and balancing:

- Dynamic balancing 320rpm
- Less than/equal to 50gm.m
- Glue the needed weights.

Axle bearings

- Taper roller cartridge type bearing
- Pre-assembled unit.
- Maintenance free-overhaul 1.2 million km.
- Sensors for detecting speed

DISC BRAKE SYSTEM

- Axle mounted disc brake
- Two discs per axle of dia 640 mm
- Inbuilt slack adjuster in brake cylinders
- 35mm brake pads

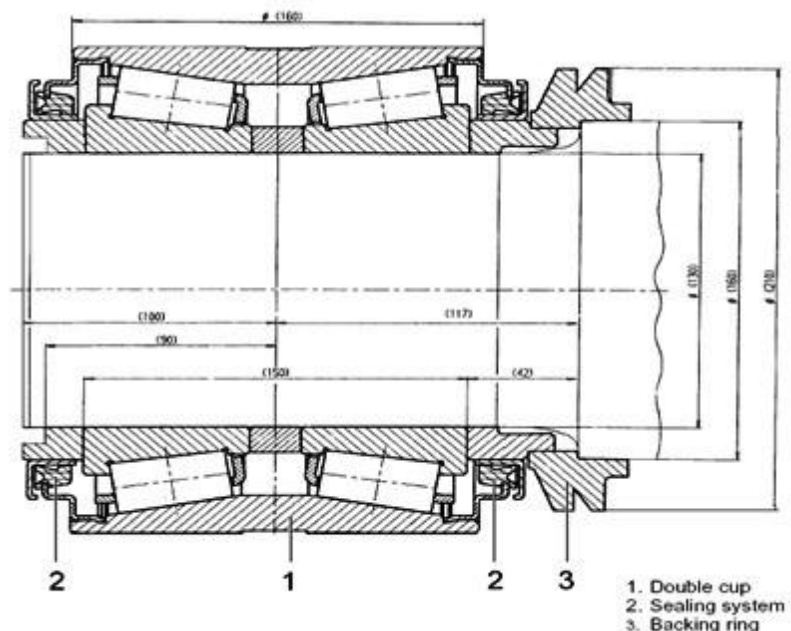


FIG. 1-4 AXLE BEARING LONGITUDINAL SECTION

SECONDARY SUSPENSION

- Nest of flexi-coil springs inner and outer, rubber spring and secondary pad
- Vertical dampers
- Lateral dampers
- Yaw dampers
- Anti-roll bar
- Anchor Links

PRINCIPLES OF FORCE TRANSMISSION

- **Vertical forces:** Body-secondary springs-bogie frame-primary springs/ball joint control arm-axles.
- **Lateral forces:** Body-Secondary Springs-Bogie Frame-Ball Joint Control Arm-Axles.
- **Longitudinal traction efforts and braking powers:** Body-Traction Centre-Traction Rods-Traction Lever-Bogie Frame-Control Arm-Axles.

Curve Negotiation: An Articulated Control Arm (SGCI TO IS:1865) Is Connecting Axle Bearing And Side Frame Through Elastic Connection, Which Will Provide Flexibility Between Axle And Side Frame.

Fiat Bogie Frame Fabrication: Sequence of Operations.

- **Stage-I:** Fitting of top flange, bottom flange, web and 6 ribs
- **Stage ii** Fitting of control arm bracket, inside stiffener, brake head bracket, bkt for vertical damper.
- **Stage iii** Robotic welding of above items.(mfr. M/S IGM ROBOTER system as Austria)
- **Stage iv:** Yaw damper assembly Bkt. & curve roll link Bkt. Tacking.
- **Stage v:** Full welding of bogie frame on manipulator
- **Stage vi:** Flame straightening of side frame.

SALIENT FEATURES:

Wheel Base	2560 mm
Diameter of wheels	915 mm(New), 845 mm(Condem)
Distance between the wheels	1600 mm
Brake disc diameter	640 mm
Bogie width	3030 mm
Bogie length	3534 mm
Bogie weight	6300 kg

Speed limit: Up to 160KMPH without any modification & up to 200KMPH with minor modification

Riding index for Chair car: VERTICAL – 2.75 & LATERAL – 2.5

Comparison of salient features of fiat bogie against I.C.F. Bogie

S.N.	I.C.F.	FIAT
1	Max operating speed=130 kmph tested speed=140 kmph	Max operating speed=160 kmph tested speed=180 kmph Potential for operation=200 kmph
2	Bogie Frame I Type	H Type Construction
3	Wheel Base =2896 Mm	Wheel base = 2560 mm
4	Wheel Dia =915 Mm	915 mm(new) 845 mm (worn)
5	Clasp Type Brake	Axle Mounted Disc Brake
6	Spherical Roller Bearing	Tapered Roller Bearing
7	Primary Single Spring	Primary nested spring =2 nos.
8	Limited Noise Control Features	Noise Controlled By Using Thick Rubber Pad
9	Secondary spring on L.S. Beam	Secondary spring directly mounted on side frame (no L.S. Beam)
10	Coach Load Is Transferred Through Side Bearer (100%)	Through bogie body connection to side frame via sec. Springs.
11	Center Pivot Transfer Traction And Shock Load	Pivot assembly on transverse beam and bracket on dome take traction/ braking, shock load.
12	Ride Index Transverse=3.5 Vertical=2.5	Transverse=2.75 vertical=2.5
13	Anti roll bar has not been provided.	Anti roll bar has been provided to curb the tendency of roll.
14	Maintenance. Reqd. 1)axle box guide arrangement 2)spherical roller bearing 3)clasp bearing 4)more pin joints	Very little maintenance. In P.O.H.
15	Weight Of Bogie=6.2 T	Weight Of Bogie=6.3 T

MAINTENANCE SCHEDULE: Periodic Maintenance Expiration And Operations Summary

In the following table the periodic expirations of the operations are summarised.

The indication in kilometers, to which said expirations make reference, is priority in comparison to those temporal. The latter are valid if the routes of the roadway correspond to the suitable period.

TIME	km (x 1000)	TYPE
daily	-	DA
1 week	8	D1
1 month	33	D2
6 months	200	D3
1 year	400	D4
2 years	800	W1
4 years	1600	W2
8 years	3200	W3
16 years	6400	R1
24 years	12800	R2

Note: 'D' stays for 'Depot', 'W' for 'Workshop', 'R' for 'Rehabilitation

Periodic maintenance expirations

In the following Table II are listed all the maintenance operations described in detail on the maintenance cards in paragraph 6.3.

MAINTENANCE OPERATION	PERIODICITY										CARD NR.
	DA	D1	D2	D3	D4	W1	W2	W3	R1	R2	
Wheels and axle check			X	X	X	X	X	X			01
Brake disc wear check			X				X				02
Axle bearings and instruments	X	X	X		X	X	X	X	X		03
Electric equipment	X	X	X						X		04
Control arm check			X	X	X		X	X			05
Primary suspension check		X			X	X	X	X	X	X	06
Primary suspension dampers check		X			X			X			07
Brake equipment check	X	X	X			X	X				08
Check hand brake equipment		X	X	X				X			09
Pneumatic equipment check		X					X		X		10
Bogie frame check			X	X	X		X			X	11
Rotation limiter check				X					X	X	12
Anti-roll bar assembly check			X	X		X	X				13
Secondary suspension check		X			X	X	X	X		X	14
Dampers check		X			X			X			15
Traction center check			X	X			X	X			16
Rubber pads check			X	X			X	X			17
Bearing running temperature check				X							18
Bearing inspection						X		X			19

Maintenance operations summary

HYBRID COACHES



Introduction: A mixed type coach where the body is utilized on LHB pattern and the bogie on ICF pattern is being utilized to reduce the original cost as well as for easy maintenance, that mixed coaches are known as HYBRID Coaches.

Reasons For Adoption Of Hybrid Coach:

- In LHB Coaches LHB coach body on German Technology and FIAT Bogie on Switzerland Technology are utilized. The total cost of LHB coaches imported was approx. 3.75 crores and cost of RCF manufactured is approx. 2.75 crores where as initial cost of Hybrid coach 1.7 crore/coach.
- Now a days these coaches are being utilized in Duronto Rakes.
- Critical maintenance of FIAT Bogie.
- Higher carrying capacity due to longer length.
- To obtain Self-Generating system and to avoid one number extra coach as power car in the rake.

Modification in ICF bogie for hybrid coach:

- Provision of Air Spring Suspension spring on secondary suspension in place of Bolster coil spring.
- Use of modified Lower Bolster plank.
- Shifting of position of vertical shock absorber on secondary suspension.
- Elimination suspension link, pin and stone on secondary suspension.
- Elimination of equalizing stay rod.
- Use of lateral shock absorber.
- Provision of Bogie mounted air brake system.

Salient Features:

- Stainless steel shell, weight is reduced by 10% per meter.
- ICF Bogie with Air Spring.
- Provision of H-Type tight lock coupler with Ant climbing feature.
- Provision of CDTs system.
- Conventional Electrics – AC-unit & control panels.
- Conventional interior furnishing.
- SS Partition frames.
- Floating floor with sound insulating board –LHB type.

- Sound Insulating paint.
- Entrance doors and windows are wider.
- Fire barrier sliding vestibule door.
- Safer due to H-Type tight lock coupler.
- Material as per upgraded Specifications.

A longer coach: LHB coaches are approximately 2-meters longer than the conventional ICF type coaches.

This means “more travel space” “increased seating capacity”, “wider bays and doorways” etc.

A lighter coach: Per meter length, weight of LHB coach is approximately “10%” lesser than the conventional coach.

This not only means lower haulage costs but also less wear and tear of the coaches and track.

Coach Shell:

- Stainless steel material
- Sidewall & roof structural members in ferritic steel
- Sidewall and end wall sheets of ferritic steel
- Roof sheet & trough floor in austenitic steel
- Irsn-41 steel for all other structural members. 6 mm sole bar.
- Structural joints
- Inter-locking joints of vertical & horizontal members

Roof –Corrugated:

- Corrugated roof sheet spot welded to z-section roof arches
- Uniform height of arches along it's length
- Roof weighs only about 1000 kg as against 1300 kg in conventional coach

Interlocking joints :

- Positive interlocking between all horizontal and vertical members
- Better strength, reduction of side wall thickness to 60mm from 90 mm, better geometrical integrity



Better Riding Quality:

- Plush interiors of international standards.
- Improved air-conditioning through better duct designing & humidity control.
- Bigger size sealed windows filled with “argon” gas for a panoramic view & heat insulation.
- Modular “oriental” & “western” style toilets with “CDTS” to avoid soiling of station premises.
- Well equipped pantry with hot cases, deep freezer, bottle coolers etc.
- Flush type swiveling berth reading light.
- Polycarbonate transparent centre tables.

Coach Interiors:

- Ergonomically designed seats as per Indian anthropometric data.
- Large windows with "good visibility".
- Aluminium luggage racks with "inbuilt reading lamps".
- Insulation against "noise".
- Use of "fire- retardant FRP" for paneling.
- "Hand-safe feature" in all automatic sliding doors.
- "Anti-skid PVC flooring".
- UIC vestibules and "auto closing vestibule door".

Furnishing of Stainless Steel Coaches :



Cabin Arrangement

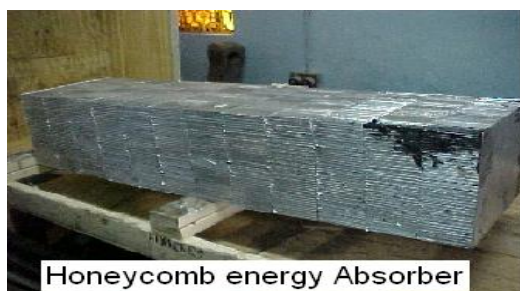


Lavatory Arrangement

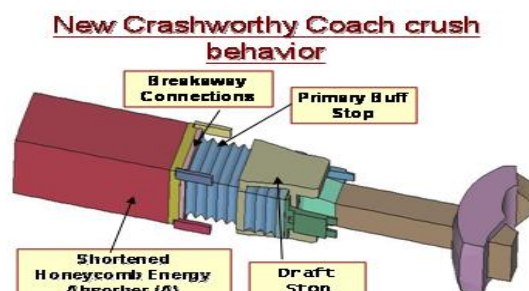


CRASH-WORTHINESS:

- In Order To Prevent Injuries And Casualties Due To Second-ARY Collisions Of Passengers With Objects And Fittings Inside The Passenger Coaches, Various Features & Modifications Have Been Incorporated Inside Coaches Like All Fitting's Nooks And Corners Eliminated And Coated With Poly - Urethane Foam
- Re-designed for energy absorbing coach ends, which collapse in a planned manner under any impact, keeping the passenger area intact.
- "frp doors" to aid collapse at door-way level.
- Buffer mounting area with a box structure, which will ensure end to end contact between the colliding coaches & improve chances of energy absorption in the coach ends.
- Modified "CBC draft gear" with crush element as "honeycomb energy absorber".



Honeycomb energy Absorber



MORE SAFE: Safety of passengers is of paramount importance, so a number of precautionary measures have been adopted in LHB design coaches, like:-

- “Four emergency exit windows” for faster passenger evacuation during emergencies.
- “Wider vestibule design” for smooth inter-coach movement.
- “Convenient to operate emergency alarm pull operation” and “fire-retardant furnishing”.
- “Tight lock centre buffer coupler” makes coaches anti-climbing.

Capacity in Stainless steel Hybrid Coaches:

Coach	No. of Seats/Berths in SS coaches	No. of Seats/Berths in Conventional
AC 2 TIER	52	46
AC 3 TIER	72	64
SCN	78	72
GS	99	90
DSL R	30 +4	20+4
AC 2T + AC 3T	24 +40= 64	24+32=56

Shell Material:

- Roof sheet trough floor – Austenitic (SS 304)
- Other member – Ferritic (SS 409M)
- Under-frame – Corten Steel
- X2 CrNi12 – Ferritic Steel (SS 409M)
- X5 CrNi1810 – Austenitic Steel (SS 304)



Air Spring: Air spring is a rubber bellow containing pressurized compressed air with an emergency rubber spring providing various suspensions characteristic to maintain a constant buffer height irrespective of the loaded condition.

Necessity for Introducing Air Spring in Hybrid Coaches:- In the passenger trains, the no. of passengers entraining (Super Dense Crush Load) into the coach cannot be controlled and hence the payload of the coaches increases from 18 tons to 34 tons. This abnormal increase of payload reduces the riding clearance between the coaches and wayside platforms and also reduces the buffer height resulting in severe hitting of the coach on the platform.

Due to the Super Dense Crush Load the bolster springs become solid, which in turn damages/breaks the coil spring resulting in discomfort to the passenger.

So to overcome the above problem on air suspension (air spring) is introduced in the secondary suspension to maintain a constant buffer height irrespective of loaded condition by varying the pressure of air inside the air spring.

Comparison with existing coil suspension:- Unlike steel spring, air springs retain their height under changing loads. The low natural frequency of air spring suspension remains virtually constant. In case of coil spring, deflection is proportionate to the load, therefore, under high pay load situation, space constraint become critical, leading to the use of stiffer springs resulting in unsatisfactory ride behavior and reduced speed potential. Air springs through their control mechanism, offer proportionate stiffness constant floor height and prospects of better ride behavior with higher speed potential.

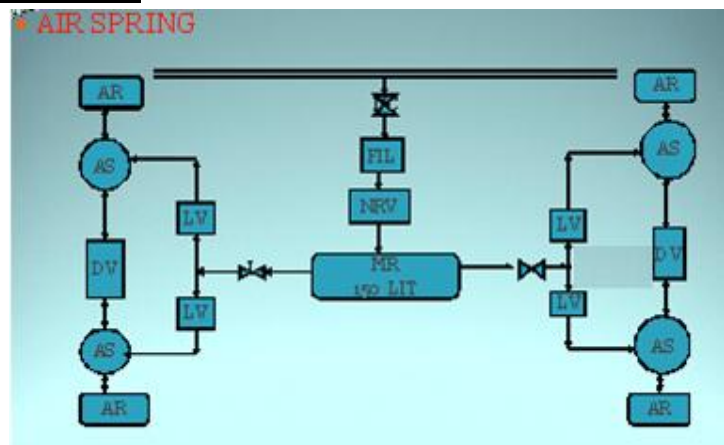
Working Principle: With changing loads, air spring reacts initially by changing the distance between air spring support and coach body. Then actuating height monitoring valve(leveling valve), either by taking the compressed air to the air spring or releasing air pressure from it to the atmosphere. This process continues until the required height is maintained. This mechanism ensures a constant floor height on coaches provided with air springs, irrespective of the load.

Main Equipments Of Air spring:

- | | |
|------------------------------------|-----------------|
| 1- Air Spring | - 04 nos./coach |
| 2- Leveling Valve | - 04 nos./coach |
| 3- Duplex Check Valve | - 02 nos./coach |
| 4- 40 Litre Auxl. Reservoir | - 04 nos./coach |
| 5- Bogie suspension Isolating cock | - 02 nos./coach |
| 6- Non-return valve | - 01nos./coach |
| 7- Coach suspension Isolating cock | - 01 nos./coach |
| 8- 150 Litre main reservoir | - 01 nos./coach |
- **M.R.** - one no. M.R. Of 150 Litre is provided for air springs to store the compressed air from F.P. This compressed air reaches to the M.R. Through an isolating cock, filter & non return valve.
 - **LV**- one leveling valve for each air spring is provided. i.e., total 04 nos. LV are used in a coach. Its main function is to control flow of compressed air to spring during loading/ unloading. One installing lever is also attached with it.
 - **DCV**- one DCV is provided between two air springs of the same bogie. It starts functioning when difference of air pressure between two air springs on the same bogie goes beyond 1.5kg/cm^2 and transfers air from one spring to the other.
 - **AR**- one AR of 40 Litre capacities is provided with each air spring. It increases the effective area of the spring and it has the same pressure as that of air spring.

The Main Components of Air Spring are as under:

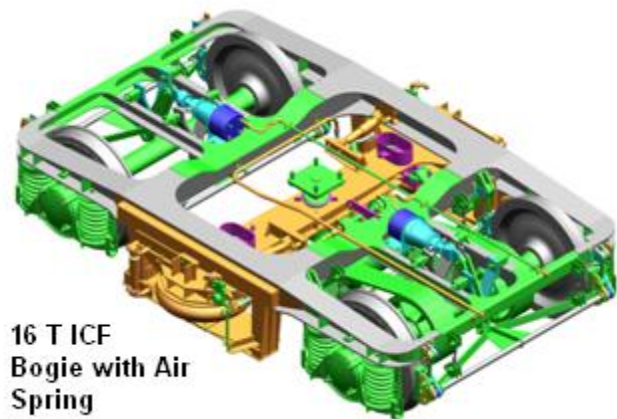
- Bellows
- Lateral shock absorbers
- Level equalizing valve with lever
- Branch pipe 20dia.
- Inner side emergency spring
- 150 liters capacity reservoir.
- 40 liters capacity auxiliary reservoir-04nos.
- Non-return valve
- Isolating cock



- Duplex check valve
- Air spring support cradle
- Air spring releasing valve

CHARACTERISTICS (AIR SPRING):

- Properties of air for cushioning effect used.
- Air Spring made up of rubber bellow & emergency spring.
- Air Spring is height controlled load leveling suspension device.
- Reduce problem of low buffer/coupler heights. Retain height under changing load thus maintain the low naturally frequency virtually constant.
- Air Springs offer a load proportionate stiffness, prospect of better ride quality and higher speed potential.
- Air spring offer soft flexibility in vertical direction by compression of air.
- Air spring absorbs lateral force by variation in effective area in lateral direction.
- Achieved reduced air consumption by designing delayed reaction Leveling Valve.



Air Suspension Bogie:

MERITS:

- Required pressure of 7bar is taken from F.P. no extra provision of pressure is required.
- Ride comfort is better than ICF.
- In ICF bogie coil springs are working on load proportionate deflection system some time which becomes too less (spring become solid) but in air spring a fixed deflection range is maintain having characteristics to vent and receive air as per requirement.
- Ride quality is the same in empty and loaded.
- With the use of control or leveling valve No. extra air is utilized.
- Constant floor height is maintained resulted more comfort to passengers.
- Springs are also able to except lateral thrust to act as flexi coil.
- On typical load (more load) ride quality is same.
- Improved reliability and reducing maintenance efforts.
- Choice to set deflection range during manufacturing.
- Passenger fatigues is reduced.
- The gap between bolster and bogie frame should be maintained 255^{+5}_{-0} mm during running which provides more deflection range in compare to coil spring provided on old ICF coaches.
- Reduced Brake Binding.

DEMERITS:

- *If the pressure is reduced more than 1.5 kg/cm² in both bellows, the automatic emergency valve will sense and there will emergency brake application resulted recharging , setting of audio visual indicating valve is required en-route also.(RDSO Trials in process)*
- *In rolling examination the position of leveling valve lever must be ensure to keep in horizontal position*
- *If lever is in vertical position, then train should be allowed with restricted speed of 60kmph or coach should be detached as decision taken by train examining staff.*
- *More educated supervisors and staff is required to provide more attention during rolling in and rolling out examination*
- *Spares leveling valve, isolating cock, duplex valve etc must be provided at en-route stations for emergency use which will be difficult to maintain.*

FAILURE INDICATION AND BRAKE APPLICATION SYSTEM
(Under Development)

Back ground: The pneumatic suspension system works under two different modes, one in the inflated condition and the other in deflated condition when there is no air cushions and load is directly transferred from the car body to the emergency rubber spring (bumper). Under deflated condition of the air spring, the maximum permissible operating speed of the coach is restricted on account of inferior riding behavior. The requirement of the foolproof arrangement in the suspension design which will apply service brake application in the train in the event of air spring failure (which will ultimately demand the reduction in the operating speed) has been felt in the above context. An emergency situation is possible due to the likely hood of spring damage, possibility of air supply failure or due to sudden rupture or bursting of bellows of air spring due to any reasons.

Objective: The objectives of proposed arrangements are as below:

1. To develop a failsafe system which can immediately and automatically reduce the speed of train by applying the brake and stop within a reasonably safe distance in the event of the air spring getting deflated due to any reason and being absolute pressure in any air spring of corresponding bogie below $1 \pm 0.15 \text{ kg/cm}^2$.
2. To facilitate for identification of affected coach through air whistling/ hissing sound and isolating air suspension of the train at a restricted speed up to the desired destination.
3. System should also be able to perform properly even in case single pipe working of the train is restored.

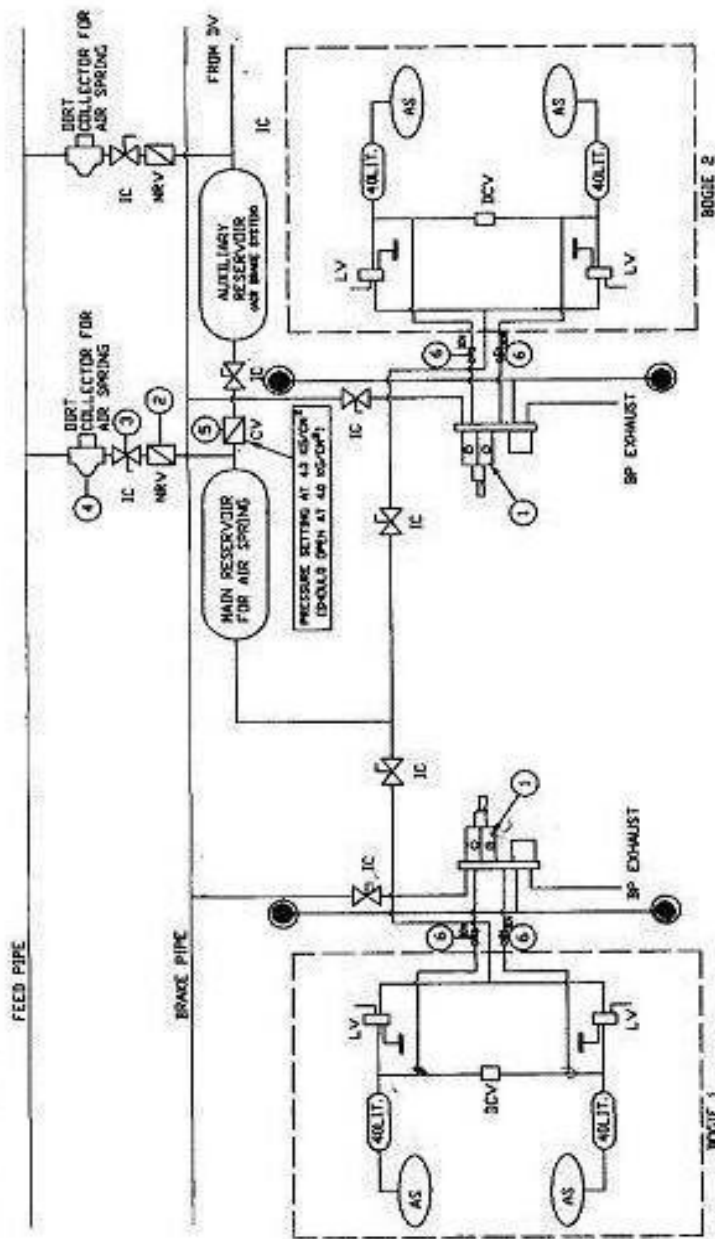
Working principle and technical requirement of the failure indication cum brake application (ASFI) valve: The failure indication cum brake application (ASFI) valve arrangement is based on the air pressure inside the air spring. Under normal operating conditions when the air spring is inflated, the air pressure inside the air bellow and auxiliary reservoir remains almost at the same level. The air pressure is minimum under tare load. The brake pipe is charged at 5 kg/cm^2 under released brake conditions and 1.6 kg/cm^2 drop in air pressure below 5 kg/cm^2 in the brake pipe shall cause service brake application in the entire rake formation.

RDSO has prepared scheme for development of air spring failure indication system for passenger coaches.

Basic Principle of ASFI (Air Spring Failure Indication) valve: In this system we can use four numbers 3- ports valves per coach (01 valve for each air spring) for service brake application in the train. One port of the valve will be connected to air spring (variable pressure between 2 to 6 kg/cm^2) and other will be connected to brake pipe (5.0 kg/cm^2 which may also vary from 5 to 0 kg/cm^2) by 10 mm bore pipe connection. Third port will be exhaust port venting to atmosphere.

Principle of operation: In the event of air spring failure, air from air spring is depleted. As soon as the air pressure in air spring bellow is reduced up to $1 \pm 0.15 \text{ kg/cm}^2$ the activation valve will connect 5.0 kg/cm^2 of brake pipe pressure. Depletion of air pressure through ASFI valve to atmosphere will cause the exhaust in BP pressure through a suitable size choke provided at exhaust port of ASFI valve to drop $1.6 \pm 0.15 \text{ kg/cm}^2$ of BP Line. This will cause service application brakes in the train. The train can be started by isolating the system by isolating cock or resetting the ASFI valve. The schematic diagram of this system is given below:-

COACH



40LIT. ADDITIONAL RESERVOIR OF 40LITER CAPACITY

- AS AIR SPRING
- DCV DUPLEX CHECK VALVE
- LV LEVELING VALVE
- INDICATOR ON COACH BODY
- ISOLATING COCK
- NON RETURN VALVE
- CHECK VALVE
- ISOLATING COCK WITH VENT FEATURE

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COACH

FEED PIPE

BRAKE PIPE

10 NB PIPE

20 NB PIPE

FROM DV

DIRT COLLECTOR FOR AIR SPRING

MAIN RESERVOIR FOR AIR SPRING

AUXILIARY RESERVOIR FOR BRAKE SYSTEM

IC (Interlocking Cock)

NRV (Non-Return Valve)

AS (Air Spring)

LV (Level Valve)

DCV (Duplex Check Valve)

BOGIE 1

BOGIE 2

PRESSURE SETTING AT 4.0 KG/CM² (SHOULD OPEN AT 4.0 KG/CM²)

ASFI VALVES

- Qty. 01 per Air spring (02 will be mounted together on same mounting plate for each bogie)
- 02 Inlet ports with female thread for 10 mm bore pipeline (i.e. 01 from BP and 01 from Air spring) shall be provided
- ASFI valve may be with or without built in indicator.

1	ASFI	AIR SPRING FAILURE INDICATION VALVE	INDIAN RAILWAYS STANDARD	CG-11037
2	NRV	NON RETURN VALVE		
3	IC	ISOLATING COCK		
4	DC	DIRT COLLECTOR		
5	CV	CHECK VALVE		
6	AS	AIR SPRING		
7	LV	LEVELING VALVE		
8	DCV	DUPLEX CHECK VALVE		

Controlled Discharge Toilet System (CDTS)

Objective: The main objective of introducing this type of lavatories in coaches of Indian Railways to have clean environment around Railway. Track within city limits, station premises and maintenance sidings. This system works on electro- pneumatic principle where in, the waste generated from the coach lavatories during run is collected in a retention tank and is disposed off far away from the city limits on meeting certain predefined conditions.

Working principles: By simply pressing a flush button a flush cycle is activated. Depressing the flush button start the flow of water into the toilet bowl and opens the slide valve connecting the toilet bowl to the waste retention tank. At the end flush cycle the waste water stop and the toilet is shield of from the retention tank by the slide/ flapper valve, which acts as an effectives stench trap, preventing odor from entering the toilet room.

The waste is stored in the retention tank until the following two parameter are satisfied; Viz.,

- 1) The train should be in acceleration & running above 30kmph and the number of flushing should be More than five.
- 2) Train should be in retardation and reducing speed of 30kmph .irrespective number of flushes.

On meeting either of above points the retention tank discharge valve opens expelling the contents of the tank at the rail side, away from the railway station and heavily populated areas.

The retention tank discharge valve remains open only for small period of time long enough to completely drain the retention tank, than closes and remain close until the above mentioned parameters are again satisfied.

The toilet system is designed to operate with low quantity and pressurized water bowl wash that covers nearly 100% of the toilet bowl area. The waste is removing from the toilet bowl and transferred to a retention tank within minimal amount of water. Water consumption is only 2.5 liters per flush cycle for the Indian style toilet. Bowl and 1.5 liters for the European style toilet bowl.

Merits:

- 1) CDTS is helpful to clean the station surroundings, nearby areas as well as maintenance siding of railway station.
- 2) Passenger will feel fresh environment stations.
- 3) International image of Indian railway will be improved.
- 4) Man power will be reduced.
- 5) Working efficiency of staff will be increased.
- 6) Water consumption will be reduced for cleaning of track at Station.
- 7) Reduction in entering of cattle in platform campus.
- 8) Good environment will attract more public resulted increased railway revenue.

Demerits:

- 1) In failure of CDTS, load in retention tank will be increased resulted in bad odors and chances of damage.
- 2) In case of failure of fail safe mode in en-route it may create tension to passenger in journey.
- 3) As per fitment location of CDTS it may be damaged by CRO. .
- 4) Due to fitment of CDTS at both ends. Staff will face problems to conduct rolling in & rolling out examination properly.

- 5) It may create trouble to examination staff to attend any repair to break gear components if there is any failure in en-route section due to fitment location and bad smelling of CDTs.

The system requirements are:

1. Compressed air (which is tapped from feed pipe).
2. Electric supply of 24 Volts

1) Intermediate Tank

- I) It is made up of 316 L stainless steel.
- II) It is designed to store maximum 50 liter when outlet valve is closed.

III) In case of extra flush, overflow is provided.

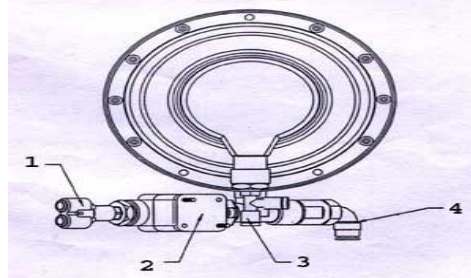
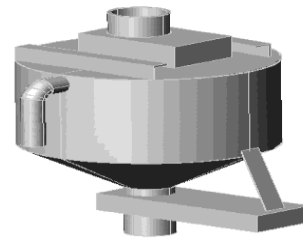
- 2) **Flapper Valve (Inlet):** It is fitted at the top end of intermediate tank. It is made up of composite/Nylon material. As soon as the flush button is operated, the disc opens vertically for a pre determined time. The continuous flow of water during this time ensures complete transfer of waste into the retention tank. The valve closes as soon as the flush cycle is completed.

- 3) **Slide Valve (Outlet):** It is fitted at the bottom of the intermediate tank. It is made up of composite material/Nylon. It is operated on getting signal from PLC.

- 4) **Programmable Logic Controller (PLC):** It is the heart of the system. It performs all the logic operations. It is programmable. PLC is made by Siemens/ Crozet. PLC works on 24 V D.C. having 8 inputs & 4 outputs. It senses the speed of the coach and no of flushes and gets all the operations done by actuating corresponding valves.

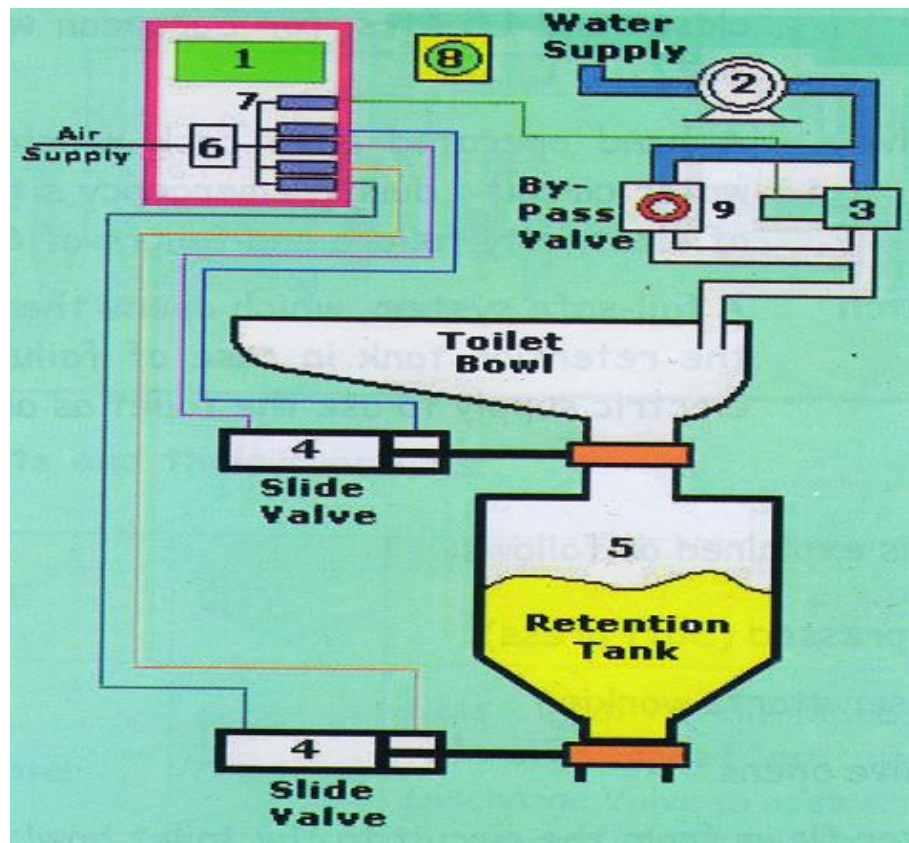
- 5) **Water Pressurizer:** It is made up of two plates of stainless steel fitted facing each other with spring and diaphragm in between. It is fitted below the coach under frame just beside retention tank. Its main function is to increase the pressure of water coming from overhead tank and send the same into lavatory basin as and when the flush button is pressed by the passenger inside the lavatory. The duration of flow is again controlled by PLC.

- 6) **Flush Valve:** It is fitted in the water line provided for the flushing through WPP and fish tail. It acts just like as valve operated pneumatically to facilitate opening and closing of flow of water as and when required.



Working cycle:

- On each flushing (Collection): Simply pressing flush button, starts the flow of water into the toilet bowl and opens the flapper valve connecting the toilet bowl to the waste retention tank. At the end of each flush cycle the wash water stops and the toilet is sealed off from the retention tank by the flapper valve, which acts as an effective stench trap, preventing odor from entering in the toilet room.
- The retention tank discharge valve remains open only for a small period of time (one minute or less) long enough to completely drain the retention tank, then closes and remains closed until the required parameters are again satisfied.



Problem Identification & Remedies

Problem	Cause	Remedy
1.No flush water	1) Valve Y3 on the control panel is defective or non-operational. 2) The control panel fails to transmit signals (the red light is not illuminated). 3) An air hose may be bent. 4) The shut-off cock on the water tank is closed. 5) No water in the tank	1) Activate valve Y3 on the control panel manually. 2) Check the electric signals and connections. 3) Check the air hoses (Activate Y3 manually). 4) Open the cock. 5) Fill the tank.
2.Continuous water falling from nozzles	Some foreign particles in water can obstruct the flush valve operation.	Remove the flush valve and clean it with a soft cloth and again put it into the line.

		
3. Flapper valve not open during flushing.	1) The inlet valve fails to open. 2) Valve Y2 on the control panel is defective or non-operational. 3) The controller (PLC) fails to transmit signals. 4) An air hose may be bent. 5) If after checking the above 4 points, it is still not working.	1) Activate the push button "Inlet"(Green button) on the control panel . 2) Activate valve Y2 on the control panel manually. 3) Check the electrical signals and connections. 4) Check the air hoses (Activate Y2 manually). 5) Change the valve block Y2.
4. Flapper wall making heavy noise/splashing water.	Adjustment of speed controller of its cylinder irregular	Adjust the speed controller as shown in picture with the help of a screw driver to control speed to avoid water splash/noise
Speed controller  		
5.The outlet valve fails to open/ Close.	1) The outlet valve fails to open. 2) Valve Y1 on the control panel is defective or non operational. 3) The controller (PLC) fails to transmit signals. 4) An air hose may be bent. 5) If after checking the above points, it still not working . 6) After doing all these it the problem still persist.	1) Activate the push button "outlet"(Back push button) on the control panel. 2) Activate valve Y1 on the control panel manually. 3) Check the electrical signals and connections. 4) Check the air hoses (Activate Y1 manually). 5) Change the valve blockY1. 6) Refer to its maintenance schedule.
6.No activity by the Pressurizer	1) The ball valve of Pressurizer closed 2) There is a failure signals in the controller. 3) Valve Y3 on the control panel is defective or non operational. 4) Water is leaking from the sides.	1) Open the ball valve. 2) Check the electric signals and connections. 3) Activate valve Y3 on the control panel manually. 4) Tight the nut bolts.

7.No water comes from wall protector	1) The flush valve fails to open. 2) Valve Y3 on the control panel is defective or non-operational. 3) The controller (PLC) fails to transmit signals. 4) An air hose may be bent.	1) Activate valve Y3 on the control panel manually. 2) Activate valve Y3 on the control panel manually. 3) Check the electrical signals and connections. 4) Check the air hoses (Activate Y3 manually).
8.The toilet basin is not evacuated	1) There is a blockage in the pipe between the basin and the intermediate tank. 2) The inlet valve fails to open. 3) Valve Y2 on the control panel is defective or non-operational. 4) The control panel fails to transmit signals. 5) An air hose may be bent.	1) Open the inlet valve by activating the push button and push or press the blockage into the intermediate tank. 2) Activate the push button "Inlet" (Green push button) on the control panel. 3) Activate valve Y2 manually. 4) Check the electric signals and connections. 5) Check the air hoses (Activate Y2 manually).

Service & Maintenance schedule of Pressurizer:

- The spring inside the pressurizer should be fitted properly.
- The pressurizer can be opened by opening the side nuts and can be properly cleaned.
- Tightening of nuts & bolts should be checked daily.
- The pressurizer should be properly fitted at define position.



Fail safe mode: Please read the operation of flapper valve as under:

- In case of loss of air and/or power, the flapper valve shall still remain in closed position.
- A push button style water by pass valve is provided on the interior of toilet room. When the passenger pressed this valve marked "No LIGHT– PRESS TO FLUSH" water flows into the toilet bowl uniformly.
- This water will continue to flow as long as the push button is pressed. When the push button is released the flow of water stops automatically.
- In such a situation the flapper disc opens partially when the push button is pressed and permits the waste to pass to the retention tank. As shown as the push button is released the flapper disc comes back to its original position.

BIODEGRADABLE TOILET

Introduction: The provision of biological toilet is being made in passenger coaches to control pollution over station & city area along with running track up to destination by means of which toilet waste is being biologically converted in to liquid form.

On trial basis it is provided in 23 newly built coaches at RCF Kapurthala. All the 23 coaches are being used in rake of 1107/ 1108(Bundelkhand Exp.) Since 18 December 2010 in JHS division under coordination of DRDE Gwalior.

Advantages of Biodegradable Toilet

1. The discharge from toilet is without fecal material. The problem of dropping of night soil on tracks is thus totally eliminated from these bio-toilets.
2. There is still one conventional toilet with normal discharge system at each end of the coach. Hence, the soiling of end walls of the coaches as well as under frame could not be eliminated completely. But there is a decrease in the accumulation of fecal muck on under frame and end walls of the coach.
3. No perceptible foul smell was noticed in the bio-toilet lavatories. Due to availability of water in P trap, the smell is not coming back into the coach toilet.

AEROBIC BIO-TOILET

The system is based on aerobic bacteria which require oxygen for disintegration of waste, CO₂ gas is harmless and odorless. The waste water is discharged to the ground through the secondary tank after chlorination then it is disinfected before discharge.

Function : Degradation of human waste into CO₂, water & carboneous matter.

Biological (Waste treatment) Tank Features:

- The waste treatment tank includes the following special design feature for application in Indian Railway condition.
- Tank Volume = 316 litre. Constructed by stainless steel with coat of special epoxy applied to the inside surface the tank.
- Patented synthetic filter media surface area in a tank = 10 Sq feet.
- Max. Capacity of water per flush (Pressurized water) = 2.5 litre / flush.
- Total flushes for each toilet per person per 24 Hrs. period is 05.
- Thus hydrologic loading is expected to be 224 litre of water + 720 ounces of normal human solid waste per day.
- The tank includes the screen tray to collect foreign objects such as bags papers, tin-can etc. This tray can be accessed from rail side Via a water tank access panel /door & cleaned periodically to remove any foreign non –bio degradable objects.
- The tank is fitted with a rinse port & rinsing device to facilitate cleaning of the integral debris tray that collects foreign objects & general tank rinsing during the recommended annual maintenance.
- An especially design flapper Valve is provided between the tank & toilet room to prevent odors.
- The tank is installed under the coach. The tank effluent discharge port is located at the bottom of the tank. The tank is designed to allow draining of the tank effluent out the view of the public eye & on to road between the rails.

- An out Position of tank primary chlorinator & primary chlorinator access cap for case of loading, chemical slugs & inspection of the content are provided. To prevent damage it has been provided with metallic cap.
- The tank is designed to accommodate 18 passengers per toilet when used with a toilet using not more than 2.5 liters water per flush.
- The tanks include two clean out ports in addition to clear/ translucent plug for visual detection of any foreign material.
- Tank discharge utilized 1.5 inches drain fitting.
- The tank is designed for human waste water and toilet paper only. All non organ materials must be removed.
- The material is a flexible thermoplastic elastomeric of Microspore – own custom/ proprietary design that is injection molded.

Micro- Culture :-

- It is a simple and effective product to enhance the system operation. It accelerates the breakdown of paper grease and other bio-degradable solids and minerals.
- Micro-Culture is a combination of naturally occurring bacterial strains that accelerates the breakdown of waste products. The organizers are non toxic, odorless and harmless to human, animal, plant marine life micro-culture bacteria are 100% active.
- Micro-culture must be used at a rate of **08 ounce** capacity **two** bags per tank every **90 days** to accelerate the biological process.
- Many users use paper, towels for wiping washed heavy paper seat cover in toilet room throws fast foods, napkins, feminine hygienic products/ news papers flash down toilet.
- The waste treatment slows down due to paper deposits. So adding micro-culture regularly will help to prevent treatment system tank back-ups.

Chlorinator assembly

- The primary chlorinator is located to the treatment tank and contains chemical slugs developed and patterned by Microspore that have been chemically and dimensionally designed for maximum effluent content and bacteria kill.
- The effluent flow through the primary chlorination and around the chemical slugs. The killing of bacteria in the effluent starts at this time. The bacteria free effluent then flow out by the discharge of the treatment tank assembly directly to the road bed atmosphere.

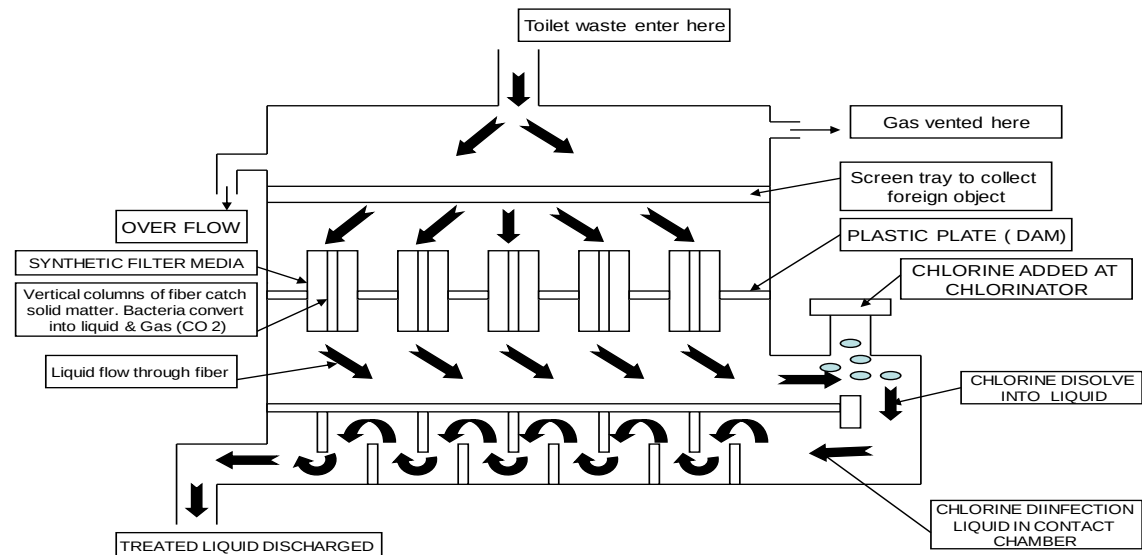
Biological Process

- The waste treatment tank, biological processes and treats the human waste and waste water into effluent drain continuously directly on to the road bed between the tracks.
- This system requires no handling of raw waste water and no manual draining.
- It requires the regular addition of the patented chemical (chlorine slug) for disinfections.
- The treatment tank utilizes microphor's new synthetic filtration media.

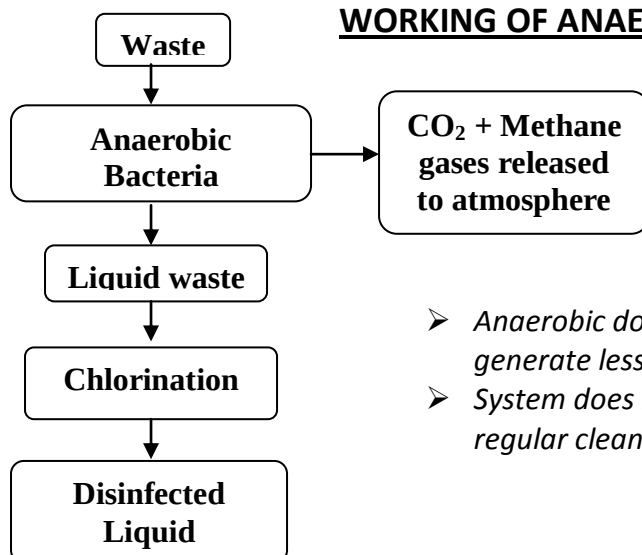
Benefits of Synthetic Media

- The new synthetic media has larger surface area than the tank body.
- The red wood bark media has an average cycle life of 5.7 years.
- Water consumption is low
- The synthetic media converts solid waste directly into gasoline and water form except foreign material.

Treatment Tank Operation



WORKING OF ANAEROBIC SYSTEM



- Anaerobic doubles its population in 6 to 8 hours; generate less bio-mass.
- System does not require Oxygen and also does not require regular cleaning

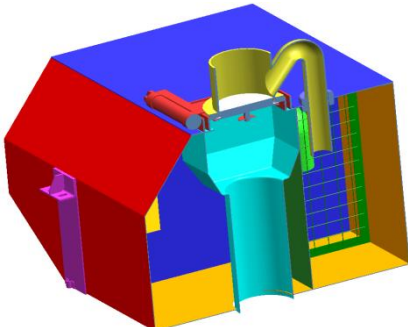
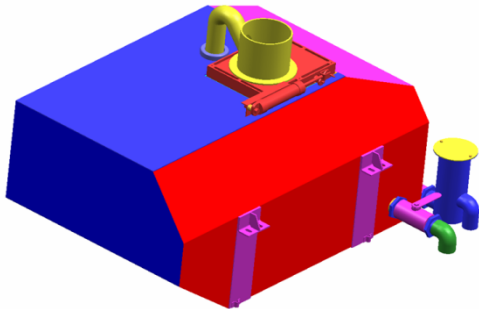
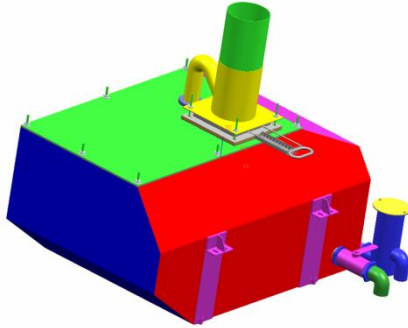
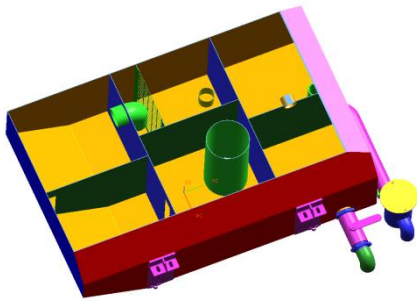
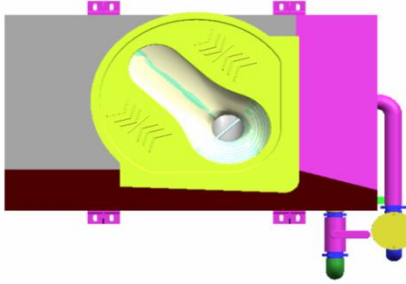
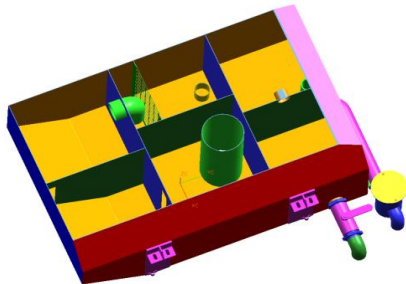
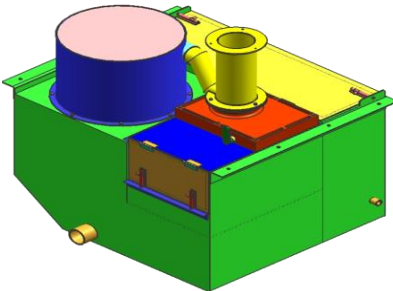
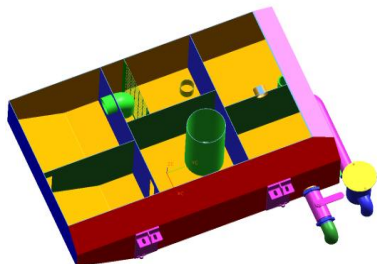
Limitations of initial design:

(1) Digester tank not properly designed causing. (2) Foul smell. (3) Lack of non-bio-degradable separation system. (4) Choking due to accumulation of non-bio-degradable

These toilet are basically of four types.

- (1) **System with flapper valve:** There is provision of P.L.C. which is operated by pushing a button which is provided in toilet room and pressurized water with opening of flapper valve is ensured.
- (2) **System with manual slider valve:** In this system there is no provision of P.L.C. and water flushing is ensured through gravity system and opening of slider valve is ensured manually.
- (3) **System with reduced Opening at inlet (50 mm):** In this system there is no provision of P.L.C. and water flushing is ensured through gravity system.

(4) System with Solid Liquid Separator: In this system there is no provision of P.L.C. and water flushing is ensured through gravity system.

System with flapper valve				
FEATURES	Pneumatics	Yes	 	
	Electrics	Yes		
	PLC	Yes		
	Flush	Pressurized		
Fail safe mode exist- chute system can be operated without dismantling of tank				
System with manual slider valve				
FEATURES	Pneumatics	no	 	
	Electrics	no		
	PLC	no		
	Flush	gravity		
Fail safe mode exist- chute system can be operated without dismantling of tank				
System with reduced opening at inlet for western style Hindware commode is proposed				
FEATURES	Pneumatics	no	 	
	Electrics	no		
	PLC	no		
	Flush	gravity		
Fail safe mode exist- chute system can be operated without dismantling of tank				
System with solid liquid separator				
FEATURES	Pneumatics	no	 	
	Electrics	no		
	PLC	no		
	Flush	gravity		
Fail safe mode exist- chute system can be operated without dismantling of tank				

Components of Biodegradable Toilet:

- 1) **Retention Tank:** The system is provided with 500 litre capacity of retention tank, which is mounted in under frame of coach body by means of four brackets below toilet room. In this system tank has been designed keeping in view of DRDE Bacteria. The fecal matter will pass through "U" trap to the tank. "U" trap bend towards inlet side of tank shall be slanted to achieve double seal and will also eliminate gas emission to the toilet room.

Entire tank is divided in to 6 no. of chambers by means of net grill. Anaerobic Bacteria those are developed by DRDE are put in all chambers. Bacteria increased their population double in 6 to 8 hours.

- 2) **Vent Hole:** In the retention tank CO₂ and CH₄ (Methane) gases are released from fecal matter. To exhaust these gases in atmosphere vent hole has been provided in the upper surface of the tank. Preferably vent hole is provided over first or last chamber of the tank.
- 3) **Flapper Valve:** One Flapper Valve is provided for water seal and to arrest foul smell. It will also act as passage to clear foreign materials such as bottles, polythene etc. This will also act as a fail safe mode in case of emergency. Since the system has a P.L.C. the flapper valve can be opened at a pre determined cycles say 8 to 10 cycles.
Note: Another version of Biological Toilet flapper valve with P.L.C. is replaced by Manual Slider Valve, which is operated from outside.
- 4) **P.L.C.:** This an electronic device which is arranged inside wall of toilet room. There is provision of rectifier with transformer to convert 110 V to 24 V D.C. for operation of P.L.C. system. In side toilet room a push button has also been provided. System is operated by simple manual pushing of button.
- 5) **Solenoid Valve:** It is provided with Flusher valve of toilet room, which is connected with P.L.C. system. During course of flushing by pushing button, it provides pressurized water through flusher pipe.
- 6) **Chlorinator Tank:** A small tank of one litre capacity is integrated beside of retention tank. It has two nos. of outlet for collecting samples for periodical testing. The tank exterior has been painted with heat shielding paint.
- 7) **Over Flow Drainage Pipe:** It is provided inside wall of tank. In case of overflow of tank disinfected liquid discharges to the track.

TESTING PROCEDURE: For testing samples are collected by DRDE(Defense Research & Development Establishment) and following parameters are checked.

- 1) P.H. Value
- 2) Total Solid
- 3) T.D.S.
- 4) C.O.D.

'H' TYPE CBC COUPLER

Coupler:

- The coupler provides a means of mechanically connecting individual adjacent vehicles in order to make a train.
- The couplers are located at both end of each vehicle.
- When connected with a coupler of an adjacent vehicle it allows the vehicle to move independently to accommodate track curvature and elevation change while remaining connected together.



Advantages of CBC:

- High hauling capacity permits to attach more No. of coaches.
- Rake fitted with CBC can be run with comparatively high speed.
- No need to provide separate Buffering Gears. Hence the incidence of Buffer interlocking is avoided.
- Coupling/Uncoupling can be done easily from track side.

"H" Type Tight Lock Coupler:

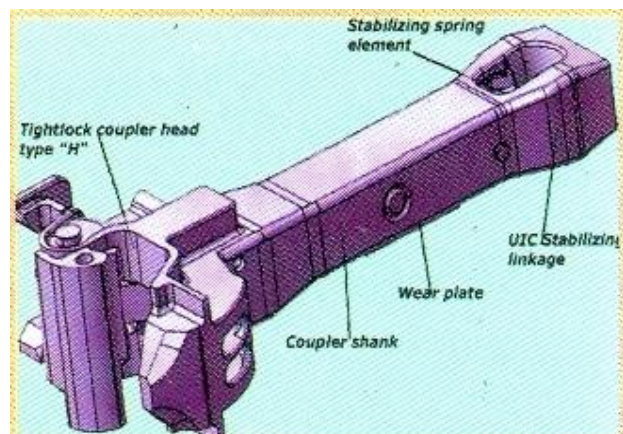
- The AAR "H" type tight lock couplers are used in LHB/ HYBRID coaches instead of screw coupler.
- The coupler is opened manually using the coupler operating rod and is closed automatically when the couplers on adjacent vehicle are mated.
- The coupler automatically locks when fully mated.

Main Parts Of "H" type CBC:

- "H" Type Tight Lock Coupler Head.
- Draft Gear.
- Supporting Device.
- Manual Uncoupling Device.

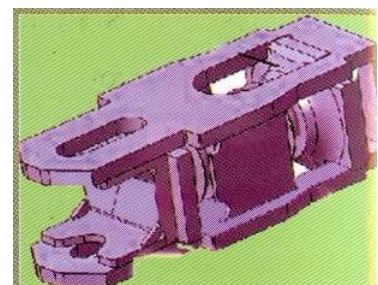
"H" Type Tight Lock Coupler Head:

- The coupler head has a shank/Draw bar casted together with the head.
- Parts like Knuckle, lock etc, are assembled in the coupler head to enable coupling and uncoupling.
- Tail end of the draw bar is provided with the UIC stabilizing link and connects to the draft gear through the central pin.
- A backlash compensation device is assembled in the shank at tail end of the coupler head.



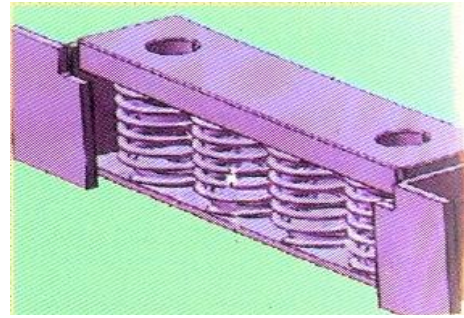
Draft Gear:

- The Draft Gear is a double acting device for energy absorption during coupling and services.
- The device is designed to fit into the draft gear pocket of the Coach and absorb the dynamic energy in both i.e. draw and buff modes.



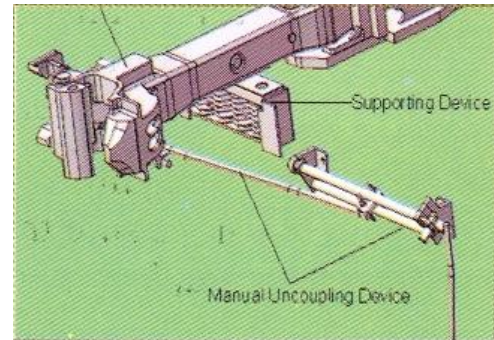
SUPPORTING DEVICE:

- Supporting device comprises of four preloaded springs.
- The device is placed on a platform and bolted to the car structure.
- The coupler head rests on the top wear plate of the supporting device and this device supports the coupler weight.
- Its main parts are- Wear plate, Frame, Compression spring, Nut & Bolts.



Manual Uncoupling device:

- The manual uncoupling device is mounted on one side near end wall of coach connecting the uncoupling mechanism on coupler head through the sliding rod.
- Handle of device is unlocked, lifted and then rotated in the clockwise direction for uncoupling.
- After coupling, locking of the handle has to be ensured to prevent unauthorized uncoupling.



TECHNICAL DATA:

- **Draft Gear capacity:**
 - Dynamic energy absorption capacity: 45 kJ (min).
 - End force : 1600 kN (max)
- **Stroke :** Tension- 58 mm, Compression- 80mm.
 - Pre load : 30 kN
- **Strength of Coupler Head with Draft Gear :**
Tensile load = 1000 kN, Compressive load = 2000kN.
Bending test of Coupler head: 300kN.
- Coupler length : 1030 ± 5mm
- Weight of coupler : 500kg.
- Gathering range: Horizontal: ± 110mm, Vertical : ± 90mm.
- Maximum horizontal swing coupler : ± 17.85 °
- Maximum vertical swing coupler : ± 7 °
- The maximum slack is restricted to 3.5 mm

Salient Features:

- Coupler head is a standard AAR 'H' type with backlash compensation device.
- 'H' type coupler provided with anti climbing feature.
- Coupling is possible under angular misalignment both horizontally & vertically.
- The coupler permits coupled trains to negotiate vertical and horizontal curves and allows rotational movement.
- Uncoupling can be achieved manually from track side by means of a combination of rod & levers.
- Draw Gear ensures cushioning effects in both buff & draft.

Check points for proper coupling:

- Knuckle and coupler head machined faces should be parallel.
- Telltale slot should be clear of Rotary Lock Lift Lever.
- Rotary Lock Lifter rib should be vertical.
- Securing Bolt of manual uncoupling device should be in locked condition. This is to prevent manual uncoupling inadvertently.



Coupled H type coupling

Check points for proper uncoupling:

- Securing Bolt should be in unlocked condition. Uncoupling device handle can be operated to uncouple.
- Rotary Lock Lift lever should be visible.
- Rotary Lock Lifter rib should not be vertical.
- Knuckle and coupler head machined faces should not be parallel.
- No oil grease should be applied on coupler parts such as Knuckle, Lock, and Rotary Lock Lifter etc.

Condition for coupling:

- Coupling is possible under any of the following conditions.
 - When both knuckles are in open condition.
 - When one knuckle is open & other is in closed condition.
- Condition will not permit coupling.
 - If both knuckles are in closed condition coupling will not take place.

AIR BRAKE SYSTEM

Air brake: The brake system in which compressed air is used in the brake cylinder for the application of brakes is called air brake.

Necessity to introduce the Air Brake in rolling stock :The existing vacuum brake has got its own limitations like brake, fading, increased application and release timings etc., In practice it is not reliable for trains running in higher altitudes due to insufficient vacuum levels in brake van and train engine.

So to overcome all these, it has become necessary to introduce Air brake system to control the speed of the train and to stop it within a reasonable distance, irrespective of length, load of the train distance covered and altitude of the train.

Advantages of air brake over vacuum brake system:

The advantage of Air brake over Vacuum brake is

- 1) Uniform brake power is possible throughout the train in airbrake but it is not possible in case of vacuum brake since the pressure drop at the rear of the train is up to 20%
- 2) The propagation rate of compressed air is 260 m/sec to 280 m/sec while in vacuum brake 60 to 80 m/sec.
- 3) The Air brake trains have potentiality to run longer than 600 meters length.
- 4) The air brake trains have potentiality to run heavier loads than 4500 tons.
- 5) Shorter braking distance.
- 6) Suitable for higher altitudes.
- 7) Compact and easy to maintain.
- 8) Consumption of spare parts is very less.
- 9) Simple brake rigging.
- 10) Quicker application and release, so better punctuality can be achieved.
- 11) Better utilization of rolling stock since less maintenance and pre departure detention.

Types of Brake System used in Indian Railway Rolling Stock:-

- 1) Single pipe graduated release & application conventional air brake system utilized in Goods Stock.
- 2) Twin pipe graduated release & application conventional system utilized in Coaching Stock.
- 3) Twin pipe graduated release & application Bogie Mounted Air Brake System at present utilized in more than 98% coaches in place of conventional twin pipe air brake system.
- 4) Twin pipe graduated release & application Axle Mounted Disc Brake System, utilized in LHB coaches made by RCF Kapurthala.

Types of Air Brake System:-

- 1) **Direct Release System :-** (100% creation & drop of pressure)
In direct release system, the brake cylinder pressure cannot be reduced in steps by increasing the brake pipe pressure in steps during release. The brakes are released immediately, as soon as releasing of brakes is initiated.
- 2) **Graduated Release System:-**
In this system the brake cylinder pressure can be reduced gradually in steps in proportion to the increase in brake pipe pressure.

NOTE: - In both types of the system brake application is directly proportional to the reduction in brake pipe pressure.

There are two types of Graduated Release Air Brake:-

1) Single Pipe Air Brake System.

2) Twin Pipe Air Brake System.

1) Single Pipe Air Brake System: -

There is only one pipe called brake pipe running from loco to the brake van in order to get continuity of air for the application & release of brakes.

i) At present running in goods stock.(Except latest developed BOX-N HL, BCN HL wagons)

ii) Releasing time is more 45 to 60 sec.

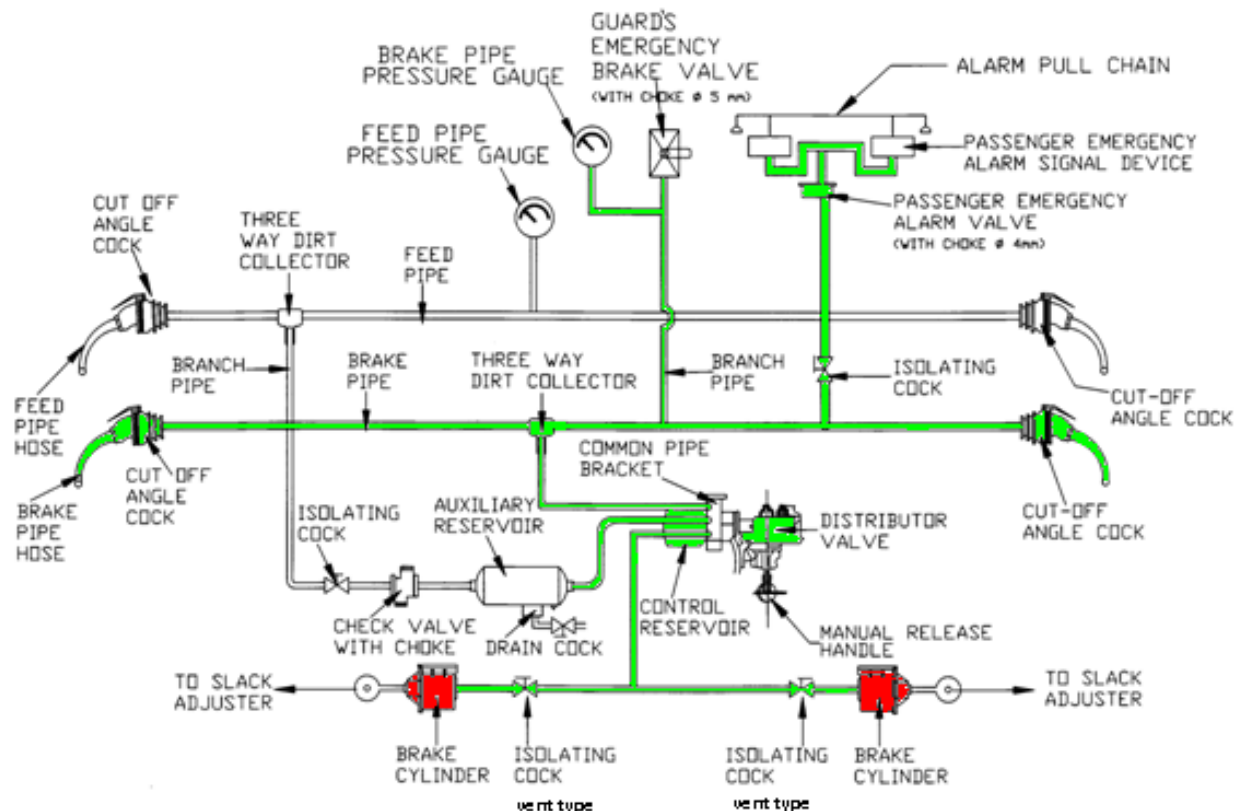
2) Twin Pipe Air Brake System :-

i) In addition to brake pipe, there is one more pipe called feed pipe , running loco to the brake van to charge the auxiliary reservoir continuously to 6 kg/cm².At present running in coaching stock and also in latest developed BOX-N HL, BCN HL wagons)

ii) Releasing time is less 15 to 20 sec

MAIN COMPONENTS OF AIR BRAKE SYSTEM(SINGLE PIPE)

- 1) Brake pipe – dia. 32 mm (for goods) & Pressure 5 kg/cm².
- 2) MU washer
- 3) Palm Coupling
- 4) Air Hose (Length: 660^{+6}_{-0} mm)
- 5) Cut Off Angle Cock
- 6) Grip Seal Joint
- 7) Branch Pipe (Dia.: 20 mm)
- 8) Isolating Cock
- 9) Dirt Collector
- 10) Common Pipe Bracket
- 11) Intermediate Piece
- 12) Distributor Valve
- 13) Control Reservoir
 - i) 6 liter (Goods): 5 kg/cm²
 - ii) 9 liter (Coaching): 5 kg/cm²
- 14) Auxiliary Reservoir
 - i) 100 liter (Goods): 5 kg/cm²
 - ii) 200 liter (Coaching): 6 kg/cm²
- 15) Manual Release Handle
- 16) Isolating Handle of DV
- 17) Drain Cock in DV
- 18) Branch Pipe for BC
- 19) Isolating Cock for BC
- 20) Brake Cylinder Dia – 355 mm (Pressure max. 3.8 ± 0.1 kg/cm²)



SCHEMATIC LAYOUT OF TWIN PIPE GRADUATED RELEASE AIR BRAKE SYSTEM

Extra Fitting For Coaching:-

- 1) Feed Pipe – Dia. 25 mm & pressure: 6 kg/cm²
- 2) Non Return Valve with Choke: dia. 3 mm
- 3) Isolating Cock & Dirt Collector for FP
- 4) Branch Pipe for PEAV & PEASD.
 - i) PEAV choke dia.: 8 mm (NEW)
 - ii) PEAV choke dia.: 4 mm (OLD)
- 5) Isolating Cock for PEAV & PEASD
- 6) Vertical Pipe for Guard Van Valve
- 7) Guard Valve with Handle & choke dia.: 8 mm
- 8) Guard Van Valve BP & FP gauge

Working Principle Of Air Brake System:

Under normal conditions the Brake pipe is charged with 5kg/cm² from the loco. The control reservoir and the Auxiliary reservoir are also charged with 5 kg/cm² from BP through Distributor valve in case of single pipe system. In twin pipe system the auxiliary reservoir is charged to 6 kg/cm² through feed pipe.

When the brake pipe is charged at 5 kg/cm² the brake cylinder is connected to exhaust through distributor valve in order to keep the brake in released position fully.

Whenever the brake pipe pressure is reduced below the CR pressure, the DV connects the auxiliary reservoir with the brake cylinder and the air from AR is sent into the brake cylinder to apply the brake. Whenever the brake pipe pressure is equal to CR pressure the DV disconnects the BC from AR and in turn connects the BC with Exhaust for the release of brakes fully.

The different processes involved in working of Air brake system:- The processes involved in working of Air brake are

- a) Charging
- b) Application
- c) Release
- d) Manual Release

During Charging:

- a) Brake pipe is charged at 5 kg/cm² by the drivers brake valve from the Loco.
- b) Feed pipe is charged with 6 kg/cm²
- c) AR is charged at 6 kg/cm² (Up to 5 kg/cm² is charged from both brake pipe and feed pipe. Beyond 5 kg/cm² & up to 6 kg/cm² it is exclusively charged from feed pipe.
- d) The CR is charged through the distributor valve at 5 kg/cm² from BP.
- e) During charging Brake cylinder is connected to exhaust through distributor valve to keep the brakes in released condition.

During Brake Application

The brake pipe is reduced in steps as given below

Sl. No.	Stages	BP pressure is reduced by
1	Minimum Reduction	0.5 to 0.8 kg/cm ²
2	Service application	0.8 to 1.0 kg/cm ²
3	Full service application	1.0 to 1.5 kg/cm ²
4	Emergency application	5.0 kg/cm ²
5	Release stage	No reduction (BP at 5.0 kg/cm ²)

When the brake pipe pressure is reduced in steps as shown above, the air from AR is sent into BC to a maximum pressure of 3.8 ± 0.1 kg/cm² during full service application as well as emergency application.

During minimum reduction and service application the admission of air from AR in to BC is directly proportional to the reduction in the BP pressure. (i.e. 2.5 times charging of BC as per reduction in BP but it is applicable only up to 1.5 kg/cm² of pressure drop in B.P.)

NOTE : Before AR is connected to BC the AR and CR are disconnected from BP and BC is also disconnected from Exhaust.

The AR is continuously charged to 6.0 kg/cm² during brake application by Feed pipe. The CR pressure should remain constant at 5.0 kg/cm². However there may be a little drop in CR pressure during brake application due to the design feature.

During The Releasing / Recharging:- During release the BP pressure is increased in steps. When the BP pressure is increased in steps the brake cylinder is disconnected from AR and in turn connected to exhaust. The air from Brake cylinder is released / vented progressively depending upon the increase in the brake pipe pressure. When the brake pipe pressure is brought to 5.0 kg/cm², air from the brake cylinder is completely exhausted and the brakes are released fully.

Manual Release:- Whenever the loco is detached BP pressure is brought to zero and brake application takes place due to the existence of CR pressure at the bottom of the main diaphragm. To release the brakes manually the hollow stem in the DV should be brought to the normal position by releasing the air from CR. To facilitate this, the release valve is given a brief pull which is provided at the bottom of the DV. During this operation the air from CR is released which in turn brings the hollow stem to the normal position to connect BC with exhaust for releasing of brakes.

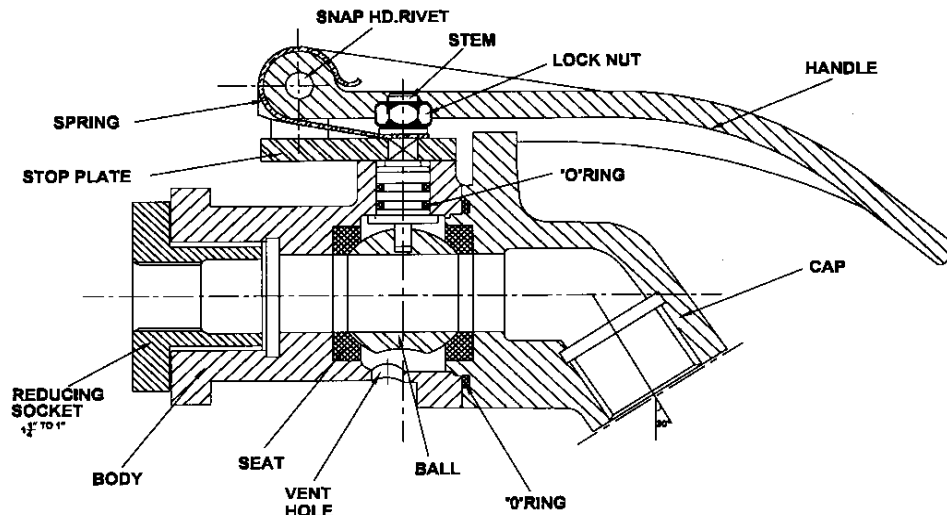
DIFFERENCE BETWEEN AIR BRAKE & VACCUME BRAKE SYSTEM:-

SI NO.	AIR BRAKE SYSTEM	VACCUM BRAKE SYSTEM
1.	Air Brake works on compressed air at 5 kg/cm^2 maintained in brake pipe.	Vacuum Brake works on atmospheric pressure at 1.03 kg/cm^2 .
2.	At the time of brake application compressed air is admitted into BC up to 3.8 kg/cm^2	In vacuum brake, air at atmospheric pressure is admitted.
3.	DV is the main functioning unit in the Air Brake System.	Vacuum Cylinder is the main functioning unit in the Vacuum Brake System.
4.	Brake application is caused by the outward movement of the Piston.	Brake application is caused by the inward movement of the Piston.
5.	BC is connected to AR during brake application & to exhaust during brake release through the DV.	VC is directly connected to the train pipe during brake application & release.
6.	No brake power fading.	There is always a brake power fading to the exhaust of 20%
7.	Uniform brake power is possible throughout the train due to the higher propagation rate of compressed air (260 to 280 m/sec.)	Uniform brake power is not possible due to the lower propagation rate of atmospheric air in the vacuum. (60 to 80 m/sec.)
8.	Air hoses are used to provide flexible connection between adjacent vehicles.	Hose pipe are used to provide flexible connection between two adjacent vehicles.
9.	Palm ends / coupling heads are used on the coupling side of air hoses.	Universal couplings are used on the coupling side of hose pipe.

10.	<i>For any reason, if the cylinder has to be made in operative, it can be conveniently done by closing the isolation cock.</i>	<i>If the cylinder has to be made inoperative, the train pipe nipple or the siphon pipe has to be dummied.</i>
11.	<i>MU washers are used to make air tight joints.</i>	<i>IR washers are used to make the air tight joints.</i>
12.	<i>Emergency braking distance is 632 m (4500 tones trailing load level track at 65 kmph speed).</i>	<i>Emergency braking distance is 1097 m (4500 tones trailing load level track at 65 kmph speed).</i>

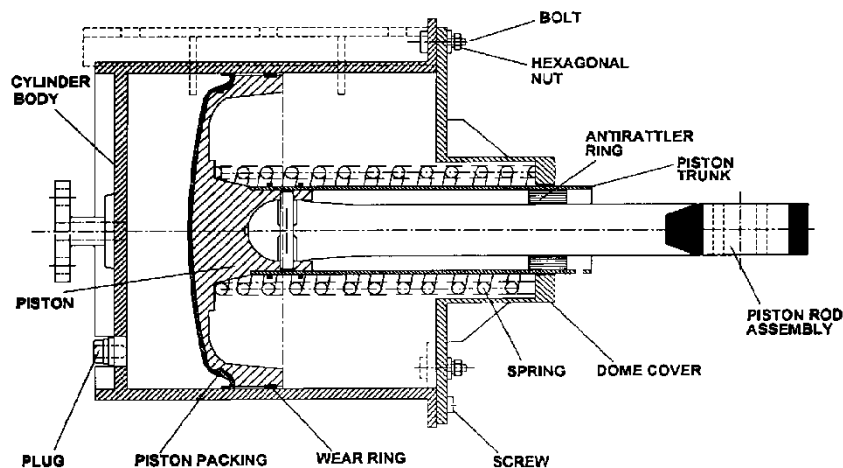
AIR BRAKE SUBASSEMBLIES

- 1) **Common Pipe Bracket** – Common Pipe Bracket is permanently mounted on the under frame of a vehicle. One face is mounted with distributor valve along with intermediate piece and other pace with control reservoir. The advantage of fitting a common pipe bracket is to remove the distributor valve for repair or replacement without disturbing the pipe connections.
- 2) **INTERMEDIATE PIECE** – Intermediate piece serves the purpose of blanking all the other parts on the common pipe bracket front face other than required for a particular make of distributor valve. Intermediate piece is mounted on the common pipe bracket face with a common gasket and the distributor valve is fastened to the intermediate piece.
- 3) **BRAKE PIPE HOSES** – In order to connect two successive wagons, the brake pipes (BP) installed on the under frame are fitted with flexible hoses. Length – min 666 mm , max – 784 mm
- 4) **BRAKE PIPE COUPLING** – To connect subsequent wagons, the hoses of BP are screwed to coupling and hose nipple by means of stainless steel ‘Bend it type clips. The coupling is designed in the form of palm end and hence also known as palm end coupling. Since a joint is formed at the coupling head, leakage may take place, through it. Therefore, it is necessary to subject the hose coupling of brake pipe to leakage test. The air brake hose coupling are provided in the brake pipe line.
- 5) **CUT OFF ANGLE COCK** – Cut of angle are provided on the air brake system to facilitate coupling and uncoupling of air hoses. When the handle is parallel to cut off angle cock, the air can easily pass through the cock. The position of the handle is known as open position. When the handle is placed perpendicular to the cock body, thereby closing the passage of air. This position of handle is known as closed position. The cut off angle cock is to be completely dismantled and overhauled during POH or when there is some specific trouble.

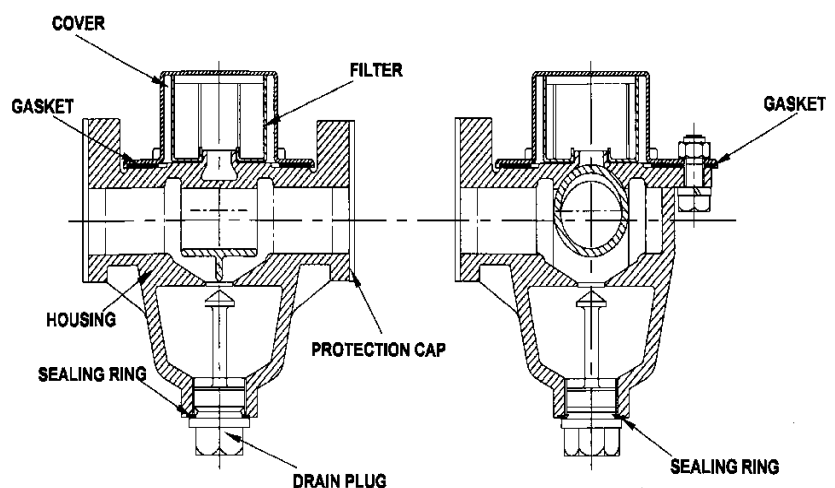


ISOLATING COCKS ARE OF 2 TYPES (I) WITH CHOKE (II) WITHOUT CHOKE

- 6) **BRAKE CYLINDER** – Brake cylinder is provided for actuating the brake rigging for the application and release of brakes. The brake cylinder receives compressed air from auxiliary reservoir after being regulated by the distributor valve and develops mechanical brake power by outward movement of its piston assembly.



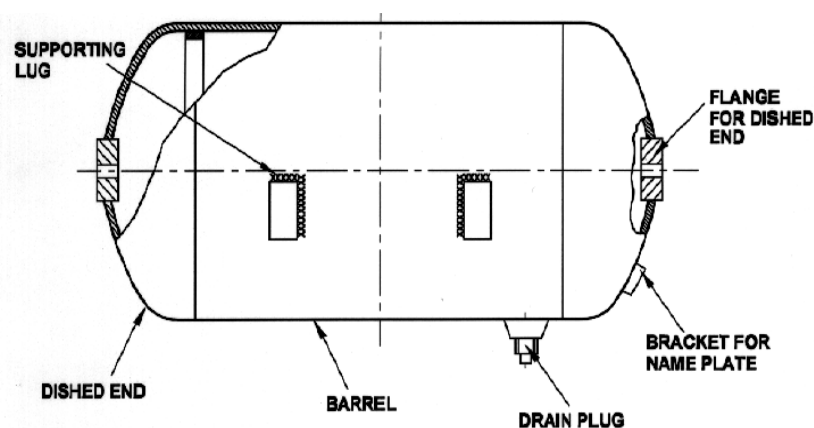
- 7) **DIRT COLLECTOR** – Dirt collectors are provided at the junction of the main pipe and branch pipe in both feed pipe and brake pipe. These are meant for removing dust, moisture and scale particles from air before it enters the DV & AR. This is achieved by centrifugal action.



8) **AUXILIARY RESERVOIR –**

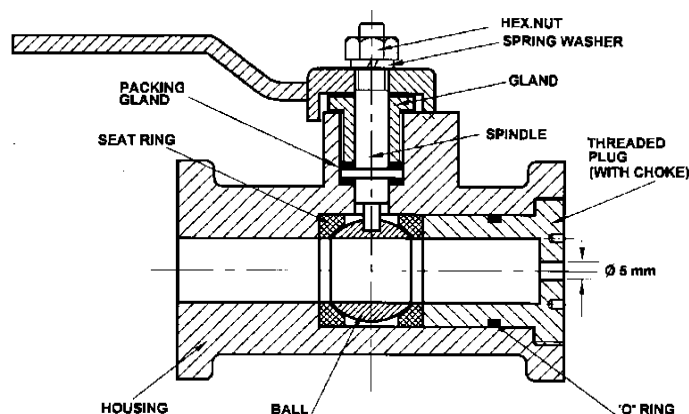
It is a pressure vessel and its function is to feed dry compressed air to the brake cylinder for application of brakes. AR is charged through brake pipe as well as feed pipe in coaching. Pressure in AR for goods single pipe – 5

kg/cm² and for Coach twin pipe 6 kg/cm². Capacity of AR in Goods, Coaches, BVZC are 100 ltrs, 200 ltrs, 75 ltrs respectively. No. of AR provided per wagon is 1 and No. of AR provided in Rajdhani Coaches per coach – 2 (2 x 100 ltrs. Is conventional) & 1 (1 x 100 in BMBC) Charging time of AR (6000 litre or one complete rake) 180 – 240 sec.



9) **GUARD'S EMERGENCY BRAKE VALVE**

- GEBV is provided in the Guards compartment. This valve provide a facility to the guard initiate brake application in case of any emergency. Dia of Guard van valve choke – 8 mm



10) **CONTROL RESERVOIR –**

The control reservoir is mounted on the under frame of coach very near to the DV and connected through a small chock and a pipe with the DV. During braking operation and releasing operation through recharging operation, the air pressure inside the control reservoir remains unaltered. Only during manual release the air of control reservoir escapes to the atmosphere through release valve of DV. Capacity of CR in per wagon and Coach are 6 litre and 9 litre respectively.

11) **DISTRIBUTOR VALVE –** DV is most important functional component of Air Brake System and is also some time referred to as the Brain of air brake system. It is connected to brake pipe AR & BC. It senses the BP pressure variations & work automatically to provide brake application as well as brake release. The distributor valve assemble consists of a valve body, a common pipe bracket and CR.

The DV mainly performs the following functions:-

- Charge the air brake system ie. CR & AR to regain pressure during normal running condition charging time (0 to 4.8 kg/cm²)

Type of DV	CR Charging		AR Charging	
	KEO	C ₃ W	KEO	C ₃ W
Old timing	160 ± 10	260 ± 20	170 ± 10	270 ± 30
New timing	185 ± 25	165 ± 20	140 ± 30	175 ± 30

- ii) *Helps in graduated brake application, when pressure in brake pipe is reduced in steps.*
- iii) *Helps in graduated brake release, when pressure in brake pipe is increased in steps.*
- iv) *Quickly propagates reduction of pressure in brake pipe throughout the length of train by arranging additional air pressure reduction locally inside the DV.*
- v) *Limits max. brake cylinder pressure up to $3.8 \pm 0.1 \text{ kg/cm}^2$ for full service/emergency application.*
- vi) *Controls the time for brakes application & brake releasing depending on service condition.*

ITEM	WAGON	COACH
<i>BC Filling time (from 0 to 3.6 kg/cm^2)</i>	<i>18 – 30 Sec.</i>	<i>3 – 5 Sec.</i>
<i>BC Release time (from 3.8 to 0.4 kg/cm^2)</i>	<i>45 – 60 Sec.</i>	<i>15 – 20 Sec.</i>

- vii) *Facilitates complete discharge of air from the air brake system manually with the help of QRV/DRV.*
- viii) *Protects overcharging of CR when brake pipe pressure is quickly increased for releasing brake*

S. No.	KEO	C₃W	Function
1	Three pressure valve	Main valve	Supply of requisite amount of pressure in brake cylinder, when B.P. pressure is reduced, also it provides passage for BC pressure to exhaust in atmospheres when BP pressure is raised.
2	U controller with U chamber	Quick service valve	The function of this part is to create initial pressure drop in brake pipe pressure by allowing a sudden entry of BP pressure into large volume bulb / U chamber at the start of brake application. This ensures rapid propagation of pressure reduction in BP through out the length of train
3	Minimum pressure limiter	Inshot valve	When application is initiated causes rapid charging of brake cylinder up to a determined pressure to overcome rigging resistances.
4	Max. pressure limiter	Limiting device	Restrict the max. BC pressure to $3.8 \pm 0.1 \text{ kg/cm}^2$
5	Check cover	Application of release chocke	Regulates application & release times.
6	A – controller	Cut off valve	Connects the B.P. to C.R. during charging & cut off the connection when B.P. pressure is reduced for brake application.
7	R – charger	AR check valve	Helps in charging the AR in addition to charging. It checks flow of F.P. into B.P. in dual pipe system & also checks back flow of air pressure when B.P. pressure is reduced for brakes application.
8	Quick release valve (QRV)	Double release valve (DRV)	Allows the brakes of a coach to be released by means of single brief pull to release wire, without any loss of air in A.R.

12) Passenger Emergency Alarm System

Passenger emergency alarm system consists of two components:

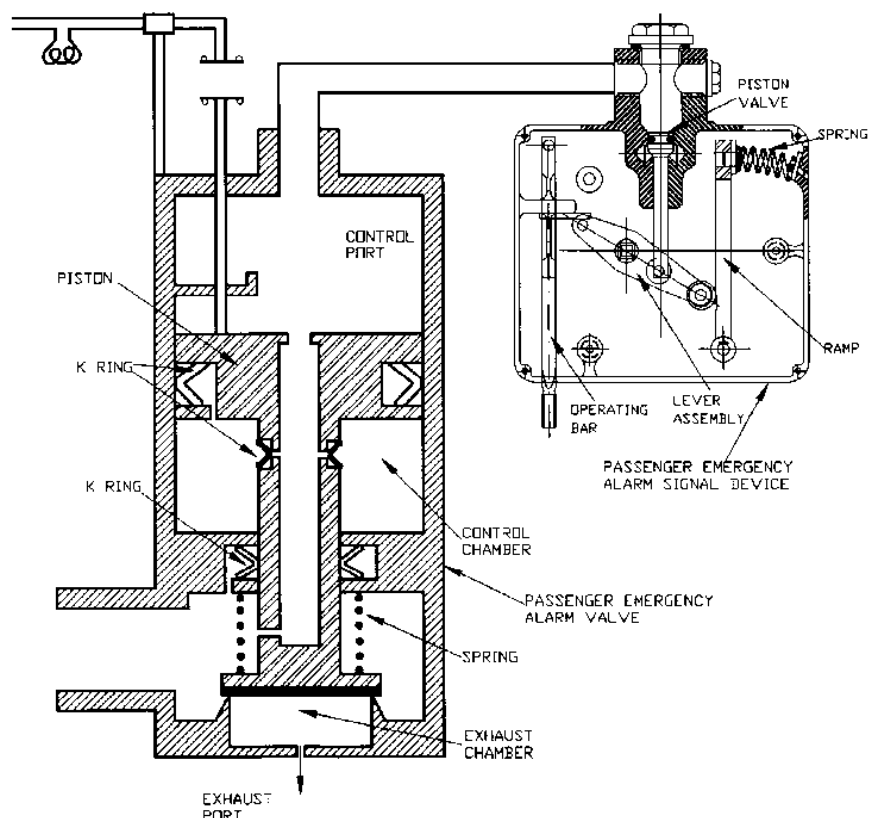
- A. Passenger Emergency Alarm Signal Device (PEASD).
- B. Passenger Emergency Alarm Valve (PEAV).

These two components in combination give an indication to the driver that any passenger is in need to stop the train. The indication is transmitted from the coach when the passenger pulls the chain

13) Passenger Emergency Alarm Signal Device

Passenger Emergency Alarm Signal Device (PEASD) is a manually operated pilot vent valve. It is operated through mechanical force exerted by pulling the alarm chain provided inside the coaches for emergency use.

The passenger emergency alarm signal device does not need any maintenance during normal service except when it is found damaged or is due for periodic overhauling.



Working: When ever chain is pulled to stop the train, the rocker arm provided in PEASD slides and actuate the pilot valve to open the atmospheric hole, as a result the BP pressure is reduced in PEASD. As the PEAV is connected in series with PEASD resulting BP pressure is also reduced in upper chamber of the PEAV. Due to variation of BP pressure in both the chamber of PEAV the main valve floats from its position & relate the lower chamber with the atmosphere after opening of the modified choke (8 mm dia.) this fast venting of pressure create reduction of pressure in BP pipe of the said coach & brakes take place to stop the train.

TROUBLE SHOOTING & REMEDIAL MEASURE

SI. NO.	CAUSE	REMEDY
1.	<i>MU (Multiple Unit) washer is damaged or displaced.</i>	<i>Change the damaged one or reset the displaced one.</i>
2	<i>Brake pipe joint leakage or Drain plug leakage.</i>	<i>Tighten all loose joint to stop leakage.</i>
3	<i>Hand brake in ON condition.</i>	<i>Check that Hand brake is in release condition.</i>
4	<i>Improper setting of empty load Device.</i>	<i>Ensure proper setting of empty load device i. e. loaded /Empty condition.</i>
5	<i>Horizontal beam's guide brackets jammed.</i>	<i>Ensure smooth working of horizontal lever.</i>
6	<i>Improper working of SAB.</i>	<i>Replace the faulty SAB with a good working SAB.</i>
7	<i>Exhaust port of DV is choked condition or DV not working properly.</i>	<i>Replace the faulty DV with a good working DV</i>
8	<i>Brake cylinder spring stuck up (commonly) or broken (rarely)</i>	<i>Release brake cylinder spring in proper position using the check not at the end of brake cylinder. Replace the broken spring with a new one.</i>
9	<i>Any part of brake rigging is in jammed condition.</i>	<i>Ensure smooth operation of all parts of brake rigging.</i>
10	<i>Brake beam stuck up in bracket beam pocket or brake beam hanger pin jammed.</i>	<i>Ensure free movement of brake beam in brake beam pockets.</i>
11	<i>Cut off angle cock is partially closed.</i>	<i>Cut of angle cock is opened fully.</i>

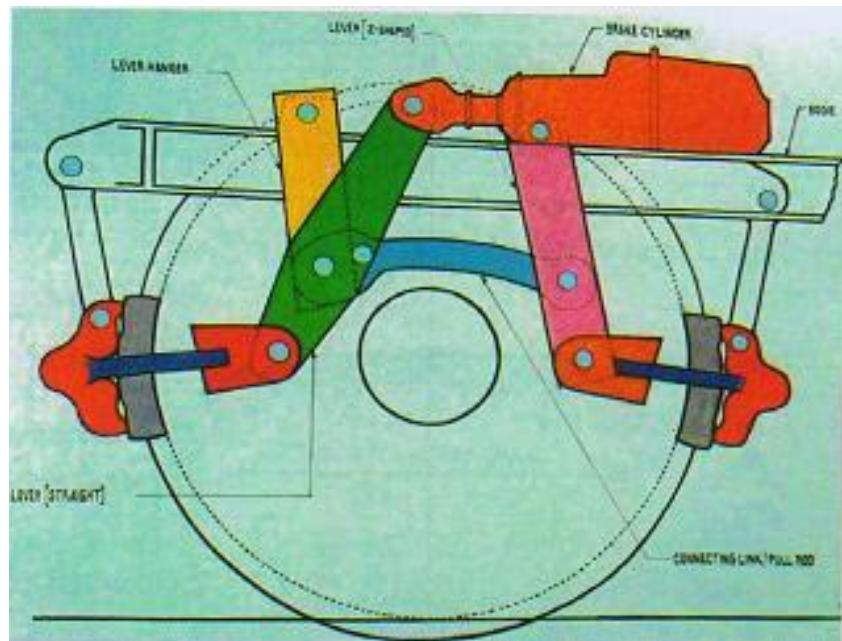
Bogie Mounted Brake System

Introduction

- In order to overcome the problem faced due to breakages and malfunctioning of SAB en-route and also due to the frequent breakages and replacement of cast iron brake blocks.
- The SAB is completely eliminated by providing the brake cylinder on the bogie itself, & frequent breakages and replacement of C.I. brake blocks are minimized by providing high friction composite 'K'-type brake blocks.

Advantages Of BMBS

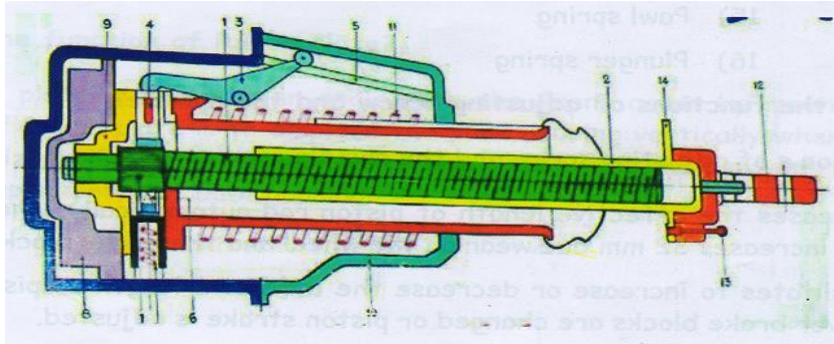
- In built single acting slack adjuster to take up a slack automatically.
- Weight is reduced and C.C. can be increased (weight is reduced 492 kg per coach as compared to conventional coaching stock)
- Number of pin joints is reduced 102 to 84.
- Fulcrum losses are reduced.
- Braking distance can be reduced at a speed of 110 km ph with 18 coaches is 800 m. (Conventional stock is 905 m.)
- Maintenance cost is low.
- Noise is reduced due to under frame mounting, SAB is eliminated.
- Mechanical efficiency is increased.
- Reliability of brake system is increased
- Speed of the train can be increased due to better controlling of train.
- Wheel Wear is reduced due to co-efficient of friction of 'K'-type brake block is 0.25.
- Mechanical advantage is increased.
- 13 T (Non-AC) 1 : 4 (in conventional) & 1 : 7.6 (In BMBC).
- 16 T (AC) 1 : 5.5 (In conventional) & 1 : 8.4 (In BMBC).
- Life of the brake block is increased. (5.5 to 6.0 times in comparison to C.I.).
- Mounting and dismounting of brake cylinder is easy during IOH & POH.
- Weight of the brake block is reduced resulted transportation and handling is easy.



Other Data of B.M.B.S. :-

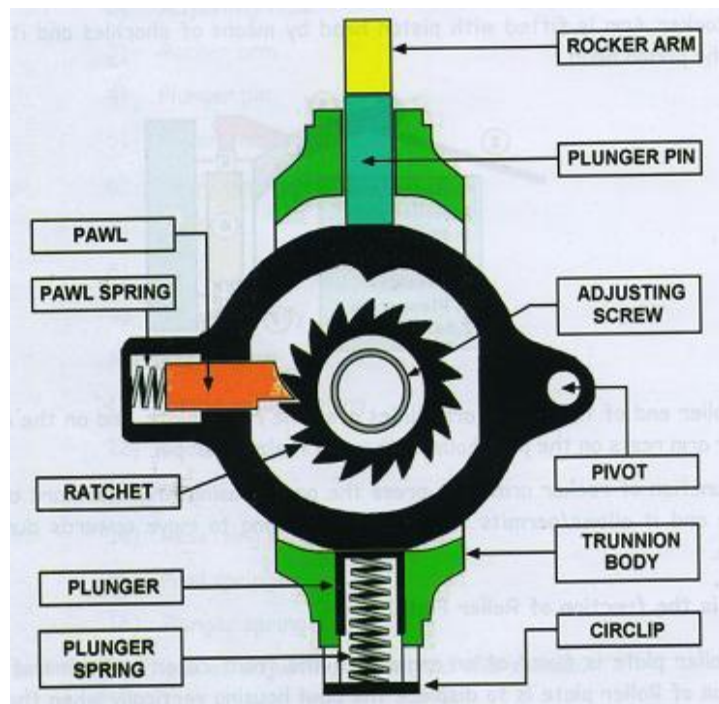
- Totally four numbers of 8" dia. (203 mm) size brake cylinders are used in place of two nos. 14" (355 mm) cylinder in convention air brake system.
- It has an inbuilt single acting slack adjusting capacity automatically to an extent of 305 mm slack adjustment whenever the piston stroke is increased more than 32 mm (clearance increased due to wear on brake block and wheel).

- These cylinders are mounted on central longitudinal members connecting the bogie transom and head stock on either side.
- Piston stroke should be maintained within 32 mm.
- The total number of teeth on adjuster ratchet is 18 nos.
- The circumferential displacement of ratchet in one stroke is by 20 degree.
- The liner displacement of adjuster tube in one stroke is only by 0.366mm.
- The liner displacement of adjuster tube in one complete rotation of ratchet is by 6.4mm.
- As a conclusion to adjust 25 mm of slack total 72 braking strokes are required.



MAIN COMPONENTS OF BMBC

1. Adjuster screw with Ratchet
2. Adjuster tube
3. Rocker arm
4. Roller plate
5. Pawl Housing Ring
6. Pawl
7. Piston
8. Trunnion Body
9. Front Cover
10. Piston Return Spring
11. Cross Head
12. Latch
13. Resetting Plate
14. Pawl Spring
15. Plunger Spring
16. Ratchet



Precautions to be followed while maintaining the BMBC.

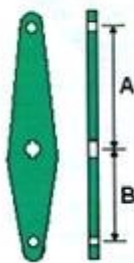
- Ensure the bogies are provided with high friction K type composite brake blocks.(as the coefficient of friction of Composite Brake Block L – Type is 0.17, K - type is 0.25 & for Cast Iron Brake Block it is 0.12).
- Ensure that floating lever, Z-arm are not interchanged between AC / Non-AC coaches.
- Ensure connecting link (Curved Pull-rod) is not interchanged between AC / Non-AC coaches.

- Whenever wheel dia. Is reduced below 839 mm, ensure the curved pull rod hole is shifted to next inner hole.
- Ensure the pull rod is not reversed.
- Ensure 38 mm packing is given in between dash pot and axle box wing whenever wheel dia. Is reduced to 839 to 813 mm.
- Whenever red mark is seen on the adjusting tubes replace all the brake blocks since further take up of clearance is not possible. If slack take up feature is not possible then adjuster tube to be extended to outside by disengaging of latch provided with resetting plate
- Snout out position should be kept on 3 O' Clock to 9 O'clock (Old 6 O' Clock to 12 O' Clock)

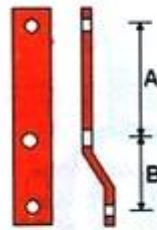
The following parts are used in the Brake Rigging.

- Lever- Straight.
- Lever - Z shapec
- Lever Hanger
- Connecting Link.

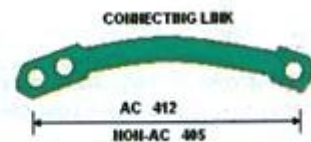
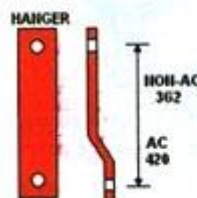
DIMENSIONS OF BRAKE RIGGING -BMBS



LEVER -STRAIGHT	A	B
NON-AC	302	158
AC	366	174



LEVER- Z SHAPED	A	B
NON-AC	256	134
AC	311	148



WORKING PRINCIPLE

- Whenever driver applied the brakes, piston is charged at 3.8 kg/cm^2 of pressure.
- Piston assembly started to move in forward direction.
- There will be no any change in the position of adjuster ratchet if the piston stroke is within 32 mm.
- If the piston stroke is exceeded more than 32 mm, in return stroke the fulcrum of the rocker ram is change resulted the pressure on plunger pin is released and pawl housing ring is started to rotate in clock direction due to release of pressure, mean while ratchet also rotate on its axis and change circumferential position by 20 degree (shifting by one teeth only) and then locked by pawl.
- Due to change of this clock wise position of ratchet ,the adjuster tube is linearly displaced outward at a rate of 0.366 mm per stroke and locked it in permanent feature resulted the gap between wheel and brake block is reduced.

FUNCTIONS OF MAIN PARTS

(1) ADJUSTING SCREW AND ADJUSTING SPINDLE

- The adjusting screw is connected with a ratchet and forms a single unit. The adjusting screw is provided with double starts thread with a pitch of $1/8"$ (3.15mm). The ratchet is provided with 18 teeth.
- The function of adjusting screw is to move the adjusting tube forward to increase the effective length of piston rod automatically and also decrease/increase the effective length of piston rod manually. When the adjusting screw completes one full rotation.

It makes adjusting tube to move forward by $2 \times \frac{1}{8}'' = \frac{1}{4}''$ (6.3mm)

- If the ratchet is moved/rotated by one teeth the adjusting screw is rotated by $\frac{360^\circ}{18} = 20^\circ$, which in turn moves the adjusting tube outward/forward by $\frac{1}{4}'' = \frac{1}{72}''$ (0.33mm). It is clear that to move the adjusting tube forward automatically by $\frac{1}{4}''$ it requires 18 return strokes or to move the adjusting tube forward by 1'' it requires 72 return strokes.

(2) ROCKER ARM:-

Rocker arm is fitted with piston head by means of shackles and it moves along with y=the piston head.

The roller end of the rocker arm slides over the roller plate and the other end of the rocker arm rests on the pawl housing through plunge pin. The function of rocker arm is to press the pawl housing downward during return strokes and it allows the pawl housing to move upwards during forward stroke.

(3) ROLLER PLATE:-

The roller plate is fixed at an angle with the front cover by means of bolts. The function of roller plate is to displace the pawl housing vertically when the rocker arm moves horizontally. It converts the linear displacement of pawl housing.

(4) PAWL HOUSING RING/PAWL:-

The pawl housing ring is fitted with pivot pin of trunnion body at one end and the other end of the pawl housing ring moves freely.

A spring loaded pawl is housed at the free end of the pawl housing.

At the bottom of the pawl housing, a spring loaded plunger is kept between trunnion body and the pawl housing to move the upward during forward stroke.

The function of pawl housing, the pawl is to move the ratchet by one tooth, whenever piston stroke exceed 32mm to increase the effective length of the piston rod during return stroke.

DIFFERENCE BETWEEN SAB & INBUILT SLACK ADJUSTER OF BMBC:

SAB	BMBC
➤ It is double acting. It can both take up & pay out the clearance automatically between the wheel and brake block.	➤ It is a single acting. It can only take up the clearance automatically between the wheel and brake block.
➤ The effective length of pull rod is decreased during take up the clearance between wheel & brake block.	➤ The effective length of piston rod is increased during take up the clearance between wheel & brake block.
➤ The effective length of pull rod is increased or decreased with reference to the control rod 'A' dimension.	➤ The effective length of piston rod is increased only when the piston stroke exceeds the pre-determined on working stroke of 60mm.
➤ It maintains a uniform piston strokes for all the cylinders throughout the formation.	➤ The piston stroke of the cylinders is not uniform throughout the formation and varies up to 60mm.

➤ Spindle is made up of triple start thread.	➤ Adjusting screw (spindle) is made up of double start thread.
➤ To adjust the slack, the length of the pull rod is increased or decreased during forward stroke.	➤ To adjust the slack the length of the piston rod is increased during return stroke.
➤ When length of the pull rod increased manually the clearance between the wheel & brake block increases.	➤ When the effective length of piston rod is increased manually, the clearance between the wheel & brake block decreases.
➤ It does not require adjustment of piston stroke every trip.	➤ Every trip the piston stroke requires to be adjusted.

DIFFERENCE BETWEEN UNDER FRAME MOUNTED & BOGIE MOUNTED BRAKE SYSTEM:

Sl. No.	Description	U/F Mounted brake System	Bogie Mounted brake System
1	Slack Adjuster	External	Internal
2	Type of Slack Adjuster	Double Acting	Single Acting
3	Capacity of Slack Adjuster	450 mm	305 mm
4	Size Brake cylinder	14 inches	08 inches
5	No. Of Cylinders	02 per coach	04 per coach
6	Brake force available on brake head	3 Tonne (Non Modified) 2.2 Tonne (Modified)	1 Tonne
7	Type of Brake Block	Low friction Composite L – Type Brake Block	High friction Composite K-Type Brake Block
8	Coefficient of Brake Block	0.16 – 0.18	0.28 – 0.30
9	Thickness of Brake Block	60 mm	50 mm
10	Piston Stroke	80 – 100 mm(Non Modified) 60 – 70 mm (Modified H/L)	Working stroke – 32 mm
11	Capacity of Truss Beam	16 Ton	13 Ton
12	Weight of Brake Block	3.06 kg	2.5 kg
13	Anti Vibration Bracket	Required	Eliminated
14	Horizontal Lever	Required	Eliminated
15	Bogie Pull Rod	Required	Eliminated
16	Life of the Brake Gear Components including Wheel	Less	More
17	Number Brake Gear Adjustments	07	02

Testing of Distributor Valve:

S. No.	Description of Test	Results Required	Observation	
1	AR Charging Time from 0 to 4.8 kg/cm ² (Main Reservoir pressure > 7.5 kg/cm ²)	270 ± 30 sec for C ₃ W (170 ± 10 sec for KE)		
2	CR Charging Time from 0 to 4.8 kg/cm ² (Main Reservoir pressure > 7.5 kg/cm ²)	260 ± 20 sec for C ₃ W (160 ± 10 sec for KE)		
3	Leakage Test (Brake Release). Check DV Leakage by Soap water only at joints.	No Leakage		
4	FULL SERVICE APPLICATION & RELEASE			
	BC filling time from 0 to 3.6 kg/cm ²	3 to 5 seconds		
	Maximum Brake Cylinder Pressure	3.8 ± 0.1 kg/cm ²		
	Leakage Test (Application). Check DV Leakage by Soap water only at joints.	No Leakage		
	BC Release time from 3.8 ± 0.1 kg/cm ² to 0.4 kg/cm ²	15 to 20 seconds		
5	OVERCHARGE PROTECTION (BP pressure 6 kg/cm ²)	CR pressure should not increase by more than 0.1 kg/cm ² in 25 sec.		
	CR overcharge reduction test. Overcharge CR to 5.7 kg/cm ² and pull double release lever for 3 seconds.	Overcharged CR should come to regime pressure of 5 kg/cm ²		
6	EMERGENCY APPLICATION		Single pipe	Twin pipe
	BC filling time from 0 to 3.6 kg/cm ²	3 to 5 seconds		
	Maximum Brake Cylinder Pressure	3.8 ± 0.1 kg/cm ²		
	Leakage Test (Emergency). Check DV Leakage by Soap water only at joints.	No Leakage		
	BC Release time from 3.8 ± 0.1 kg/cm ² to 0.4 kg/cm ²	15 to 20 seconds		
7	SENSITIVITY & INSENSITIVITY			
	BP pressure drop at the rate of 0.6 kg/cm ² in 6 Seconds	Brake should start applying within 1 sec.		
	With a pressure drop stopped immediately after the operation of QRV.	Brakes must remain applied.		
	BP pressure drop at the rate of 0.3kg/cm ² in 60 Seconds	Brakes must not apply.		

8	REFEEDING		
	Create leak in BC through a 2 mm choke	BC pressure should decrease initially but re-feeding should be available and BC pressure should get stabilized at some pressure	
9	GRADUATED APPLICATION Decrease BP pressure in steps as below BP Pressure (Kg/cm ²) 4.8 4.6 4.4 4.2 4.0 3.8 3.6		B. C. Pressure
	Continue Graduated Application until max. BC Pressure is obtained	BP pressure drop must be between 1.4 and 1.6 kg/cm ²	
	BP Pressure at maximum brake application	BP pressure drop must be between 3.4 and 3.7 kg/cm ²	
10	GRADUATED RELEASE Increase BP pressure in steps as below BP Pressure (Kg/cm ²) 3.6 3.8 4.0 4.2 4.4 4.6 4.8		BC Pressure
	Check BP Pressure when BC pressure is 0.4 kg/cm ² (Recharging pressure to release BC Fully)	4.85 kg/cm ² approx.	
11	QUICK RELEASE TEST		
	Apply emergency brake & pull briefly the double release valve lever	BC & CR are automatically exhausted to zero	

	CR check valve reset test. Start recharging of the system	CR should be isolated from atmosphere when BP pressure exceeds 0.2 kg/cm ²	
	Twin pipe operation Repeat the test : i) Full service application & Release. ii) Emergency application.	Fill in column against test nos. 3 & 6	

AIR BRAKE TESTING:-

- 1) Engine Testing
- 2) Continuity Test
- 3) Rake Test
- 4) SWTR / SCTR

1) ENGINE TESTING :- (RDSO – Pamphlets 9408)

With the use of 7.5 mm diameter test plate (master test plate gauge) .

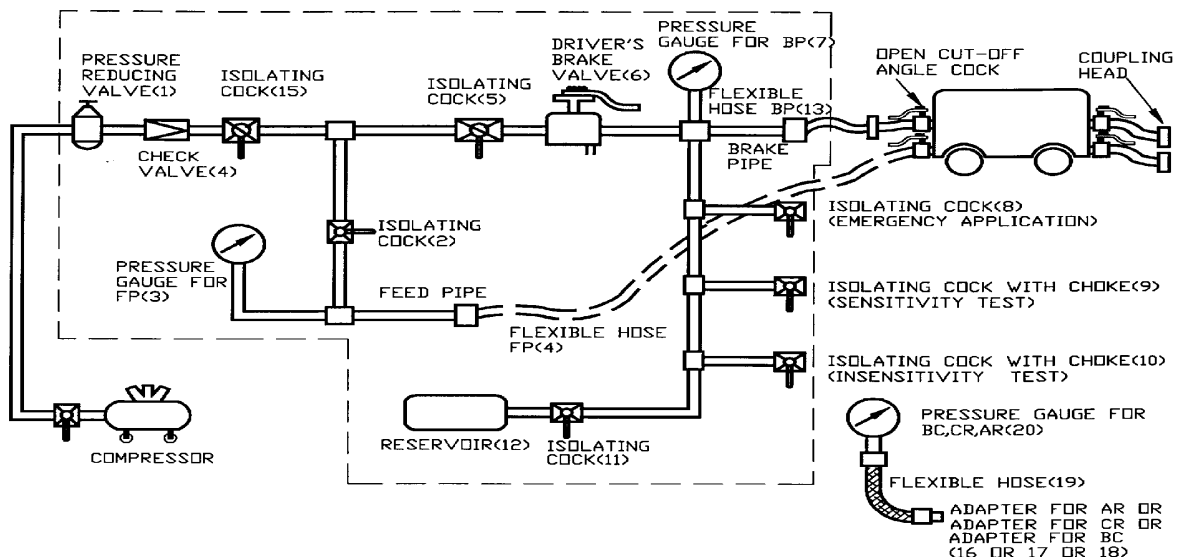
- **Object:-** To check the leakage in engine & compression capacity of engine.
- **When to be Conducted:-** If there is any dispute between C & W and Loco department regarding creation of requisite pressure in engine as well as in the rake.
- **Procedure:-**
 - i) Detach the engine from rake.
 - ii) Ensure –
 - a) MR pressure = 8 to 10 kg/cm²
 - b) BP pressure = 5 kg / cm²
 - c) FP pressure = 6 kg/cm²
 - iii) Ensure pressure in BP at 5 kg/cm² & open the cut off angle cock from the both sides (front & rear) to drop the BP pressure suddenly to wash out the system.
 - iv) Close the angle cock & charge the BP again at 5 kg/cm².
 - v) Apply master test plate gauge with rear side of the BP hose palm.
 - vi) Open the cut off angle cock of same BP hose & ensure the pressure in master test plate gauge. After 60 sec not to be more than 1 kg/cm² or gauge reading should not be less than 4 kg/cm².
 - vii) If drop is more than 1 kg/cm², it indicates that said engine is leaking or having low compression capacity.
 - viii) Engine must be detached from the rake & other engine is required for train operation.
 - ix) Same procedure can be adopted to check the leakage in FP.
- **Joint Committee Members:-**
 - i) On examination station – LI + CWI + Dy. SS
 - ii) On road side station – Driver + Guard + TXR
 - iii) In Goods yard – SSE/ SE + LI + AYM / CYM.

2) CONTINUITY TEST:-

To be conducted by Guard & Driver.

- **Object:-** To check the continuous flow of air from engine to last vehicle (or any floating obstruction)
- **When To Be Conducted:-** Before departure of primary / secondary rake from platforms or if any attachment / Detachment of engine or rolling stock in the road side station.
- **Procedure:-**
 - i) After completion of rake on plat form, before departure guard & driver jointly conducted the above test to ensure flow of air.
 - ii) At first guard & driver will communicate with walkie- talkie or telephone to perform the continuity test.
 - iii) After that guard will drop 1.0 kg/cm² of BP pressure from last vehicle & ask driver for same pressure drop is sensed in engine.
 - iv) If yes, now driver will drop total amount of BP pressure in engine & same is asked by the guard.
 - v) After drop assistant driver & guard will come down from the train and ensure brake application especially in at least two or three vehicles near to engine and rear brake van.
 - vi) If brakes are applied same should be released (in two or three vehicles) & communicate each other.
 - vii) At last requisite pressure is created in engine and brake van & train is ready to go.

3) SWTR / SCTR:-



- **When To Be Conducted:-** Single wagon / coach test is done only , if there is any major repair / replacement is carried out with air brake components or during schedules that is IOH or POH / ROH. (Except air hose, MU washer) if any of the part is repaired or replaced SWTR or SCTR is done).

➤ **Necessary Equipments:-**

- i) Test rig fitted with all necessary equipment.
- ii) Dummy palm with gauge (separate for BP & FP).
- iii) CR, AR, BC are tested with pressure gauge with flexible hose & adopter.
- iv) Stop Watch.
- v) Measuring scale.
- vi) Teflon tap & soap solution.
- vii) Tool kit with different types of tools.
- viii) Compressor (minimum operating pressure is 7 kg/cm^2)

Test Procedure:-

- i) Visual examination.
- ii) Adjustment of 'A' dimension & piston stroke for correct brake power application.
- iii) At rear end, dummy palm end with gauge is used.
- iv) Attach & join the coach with the test rig.
- v) Join test rig with compressor.
- vi) Connect the pressure reducing valve at compressor end with test rig.

Proforma for SWTR/ SCTR

Coach No. _____

Type Of DV _____

BP Pressure _____

FP Pressure _____

Sl. No.	Checks	Specified	Result (Actual)
1.	Leakage		
	a) BP pressure	$0.25 \text{ kg/cm}^2/\text{minute (max.)}$	
	b) FP pressure	$0.25 \text{ kg/cm}^2/\text{minute (max.)}$	
2.	a) Brake Cylinder Filling Time (from 0 – 3.6 kg/cm^2)	3 to 5 sec. (Coaching) 18 to 30 sec (Goods)	
	b) Maximum BC Pressure	$3.8 \pm 0.1 \text{ kg/cm}^2$	
3.	Brake Cylinder Release Time (From $3.8 - 0.4 \text{ kg/cm}^2$)	15 to 20 sec. (coaching), 45 to 60 sec. (goods)	
4.	Sensitivity & Insensitivity		
	a) Brake application when BP pressure is reduced at the rate of 0.6 kg/cm^2 in 6 sec.	Brake should be applied	
	b) Brake application when BP is reduced at the rate of 0.3 kg/cm^2 in 60 sec.	Brake should not be applied	
5.	Emergency Brake Application & Release		
	a) Maximum brake cylinder pressure	3.8 kg/cm^2	
	b) Manual release of brake cylinder after emergency	Brake cylinder get fully released	

6.	<i>Graduated Application & Release</i>		
	<i>a) Application</i>	<i>BP pressure decrease in steps & BC pressure builds up in steps</i>	
	<i>b) Leakage in BC pressure in 5 min. after emergency application</i>	<i>0.1 kg/cm² in 5min.</i>	
	<i>c) Release</i>	<i>BP pressure increases in steps & BC pressure reduced in steps</i>	
7.	<i>Piston stroke</i>	<i>For Chg. :- 90 ± 10 mm</i>	
		<i>For Goods:-</i>	
		<i>Empty:- 85 ± 10 mm</i>	
		<i>Loaded:- 130 ± 10 mm</i>	
8.	<i>Passenger Alarm System</i>		
	<i>a) Pull the Handle</i>	<i>i) Air to exhaust from PEAV & PEASD</i>	
		<i>ii) Brake should apply</i>	
		<i>iii) Coach indication light should glow</i>	
	<i>b) Reset PEASD</i>	<i>Exhaust of air should stop & indicating light should extinguish</i>	
9.	<i>Guard's Emergency Valve Test</i>		
	<i>1. Operate guard's valve handle</i>	<i>i) BP to exhaust</i>	
		<i>ii) Brake should apply</i>	
	<i>2. Normalize the handle</i>	<i>Air exhaust to stop</i>	
	<i>3. Testing of BP & FP gauge in guard's compartment</i>	<i>Gauge should register variation in BP& FP pressure</i>	

Signature:- - JE/ SSE

Testing Supervisor:-

4) RAKE TESTING:-

The different activities / tests to be performed during rake test-

- A Carry out visual examination*
- B Prepare of test rig for rake test*
- C Leakage, service application & release test*

Proforma For Rake Test

Coach No. _____

Type Of DV _____

BP Pressure _____

FP Pressure _____

Sl. No.	Checks	Specified	Result (Actual)
1.	Leakage		
	c) BP pressure	0.25 kg/cm ² /min (max.)	
	d) FP pressure	0.25 kg/cm ² /min (max.)	
2.	Service Application & Release Test		
	a) Brake application when BP pressure reduced to 1.5 kg/cm ²	Brake should apply	
	b) Observe piston stroke of Brake Cylinder	Piston in applied position	
	c) Record the piston stroke i) For Coaching , ii) For Goods – (Empty), (Loaded)	90 ± 10 mm , 85 ± 10 mm , 130 ± 10 mm	

OTHER DATA

1. Application time for the Coaching stock – 3 to 5 sec.
2. Application time for the Goods stock – 18 to 30 sec.
3. Releasing time for the Coaching stock – 15 to 20 sec.
4. Releasing time for the Goods stock – 45 to 60 sec.
5. Charging time of AR (6000 litre) – 180 to 240 sec.
6. Dia. Of BP and FP in Coaching – 25 mm
7. Dia. Of BP in Goods – 32 mm
8. Dia. Of Branch pipe in Coaching and Goods – 20 mm
9. Dia. Of Branch pipe from PEAV to PEASD – 10 mm
10. Dia. Of BC in BVZC – 305 mm
11. Dia. Of BC Goods & Coach – 355 mm
12. Dia. Of BMBC – 203 mm
13. Dia. Of BC in Disc brake system – 10" (256 mm)
14. Dia. Of Disc in Disc brake system – 640 mm
15. Capacity of CR in Goods – 6 liter
16. Capacity of CR in Coaching – 9 liter
17. Capacity of AR for goods – 100 liter
18. Capacity of AR for Coaching – 200 liter
19. Capacity of AR in BVZC – 75 liter
20. Pressure in B.P. – 5 kg / cm²
21. Pressure in F. P. – 6 kg/cm²
22. Pressure in CR – 5 kg/cm²
23. Pressure in AR for Goods single pipe – 5 kg/cm²
24. Pressure in AR for Coach twin pipe – 6 kg/cm²

25. Pressure in BC – $3.8 \pm 0.1 \text{ kg/cm}^2$
26. Number of dirt collector in Coach – 2
27. Number of dirt collector in Goods – 1
28. Number of isolating cock in a coach – 5 (max)
29. Number of isolating cock in a wagon – 1 (max)
30. Piston stroke Goods – $85 \pm 10 \text{ mm}$ (Empty) $130 \pm 10 \text{ mm}$ (Loaded)
31. Piston stroke coaching(ICF) – $90 \pm 10 \text{ mm}$
32. Piston stroke Coaching(LHB) – 10 to 12 mm
33. A – dimension for 13 t – $16 \pm 2 \text{ mm}$ 16 t (Coaches) – $22 \pm 2 \text{ mm}$
34. A – dimension in BOXN – $70 \pm 2 \text{ mm}$ BOX – $50 \pm 2 \text{ mm}$
35. E – dimension in Coaching – $375 \pm 25 \text{ mm}$, Goods – 555 - 575 mm
36. Dia. Of NRV – 3 mm
37. Dia. Of PEAV – 8 mm
38. Dia. Of Guard Van Valve – 8 mm
39. Max. Pressure of MR – 8 to 10 kg/cm^2
40. No. of cut off angle cock per wagon – 2
41. No. of cut off angle cock per Coach – 4
42. No. of PEAV per Coach – 1
43. No. of PEASD per Coach – 2
44. No. of Guard van valve – 1
45. Position of isolating cock –
 - A. Vertical - DV is working
 - B. Horizontal - DV isolated
46. Minimum time required for pulling manual release handle of DV – 10 sec. at 15^0
47. Schedule – Monthly for air brake coaching
 - 'J' - Manual release, Micro switch testing
 - 'K' – Alarm chain test, Guard van valve testing – 2 month

BRAKE CYLINDER DIAMETER

- a. Bogie Mounted Brake Cylinder – 203 mm (8")
- b. BVZC (Brake Van) – 305 mm (12")
- c. Conventional trolley Wagon/Coach) – 355 mm (14")
- d. Wagon BOXN HL – 300 mm (12")
- e. LHB – 255 mm (10")

PISTON STROKE

Type of Wagon	PISTON STROKE (mm)	
	Empty	Loaded
BOXN, BCN, BCNA BRN, BTPGLN	$85 \pm 10 \text{ mm}$	$130 \pm 10 \text{ mm}$
BOX, BOI, BCX, UIC Trolley	130 mm	180 mm
BTPN	$87 \pm 10 \text{ mm}$	$117 \pm 10 \text{ mm}$
BVZC (Brake Van)	$70 \pm 10 \text{ mm}$	----
BOBRN / BOBR	$100 \pm 10 \text{ mm}$	$110 \pm 10 \text{ mm}$
BLC	$95 \pm 10 \text{ mm}$	$120 \pm 10 \text{ mm}$

Charging time for Rake: 10 to 12 min* (Coaching)

18 to 22 min* (Goods)

BRAKE BINDING

BRAKE BINDING: It is the phenomenon of binding the wheels by the brake gear system or when brake block is in contact with wheel tread while the driver's brake valve is kept in released position. Hence undesired braking of wheels by brake blocks is called Brake Binding.

Implication of Brake Binding in the Railway System:

It has the direct bearing on the punctuality of trains. Nearly 30% cases of C&W punctuality losses were only on account of "Brake Binding". Besides punctuality loss, other consequences of brake binding on the Railway System are as under:-

1. Damage to the wheels & sometimes coach detachments resulting revenue loss.
2. Harassment to the passengers due to coach detachments resulting in their stranding in mid-section or change of coach.
3. Poor Riding quality.
4. Damage to the track.
5. Damage to the Rolling Stock components such as Roller Bearings, springs etc.
6. More tractive force needed to the locomotive.
7. Unpleasant sound to the passengers.

Following are the factors causing "Brake Binding" in the coaches:-

1. C&W:

1. Mal function of DV causing late release or leakage.
2. Improper mounting of DV on common pipe bracket.
3. Mal functioning of SAB.
4. Isolating cock not working.
5. Leakage in the brake system.
6. Improper release of the rake while attaching/detaching/loco change by TXR staff at TXR stations.
7. Defects in brake cylinder.
8. Defects in dirt collector.
9. Defects in brake gear.

2. LOCO & ITS OPERATION:

1. Not giving proper release time after application.
2. Variation of pressure/vacuum level during run/loco change.
3. Malfunctioning of air flow meter resulting in non indication of ACP/leakage in brake system.
4. Not stopping and resetting ACP/leakage after noticing air leakage through AFI.
5. Inadequate pressure/vacuum creation/maintenance capacity of locomotive resulting in higher release time of the brakes.
6. Driver misreporting brake binding to camouflage some other failure.
7. Improper release of the rake while attaching/detaching/loco change by driver at non-TXR station.

3. OPERATING:

1. Guard misreporting brake binding to camouflage some other failure.
2. Improper release of the rake while attaching/detaching/loco change by guard at non-TXR station.
3. Attachment of unexamined/untested stock after train maintenance and testing, especially VPU's and other special stock.
4. Not locking SLR guard cabin resulting in unauthorized persons traveling and tempering with Guard valve and hand brake.

4. LAW & ORDER

1. Frequent ACP operation.
2. Operations of DV handle for brake application during run.
3. Closing of angle cock in the coach ends.
4. Operation of hand brake from SLR by unauthorized occupants.

5. MISCELLANEOUS:

1. Hitting by ballast/foreign material resulting in breakage of dirt collector/ drain plug of AR/other brake system components.
2. Cattle run over resulting in damage to brake system.

Following measures should be taken to contain brake binding incidences:-

1. Ensure proper maintenance of DV

- Check holes of common pipe bracket are one-to-one matched with the corresponding location on the Common Pipe Bracket (CPB) seal.
- Check collars provided on the CPB seals are uniformly projected.
- Use small quantity of hard grease on the CPB before positioning of the seal to avoid displacement while mounting.
- Check that release choke is clear.
- Overhaul/Repair distributor valve with intermediate flange as a single unit.
- Check holes of Intermediate flange seal are one-to-one matched with respective location in the distributor valve.
- Check collars provided on the seal are uniformly projected.
- Check the valve is free from rubber bulging or debonded for valve plate (10) in KE type or valve (37) in C₃W type.
- Check for smooth movement of hollow stem in its guide during assembly. This may be ascertained by:
 - (i). Movement of hollow stem. Check that hollow stem is not bent.
 - (ii). Check that outer surface of the hollow stem is polished.
 - (iii). Apply light coat of thin grease on the hollow stem.
- Check the condition of CPB filter. This is ascertained by:
 - (i). Clean CPB filter in every schedule.
 - (ii). Clean vertical passage in CPB.
- Identification of late releasing DV's during maintenance.
- Locking of Distributor Valve isolating handle.
- In the POH phosphating and rewaxing of the Distributor Valve Body to be done to avoid corrosion.

2. Proper inspection of dirt collector

- Clean Dirt collector in every schedule.
- Provision of Dirt Collector cover.

3. Proper inspection of brake regulator

- Identification of sluggish Slack Adjuster.
- Check that the barrel has no dent.
- Confirm that Traction sleeve passes smoothly inside the entire length barrel during assembly.
- Check axial play of the Adjuster nut with the spindle. Since actual measurement is difficult, excessive play must be felt while assembly.
- Make sure that the adjuster nut falls freely along the spindle when held vertical.

4. Brake gear system

- Adjust end pull rod hole and maintain length of pull rod such that the Equalising levers be in near vertical in brake applied position.
- Ensure that the Z-lever pins used with Equalising levers are polished.
- Ensure the corresponding bushes are without any oblong deformation.
- Radial clearance is maintained within 0.75 mm.
- Ensure that the Actuating rod is not twisted. The straightness is realized by smooth fitment of the Actuating rod with the both Equalising levers with the pins.

5. Brake Cylinder

- Identification of sluggish brake cylinders during maintenance.
- Breathing passage clearing can be ascertained by avoiding greasing of Trunk during assembly and cleaning annular space between Trunk and Front cover

6. Locomotive

- Check for fluctuation in BP pressure.
- Check for moisture/oil/dirt content in compressed air.
- Dynamic Brakes of the locomotive should be operative.
- Ensure proper working of air flow indicator and audio visual system.
- Locked axle of the locomotive is avoided.

7. Train Handling

- Wait for one minute at least after moving brake valve handle to release position after emergency application.
- Avoid extended application of the pressure surge to avoid over charging.
- Keep close watch in the air flow indicator to monitor incidence of excess leakage.
- Keep close watch in the Ampere meter reading for high current.
- Keep vigilant on tampering DV isolating handle position.
- Drain condensate from air reservoir, oil separator and filter.
- Open drain cock slowly so that condensate is carried out with the air.

8. Modification in coaches

- Brake Binding Indicator should be provided on every coach.
- Hand Brake application indicator should be provided on every SLR.

9. ACP system

- Oversize choke of the ACP be eliminated.
- Resetting key should be welded on the coach itself to properly reset the ACP.
- Proper releasing of coaches should be ensured after every ACP on the adjoining coaches also.

10. Training to the staff – C&W as well as operating & loco staff.

11. BP/FP pipes

- The B.P. & F.P. should be secured with 'T' bolts to avoid decoupling en-route.
- Proper examination of feed pipe & brake pipes during maintenance.

12. System Improvement

- Introduction of Rolling out Examination to detect the coaches which are going in Brake binding condition. Also Rolling-In-Examination be ensured.
- Moisture separator/Air Dryer should be provided in all the locomotives & R.T.R pipe lines to avoid ingress of moisture in the air Brake system.
- Proper releasing of rakes on the platform and on washing lines after maintenance.
- Pre-tested slack Adjuster and Distributor Valves should be replaced on the coaches,
- Coaches attached from the yard should be tested with the rake on W/Line.

13. Others

- Provisions of Quick Release pull rods.
- Pressure relief valve on the pipe line between Distributor Valve & the brake cylinder should be fitted so that the excessive pressure beyond 308 kg/cm² is avoided.
- Pad locking of Guard cabin of the front SLR should be ensured by the Operating.

Trouble shooting & the remedial measures

1) Leakage rate of the much higher than the permitted value of 0.2 kg/cm²/min in BP line.

CAUSES	REMEDY
Locate the defective coach, handle of the angle cock (one or more) not completely in 'ON' position.	Bring the angle cock handle to open position.
Angle cock leaking /defective, pipe threads & check nut joints leaking. Hose pipe cracked or coupling improper.	Tighten angle cock properly. Change hose coupling & angle cock, if defective.
M.U. washer at the coupling head defective.	Replace M.U. washer.
Drain plug of dirt collector loose/ leaking/ its seal damaged.	Change seal & tighten the drain plug.
Common pipe bracket flange joints or CPB filter cover loose/ seal defective.	Replace sealing ring or seal & properly tighten flange & filter cover.
DV to CPB studs loose/ seal defective.	Tighten stud nuts. Replace if defective.
Joints at PEAV branch pipe leaking.	Replace seals of flange joints & tighten joints properly.
One or more PEASD is not reset after operation.	Reset PEASD & observe. If hissing sound is heard from PEASD, replace it.
Guards emergency valve partially in opened condition or joints of branch pipe of pressure gauge leaking.	Keep the valve closed. Tighten the joints properly.

2) Leakage rate of the much higher than the permitted value of 0.2kg/cm²/min in FP line.

CAUSES	REMEDY
Flange joints of branch pipe to isolating cock or check valve loose. Seal of joint defective.	Tighten the joints. Replace sealing ring, if leakage still persists.
Locate the defective coach, handle of the angle cock (one or more) not completely in 'ON' position.	Bring the angle cock handle to open position.
Angle cock leaking /defective, pipe threads & check nut joints leaking. Hose pipe cracked or coupling improper.	Tighten angle cock properly. Change hose coupling & angle cock, if defective.
M.U. washer at the coupling head defective.	Replace M.U. washer.
Drain plug of dirt collector loose/ leaking/ its seal damaged.	Change seal & tighten the drain plug.
Drain plug of AR or drain cock loose/ leaking/ defective.	Identify defect & replace sealing washer. Tighten the joints on FP line.

3) Both the Brake Cylinder Pistons of One Coach not coming out on Brake Application.

System of the coach is not charged due to the dirt collector on BP clogged closing the BP supply.	Clean the dirt collector.
DV isolating cock kept close or brake cylinder isolating cock closed.	Open the isolating cocks of DV & BC.
BP pressure through C.P. bracket path clogged & air not reaching to DV.	Clean the holes of CPB.
Drain plug of CR loose or CR not well sealed to CPB.	Replace seal & tighten the CR drain plug.
Joint seal between CPB & sandwich piece or between sandwich piece & DV defective/cracked/loose.	Replace the joints if found defective.
DV defective.	Replace DV.

4) Brake Cylinder Piston Stroke of One or More Coaches is too less or in Excess.

Dimension 'A' of the SAB not properly set.	Set 'A' dimension properly.
Brake rigging horizontal lever, floating lever of equalizing lever jammed.	Rigging by lubrication.
Slack adjuster not functioning / defective.	Replace SAB.
Brake cylinder piston assembly is sticky.	Replace brake cylinder.

5) Brake cylinder piston of one or more coaches not returning back on releasing brake.

Dirt collector of related coach clogged	Clean dirt collector filter.
Cock of the brake cylinder branch pipe partially closed	Open the cock fully
BC piston jammed or brake rigging gears not moving freely	BC return spring lost its stiffness. Replace the BC. Agitate levers and lubricate.
Excess release time of DV	Rectify the leak in BP line with Teflon tape. BP release choke is closed with dirt, clean the released choke.
DV defective	Replace DV

6) Only one brake cylinder piston of a coach not returning back.

CAUSES	REMEDY
Brake rigging of the relative side jammed	Lubricate the lever joints
Isolating cock on BC branch pipe partially closed	Open isolating cock fully
BC piston assembly movement not smooth or return spring defective	Return spring is weak, replace the brake cylinder
Slack adjuster not functioning properly	Replace slack adjuster

7) Alarm chain system on chain pulling showing no response of either hissing sound or partial brake application.

Mechanical gearing of the chain line connected to operate PEASD defective	Correct the mechanism or lubricate it
Isolating cock of PEAV branch pipe may be closed or defective	Open isolating cock fully. If isolating cock is defective, replace.
PEASD defective	Replace it
PEAV defective	Replace PEAV

8) Brake cylinder air draining time is too long i.e., time taken is more than 20 seconds, piston is not returning to its original position

<i>Release choke of DV clogged</i>	<i>Replace Dv</i>
<i>Return spring inside the brake cylinder lost its stiffness</i>	<i>Identify the defective BC and replace it</i>
<i>Brake rigging gear jammed</i>	<i>Lubricate the levers of the brake rigging</i>
<i>BC wear ring and piston packing creating trouble during return movement of the piston assembly</i>	<i>Replace BC</i>
<i>BC isolating cock partially closed</i>	<i>Check isolating cock and open it fully</i>

9) Alarm chain system is not working even after chain pulling i.e. no signs of brake application.

<i>Chain pulling mechanism is sticking somewhere and the pulling force is not causing PEASD to operate</i>	<i>Identify the defect causing chain pulling force to be ineffective and attend to it.</i>
<i>Isolating cock of PEAV branch pipe may be closed or defective</i>	<i>Open isolating cock fully. If isolating cock is defective, replace.</i>
<i>PEASD or PEAV not functioning properly. Also no hissing sound is heard for one or both the equipments</i>	<i>Replace the defective one or both if the system is still inoperative</i>

10) Piston movement is not smooth rather sticky or jerky in application and release.

CAUSES	REMEDY
<i>Piston rubber packing is running dry i.e. without grease or its lip turned backwards.</i>	<i>Open the cylinder cover. Clean and apply grease. Assemble it properly.</i>
<i>Piston return compression spring is deformed, creating trouble in motion</i>	<i>Check the spring. If it is deformed, otherwise change it.</i>
<i>8 Nos. Hex. Nut M-12 not equally tightened</i>	<i>Tighten the Hex nuts equally</i>
<i>Piston assembly with guide tube bent or damaged</i>	<i>Change the piston with guide tube if bent or damaged</i>
<i>Spring lost its design stiffness</i>	<i>Replace the spring</i>

11) Regular hissing sound from PEAV is appearing during charging of the system or system is not charging appropriately.

<i>Piston rod sub assembly such as Rubber Vulcanized seat is defective</i>	<i>Change the piston rod</i>
<i>Scratches on bottom bush seat</i>	<i>Try to remove the scratches/ aberrations by polishing with fine emery paper or gentle lapping</i>
<i>Defective compression spring on piston rod</i>	<i>Change the spring</i>
<i>Scratches on inner surface of the brush is obstructing the free movement of plunger</i>	<i>Remove scratches by polishing with fine emery paper to make free movement</i>
<i>'O' ring sealing the top cover is defective</i>	<i>Change the 'O' ring</i>
<i>'K' rings fitted external on the piece and/or 'K' ring fitted internal on position is/are defective</i>	<i>Check and change defective 'K' ring(s).</i>

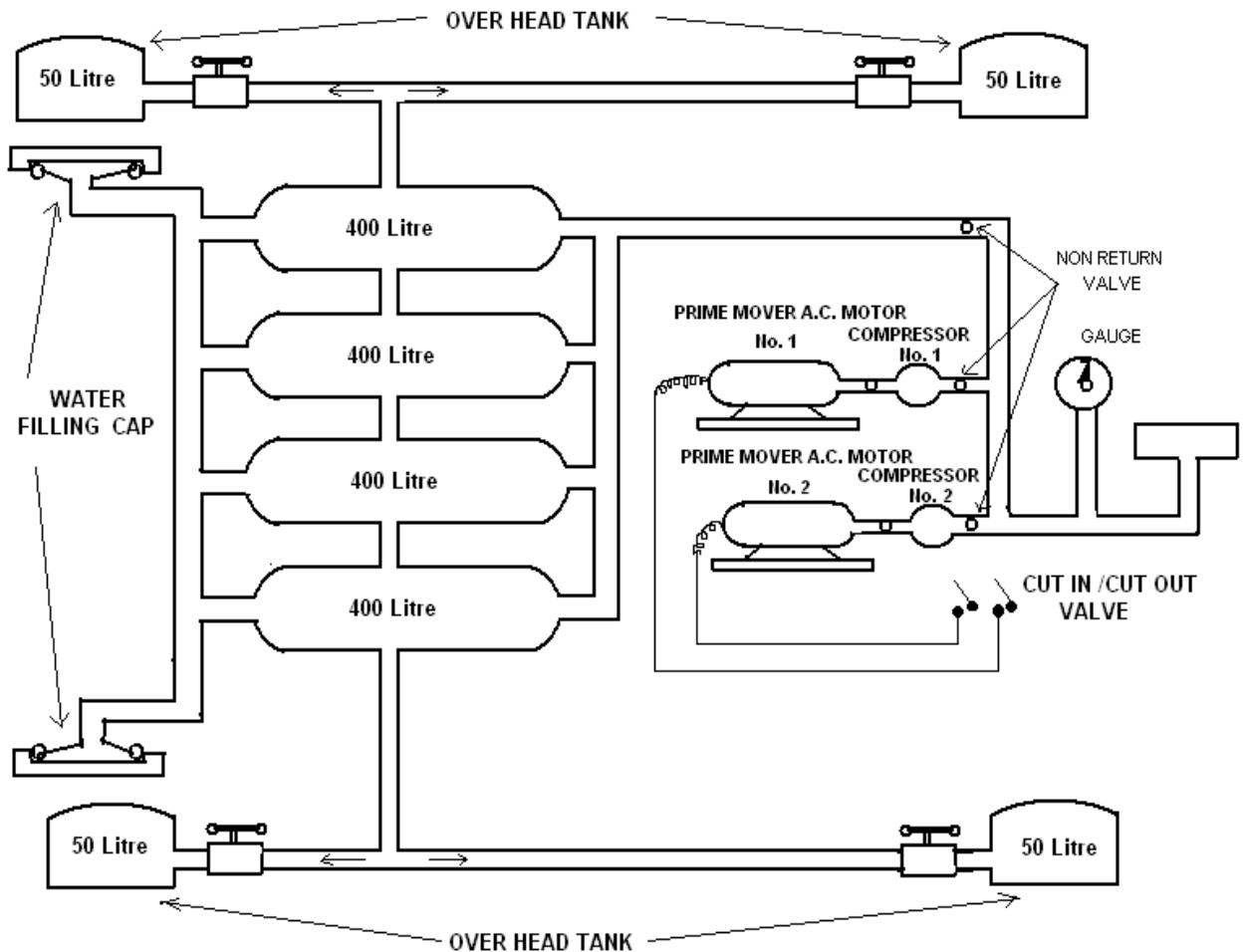
12) No hissing sound from PEAV specially on pulling the pressure rod of PEASD and does not show any brake pressure reduction effect.

<i>'K' ring external on piston rod subassembly is defective</i>	<i>Change the 'K' ring</i>
<i>Nylock nut on the piston rod subassembly is loose</i>	<i>Tighten the nylock nut</i>
<i>Piston rod assembly including piston movement is jammed</i>	<i>Dismantle the assembly & ensure smooth movement</i>
<i>Drain choke of 8 mm clogged</i>	<i>Clean the drain choke</i>

13) On resetting PEASD, hissing sound from PEAV not disappearing.

<i>Piston rod assembly is not moving smoothly</i>	<i>Smoothen the movement of piston rod assembly</i>
<i>Causes mentioned at 11</i>	<i>Remedies mentioned at 11</i>
<i>Leakage observed from nipple adapter threads due to undersize threads or worn out/loose nipple</i>	<i>Change the nipple adapter. Use Teflon tape on threads to get tightening of nuts</i>
<i>Leakage observed at top cover joints due to loose hexagonal head studs</i>	<i>Tighten the studs fully</i>
<i>Leakage observed at top cover joints due to defective 'O' ring</i>	<i>Change the 'O' ring</i>

WATER RAISING APPARATUS



1. **Introduction:-** Earlier, the A.C. equipment such as compressors, condenser, evaporators etc. in coaches were mounted in the under frame. These are now getting mounted in the roof which occupied the space provided for overhead water tanks. These coaches are called Roof Mounted AC Coaches.
2. **Overhead and Under-Slung Water Tanks:-** In roof mounted Ac coaches only a small overhead tank with a capacity of about 50 litre is fitted over the ceiling of each toilet from which pipe connections are taken for the taps, in each of the toilets. Water is filled and stored in 4 larger capacity water tanks of cylindrical shape mounted on the under frame, each having a capacity of 400 litre. These tanks are called "Under-Slung Water Tanks".
3. **The need for Water Raising Apparatus:-** In the conventional overhead water tank system, the water from the tank was supplied to the taps by gravitational method. The water in under slung water tank however needs to be pumped up to the smaller capacity overhead water tanks continuously for the use of passengers, without any interruption.
4. **Water Raising Apparatus:-** This is a small air compressor run by electrical motor. This apparatus raises or pushes water from under-slung water tank to the smaller overhead water tanks by using compressed air, as and when the overhead tank needs to be filled.

5. **Components of WRA:** -

- | | | |
|---|---|--------|
| a. Prime mover – Small AC induction motor | - | 2 nos. |
| b. Air compressor – Reciprocating type | - | 2 nos. |
| c. Cut off, switch-on-arrangement | - | 1 no. |
| d. Pressure gauge | - | 1 no. |
| e. Non return valves | - | 3 nos. |
| f. Under Slung Water tanks(400 litre capacity) | - | 4 nos. |
| g. Side filling pipes | - | 2 nos. |
| h. Air Tight closing caps with rubber lining | - | 2 nos. |
| i. Over head tanks (each 50 litres capacity) | - | 4 nos. |
| j. Air pressure release valves | - | 4 nos. |
- (2 in toilets, 2 in under frame)

6. **Layout of WRA system:** - The under slung water tanks can be positioned in two ways. In one type 4 tanks are mounted across in one line laterally with the coach.

In another, 2 tanks are mounted on one side longitudinally and 2 tanks on another side of the coach. These water tanks are invariably interconnected by 2" dia pipes in both configurations.

7. **Water tank configuration under the coach:** - The layout of the under slung water tanks is shown in above schematic diagram.

8. **Method of filling of water in coaches fitted with WRA:** - Before filling water in the under slung tank, the WRA compressor should be switched off. This should be done about 10 minutes before the train reaches a watering station. In case of filling in pipe lines, the compressor should be switched off well in advance. After releasing the air pressure in water tanks by pulling the air release valves, the side filling pipe flap has to be opened carefully.

Water is filled through side filling pipe. When water is fully filled in all the under slung tanks, the excess water will flow through a pipe of smaller diameter which is connected to the side filling pipe.

This indicates that the tanks are completely filled up. The side filling pipes have rubber washers to engage tightly over the machined face of the side filling pipe. After filling the water tanks, the side filling pipe flap has to be closed firmly to avoid air leakages. This is very important since the WRA system works under compressed air pressure built into the under slung tanks.

9. **Functioning of WRA:** - After filling the under slung tanks, the WRA motor must be switched on, by a switch located inside the panel board near door way. The motor powers the compressor, which sucks the atmosphere air through a small filter, compress the air and send it to main tanks through the delivery pipe.

10. **Non – Return Valves:** - there are 2 motors and two compressors operated individually by separate switches. Even if 1 compressor fails, the system can be operated through another compressor. 2 non- return valves are fitted just after the delivery pipe of each compressor. Both the delivery pipes are connected with a Tee joint and one more non return valve is fitted after the Tee joint .Thus there are 3 non return valves in the system altogether. These non return valves allows compressed air from the delivery pipe of the compressor to the under slung water tanks, only in one direction and prevent the back flow of either the air or water from the tank to the compressor. These Non return Valves act as safety valves to the compressor preventing the entry of water into the compressor.

11. **Pumping up the water:** - The compressed air thus passes through the non-return valves and pushes the water from the under slung water tank. The under slung water tanks are connected to the over head tanks. When the pressure of the air reaches the required level, water from the under slug tank is pushed to the over head tanks. When the outer head water tanks get fully filled, the air pressure continues to increase. This can be read by the gauge in the panel board. Since the water is in a closed circuit (due to continuous running of WRA pump) the pressure in the system slowly builds-up and reaches 0.75 kg/cm^2 .The cut off switch then comes into operation and cut-off the compressor motor. If there is no cut off switch the pressure will rise beyond the safe limits leading to bursting of pipe lines and joints.
12. **Cut-OFF /ON Switch Arrangement:-** As the passengers use the water and water in the over head tank gets depleted, the pressure in the system drops slowly. When the pressure goes below 0.3 kg/cm^2 , the compressor motor is switched on by the cut- off switch on relay and WRA starts functioning once again and the pressure builds up again. Thus, depending upon the air pressure in the system, the compressor gets switched on or off pumping the water to the over head tank as necessary.
13. **En-Route Failures and Troubleshooting:** - Failure of the WRA leads to a situation of no water in wash basins or toilets and causes lot of complaints amongst passengers and embarrassment to accompanying railway staff. Troubleshooting in such case should be swift and effective. The easier checks must be done before making elaborate dismantling.

Troubleshooting of WRA Arrangement on RCF make Coaches

1. Coach not taking water: -

- (i). Put off the WRA switch and remove other end WRA cap to allow the air to come out. Now water should flow into coach, if not then,
- (ii). Open the stud of Non Return Valve 1 or 2(from which side watering pipe is inserted), lift the valve and again tighten it by ensuring the proper seating. Now water shall enter into the coach.

2. Water not available on one end of Toilet: -

- (i). There are two air release valves cocks one on each end of the toilet (diagonally opposite). Release the air lock by pulling its handle downward. The water should start coming. If not then,
- (ii). There are 2 isolating cocks each in both end of the toilet diagonally opposite on the roof in covered condition. Open the cover, check the valve and open the same if found closed. Check these valves for chocking. If chocked then clean it. Water shall start flowing.

3. Water not available in all toilets: -

This will happen if WRA is not build up pressure.

- (i). Check for any leakage of air or water in the tank and pipe lines and attend it. The pressure will build up. If not,
- (ii). Call electrical staff for checking the compressor of WRA.

4. Leakage in the tank no 1 or 2 or piping (except outlet pipe) encountered en-route: -

Problem in the leakage of water tank 1 or 2 or any pipe on the side of water tank one and two (except outlet pipe). Close the isolating cock 1, 2, and 4. Water will come in all 4 toilets if the same is available in the tanks. On en- route station fill the water in tank no 3 and 4 only.

5. Leakage in tank no 3 or 4 or pipe line of that side (except outlet pipe encountered en-route): -Close the isolating cock 1, 2 and 3. Water will come in all 4 toilets. On the next station arrange the filling of water only in tank no 1 and 2.

Precaution: - The air pressure must not increase beyond 0.7 kg/cm² otherwise the rubber hose provided in the toilet may burst and the water leakage will start.

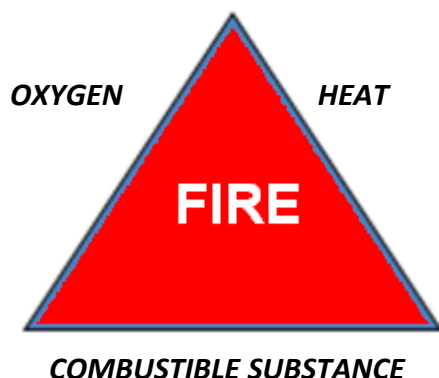
Advantages of under slung water tanks over the conventional tanks (overhead tanks): -

1. Capacity more, instead of 1100 litres, now it is 1800 litres in case under slung water tanks.
2. Riding capacity is smooth, due to weight come down from top to bottom.
3. Maintenance and wastage is less than conventional.
4. To protect the valuable materials during the course of unusualness at the time of rolling and work as Shield/Guard
5. Easy to fill water to water slung tank with low pressure at high drain with gravitational force.
6. If any tank leakage from system that may be isolated from other three tanks.
7. Due to increase of water tank volume; watering station distance more and watering time reduce for gravitation force filling.

FIRE

Definition of Fire Fire is a chemical chain reaction, in presence of correct proportion of oxygen, heat and fuel.

Fire Triangle Fire is a chemical reaction between a fuel and oxygen in presence of heat. In other words, it may be stated that three things are essential for fire which can be represented by the three arms of a triangle, viz. Heat, combustible substance and the supporter of combustion or oxygen. A fire can not take place in absence of any one of these three factors.



TYPES OF FIRE (CLASSIFICATION) :-

CLASS OF FIRE	TYPE OF EXTINGUISHER USED
CLASS 'A' <i>Fires involving solid combustible materials of organic nature such as wood, paper, rubber, plastic etc. where the cooling effect of water is essential for extinction of fires .</i>	Water expelling type extinguishers.
CLASS 'B' <i>Fires involving flammable liquids or liquefiable solids, where a blanketing effect is essential.</i>	Foam, dry powder, vaporizing liquid, and carbon dioxide extinguishers.
CLASS 'C' <i>Fires involving flammable gases under pressure including liquefied gases, where it is necessary to inhibit the burning gas at fast rate with an inert gas, powder or vaporizing liquid for extinguishers.</i>	Dry powder, vaporizing liquid and carbon dioxide extinguishers.
CLASS 'D' <i>Fires involving combustible metals, such as magnesium, aluminum, zinc, sodium, potassium etc, when the burning metals are reactive to water and water containing agents and in certain cases carbon dioxide, halogenated hydrocarbons and ordinary dry powders. These fires required special media and techniques to extinguishers.</i>	Extinguishers with special dry powders for metal fires.

METHODS OF EXTINGUISHING FIRE :

- A) **STARVATION** – means removing records, furniture and other combustible substances away from the fire
- B) **SMOTHERING** – means prevention of Oxygen, access of fresh air into fire, using sand, foam etc., to cover fire.
- C) **COOLING** – means reducing the degree of temperature or heat.

GENERAL CAUSES OF FIRE IN RAILWAYS:

- (i). Carrying stoves, sigris, gas cylinders, kerosene oil, petrol fireworks etc. in passenger's compartment.
- (ii). Making fire/using fire near paper, wood, petrol or such inflammable articles
- (iii). Lighted match sticks, cigarette ends carelessly thrown.
- (iv). Short circuits in electrical wirings.
- (v). Using naked light during authority token delivery, to the driver, shunting of inflammable loads, sealing of inflammable wagons.
- (vi). Use of open fire, smoking near gas/petrol tank.
- (vii). Non-removal of garbage from pantry car/coaches.

Everyone should take all possible precautions to keep away from doing the above mistakes so that possibility of breakage out of fire can be minimized.

Types of fire extinguisher

- **Water Fire Extinguishers:** The cheapest and most widely used fire extinguishers. Used for Class A fires. Not suitable for Class B (Liquid) fires, or where electricity is involved.
- **Foam Fire Extinguishers:** More expensive than water, but more versatile. These are used for Class A & B fires. Foam spray extinguishers are not recommended for fires involving electricity, but are safer than water if inadvertently sprayed onto live electrical apparatus.
- **Dry Powder Fire Extinguishers:** Often termed the 'multi-purpose' extinguisher, as it can be used on classes A, B & C fires. Best for running liquid fires (Class B). Will efficiently extinguish Class C gas fires, BUT BEWARE, IT CAN BE DANGEROUS TO EXTINGUISH A GAS FIRE WITHOUT FIRST ISOLATING THE GAS SUPPLY. Special powders are available for class D metal fires.
- **CO₂ Fire Extinguishers:** Carbon Dioxide is ideal for fires involving electrical apparatus, and will also extinguish class B liquid fires, but has NO POST FIRE SECURITY and the fire could re-ignite.
- **Wet chemical** : Specialist extinguisher for class F fires.
- **For Metal Fires:** A specialist fire extinguisher for use on Class D fires – metal fires such as sodium, lithium, manganese and aluminium when in the form of swarf or turnings.

Types of fire extinguisher labels

- **Class A Extinguishers** will put out fires in ordinary combustibles, such as wood and paper. The numerical rating for this class of fire extinguisher refers to the amount of water the fire extinguisher holds and the amount of fire it will extinguish.



- **Class B Extinguishers** should be used on fires involving flammable liquids, such as grease, gasoline, oil, etc. The numerical rating for this class of fire extinguisher states the approximate number of square feet of a flammable liquid fire that a non-expert person can expect to extinguish.



- **Class C Extinguishers** are suitable for use on electrically energized fires. This class of fire extinguishers does not have a numerical rating. The presence of the letter "C" indicates that the extinguishing agent is non-conductive.



- **Class D Extinguishers** are designed for use on flammable metals and are often specific for the type of metal in question. There is no picture designator for Class D extinguishers. These extinguishers generally have no rating nor are they given a multi-purpose rating for use on other types of fires.



Duties of Members of Fire Brigade during fire in trains:- It must be remembered that no fireman should go on or near the track unless it is essential. The officer-in-charge should, on arrival:-

- Assess the situation; decide whether immediate action is necessary and set up safety procedures as quickly as possible.
- Identify accurately, the location of the incident to this bridge control. They will alert Railway Traffic Control to the brigade's presence on or near the track.
- Request, through brigade control, the measures he wishes the railway authorities to take and ensure by confirmation that it has been done.
- Inform brigade control of the action he is taking.
- Exercise rigid control of all personnel in the danger area, including IR staff and be ready to change the disposition of men, appliances and equipments to ensure the greatest safety under all circumstances.
- Make sure all fire brigade personnel are wearing conspicuity jackets.

Fire In Coach:- The following precautions should be taken immediately:-

- First of all pull the alarm chain to stop the running train.
- Switch off all lights and fans from the junction box.
- Cut off electrical connection from next coach through EFT at both the ends.
- Cut off battery fuse/ terminals.
- After stopping of train, coach should be isolated and detached immediately to avoid spreading of fire to adjacent coaches.
- Disconnect the AC generator (Alternator) to remain the generator dead.
- Then apply necessary fire extinguishing methods on the coach.

Isolation/ Detachment Of Burning Coach:-

- Guard / Asstt. Driver will detach the burning Coach from train.
- Coach Attendant & Ticket checking staff on duty will also assist them.
- All Railway staff traveling by same train has also to assist.
- Before uncoupling the Coach lift up the Vestibule Fall plate.
- Close the cut of angle cock of the loco side and Guard side Coach.

6. *Uncouple the coupler of rear coach and pull the culprit coach, safe distance ahead.*
7. *Uncouple the coupler of the front coach and pull the front side rack away from culprit coach, safe distance ahead.*
8. *Driver will intimate to Power Controller giving full details of fire.*
9. *Guard will intimate to Chief Controller giving full details of fire*
10. *Conductor will intimate to Commercial Controller giving full details of fire.*
11. *Then take necessary Fire extinguisher action from available resources.*

Provision Of Fire Extinguishers In Train:- Fire extinguishers (DCP Type of 5 kg capacity for BC class fire) are provided in every guard compartment, Locomotives and all types of AC Coaches at following locations:-

- SLR - 02 nos. in Guards cabin.
- Diesel Loco - 04 nos. in Drivers cabin
- AC Loco - 04 nos. (02 nos. each in both Driver cabin)
- AC Coaches - 02 nos. each near electrical panel.

Display Plate Of “Prevent Fire”:- Inside the train compartment

- *Do not throw lighted articles.*
- *Do not carry explosive articles.*
- *Do not carry inflammable articles.*
- *Do not light up stove or sigri.*
- *Offenders are punishable with imprisonment and up to three years & fine up to Rupees Three Thousand.*
- *Help Railways- Reach you safely.*

Provision Of Fire Retardent Material In Coaches:- Fire retardant curtains (Roller blinds), Enamels, different gel coatings, mica and PVC in Flooring and berths, are being used in coaches to reduce the gravity of fire.

Pantry Car Fire Safety Procedure:- Sleeping cars and catering units carry LPG cylinders, for which there should be a main supply shut off valve. If it is possible to isolate a burning vehicle from the rest of train, the attendant should ask the train crew to do this.

Fire Safety Procedure:-

1. *Valve Leak & Body leak cylinders are remove before installing to manifold on train.*
2. *All electrical fitting in the area are flameproof type.*
3. *To prevent generation of sparks due to following of cylinder etc. the floor and supporting structure are given a coating of antistatic mastic.*
4. *The storage of cylinders are protected with spray system having density of 20 ltr. Per min/mtr²*
5. *To protect this manifold from high temperature in summer condition a cooling water ring is provided on top of the operating manifold.*
6. *For First Aid fire fighting extinguishers of dry chemical type and carbon dioxide type are provided.*
7. *To direct leakage of LPG a gas detection systems is provided having sensors all over the pantry car are and storage.*
8. *All LPG pipeline from manifold are provided with pressure release valve.*

FIRE – DO's & DON'T's :

AVOID :

- (i). *Smoking or lighting a flame where it is prohibited;*
- (ii). *Throwing of oil soaked jute/cloth carelessly;*
- (iii). *Throwing of burning match-stick bidi/cigarettes carelessly;*
- (iv). *Do not overload electric circuits in a wagon;*
- (v). *Do not allow running of hot axle wagon;*
- (vi). *Do not keep explosive materials along with combustible materials;*
- (vii). *Carrying of gas stoves, kerosene oil, gas cylinders, kerosene jars, petrol or other inflammable articles in passenger carrying trains, should be restricted.*

OBEY :

- (i). *Be speedy but do not get nervous.*
- (ii). *Locate the fire extinguishing substances viz. water bucket with water/sand, fire extinguishers etc.*
- (iii). *Raise alarm.*
- (iv). *Inform all concerned to assist in extinguishing the fire.*
- (v). *Segregate/remove all valuable properties.*
- (vi). *Keep fire extinguisher/water buckets in readiness.*
- (vii). *Arrange to isolate the source of fire if possible.*
- (viii). *Ascertain the type of fire viz. – dry, oil, gaseous, electric and use the right type of extinguishers.*
- (ix). *In case of the fire from electricity, switch "OFF" the source.*
- (x). *Do not stand against the wind.*

IMPORTANT :

- (i). *Driver, guard passing a section if notices fire of serious nature in a village endangering life, may stop his train out of course to the next station and shall inform station master about the fact with KM/Location.*
- (ii). *In case of total interruption of communication or as the case may be a train can be stopped out of course to communicate information asking assistance.*
- (iii). *Driver, guard passing through or entering a station if notices fire, need not stop out of course. Stopping trains where fire may cause damage to train/passenger/goods, shall control their train suitably so that fire can not harm vehicle/wagon and shall take protection required as per GR 6.03.*
- (iv). *But if they notice fire on approaching/passing train on adjacent line shall switch on flasher light, whistle continuously and exhibit red hand signal till the affected train is passing. They shall stop their train immediately to the spot where field telephone socket is provided or to the station whichever comes first to communicate first fire information to control.*
- (v). *When the fire is of serious nature, nearby municipal, private or Railway fire brigade may be called without delay.*

- (vi). The occurrence of a fire shall be informed to the nearest station by most expeditious means and the Station Master on receipt of such information shall relay control, IOW, Divisional Security Officer and other agencies.

FIRE – WHAT TO DO WHEN: In the event of a vehicle on a train being on fire

- (i). The train shall be stopped.
- (ii). The burning vehicle to be isolated, a distance of not less than 45 meters being left between it and the other vehicles of the train.
- (iii). The train shall be protected in accordance with GR 6.03, if not protected by fixed signals.
- (iv). Every effort shall be made to extinguish the fire and to save the wagon labels, seals and the contents of the vehicle.
- (v). Earth or sand, if available shall also be used.
- (vi). In case of fire is discovered when the train is near a tank or a watering station, the guard and driver shall use discretion to proceed there, but no such attempt shall be made until the portion of the train in rear of burning vehicle has been detached.

When a person's clothing catch fire

- (i). Approach him holding with the nearest available wrap in front of you.
- (ii). Wrap it round him.
- (iii). Lay him flat on the floor, smothering the flames.
- (iv). On no account should be rush into the open air.
- (v). Call for assistance.

Fire caused by petrol or any other inflammable liquids, acids or gases

- (i). Segregate the affected wagon, coach or area involved.
- (ii). On opening a wagon do not enter immediately. You would, thus, avoid fumes which may be dangerous.
- (iii). Use D C P Type fire extinguishers and sand and not water or soda acid type fire extinguishers.
- (iv). Do not bring naked lights near the site of fire.
- (v). Inform the nearest Railway or civil fire stations intimating that the fire has been caused by petrol or any other inflammable liquids or gasses or acids.

Correctly Use A Fire Extinguisher:- There are four important steps you must know to correctly use a fire extinguisher.

The PASS method can help you to easily remember those steps.

PASS

- Pull the pin.
- Aim the extinguisher or nozzle at the base of fire.
- Squeeze the handle and release the extinguishing agent.
- Sweep the extinguisher from side to side across the base of the fire until it appears to be



FREIGHT STOCK

The term freight stock means all rolling stock other than coaching stock and locomotives irrespective of contents and whether attached to a passenger or goods train. The term wagon is applicable only to freight stock. Indian railways have different types of wagons to transport different commodities like coal, cement, fertilizers, ores, food grains, petroleum products, iron and steel and other finished products. To cater for the transportation requirements various types of freight stocks having different features in use are classified as under.

(1) According to it's Under Gear

- a) Four Wheeled wagon
 - i) Conventional Wagon.
 - ii) Modified Wagon
 - iii) Tank Wagon
- b) Bogie stock wagon
 - i) Diamond frame bogie
 - ii) Cast steel bogie
 - iii) UIC fabricated bogie
 - iv) CASNUB Bogie

(2) According to Its Utility

- a) Open wagon
- b) Covered Wagon
- c) Flat Wagon
- d) Hopper Wagon
- e) Well Wagon
- f) Container Wagon
- g) Tank Wagon
- h) Explosive Wagon
- i) Brake Van

POH INTERVAL OF AIR BRAKE STOCK

Sr. No	Stock	POH (Years)	
		First	Subsequent
1	BOXN, BRN, BPTH, BOST BOXNHS, BOXNHA, BOY (newly built with Casnub bogie)	6	4.5
2	BCN/BCNA, BTPN, BLCA/B BCCN/R, BOBR, BOBRN, BFNS, BCNAHS	6	6
3	BTPGLN, BTAL/BTALN	4	4
4	BOY (with Amsted bogie)	3	3
5	BVZC, BVZI	2	2

POH INTERVAL OF VACUUM BRAKE STOCK

Sr. No	Stock	POH (Years)	
		First	Subsequent
1	BOX, BCX, BRH, BOXK, BOXKH, BFK, BFKL, BOBY	4.5	4.5
2	BOBS	3	3
3	BTPGL, BOI, BRS	4	4

4	<i>TX, THA and all other 4Wh. Tank wagons</i>	2	2
5	<i>Bogie well wagons (BWT, BWS, BWH, etc)</i>	4.5	4.5
6	<i>BOB (PB), BOBC (PB) and other deptt hopper wagons and other bogie wagons with PB</i>	4.5	4.5
7	<i>Brake Vans</i>	2	2
8	<i>All other wagons other than listed above</i>	2	2

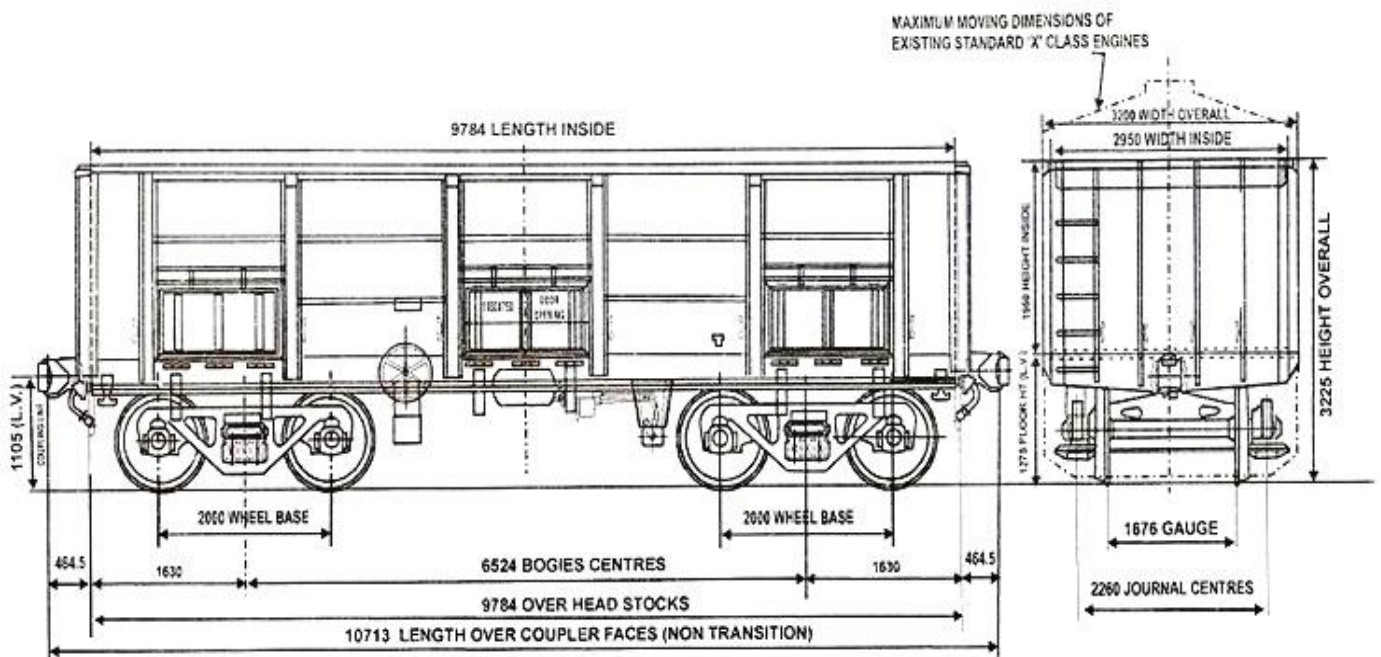
ROH INTERVAL OF AIR BRAKE STOCK

S. No.	Stock	ROH (Months)
1	<i>BOXN, BRN, BTPH, BOST, BOXNHS, BOXNHA, BCN/BCNA, BTPN, BLCA/B, BCCN/R, BOBR, BTPGLN, BOBRN, BFNS, BCNAHS, BTAL/BTALN</i>	18
2	<i>BOY (with Amsted bogie)</i>	18

ROH INTERVAL OF VACUUM BRAKE STOCK

S. No.	Stock	ROH (Months)
1	<i>BOX, BCX, BRH, BOXK, BOXKH, BFK, BFKI, BOBY, BOBS</i>	18
2	<i>BTPGL, BOI, BRS</i>	24
3	<i>Bogie well wagons (BWT, BWS, etc)</i>	18
4	<i>BOB (PB), BOBC (PB) and other deptt hopper and other bogie wagons with PB (BFR etc.)</i>	18
5	<i>All other wagons other than listed above</i>	18

‘BOX’N WAGON



SALIENT FEATURES OF 'BOX'N WAGON

Sl. No.	Particulars	Description
1.	Estimated tare weight	22.47T
2.	Payload	58.81T
3.	Gross load	81.28T
4.	Axle load	20.32T
5.	Ratio of pay load to tare weight	2.72
6.	Track load density (gross)	7.59T/m
7.	Floor area	20.87 sq m
8.	Volumetric capacity	56.29 cubic m
9.	Overall width	3200mm
10.	Inside width	2950mm
11.	Journal center	2260mm
12.	Length over center buffer faces (NT)	10713mm
13.	Length over headstock	9784mm
14.	Bogie center's	6524mm
15.	Wheel diameter on tread (new /condemning)	1000mm / 906mm
16.	Wheel base	2000 $\pm$ 5 mm
17.	Height of CBC from rail level	1105mm
18.	Distance between side bearers	1474mm
19.	Type of side bearers	Roller type
20.	Type of center pivot	IRS spherical
21.	Type of brake beam	Unit type fabricated brake beam
22.	Suspension details	Long travel helical spring suspension
23.	Type of roller bearing	Std. AAR CTRB suitable for 152.4 x 276.4 mm (6"x11" wide jaw)
24.	Type of bogie	CASNUB 22W
25.	'A' dimension	70 $\pm$ ₀ ² mm

Detailed Salient Features Of BOXN Wagon:-

1. No. of doors in BOXN wagon – 2 on each side (at both ends.)
2. DRAW & BUFFING GEAR:-
 - High capacity, high friction draft gear used.
 - High tensile straight CBC used.
 - CBC coupler working capacity – 120 ton.
 - Proof load – 170 ton.
 - Hauling capacity in 1:100. In up gradient – 9000 ton.
3. BOGIE:-
 - CASNUB bogie used.
 - Axle load capacity 22.9 ton.
 - Secondary suspension, helical spring used.
 - Side frames are joined through spring plank & bolster.
 - UIC spherical type centre pivot fitted [except 22 W, 22 W(R)].

- Metal bonded constant contact rubber pad side bearer used.
- Fit for 90 kmph, 100 kmph for HS.
- Maintenance cost is less.
- Head stock with diagonal bar is removed.

4. BRAKE GEAR:-

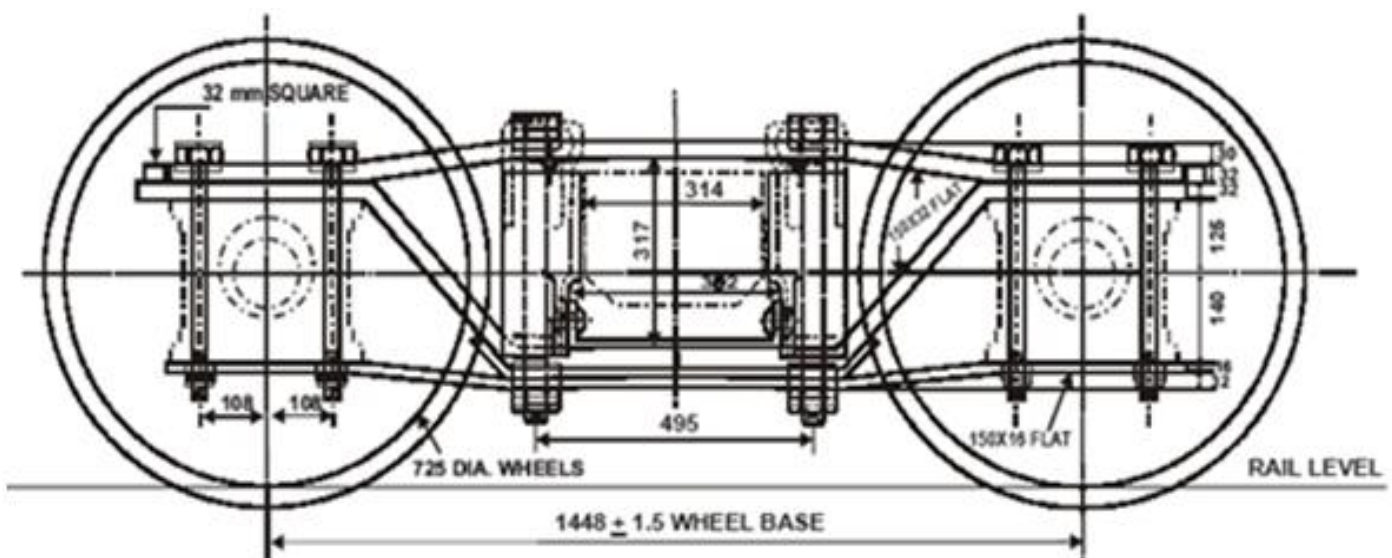
- Single pipe air brake system.
- SAB, empty load device used.
- Clearance between wheel & brake block is maintained through SAB.
- Pocket liner type brake beam.
- No. of brake beam – 4.
- No. of brake block – 8.

5. RUNNING GEAR:-

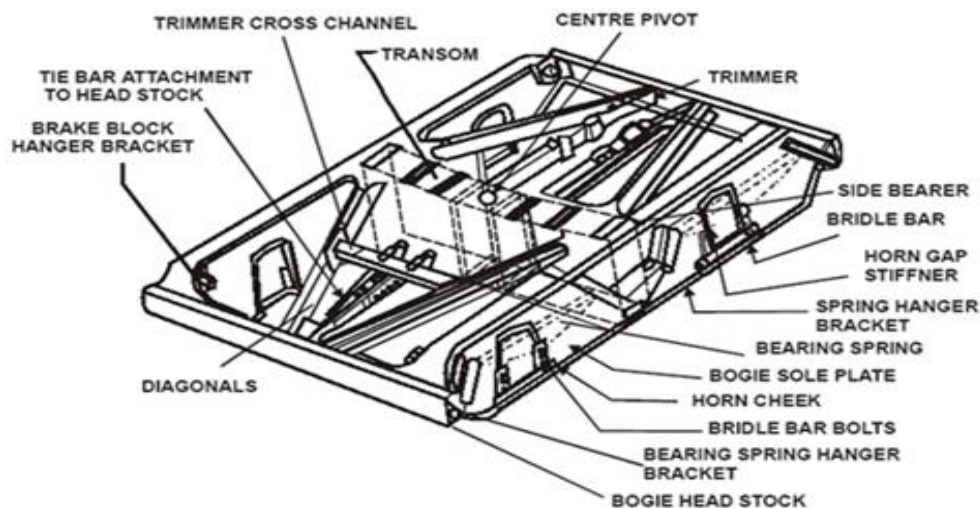
- Axle load – 22.9 ton.
- Axle Bearing – CTRB
- New wheel dia. – 1000 mm.
- Condemning dia. – 906 mm.
- Journal centre – 2260 mm.
- Dia. Of journal – 144.5 mm.

BOGIES

- | | |
|---------------------------|----------------------------------|
| I) Diamond frame bogie | – Used in BOB wagons |
| II) Cast steel bogie | – Used in BWT wagons |
| III) UIC fabricated bogie | – Used in VAC. Brake Bogie Stock |
| IV) CASNUB Bogie | – Used in Air Brake Bogie Stock |



Diamond Frame Bogie Wagons



Isometric view of UIC Bogie

This bogie consists of the following important parts:-

1. Bogie frame including integral bolster, head stock and trimmers.
2. Centre pivot arrangement.
3. Side Bearer arrangement.
4. Spring suspension arrangement.
5. Horn Cheeks.
6. Roller Bearing axle boxes.
7. Wheel sets.
8. Bogie Brake gear.

COMPARATIVE STUDY BETWEEN UIC AND CASNUB BOGIES

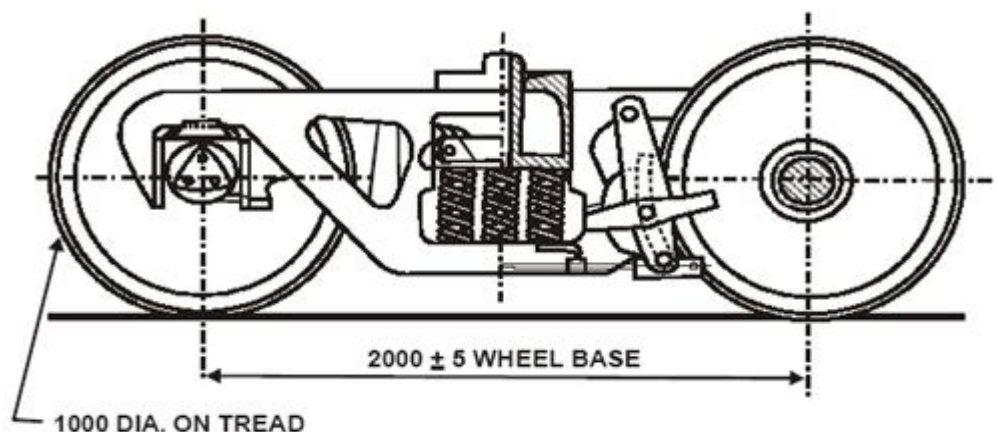
Sl. No.	UIC BOGIE	CASNUB BOGIE
1.	It is introduced in 1962.	It is introduced in 1972-73.
2.	It is a fabricated bogie.	It is a cast steel bogie.
3.	Weight: 5.5 to 6.0 T.	Weight: 5.35 to 5.45 T.
4.	It comprises sole bar, head stock, longitudinal bar, diagonals, etc.	There is a permanent departure of diagonals and head stock.
5.	It is designed to take axle load of 20.3T.	Except CASNUB 22 HS bogie (20.3 T), all version of CASNUB are designed to take axle load of 22.9 T.
6.	It is utilized in vacuum brake stock i.e. BOXC, BCXC, BRHT.	It is utilized in air brake bogie stock i.e. BOXN, BCN, BRN.
7.	It comprises two no's of wheel set fitted with cylindrical roller bearing.	It comprises two no's of wheel set fitted with cartridge tapered roller bearing.
8.	Axle capacity of cylindrical roller bearing wheel set is 20.3 T.	Axle capacity of cartridge tapered roller bearing wheel set is 22.9T.
9.	Wheel diameter 1000 mm (new) & 860 mm (condemn).	Wheel diameter 1000 mm (new) & 906 mm (condemn).
10.	There is a provision of laminated leaf spring primary suspension.	There is a provision of helical spring in secondary suspension .
11.	There is a provision of bogie transom fitted with UIC bogie spherical type pivot.	There is a provision of floating bolster with UIC & IRS pivot.

12.	Each bogie is fitted with 4 nos. of brake beam suspended with hangers.	There is a provision of 2 nos. of brake beam per bogie sliding in pocket type suspended with hanger.
13.	There is a provision of clasp type brake rigging arrangement	There is a provision of single block type brake rigging arrangement.
14.	There is a provision of metal plate side bearer arrangement.	There is a provision of metal bonded rubber pads side bearer arrangement.
15.	There is a provision of total 36 types of pins.	Only 10 types of pins are used.
16.	There is a provision of 4 nos. vertical floating lever in brake rigging.	Only 2 nos. of vertical floating lever are provided.
17.	Crown packing of 10mm, 20mm, 35mm, and 45mm are used for adjustment of CBC height.	Packing plate of 12mm and 37mm are utilized for adjusting CBC height.

CASNUB BOGIE

CASNUB BOGIE: CASNUB BOGIE means Cast Steel Bogie equipped with snubber spring .

Bogie comprises of two cast steel side frames and a floating bolster. The bolster is supported on the side frame through two nests of spring, which also incorporate a load proportional friction damping .the side frames are connected by a fabricated mild steel spring plank to maintain the squareness of the bogie



History:

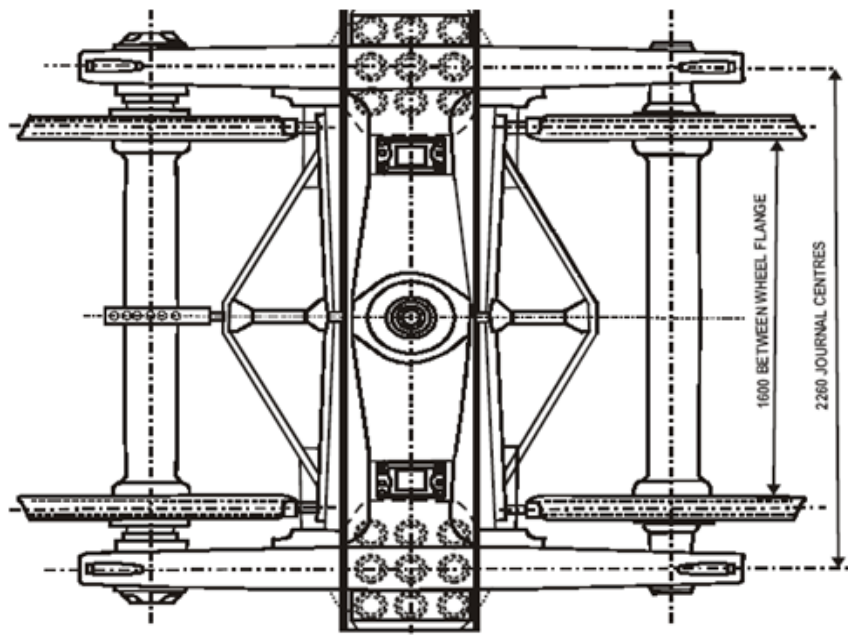
- The CASNUB bogie was first tested in 1972 under BOI wagon and was found safe to run at test speed up to 110 kmph. The test result has been published in RDSO's mechanical engineering report no. M – 265.
- In 1981 trials were again undertaken on these bogies under BOXN wagon to maintain the main line standard and its behavior was well within safety limits up to 90 kmph speed and was designated as CASNUB 22W . These were modified as CASNUB 22W (M) mainly to take care of high wheel wear reported on earlier versions. Subsequently CANUUB 22 NL (Narrow jaw) and CASNUB 22 NLB (Narrow jaw with fish belly bolster) version were introduced.
- CASNUB 22 HS bogies have been developed for high speed operation with maximum permitted speed up to 100kmph.

SALIENT FEATURES OF CASNUB BOGIE:

<i>Sl.No.</i>	<i>ITEMS</i>	<i>DESCREPTION</i>
1	Gauge	1676 mm
2	Axle load	22.9 T
3	Wheel diameter	1000 mm new, 956 mm new for 22W (retrofitted)
4	Wheel base	2000 mm
5	Type of roller bearing	Slandered AAR cartridge bearing
6	DISTANCE between journal centers	2260 mm
7	Distance between side bearer	1474 mm
8	Type of side bearer	a) Roller type clearance for CASNUB 22 W. b) Spring loaded constant contact for CASNUB 22 HS. c) Constant Contact type (metal bonded rubber pads for others. d) P U pad (Poly Urethane) for BOXN M1 & BOXNEL wagon
9	Anti rotation feature	Anti rotation lugs have been provided between bogie bolster and side frame.
10	Type of brake beam	Unit type cast steel brake beam suspended from side frames bracket CASNUB 22 W (M)
11	Center pivot	1) IRS spherical type for 22W only. 2) Spherical type (UIC concept) for others.
12	Suspension details	Long travel Helical Springs.
13	Elastomeric Pad	Elastomeric Pad has been provided between adopters and side frames pedestal roof to reduce wheel flange wear.

BOGIE COMPONENTS

- Side frame with friction plates
- Spring plank
- Bolster with wear liners
- Friction Shoes
- Load bearing spring
- Center pivot top & bottom
- Bolster pivot pin
- Side bearer
- Wheel sets with bearing units
- Side frame key with Key bolt, nut, spring washer & split pin.
- Elastomeric pads.
- Bogie brake gear components.
- Packing piece for CBC height adjustment.



BOGIE GENERAL ARRANGEMENT CASNUB 22 W

Versions of CASNUB Bogie:

CASNUB 22 W (MARK I)

- Introduced in 1972
- Weight 5.35 T
- Provision of adapter retainer bolt in outer jaw
- E M pad not provided
- Provision of wide jaw adapter
- Provision of floating bolster
- IRS pivot riveted and welded or casted
- Clearance roller type side bearer
- Sliding pocket type brake beam
- Fitted with CTRB

CASNUB 22W RETROFITTED

- Introduced in 1990
- Weight 5.35 T
- Provision of adapter retainer bolt in outer jaw
- E M pad provided
- Provision of wide jaw adapter with 129.5 mm thickness
- Provision of floating bolster
- IRS pivot riveted and welded or casted
- Constant contact metal bonded rubber pad in side bearer
- Sliding pocket type brake beam
- Fitted with CTRB

CASNUB 22 NL

- Introduced in 1989-90
- Weight 5.5 T
- Narrow jaw adapter provided
- Adapter retainer bolt not provided

- *E M pad provided*
- *Provision of floating bolster*
- *Pivot riveted and welded*
- *Constant contact metal bonded rubber pad in side bearer*
- *Sliding packet type brake beam*
- *Fitted with CTRB*

CASNUB 22 NLB

- *Introduced in 1990-91*
- *Weight 5.4 T*
- *Narrow jaw adapter provided*
- *Adapter retainer bolt not provided*
- *E M pad provided*
- *Fish belly shaped bolster*
- *Weight reduced*
- *Spherical type centre pivot*
- *Constant contact type mental bonded rubber pads in side bearer*
- *Sliding pocket type bake beam*
- *Fitted with CTRB*

CASNUB 22 HS

- *Introduced in 1993*
- *Weight 5.4 T*
- *Narrow jaw adapter provided*
- *Adapter retainer bolt not provided*
- *E M pad provided*
- *Spherical type centre pivot*
- *Spring loaded side bearer*
- *Sliding pocket type bake beam*
- *Fitted with CTRB*

These bogies are used in BOXN, BCN, BCNA, BRN, BTPN, BTPGLN, BOBR, BOBRN, BOBY, BOBYN, BFK.

Detail of Load and Snubber Spring Arrangement

Axle load	No. Of spring required				
	Outer	Inner	Snubber	Per bogie	Per Wagon
22.90 ton HA	7	5	2	28 nos.	56 nos.
20.3 ton	6	4	2	24 nos.	48 nos.
16.2 ton	4	4	2	20 nos.	40 nos.
20.3 ton (H.S.Bogie)	7	6	2	30 nos.	60 nos.

Free Height of Outer, Inner and Snubber Springs in Bogies

Bogie	Spring	New in mm.	Condem in mm.	variation in mm
All Version except 22 HS	Outer	260	245	15
	Inner	262	247	15
	Snubber	294	279	15
Casnub HS	Outer	260	245	15
	Inner	243	228	15
	Snubber	293	278	15

(Mark-I) and (Mark-II) CASNUB Bogies

Sl. No.	CASNUB 22 W (Mark-I)	CASNUB 22 W (Mark-II)
1	Introduced in 1972	Introduced in 1986
2	Weight 5.35 T	5.45 T
3	Side frame –straight and wide jaw opening, there is provision of adopter retainer bolt in outer jaw.	Modified wide jaw opening and designed to accommodate Elastomeric pad. Camber is provided in side frame of 7 degree. Provision of adapter retainer bolt in outer jaw.
4	There is no provision of Elastomeric pad.	E M pads are provided.
5	There is provision of wide jaw adapter of 48.5 mm thickness.	There is provision of wide jaw adapter of 48.5 mm thickness.
6	There is provision of floating bolster suitable for fitment of IRS pivot .Pivot may be welded and riveted with bolster or may be casted with bolster.	Bolster has been modified to suit fitment of spherical bottom pivot.
7	There is provision of clearance roller type side bearer.	There is provision of constant contact metal bounded rubber pads in the side bearer.
8	There is provision of sliding pocket type brake beam.	There is provision of hanger type brake beam which is suspended with hanger.
9	Bogie is designed to take axle load of 22.9 T axle fitted with CTRB. Its new wheel diameter is 1000 mm and condemning 906 mm.	Bogie is designed to take axle load of 22.9 T axle fitted with CTRB. Its new wheel diameter is 1000 mm & condemning 906 mm.
10	Initially spring plank of structural steel was provided which was later on changed to flanging quality steel. Initially its height was 60 mm and subsequently increased to 75 mm provided with 100 mm pitch hole at the centre which was later on closed and longer opening measuring 200 mm on either side of centre line provided.	Initially spring plank of structural steel was provided which was later on changed to flanging quality steel. Initially its height was 60 mm & subsequently increased to 75 mm provided with 130 mm pitch hole at the centre which was later on closed and longer opening measuring 200 mm on either side of centre line provided.

CASNUB 22 (W) R Bogie:- In 1990 CASNUB 22 (W) Bogies were provided with 46mm thick EM pad between pedestal crown and adopter to reduce the developing of false flange in wheel sets, for that the following modification were adopted in CASNUB 22 (W) Bogie.

- 1) Use of 46mm thick EM pad.
- 2) Reduction in Height of adopter by 23 mm (from 152.5 mm to 129.5 mm).
- 3) Used of New Wheel in dia. Of 956 mm.
- 4) Position of retainer bolt is kept below by 32mm.
- 5) Position of S.F.KEY is reversed on its location.

- 6) Provision of floating bolster.
- 7) IRS pivot riveted and welded or casted.
- 8) Constant contact metal bonded rubber pad in side bearer.
- 9) Sliding packet type brake beam.
- 10) Weight 5.35 T.

CBC Height Adjustment In CASNUB Bogie Fitted Stock:- As per Wheel Dia. Wear 12 mm and 37 mm thick metal packing are utilized as under:-

Use of 12 mm. thick packing

S.N.	TYPES OF BOGIE	WHEEL DIA	LOCATION OF PACKING
1	CASNUB 22 W	948 mm	Between Axle Box adopter and Pedestal Crown.
2	CASNUB 22 W (R)	910 mm	Between Axle Box Adopter EM Pad.
3	CASNUB 22 W (M) & Others	954 mm	Between Axle Box Adopter EM Pad.

Use of 37 mm. thick packing

S.N.	TYPES OF BOGIE	WHEEL DIA	LOCATION OF PACKING
1	CASNUB 22 W	924 mm	Between Axle Box adopter and Pedestal Crown.
2	CASNUB 22 W (M) & Others	930 mm	Between EM pad and Pedestal Crown.

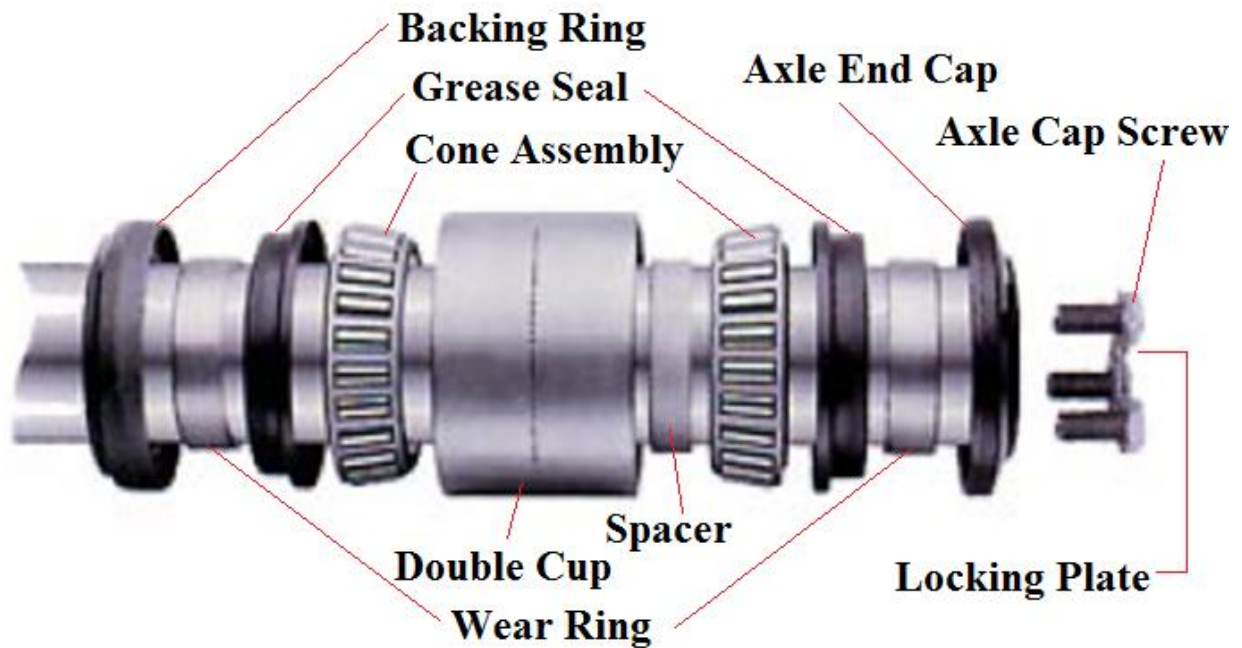
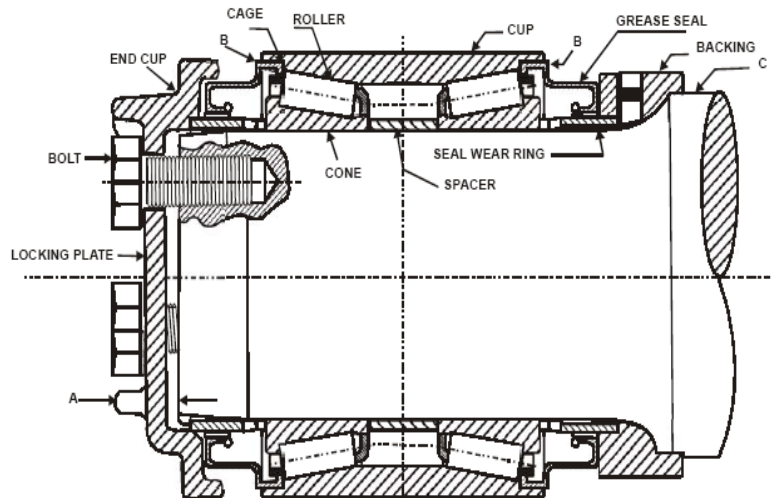
Note: 37 mm thick Packing is not utilized in CASNUB 22 W (R) Bogie.

Adjustment of Brake Gear Pin in Bogie

Wheel Dia.	1000-982 mm	981-963 mm	962-944 mm	943-925 mm	924-906 mm
Hole to be used for Brake Gear Adjustment	'A'	'B'	'C'	'D'	'E'

(CTRB) CARTRIDGE TAPERED ROLLER BEARING:

	Name of parts	No. Off
1.	Cone assembly.	- 2
2.	Cup (consisting 24 nos. roller & cage)	- 1
3.	Spacer	- 1
4.	Wear ring	- 2
5.	Grease seal	- 2
6.	End cap	- 1
7.	Axle cap screw	- 3
8.	Locking plate	- 1
9.	Backing ring	- 1



Defects in CTRB:

a) Double cup:

- Stain discoloration, corrosion, pitting and rust.
- Flaking/spilling.
- Beveling.
- Pitting.
- Electric burns.
- Cracks & fractures.
- Decrease in outer diameter.
- Increase in counter bore.

- b) *Cone assembly:*
- *Stain discoloration, corrosion, pitting and rust.*
 - *Flaking/spalling.*
 - *Smearing & peeling.*
 - *Wear & tear of Cage.*
 - *Pitting marks on roller surface.*
 - *The clearance between cage pocket & roller is more than 1.5mm.*
 - *The clearance between cage flange & inner ring is more than 2.3mm.*
 - *Internal diameter of cage more than 144.4879mm.*
- c) *Wear ring:*
- *Breakage in contact route of lip.*
 - *Scratches or cracks on out face.*
- d) *Grease seal:*
- *Hardened, cracked or cut seal lip.*
- e) *Backing ring:*
- *Pitting marks*
 - *Cracked fractured or heavy corrosion.*
 - *Loose wear ring in counter bore of locking ring.*
 - *Increase internal diameter i.e. more than 178.56mm.*
- AAR approved grease Qty/ Bearing – 455 $\pm$ 30 grams.
- Lateral play fixture – 0.51 to 0.66mm.
- Torque for axle end cap – 40 kgm.
- Bearing mounting pressure – 55 $\pm$ 5 Ton.

Defects, their cause & remedial action:

<i>Sl. No.</i>	<i>Damaged condition</i>	<i>Possible cause</i>	<i>Remedial measure</i>
1.	<i>Flaking</i>	<i>Entry of foreign particles, water & poor lubrication.</i>	<i>Improve the sealing mechanism, use the lubricant with proper viscosity.</i>
2.	<i>Peeling</i>	<i>Rough surface due to poor lubrication, entry of debris into lubricant.</i>	<i>Use proper lubricant, Improve the sealing mechanism.</i>
3.	<i>Scoring</i>	<i>Excessive load, shaft bending.</i>	<i>Check the size of load, check the precision of shaft.</i>
4.	<i>Smearing</i>	<i>High speed & light load, sudden acceleration / deceleration.</i>	<i>Improved the preload, check bearing clearances.</i>
5.	<i>Fracture</i>	<i>Impact during mounting, poor handling.</i>	<i>Improved mounting methods, provide enough back up & support for bearing rib.</i>
6.	<i>Cracks</i>	<i>Heat generation due to creep, excessive interference.</i>	<i>Correct the interference, use & appropriate shaft shape.</i>
7.	<i>Cage damage</i>	<i>Poor mounting, excessive rotation speed, shock & large vibration.</i>	<i>Check mounting method, check the rotation & load condition, reduce the vibration.</i>

8.	<i>Denting</i>	<i>Debris caught in the surface, shock during transport or mounting.</i>	<i>Wash the housing; improve the mounting & handling methods.</i>
9.	<i>Pitting</i>	<i>Exposure to moisture in the atmosphere, poor lubrication</i>	<i>Filter lubricating oil, improve sealing method.</i>
10.	<i>Wear</i>	<i>Progression from rust & electrical corrosion, sliding due to irregular motion of rolling elements.</i>	<i>Improve the sealing methods, prevent misalignment.</i>
11.	<i>Fretting</i>	<i>Vibration with small amplitude, in sufficient interference.</i>	<i>Check the interference fit, apply a film of lubricant to the fitting surface.</i>
12.	<i>False brinelling</i>	<i>Oscillation & vibration of a stationary bearing during transporting, oscillation motion with the small amplitude.</i>	<i>Secured the shaft & housing during transporting, reduce vibration while preloading.</i>
13.	<i>Creep</i>	<i>Insufficient interference or loose fit, insufficient sleeve tightening.</i>	<i>Check the interference & prevent rotation, correct sleeve tightening, tighten the raceway ring side race, apply adhesive to the fitting surface.</i>
14.	<i>Seizure</i>	<i>Excessive small internal clearance, poor precision of shaft & housing.</i>	<i>Check precision of shaft & housing. Study preload, bearing clearances & fittings.</i>
15.	<i>Electric corrosion</i>	<i>Electric potential difference between inner & outer rings.</i>	<i>Design electric circuit which prevents current flow through the bearings, insulation of the bearing.</i>
16.	<i>Rust & corrosion</i>	<i>Entry of corrosive gas & water, poor rust prevention treatment</i>	<i>Improve the sealing methods, anti-rust treatment for periods of long running</i>
17.	<i>Mounting flaws</i>	<i>Inclination of inner & outer rings during mounting or dismounting, shock load during mounting or dismounting.</i>	<i>Use appropriate jig & tool, avoid a shock load by use of a press machine, center the relative mating parts during mountings.</i>
18.	<i>Discoloration</i>	<i>Poor lubrication, oil stain due to reaction with lubricant.</i>	<i>Improve lubrication methods.</i>

CENTRE BUFFER COUPLER

CBC is combined unit of Draw and Buffing Gear, located at the centre of Body Head Stock.

- i. Used as Draw Gear.*
- ii. Used to transmit buffing force.*

ADVANTAGES OF CBC:

- *Coupler and buffing gear are both located together at the centre of the wagon.*
- *Coupling action between wagon is automatic so that more safer for operation.*
- *Hauling capacity increased.*
- *Maintenance cost is less than conventional Draw gear.*
- *Hauling capacity of conventional draw gear is- 2200 T*
- *Hauling capacity of CBC -7000 T to 9000T.*

TYPES OF CBC ON THE BASIS OF OPERATION:

- (i) Transition type CBC. (ii) Straight type CBC.*

MAIN COMPONENTS OF CBC

A. COUPLER BODY:

- | | |
|---------------------------|--------------------------------------|
| <i>i. Coupler head.</i> | <i>iv. Coupler guard arm.</i> |
| <i>ii. Coupler shank.</i> | <i>v. Coupler lock chamber.</i> |
| <i>iii. Coupler tail.</i> | <i>vi. Coupler shank wears plate</i> |

B KNUCKLE:

- | | |
|---------------------------------|--------------------------------------|
| <i>i. Knuckle nose.</i> | <i>iv. Knuckle thrower / Kicker.</i> |
| <i>ii. Knuckles lock face.</i> | <i>v. Knuckle pin.</i> |
| <i>iii. Knuckle front face.</i> | |

C. STRIKER CASTING

D. STRIKER CASTING WEAR PLATE

E. LOCKING PIECE

F. ROTARY LEVER ASSEMBLY

- (i) lever hook. (ii) lever contraction (iii) toggle.*

G CBC OPERATING HANDLE

H CBC OPERATING HANDLE BRACKET

I CBC OPERATING HANDLE WEAR PIECE

J YOKE PIN SUPPORT PLATE

K YOKE SUPPORT PLATE

L TRANSITION COUPLING

- (i) Clevis. (ii) Clevis pin.*

M FRONT FOLLOWER

N BACK STOPPER

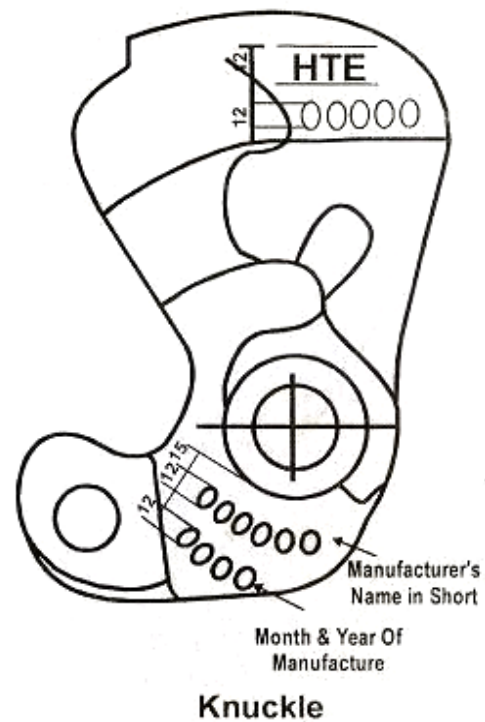
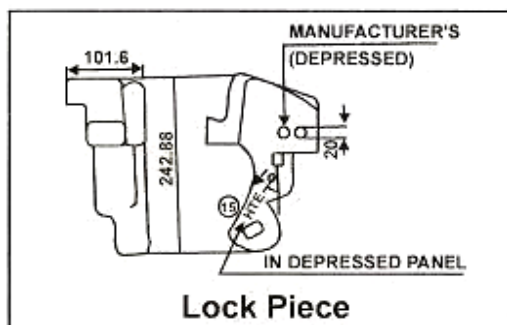
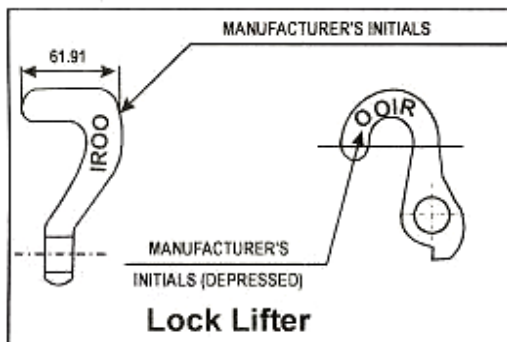
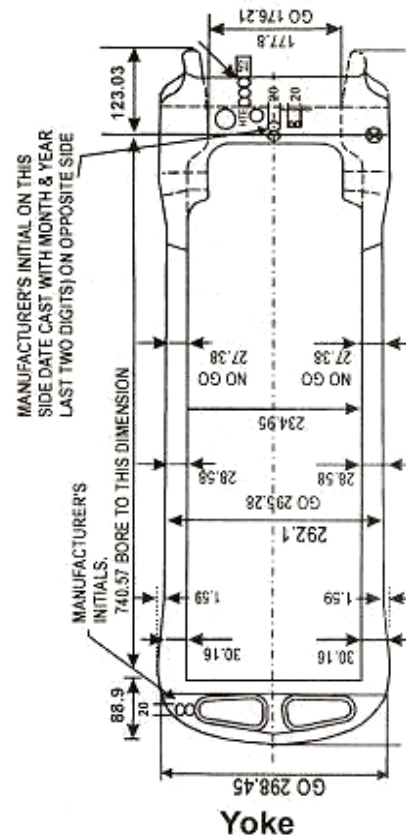
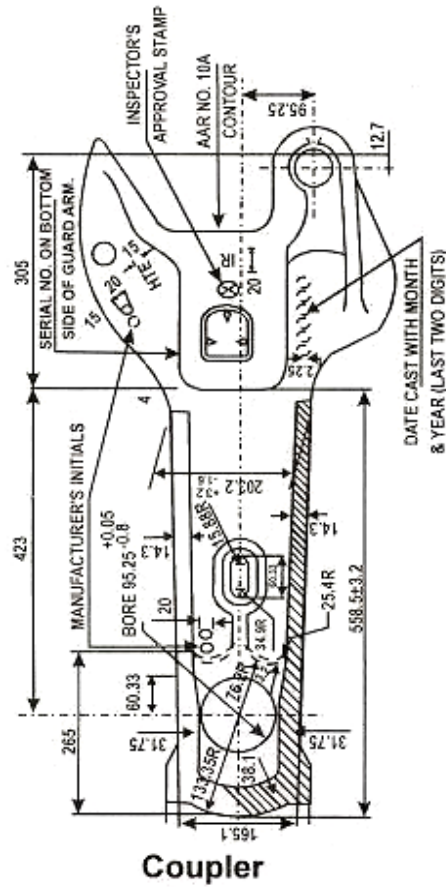
O YOKE, YOKE PIN

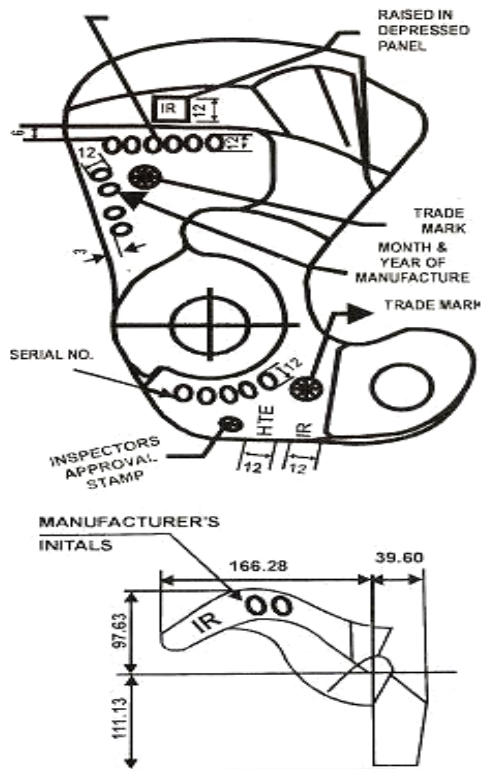
P DRAFT GEAR

- i. HR-40 [All vacuum brake bogie goods stock]*
- ii. MK-50[Air brake bogie goods stock]*
- iii. RF-361 [-----do-----]*
- iv. SL-76 [-----do-----]*

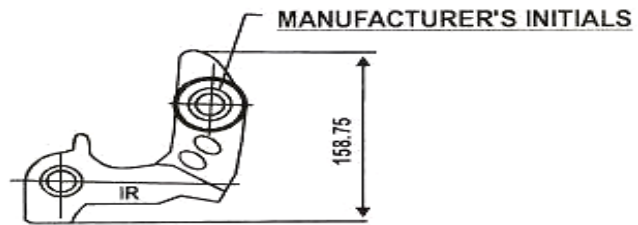
Q BEARING PIECE: Size 16 x 16 x 270- mm (Modified) with 16x16x210-mm

MARKING ON CBC (MANUFACTURERS INITIAL)





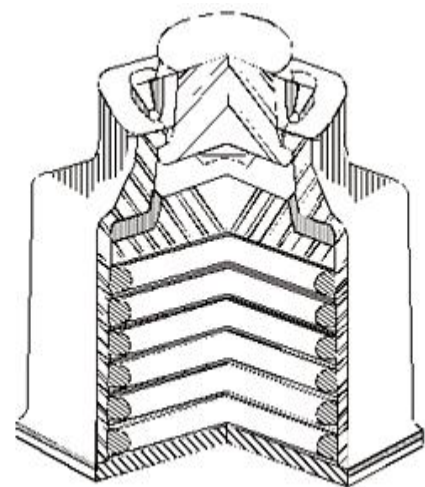
Marking of Parts



S.No.	Manufacturer's Name	In Short
1	Burn Standard Company Ltd.	BURN
2	Bhartia Electric Steel Company Ltd.	BESCO
3	Bhilai Engineering Corporation Ltd.	BECO
4	Hindustan Development Corp. Ltd.	HDC
5	Mukand Ltd.	MUKUND
6	Orient Steel Industries Ltd.	Orient
7	Sri Ranga Alloys Ltd.	Ranga
8	Titagarh Industries Ltd.	TITA
9	Texmaco Ltd.	TEXMA
10	Datre Corporation Ltd.	DCL
11	Raneka Industries Ltd.	RIL
12	Braithwaite & Co. Ltd.	BWT

Types of CBC

- i. Standard AAR coupler: HT (high tension)
 - ii. Alliance II coupler (Now only in 4 wheeler Tank Wagon)
1. STANDARD AAR (NHT) COUPLER:
 - Broad gauge goods stock
 - Used in vacuum brake bogie stock and brake van
 - Working capacity-86T.
 - Full proof load-132T.
 - Braking capacity-251T.
 - Hauling capacity in 1:100 up gradient-6500T-7000T.
 - Draft gear utilized-HR-40.1.
 - Buffing capacity-6200 Kg.
 - Long case buffer used with transition CBC.
 2. STANDARD AAR (HT) COUPLER:
 - Upgraded version of standard AAR coupler.
 - Used in air brake broad gauge goods stock.
 - Buffer not used since it is a straight CBC type.
 - Black color paint.
 - Working capacity-120T.
 - Full proof load-170T.
 - Braking capacity-295T.
 - Hauling capacity in 1:100 up gradient-9000T.
 - High capacity friction draft gear is used.



RF - 361 DRAFT GEAR

Standard AAR coupler types

NHT (Non High Tensile) → HT (10/87) (High Tensile) → HTE (5/92) → HTEA (2008)

TRAIN PARTING

When a train after formation, during run, shunting operation or while starting / stopping divides itself into two or more parts, is termed as train parting. This is termed as "J" Class Accident. In such cases the driver should keep the front portion moving as far as possible until rear portion has stopped to avoid any possibility of collision between the two parts of the train and cause of derailment. Guard should immediately try to stop the rear portion of the train by applying the hand brakes to avoid collision with front portion. Then driver & guard shall act accordingly to provision of G & SR to clear the block section in case of train parting occurred in the block section.

THE REASON OF TRAIN PARTING

Incidence of train parting is an accident as per accident manual. The train parting takes place mainly due to following reasons:-

a) Operational reason

Such as bad engineman ship by driver Brake binding due to emergency application of the brakes. Bad driving technique such as fast notching up of locomotive, sudden application of brakes, bad driving on gradient, improper road knowledge etc. can also contribute to train parting.

b) Due to defective signal

Train parting mostly occurs on up gradient followed with down gradient. The most important location where the cases are more is near the home signal where drivers while starting from home signal after stopping are not ensuring releasing of brake resulted shock load on knuckles.

c) Poor Maintenance of P Way Track

Analysis of section wise occurrence of train parting on railway indicates that bad section can be identified such as poor rail joints, mud accumulated track create uneven height with CBC coupler resulted train parting due to vertical slipping of knuckle and driver should be counseled for pre drive techniques.

d) Improper maintenance of rolling stock

Train parting takes place in goods train due to following improper maintenance of rolling stock.

Causes & their remedial action for AAR CBC:

Sl. No.	Defects	Causes	Remedial Action
1.	Excessive wear of knuckle nose.	Improper maintenance.	CBC Knuckle nose wear gauge No.- 2 must be apply at any schedule of maintenance.
2.	Expansion of guard arm.	Ineffective anti-creep mechanism.	During ROH & other repairs it should be checked by CBC gauge No.- 2.
3.	Excessive CBC drooping.	Excessive wear on shank wear plate/ missing of shank wear plate.	It should be replaced in sick line & yard examination
4.	Cracks on Coupler Body.	Defective material/ Weak draft gear.	It should be replaced during POH & ROH
5.	Cracked knuckle	More Slack/ Weak draft gear.	It should be replaced during POH & ROH

6.	<i>Bent operating handle.</i>	<i>Excessive projection of operating handle more than 75 °.</i>	<i>It should be checked and replaced by examining staff in sick line and yard.</i>
7.	<i>Dropping of Yoke pin support plate.</i>	<i>Loose & deficient rivets.</i>	<i>It should be secured properly by rivets</i>
8.	<i>Improper/ Partial locking of knuckles</i>	<i>Locking piece dislogged.</i>	<i>Ensure proper locking of knuckles by examining staff</i>
9.	<i>Breakage of front & rear stoppers.</i>	<i>Dead or weak draft gear.</i>	<i>Check the damaged draft gear.</i>
10.	<i>Deficiency of lock lifting components</i>	<i>Defective and bent operating gear.</i>	<i>Ensure that all the components lock lift assembly is intact.</i>
11.	<i>Excessive wear on the lock and knuckle at locking face.</i>	<i>Ineffective anti creep mechanism. Lock seat worn out or lug broken</i>	<i>During ROH & other repairs it should be checked by CBC gauge No.- 2.</i>
12.	<i>Stretched knuckle.</i>	<i>Defective or sudden draw and buffing forces.</i>	<i>During ROH & other repairs it should be checked by CBC gauge No.- 1.</i>
13.	<i>Dropping of the operating handle on run.</i>	<i>Improper securing of bearing piece.</i>	<i>It should be ensured at every attention by examining staff.</i>
14.	<i>Breakage of the operating lever support bracket</i>	<i>Bent/ Damaged operating lever and loose riveting of support plate.</i>	<i>Ensure proper riveting & bent lever to be replaced.</i>
15.	<i>Excessive wear of the yoke pin support plate.</i>	<i>Yoke pin hole elongated.</i>	<i>Check any excessive rubbing marks on the bottom of the yoke.</i>
16.	<i>Breakage/ Perished rubber pads in draft gear.</i>	<i>Excessive play in draft gear.</i>	<i>Proper attention during POH & ensure proper fitment of rubber pads & check for damaged draft gear by examining staff.</i>
17.	<i>Failure of anti-creep mechanism.</i>	<i>Excessive wear of CBC components.</i>	<i>Ensure all the components of CBC body/ lock lift assembly are intact.</i>
18.	<i>Self rotating CBC Operating Handle</i>	<i>Anti rotation lug not available or non standard. (Standard size 270 X 16 X 16mm). CBC operating handle bearing piece slot more wear and tear (Standard size 17.5 X 17.5 mm</i>	

e) Failure on A/C of Commercial Department

- Over-Loading (Beyond carrying capacity) :
If any particular wagon is over-loaded beyond its carrying capacity will lead to difference in buffer height between two wagons (more than 75 mm), it may cause disengagement of knuckle in locked condition i.e. vertical slipping of knuckles.
- Un-Even Loading: Due to un-even loading of any particular wagon it may lead to uneven CBC height on either end of the wagon & may cause in train parting.

f) Miscellaneous

- Due to handling of operating handle by miscreant
- Grazing of operating handle with platform while running.

ACTION TO BE TAKEN BY THE FIELD STAFF: -

1. Parting of train takes place due to breakage of CBC or screw coupling during running of train. This is a dangerous situation for safety of trains therefore precaution to be taken by crew as mentioned below
 - I. **To keep away front portion of train running:** On noticing that train has parted, the driver of the train must try to take the front portion of the train as much ahead as possible. This may be done till the rear portion of the train comes to stop. This will avoid any chance of the collision in between both the portion of parted train. For taking ahead the front portion driver may take notches with the permissible amount of traction current.
 - II. **To stop rear portion of the train:** After noticing the train parting, Guard of the train will apply hand brakes to stop the rear portion of the train as soon as possible.
 - III. **Exchange of signal:** After noticing, Driver may confirm the train parting as follows
 - 1) Experiencing sudden drop of air pressure
 - 2) Looking back and finding out that the train is parted in two portions.
 - iv) **Re coupling of both portion of the train after stoppage of both portion of train:** guard of the train should secure the rear portion of load by applying the hand brakes properly. After bringing both portions together the following step to be taken.
 - Intimation to section controller about parting of train
 - After coupling of both CBC's there should be no gap (lock should proper engage)
 - Connection of air hoses and preparation of pressure
 - Conduct continuity test.
 - Start the train to reach up to next station
 - Repeat the case to station master/SCOR in following details
 - Engine Number
 - Train load
 - Driver's & Guard's Name.
 - Location of wagon evolved in train parting.
 - BPC issuing station and date.
 - Kilometer on track.
 - Condition of gradient.
 - Train parting and load coupling time
 - Wagon particulars.
 - Probable cause of train parting etc.

v) **Clearing of load in two Portion:**

After parting of train if it is not possible to clear the load in one hook due to damage of CBC or its component, the parted load to be cleared in two portions. Front portion should be cleared by the driver up to next incoming station & rear portion can be attached and cleared with another engine after attaching with rear most wagons.

ROH of BOX N Wagon:

ROH of BOX 'N' wagon is carried out at selected major sick line where all ROH facilities for BOX 'N' wagon is available at a definite interval.

The First POH of BOX 'N' wagon is carried out at a interval of 6 years/(72 months) & subsequent POH periodicity is reduced to 4 ½ years/(54 months) accordingly railway board/RDSO has decided to carry out 3 ROH after a interval of 18 months before first POH & 2 no. of ROH in between subsequent POH.

The complete wagon has been broken down into 5 major sub systems:

1. The body including under frame, super structure & brake rigging.
2. CASNUB bogie.
3. Air brake system.
4. Center buffer coupler
5. Wheel, axle and bearing.

Each sub system is further split into individual items and the particular work to be done in respect of each item is indicated under the different ROH schedules.

BODY

1. UNDER FRAME :-
Sole bar: Scrap the portion of sole bar at doorways, clean and apply primer paint followed by Top Coat
2. SIDE WALL :-
SKIRTING: Check and patch if corroded apply thin primer and top coat on the patch.
 - SIDE DOORS: Check damage and repair, clean and lubricate hinges.
 - SIDE PILLARS: Check cracks at the base and repair.
3. UNDER GEAR:
 - BRAKE LINKAGES: Check free movement on SWTR test.
 - HAND BRAKE: Check proper working.

CASNUB BOGIE

4. BOLSTER:-
 - POCKET SLOPE LINER: change liner if thickness less than 5mm.
 - ROTATION STOP LUGS: Provide liner (thickness to suit) if dimensions less than 514mm.
 - INNER COLOUMN GIB: Provide liner (thickness to suit) if dimensions more than 142mm.
 - LAND SURFACE: Provide liner (thickness to suit) if dimensions less than 442mm.
 - OUTER COLOUMN GIB: Renew by welding if dimension more than 241mm.
5. SIDE FRAME
 - COLOUMN FRICTION LINER: Change liner if dimension more than 445mm.
 - COLOUMN SIDE: Provide liner (thickness to suit) if dimension less than 209mm.
 - ANTI ROTATION LUGS: Provide liner (thickness to suit) if dimension more than 526mm.
 - KEY SEAT TO PEDESTAL 22 W: Provide liner (thickness to suit) if dimension more than 276mm.

- CROWN ROOF 22 W (M): Provide liner (thickness to suit) if dimension more than 321mm.
- CROWN ROOF 22 NL: Provide liner (thickness to suit) if dimension more than 326mm.
- PEDESTAL CROWN SIDES: Renew by welding if dimension less than 147mm.
- PEDESTAL JAW 22 W: Provide liner (thickness to suit) if dimension more than 275mm.
- PEDESTAL JAW 22 W (M): Provide liner (thickness to suit) if dimension more than 283mm.
- PEDESTAL JAW 22 NL-S: Provide liner (thickness to suit) if dimension more than 195mm.
- PEDESTAL JAW 22 NL-L: Provide liner (thickness to suit) if dimension more than 241mm.
- PEDESTAL SIDES 22 W: Provide liner (thickness to suit) if dimension more than 102mm.
- PEDESTAL SIDES 22 W (M): Provide liner (thickness to suit) if dimension more than 195mm.
- PEDESTAL SIDES 22 NL: Provide liner (thickness to suit) if dimension more than 78mm.

6. WEDGE:

- SLOPE SURFACE: Renew by welding if dimension less than 7mm.
- VERTICAL SURFACE: If vertical surface from the centre line of spigot less than 56mm, provide liner of 6mm thickness.

7. CENTRE PIVOT (BOTTOM)

- VERTICAL SIDE 22 W: Renew by welding if wear more than 4mm.
- VERTICAL SIDE 22 W (M): Renew by welding if wear more than 3mm.
- VERTICAL SIDE 22 W NL: Renew by welding if wear more than 3mm.
- SEAT 22 W: Renew by welding if wear more than 3mm.
- SEAT 22 W (M): Renew by welding if wear more than 3mm.
- SEAT 22 NL: Renew by welding if wear more than 3mm.

8. COIL SPRING:

- OUTER: Group and use in sets. Replace if free height less than 245mm.
- INNER: Group and use in sets. Replace if free height less than 247mm.
- SNUBBER: Group and use in sets. Replace if free height less than 279mm.

9. BOGIE BRAKE GEAR:

- PINS & BUSHES: Change if clearance is more than 1.5mm.

AIR BRAKE SYSTEM

10. DISTRIBUTER VALVE:

- D.V.: Test on SWTR.
- D.V. ISOLATING COCK: Examine operation.
- D.V. RELEASE VALVE: Examine operation.
- D.V. P4AG FILTER: Clean

11. BRAKE CYLINDER:

- FILTERS OF ESCORTS AND RPIL MAKE: Clean.
- BRAKE CYLINDER OF GREYSHAM AND WSF MAKE: Lubricate.

12. CUT OFF ANGLE COCK:

- ANGLE COCK: Examine and lubricate.
- RUBBER SEALS: Change.

13. DIRT COLLECTOR:

- DIRT COLLECTOR: Clean
- SEALING RING: Change

14. **RESERVOIR:**
 - *AR & CR: Drain*
 - *SEALING RING: Change*
15. **HOSE COUPLING:**
 - *HOSE AND COUPLING: Examine*
 - *GASKET (MU WASHER) : Change*
16. **METAL PIPES AND JOINTS:**
 - *PIPE AND JOINTS: Examine leakage and repair.*
 - *SEALS (20MM & 32MM PIPE) : Change.*
17. **SLACK ADJUSTER:**
 - *SLACK ADJUSTER: Test functioning, repair if required.*
 - *A DIMENSION: Adjust.*
 - *M20 ANCHOR PIN NUT: Ensure securing by welding to pin.*
18. **AIR BRAKE SYSTEM:**
 - *BRAKE SYSTEM: Test on SWTR as per procedure.*
 - *BRAKE BLOCK: Ensure standard key split pin and all new brake block.*

CENTRE BUFFER COUPLER

19. **CBC BODY:**
 - *COUPLER BODY: Replace on condition.*
 - *CBC CONTOUR: Examine, replace if required.*
 - *SHANK WEAR PLATE: Replace on condition.*
20. **KNUCKLE:**
 - *NOSE: Replace if wear more than 4.3mm with H.T.E. knuckle. Knuckle with nose wear more than 4.3mm and less than 9.0mm can be used in yard replacement.*
 - *KNUCKLE PIN: Replace on condition.*
 - *KNUCKLE STRETCH: Examine, replace if required.*
21. **STRIKER CASTING:**
 - *WEAR PLATE: Replace*
 - *STRIKER CASTING: Replace on condition.*
22. **COUPLER MECHANISM:**
 - *ANTI CREEP PROTECTION: Examine and repair.*
 - *LOCK LIFT ASSEMBLY: Examine.*
 - *OPERATING MECHANISM: Examine.*
 - *LOCK: Examine.*
23. **DRAFT GEAR:**
 - *SLACK: Measure and take corrective action.*

24. GENERAL:

- YOKE PIN SUPPORT: Replace on condition.
- BUFFER HEIGHT: Examine and correct if required.

WHEEL, AXLE AND BEARING

25. AXLE:

- ULTRASONIC TESTING: To be carried out at every ROH and reject if fails.
- DEEP NOTCHES DUE TO GRAGING OF PULL ROD: Reject if depth is more than 5mm.
- AXLE AND HOLES: Clean and lubricate in case end cover is opened.

26. WHEEL:

- TREAD PROFILE: check with tyre defect gauge.
- HEIGHT OF FLANGE: If height more than 31mm do not use in ROH.
- SMOOTH FLANGE: If flange not completely smooth in region 'A', then do not use in ROH.

27. BEARINGS:

- Cup: Rotate the bearing for unusual sound. Check cup for crack/chipping.
- SEAL: Check seal for external damage and dent.
- BACKING RING: Check backing ring for looseness and vent fitting on backing ring with vent hole (the vent fitting should be intact or the vent hole should be plugged).
- LOCKING PLATE: Use new locking plate whenever end cover is opened.
- AXLE END CAP SCREW: Clean and lubricate in case end cover is opened.
- LOAD ZONE CHANGE: Change load zone area of the cup while lowering bogie side frame.

28. ADAPTER:

- CROWN SURFACE: Replace if worn to relief depth.
- SIDE LUG: Replace/reverse and use.
- THRUST SHOULDER: Replace if depth exceeds 0.7mm.
- MACHINED RELIEF: Replace if depth exceeds 0.8mm.

SUMMARY ROH of AIR BRAKE WAGON WITH CASNUB BOGIE:-

BOXN wagons are to be given Routine Over-Haul (ROH) normally after every 18 month at the nominated sick line/wagon depot, where proper facilities are provided.

The ROH schedule is as follows:

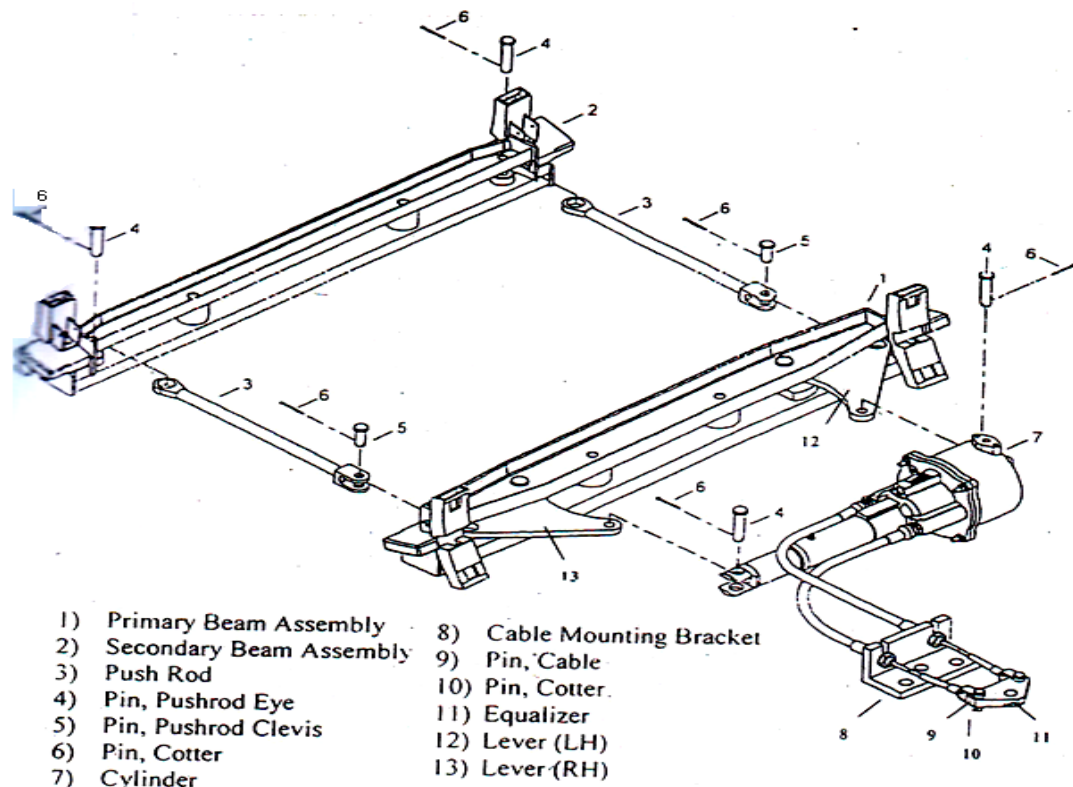
- a) Lift the body, keep it on trestles & run out bogies.
 - b) Strip bogie component for examination & repair as below:-
 - Strip spring & spring suspension arrangement including snubbing device. Check springs for free height & other defects. Replace where required.
 - Examine bogie frame. Check frame alignment as per RDSO instruction.
 - Examine pivot for welding defects/cracks/abnormal depth due to wear. Replace where necessary & lubricate with graphite flakes to IS: 495 in dry condition.
 - c) Strip brake gear levers & rods for examination of worn outs/ damaged parts.
 - d) The equipment shall be given attention in accordance with the maintenance manual issued by respective air brake equipment manufacturer:-
 - i) Cleaning of strainer discs.
 - ii) Lubrication of brake cylinders/ cleaning of its strainer.
 - iii) Check for easy operation of isolating cocks & anti pilferage device of distributor valve cut off angle cock, manual quick release valve & isolating cock.
 - iv) Draining of auxiliary reservoir.
 - v) Checking of hose coupling for serviceability.
 - vi) Cleaning of a strainer.
 - vii) Dirt collector to be cleaned.
 - viii) Leakage in pipe & joints to be checked.
 - ix) After carrying out above items of work the wagon shall be tested for proper function of air brake system with single car test device in accordance with the procedure indicated below:-
 - Connect the BP coupling of single car test rig to the corresponding coupling of the wagon to be tested. The couplings of the other end of the wagon to be closed with dummy couplings heads. Fix pressure gauge on the brake cylinder.
 - The single car test device should now be coupled to the main line of a compressor through a pipe.
 - Carry out the preparation & testing in accordance with the procedure given in the manufacturer's maintenance manual & record the reading in the test Performa.
- For passing the wagon, all the parameters shall be within specified limits.
- e) Clean horizontal levers, hand brakes & gears & lubricate.
 - f) Examine head stock for damage, bend / cracks.
 - g) Refit brake gears levers & rods, Lubricate pins & other equipment & apply graphite to horizontal levers of empty load box.
 - h) Replace worn out brake blocks. Check wheel profile, turn the wheel as needed. UST of axle to be carried out & turning of wheel to worn wheel profile during ROH.
 - i) All the wheels are to be checked ultrasonically & axle box cap bolts are to be tightened up by torque wrench with proper torque & in no case old locking plates are to be reused.
 - j) CBC knuckles are to be checked by contour gauge, anti creep / articulated rotary operation of locking arrangement to be checked.
 - k) Manual adjustment of brake gear to be done in accordance with wheel diameter.
 - l) Modification works are to be attended as issued by RDSO from time to time.

Bogie Mounted Brake System

NEED FOR ADOPTION OF BMBS IN FREIGHT STOCK:

- 1) In order to overcome the problem faced in conventional bogie breakage, missing components and malfunctioning of SAB. In BMBS fitted wagons, Bk. Cylinder is equipped with in built SAB.
- 2) In conventional BOXN/BCN wagons due to frequent theft of control rod, main pull Rod, End pull rod & Bk. Rigging parts in tripling operation of wagons causing poor brake power in Rakes, resulting Detentions of wagons. In BMBS, wagons these Items are eliminated.
- 3) In order to Improve Mechanical efficiency of brake power.
- 4) In a Normal Rake of BOXN wagon Fitted with conventional type BK system there was around 18% Brake fading after a trial run of 1000 Km. But In case of BMBS fitted Rakes No brake fading is observed after around 5000 km Trial Run.
- 5) Braking distance of conventional Rake of 58 +1 wagons of BOXN is 1350 m. However in BMBS wagons it is 880 m (approx).
- 6) In conventional wagons setting of 'A' Dimension was a major problem. If any change made in sleeve nuts of E/L device that may create to heavy loss of wagon creating brake binding / wheel skidding. In order to overcome this, BMBS fitted wagons came into field which contains in built slack adjuster.
- 7) In case of BMBS system Automatic Load sensing Device is used which Gives 2.2Kg/cm^2 Pressure to brake cylinder during empty condition 3.8Kg/cm^2 pressure during loaded for avoiding the above Problems.

Salient Feature Of Bogie Mounted Brake System Used On CASNUB Bogie



- *Use of two no's. of 10" dia brake cylinders with automatic load sensing device.*
- *Use of 58 mm thick K type non asbestos composite brake block.*
- *A constant 56 mm piston stroke is required to maintain uniform and efficient braking performance.*
- *Application of loads is at the ends the brake beam instead of centre resulted no chances of bending of brake beams.*
- *Cylinder is with double acting slack adjustment feature with constant piston stroke resulting in uniform brake performances even as the brake shoes and wheels wears.*
- *Total slack adjusting capacity is 500 mm including brake block wear, wheel wear and all clearances.*
- *All cylinders are equipped with an automatic piston stroke indicator.*
- *Replacement of the brake head is quickly and easy by removal by only one pin.*
- *Provision of mechanical hand brake system with the use of two no's stainless hand brake cables pulled through standard hand brake rigging.*
- *Easy fitment of bogie mounted cylinder on any standard bogies without making any modifications.*
- *Cylinder is mounted parallel to the brake beam and transfers forces through the bell cranks.*
- *Improves the efficiency and alignments of the braking forces with the wheels which reduced the wear of shoes and wheels.*
- *Use of pressure regulated load sensing device, installed between body and bogie under frame act on 15 psi to maintain normal sensor arm travel of 95 mm.*

WORKING PRINCIPLE OF BMBS :

During application, the air is introduced into the brake cylinder which forces out the piston along the ram Assembly. The brake cylinder is floating in nature; as a result the brake cylinder extends equally on both the sides. The extension of brake cylinder causes the rotation of the Bell crank Levers on their pivot, which is on Primary brake beam and forces the push rod to move towards the secondary beam this movement causes the secondary brake beam to move towards the wheel and apply force on the wheels. Simultaneously a Reaction force is developed which causes the primary break beam along with Levers and brake cylinder to move towards the wheel. The primary brake beam continues to move until it touches the wheels and apply force on the Wheels.

When the Brakes are released the air from the brake cylinder is exhausted to the atmosphere through the Distributor value. The return spring inside the brake cylinders pushes the piston along with the ram assembly back to its original position. The bell crank levers rotate back causing the beams to move back to their earlier positions. The Brake cylinder is equipped with a double acting slack adjuster. If there is any wear in Brake shoe/ wheel or any slackness in the structure, it will be automatically compensated by the built in slack adjuster which pays out & fill the gap.

DISADVANTAGES OF BMBS USED IN GOODS STOCK:

- More wear & tear is noticed near eye hole in push rod.
- Bk. Cylinder fails working even if there is a very small defect noticed in distributor valve/ if DV releasing time slightly more/ if APM failed.
- The system requires special parts for working such as push rod, bell crank lever, hand brake cable etc, which are not easily available.
- This type of wagon can not be mixed with a normal rake of conventional type.
- High skilled worker are required for maintenance of these wagons.

BRAKE BLOCK FORCES

S.no.	Wagon	Position	Cylinder pressure (kg/cm²)	Brake block force (ton)	Brake percentage (%)
1.	BOX N HS	Empty	2.2	1.22	43.41
		Loaded	3.8	2.11	20.78
2.	BOBRN	Empty	2.2	1.22	38.15
		Loaded	3.8	2.11	20.76

HAND BRAKE FORCE

S.no.	Wagon	Force on wheel approx.(kg)	Brake block forces(ton)	Brake percentage (%)
1.	BOX N HS	25	11.65	14.35
2.	BOBRN	25	11.65	14.35

Composite Brake Block

Initially cast iron brake block were introduced in rolling stock as a brake rigging component. To overcome the problem of frequent breakage of brake block, reduction of braking force during braking out at high speed & also frequent replacement due to wear of cast iron brake blocks, composite brake block are introduced.

These are of two types:

1. L type composite brake block ($\mu = 0.14$ to 0.17): Used in air brake freight stock & conventional coaching stock (Under frame mounted brake cylinder)
2. K type composite brake block. ($\mu = 0.25$): Used in Bogie mounted Brake system in coaching & freight stock. (Where μ is coefficient of friction)

Main Features:

1. More economical and more reliable.
2. At high speed characteristic of composite brake block coefficient of friction μ is reduced, resulted it can be utilized only with the train having speed up to 110 kmph.
3. No chance of theft.
4. Life of composite brake block is 4 to 5 times of cast iron brake block, hence maintenance cost is reduced.
5. Too much light weight in comparison to cast iron brake block.
6. Easy handling.
7. Coefficient of friction (for L- type is 0.17 & for K-type is 0.25) is more than that of cast iron brake block (0.12).
8. Condemning limit-12mm.

The main disadvantage of composite brake block is that it does not transmit/absorbs the heat generated while braking action resulting it is not suited for trains having speed of more than 110 kmph.

	Brake System	Brake Block	Width	Thickness New	Thickness Condemning
Coaching	BMBS	K – Type	76 mm	49 mm	12 mm
Goods	U/F Mounted	L – Type	85 mm	60 mm	12 mm
	BMBS	K – Type	85 mm	58 mm	10 mm

TANK WAGON



Sr. No.	Consignment	Code
1.	Petroleum & highly inflammable products	TP, TK, BTPN
2.	Raw petroleum products (Black Oil)	TL
3.	Vegetable oils	TV
4.	Bitumen	TB
5.	Molasses	TM
6.	Petroleum gases (LPG)	TPGL, BTPGLN
7.	Others	TAL, TCS

CLASSIFICATION OF TANK WAGONS

- Tanks as pressure vessels.
- Tanks for corrosive liquids.
- Tanks for petrol and other highly inflammable products.
- Tanks for middle distillates of petroleum and other products.

DESIGN FEATURE

The design of the under frame of 4 wheeler and 8 wheelers wagon is generally similar to that of other IRS wagons except that a pair of saddles is provided on the under frame at each end for mounting the barrel.

BARREL MOUNTING

S. N.	MOUNTING S/FITTINGS	FUNCTIONS
1	Safety Valve	The safety valve is provided to prevent building up of excess pressure inside the barrel. It is fitted on the barrel either on the diaphragm plate inside the dome or on a separate opening on the barrel. This is provided on highly inflammable liquid such as petrol., aviation sprit .etc.
2	Relief valve	It is a spring loaded valve fitted on the barrel on tanks for corrosive liquids. Its main function is to release built up pressure, if it exceeds the working pressure limit.
3	Safety vent	This consists of frangible disc (lead or any approved material not affected by loading), which ruptures at specified pressure. It is an additional safety fitting to safeguard against the failure of the relief valve. When the built up pressure exceed the working pressure of the relief valve and the latter fails to function for any reason the frangible disc of this safety vent ruptures to release the pressure.

4	Compressed air valve	<i>It is provided on tank from which the contents are unloaded by compressed air. Its main function is to control the rate of discharge by controlling the rate of air admission.</i>
5	Vapour Extractor cock	<i>Its function is to extract vapour from the tank while filling.</i>
6	Master valve	<i>It is a gravity discharge valve fitted with a hand wheel in the dome for manual operation.</i>
7	Bottom Discharge valve	<i>BG-4 wheeler Bottom discharge valve are provided with the single bottom discharge valve situated underneath the master valve while on BG/MG -8 wheeler stock two bottom discharge valve are fitted one on either side and connected with the master valve through a 'T' pipe. The main function of the valve is to control the flow of the contents and also to serve as an additional safety stock in case the master valve fails or breaks. The bottom discharge valve opening are also provided with blank flanges to be used with 2mm compressed asbestos fiber jointing material to serve as further check on accidental leakage of contents</i>

Definition of Steam Cleaning:-

- *It is cleaning of tank barrel from inside with the help of high pressure steam.*

Need Of Steam Cleaning:-

- *For safe entry of staff in tank barrel before repairs to be carried out.*
- *Up gradation of tank wagons for loading.*

Procedure of Steam Cleaning for Pressure Vessels, Petroleum & Other highly inflammable Products:-

- *The tanks requiring steam cleaning should be placed as near the steam supply line as possible & protected against any movement. Berthing siding should be completely isolated from all other traffic.*
- *Tanks as pressure vessels, tanks for petroleum, other highly inflammable products, vegetables oils, bitumen, coal tar & molasses are cleaned by steam. In case of pressure vessels, it should be ensured that entire gas has been discharged to the atmosphere. After ensuring that the tank barrel is no longer under pressure. The following sequence should be followed*
- *Remove the manhole cover together with manhole housing, valves etc. and leave the tank exposed to atmosphere for 24 hours.*
- *Entry of staff in the tank barrel should be strictly prohibited and signs with suitable legends displayed at a reasonable distances away from the tanks to be steam cleaned.*
- *Insert pipe through manhole and ensure the interior of barrel is thoroughly steamed from inside, the steam pipe should be provided with a "T" connection at its lower end and so directed as to blow steam towards both ends.*
- *Remove condensed steam collected in the tank barrel & keeps the barrel exposed to atmosphere for another 24 hours.*
- *The tests that should be conducted to ensure the tanks are free from contamination gases of the contents.*

Testing Procedure of Tank Wagons

- (i) **Ammonia Tank Barrel:** Fill the tank barrel with water and take a specimen of the same in a clean glass bottle since ammonia is readily soluble in water. The specimen of water should be tested for any traces of ammonia with red litmus paper. Any trace of ammonia in water would turn red litmus blue. Another very sensitive method known as Nessler's test may be applied to find out if the specimen of water contains any traces of ammonia. In this test, the reagent used is a solution of potassium mercuric iodide with potassium hydroxide.

This reagent gives a brown colour when mixed with the specimen of water containing even a minute trace of ammonia. If ammonia is detected, empty out the tank barrel and refill with fresh water. This process may be repeated till the tank barrel is free from ammonia traces completely.

- (ii) **Chlorine Tank Wagons:** Fill the tank barrel with water and take a specimen of the same in a clean glass bottle. Since chlorine is readily soluble in water, specimen of water taken out should be tested for any traces of chlorine. Any trace of chlorine in water would have a bleaching effect on coloured litmus paper. If chlorine is present, the tank should be repeatedly emptied and refilled with fresh water till free from chlorine traces completely.

- (iii) **LPG Tank Wagons:** A clean bottle filled with fresh water is lowered through the manhole. A string is attached to the bottom of the bottle before lowering. Tilt the bottle at the bottom of the tank to allow its water to flow out and the gas in the tank to take its place. The bottle should be left in this position for about 5 minutes and then withdrawn away from the tank. A lighted match stick should then be brought near the mouth of the bottle or applied to the air or gas inside and bottle. If there is no flame the tank is free from injurious gas. But, in case it gives out a flame, the tank should again be steam cleaned. After ascertaining that there is no trace of gas in the barrel the tank should be dried out by blowing in hot compressed air before proceeding with inspection and repairs.

Precautions During Loading And Unloading

- Tank wagon for inflammable liquids shall be in good condition.
- In filling tank wagon, an air space of not less than 5% of the capacity of the tank shall be left. (2.5% in case of HSD, Furnace oil, kerosene oil etc)
- All inlets and outlets shall be securely closed .
- Ensure all tank fittings are in good working condition.
- Ensure that bottom discharge valve is fully closed and fitted with a blank flange and gasket.
- Make sure that the dome cover is closed after loading.
- Don't allow unauthorized persons to operate valves
- Don't allow tank wagons to move from loading / unloading points unless the tank fittings are properly refitted and dome cover closed.

Precautions During Shunting of loaded & leaky tank wagon

- Shunting speed shall not exceed 8 kmph.
- Tank wagon when left in siding must be placed clear off the fouling marks to avoid side collision and must have hand brakes 'ON'.
- Ensure buffer wagons are placed between engine and cargo.
- In case of leaky wagon has to be moved in the yard, it must be separated from the shunting engine/pilot by at least ten wagons not containing goods of an inflammable, dangerous or explosive.

- Don't allow hump/loose shunting.
- Loaded wagon shall be stabled in cool place as far as possible.

Precautions to be followed:

When leakage is found from the Chlorine and Ammonia tanks.

- Chlorine and Ammonia gases are poisonous and have a characteristic pungent odour which gives warning of their presence in the atmosphere before dangerous concentrations are attained.
- In case of chlorine, the greenish yellow color of the gas makes it visible when high concentrations are present.
- In the case of ammonia, if sufficient concentrations of the gas are present in the atmosphere, it will irritate the eyes and the respiratory system.
- As such in the event of leakage, all present in the vicinity should be warned to keep on the windward side of the tank.

When leakage is found from the highly inflammable gas tanks.

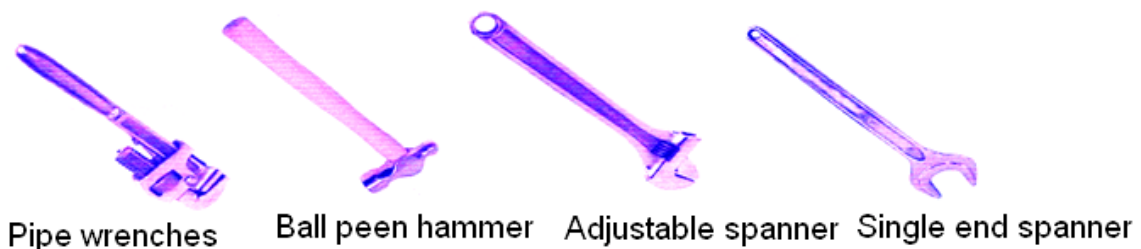
- All the flames or fires near it should be extinguished or removed.
- Smoking should not be allowed.
- Spectators should be kept away.
- Only battery operated torches or incandescent electric lights with gas proof sockets should be used.
- Oil lanterns or signal lamp used for signaling must be kept away.
- The steam engine available if any should be moved away from the site.
- The leaky tank wagon should be removed as quickly as possible to an open area where the escaping gas will be less hazardous.
- Earth should be spread over any surface on which the LPG has leaked out in the liquid form.
- Call the company concerned for further attention.

Protective/Safety Equipments

- Self contained respiratory equipment for attending major leaks.
- PVC full body covered suit, rubber gloves, gum boots, rubber goggles, helmets.
- Fire suit.
- Explosive meter.
- Canister type gas mask for minor leaks.
- Teflon tape and M-seal.
- Wooden dummies.

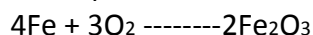
Non Sparking Tools

Material:- Aluminum bronze



Corrosion in wagons

Definition: corrosion is a chemical process of oxidation with metal to its surroundings, covering it into metal oxide, carbonates, hydroxide and sulfides. Oxidation takes place only when steel surface exposed to atmosphere or moisture. Chemical reaction is as follows:



Since mild steel is main building material for wagons corrosion in wagons is unavoidable. In wagons corrosion mainly occurs in wagon floor and roof.

Reasons of Corrosion in wagons:-

- 1) Water logging in cavities and overlaps. If wagon are not cleaned properly after unloading, the accumulated dust and refuse retain moisture for a prolonged period and speeds up corrosion.
- 2) Contact of panels with corrosive consignments such as salt and fertilizers.
- 3) Spillage of corrosive fluids due to defective packing or rough handling.
- 4) Escape of corrosive vapors from consignment such as acids.
- 5) Inadequate protection from weathering because of poor painting or poor surface preparation
- 6) Usually only exterior of wagons are painted and interior is left unattended. Corrosion develops from inside rather than outside. Incorrect fitness of side panels.
- 7) Seepage of water at corners and ends due to water accumulation on floor.
- 8) In sufficient surface preparation before welding.
- 9) Accumulation of water on floor in open wagons like BOXN, BOXNHS, BOXC etc.
- 10) Ingress of water near sole of wagon and near door.

Anti corrosive measures in wagons

- 1) Ensure thorough cleaning of wagons after unloading
- 2) Ensure proper cleaning, surface preparation and painting during POH, ROH and sick line attention.
- 3) During panel patching and welding ensure that the surfaces in contact are well fitted to avoid water pockets.
- 4) Use copper bearing steel (IS-2062) for body panels.
Use of stainless steel IRSM-44 in BOXN R in wagon body
- 5) Use of stainless steel IRSM-44 in under frame & wagon body of newly built BOXN HL/BCN H .
- 6) Only standard panel patches to be used for body repairs. Before fitment panel patches to be given two coats of zinc chromate primer after proper surface cleaning.
- 7) All covered wagons are to be made water tight during POH and ROH.
- 8) Follow proper painting schedule during POH and ROH.

Latest Anti corrosion Modification:

1. Plugging of Holes provided on Floors.
2. Use of dip Trip Plate on Sole Bar at the location of Doors.
3. Provision of Box Section Stiffener above Doors.
4. Provision of Crib Angle at bottom side of the Floor.
5. Provision of Stiffener from inside and outside at the top corner of the Wagon Body.

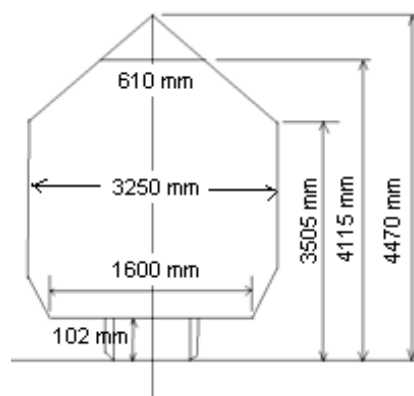
ODC: - OVER DIAMENSIONAL CONSIGNMENT

1. **INTRODUCTION:-** The transport of over dimensional consignment has increased considerably in recent years. It has therefore become necessary to give wide publicity to the procedure in vogue on the railway for arranging movement of such consignment. This circular should be read in condition with the railway boards instructions issued from time to time & should be understood by all the staff responsible for movement of oversize consignment.



2. MAXIMUM MOVING DIMENSIONS:-

- | | |
|---|-----------|
| i) Overall height at top centre | : 4115 mm |
| ii) Overall height of OHE from rail level | : 4470 mm |
| iii) Overall height at the side | : 3505 mm |
| iv) Overall width of eight wheeler | : 3250 mm |
| v) Overall width at maximum level at top center | : 610 mm |
| vi) Floor height | : 102 mm |
| vii) Overall width at bottom | : 1600 mm |



3. **DEFINITION:-** An ODC is found, when consignment which is loaded on ordinary open wagon, would infringe the maximum moving dimension of the gauge over which is to be moved.

4. TYPES OF CLEARANCES:- There are two types of clearances:-

- i) Net clearance
- ii) Gross clearance

Net Clearance: - The maximum clearance between consignment & fixed structure on running condition will be known as "Net Clearance".

Gross Clearance: - The net clearance between consignment & fixed structure in stationary condition is known as "Gross Clearance".

NOTE: Net Clearance is less than Gross Clearance.

5. CLASSIFICATION OF ODC:- The ODC has been classified in following three categories:-

- I) A- class ODC
- II) B- class ODC
- III) C- class ODC

I) A- Class ODC :- (Permitted Out of Gauge): The consignment load which infringes the maximum moving dimension, but do not infringe any fixed structure en-route by net clearance of 150 mm & above & gross clearance 230 mm & above.

- i) Speed restriction : 40 kmph
- ii) Movement : During day & night
- iii) Escorting : Not required

II) B- Class ODC:- (Exceptional Out of Gauge): The load which infringe the maximum moving dimension, but do not infringe any fixed structure en-route by net clearance of more than 75 mm & less than 150 mm & gross clearance of more than 150 mm & less than 230 mm.

- i) Speed restriction : 40 kmph
- ii) Movement : During day & night
- iii) Escorting : During Night
- iv) Escorting Staff : PWI, TI & CWI

III) C- Class ODC:- (Extra ordinary Out of Gauge): The load which infringe the maximum moving dimension, but do not infringe any fixed structure by a net clearance of less than 75 mm & gross clearance by less than 150 mm.

- i) Speed Restriction : 15 kmph
- ii) Movement : During day only
- iii) Escorting : required
- iv) Escorting Staff : JE (P-way), TI, JE (OHE)

Note: On the basis of gross clearance C-Class ODC is further divided in two categories;

1. If net clearance is more than zero but less than 75 mm then speed: 15 kmph only in day and must be escorted by TXR, PWI, LI, and TI etc.

2. If net clearance is zero then speed: dead slow, only in day and must be escorted by TXR, PWI, LI and TI etc.

6. SPECIAL INSTRUCTIONS FOR THE MOVEMENT OF ELECTRIC SECTION

1. AC Traction –

Sl. No.	Gap Between Contact Wire & ODC	Speed	Power
1.	390 mm & above	40 kmph	ON
2.	Less than 390 mm & till 340 mm	15 kmph	ON
3.	Less than 340 mm & till 100 mm	15 kmph	OFF

2. DC Traction –

Sl. No.	Gap Between Contact Wire & ODC	Speed	Power
1.	245 mm & above	40 kmph	ON
2.	Between 245mm & 170 mm	15 kmph	ON
3.	Between 170 mm & 75 mm	15 kmph	OFF

Note: If Clearance is between OHE wire & consignment is less than 100mm in AC traction area and less than 75mm in DC traction areas movement of ODC is not possible.

7. PROCEDURE FOR VARIFICATION IN CE's OFFICE:-

The following particulars shall be furnished in duplicate by the COM's office to the Chief Engineer's office in respect of nay over sized consignment.

- i) Length of consignment*
- ii) Height of consignment at top center & side*
- iii) Width of the consignment at top & bottom*
- iv) Weight of consignment*
- v) Booking station*
- vi) Destination station*
- vii) The route by which the consignment is to be booked*

In case of old consignment a sketch showing end & front elevation with the complete dimension Length, width, Height & Weight shall be sent along with the application in duplicate.

- i) Each division will send up to date rolling diagrams showing the moving dimension on the division by 30th September of every year to the Chief Engineer's Office. Clearance of OHE structure (height of contact wire, horizontal distance of OHE, columns etc.) shall be submitted by the traction branch of the division to the chief engineer's office. Thereafter CE's office must be advised to receive the diagram whenever the clearances etc. are affected to rising of track, construction of new structure & alternation of existing structure etc.*
- ii) Construction organization shall submit to CE's office roll diagram showing supposed clearance before taking any work likely to affect the moving dimension.*
- iii) While under taking any construction work, such as extension of existing platform, construction of new platform, shelters, road over bridge etc. Whether by open line or construction branch, it must be ensured that the clearance as shown in the Chief Bridge Engineer.*

8. SANCTIONING BY CHIEF ENGINEER's OFFICE:-

After verifying the particulars of the consignment vis-à-vis the moving dimension, sectional movement of the consignment will be communicated by the chief engineer's office to the COM's office. Such cases which are not within the powers of the chief engineer's office shall be submitted to CRS for sanction by the chief bridge engineer & as soon as he sanction for the movement is received from the CRS, the same shall be communicated to the COP by the chief engineer's office.

In each case section will specify the speed restrictions to be observed, the track & structures to be avoided, lowering of track, lifting of over head electrical equipment, shutting off of power etc. as necessary. COM's office will convey the sanction to the divisions & the division concerned shall be issue massage to all concerned official by wire & in case the movement over a particular division is through locations where restrictive unavoidable structures or over head equipment are located, in addition will send conformation copies & obtain acknowledgment. If acknowledgement is not received in time from any of the concerned officials it shall be the duty of the divisional control office to obtain in on telephone & recode it in the control diary to ensure that the concerned staffs are aware of the condition of movement & their respective duties.

BLC

BLC WAGON & CONTAINER SERVICE INTRODUCTION

- BLC Wagon (Bogie Low Platform Container Wagon) are introduced to overcome the ODC restriction in respect of height to accommodate 9'6" height (international container) from BFKI type wagon in lieu of 8'6" height container which were become ODC in BFKI type wagons.
- The length of A car is 13625 mm while the length of B car is 12212 mm. The coupler of A car for attaching to loco or other stock is at 1105mm. The coupler in the B car at both ends is at 845 mm from rail level. The wagons are equipped with Air Brake. The diameter of new wheel is 840 mm and the condemning size is 780 mm.

SPECIAL FEATURES

- Most Important feature is five wagons make one unit consisting two 'A' cars having High Tensile E grade CBC Coupler at its outer end & slack less draw bar at inner end and three 'B' cars are attached with 'slack less draw bar' to each other.
- One rake consists of 9 units + 1 brake van
- Construction of under frame is lighter, welded skeleton design, to get optimum tare to pay load.
- Center pivot bottom is an integral part of bolster, provided with ring liner of 5 mm thickness.
- Use of spring loaded side bearer two on each trolley in lieu of CC Pad, which carry 90% of load in empty condition. Now a day's P.U. based Side bearers are introduced.
- Manufacturer – M/S Hindustan Development Corporation or TEXMACO
- Owner, CONCOR (Container Corporation of India)

Comparison between BFKI & BLC Wagon

Sl.No	Features	BLC	BFKI
01	Wagon unit	Consists of 5 Wagons. 2 Nos of A-Car and 3 Nos of B-Cars.	Single Wagon.
02	Platform height	1009 mm	1269 mm
03	Length of wagon over head stock	A-Car - 13625 mm B-Car - 12212 mm	13716 mm
04	Tare weight	A-Car - 19.25 tons B-Car - 18.50 tons	20.5 tons
05	Pay load	61 tons	48 tons
06	Wheel diameter	Max - 840 mm Cond - 780 mm	Max - 1000 mm Cond- 906 mm
07	Permissible speed	100 KMPH	75 KMPH
08	Coupler	CBC and Slack less draw bar	CBC
09	Empty load device	Two stage automatic Pneumatically operated	Manually operated.
10	Suspension	Two stage suspension	Single suspension
11	Twist Lock	Automatic	Manual
12	Rake carrying capacity	45 Wagons	42 Wagons

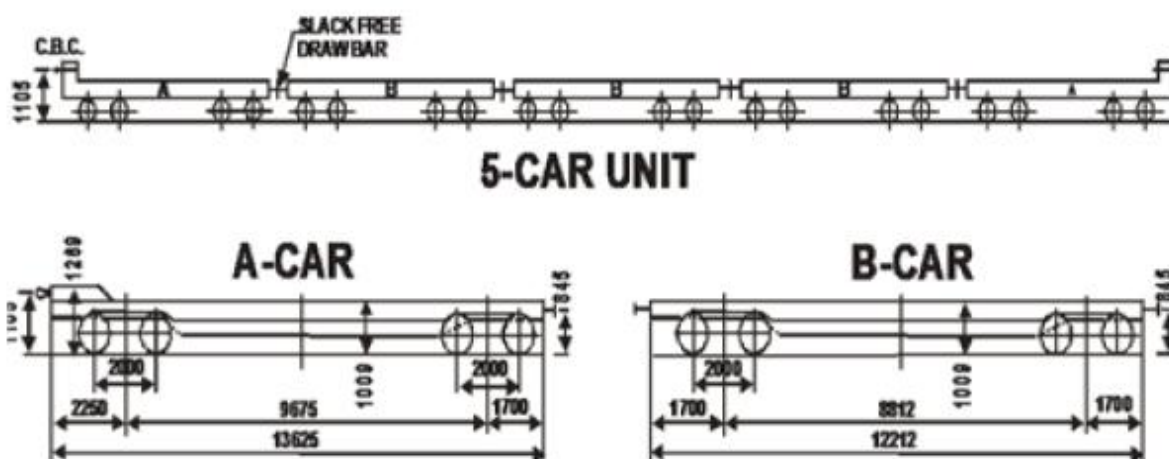
IMPORTANT DIMENSIONS

- Floor height from Rail level : 1009 mm
- Slack less drawbar height : 845 mm
- Wheel Diameter - : 840 mm (new)
: 780 mm (cond.)
- Axle Load - : 20.32 T
- Tare Weight - 'A' car - : 19.10 T
 'B' car - : 18 T
- Carrying capacity (CC) - : 61 T
- Length of 'A' car : 13625 mm
- Length of 'B' car : 12212 mm
- Length of one unit : 69 metre (One unit = 2BLCA + 3BLCB)
- Length of one rake : 619 metre
- Piston Stroke : 95 ±10 mm in empty
: 120 ±10 mm in loaded
- 'A' dimension : 70 to 72 mm
- BC pressure : 2.2 kg/cm² in empty
: 3.8 kg/cm² in loaded
- Maximum Speed : 100 KMPH



Bolster Spring Free Height & Arrangement

Type of Spring	New	Cond.	Springs in one nest
Outer	260mm	245 mm	7 nos.
Inner	243 mm	228 mm	6 nos.
Snubber	288 mm	273 mm	2 nos.



1. UNDERFRAME ALL WELDED CONSTRUCTION
2. BOGIE CAST STEEL
3. COUPLINGS A - CAR : RAISED END AAR E- TYPE CBC (NON-TRANSITION)
OTHER END - SLACK FREE DRAW BAR
B-CAR : SLACK FREE DRAW BAR
4. BRAKES AIR BRAKE & HAND SCREW
5. WHEEL 840 mm DIA. (NEW) 780 mm DIA. (FULLY WORN)



Slack less Draw Bar



Load Sensing Device

Load Sensing Device

- Use of LSD (Load Sensing Device) In lieu of E/Loaded device which automatically controls brake rigging system in empty and loaded condition.
- LSD fitted on each trolley at bolster with a bracket and a special bracket at lower plank in such a way that even when one side of body is loaded with 20 feet cont. LSD plunger is pressed bracket and LSD works at 42 T loaded condition
- A gap of 16 mm is maintained between plunger and plate.
- Special feature of LSD is spring dumper inside which shows delayed response for opening the valve as it works after actual loading/ unloading on wagon not on temporary deflection of trolley

Wheel Dia.	EPR hole
840 to 829	P
828 to 817	Q
816 to 805	R
804 to 793	S
792 to 780	T

Pressure Reducing Valve

It is used in circuit to reduce the air pressure tending towards one LSD valve only 4 kg/cm^2 instead of 5 kg/cm^2 to facilitate the working of Double Check Valve. It is connected to the inlet of one LSD valve

Double Check Valve

It is used between two LSD circuited to divert only one LSD valve pressure to C_3W_2 DV closing the port through shuttle inside it.

Automatic Twist Lock

8 nos. Automatic twist locks have been provided to secure container on wagon body. It Consists of following parts:-

- Twist lock head
- Twist lock upper housing
- Torsion spring
- Spindle
- Twist lock bottom housing (welded with wagon body)
- Torex screw
- Locator Pin



Function: - While loading container on wagon, Twist Lock head with a pressure of minimum 600 kg, rotates slightly and again come back to its original position and locks the container.

Then at the time of unloading it again rotates at opposite direction at the pressure of minimum 1000 kg.

Description & Feature of Important Parts of BLC Wagon:-

1. **Automatic Twist Lock :-** Each BLC wagon have eight twist locks, four at each corner two each in the Middle on either side, so that two sizes of container 20' & 40' can be loaded. 1000 kg of force is required at locks to unlock the container and 600 kg force is required to load the container automatically. The automatic twist lock is the compact container locating and securing device fitted at the locating points of under frame sole bar. The operation of the twist lock in securing ISO container wagon begins with placement container by loader. The container is positioned so that the bottom apertures of the four corner fittings are above the twist lock's twist head. The container is then lowered down. The weight of container causes the twist head to rotate and the corner fittings slides past the twist head, which rotates back in place engaging the corner fittings, and the containers are secured.
2. **Load Sensing Device (LSD):-** The increase of train speed as well as introduction of heavy hauled trains, a general need was felt to discard earlier adopted manually operated empty load boxes for wagon brake application. The system of manual device was to adjust leverage ratio to obtain desired braking force depending on empty load situation of the wagon. These system was time consuming and cumbersome. In order to simplify the system, it was felt that instead of adjusting leverage of brake cylinder pressure itself can be adjusted by adopting a suitable pneumatic device. The two stage automatic load sensing device along with C3W2type DV with empty load function providing two ranges of brake cylinder pressure was therefore introduced.
3. **Slack Less Draw Bar (SDB) Provided With Mini Draft Gear:-** The std. CBC with draft gear has 3" of displacement / compression at each end of the CBC. Thus the total displacement will be as much as 6" per wagon.
This enormous slack of 6" per wagon is detrimental due to, 1) as there is increased impact; the life structure will be reduced. But In the mini draft gear slack in the draft gear only about 1.5" causing smooth riding of wagon .mini draft gear has a special feature that it adjusts its slackness automatically. When there is development of slack during service due to wear and impact. The basic object of using this system is reduced the amount of free slack within the train as compared to normal CBC. The reduction of this free slack provides many benefits over conventional coupling, such as
 - a) Reduction of impact forces that act upon the car structures as result this free slack.
 - b) Reduced maintenance cost as result of induced impact loading on the wagon.
 - c) Reduction of acceleration, deceleration, and impact loading hat act upon the loading this helps to minimize the damage

MAINTENANCE PATTERN OF BLC WAGON:-

1. BLC wagon is maintenance free or requires very less maintenance. It is subjected to maintain for in CLOSED CIRCUIT RAKES.
2. The BPC is valid for 6000km OR 30days for 'A' Class & 7500km OR 35 days for 'A' SPL Class yard whichever is earlier.

Following are other condition for validity of the BPC.

1. If the rake stabled in the train examination yard for more than 24 hrs or more the rake is subjected safe to run examination and same BPC is revalidated for remaining period of the BPC.
2. If one unit i.e. five wagons of the CC rake is detached/attached this is not INVALID.
3. If the km /days of the BPC is completed before reaching the base depot, the validity of the BPC is extended for 7 days for one time unloading only. If it is in loaded condition after unloading the rake is sent in empty condition to base depot.
4. If km of run is not logged on BPC by crew, the BPC valid for 15 days from the date of issue of BPC.

PROBLEM FACED DURING MAINTANENCE AND OPERATION OF BLC RAKE:-

1. Due to low platform container, it is trouble to repair and testing of brake power properly and consuming more time.
2. Provision of LSD, the plunger accumulating dust / cement and due to such problem, it is necessary to pay more attention for avoiding mall functioning of LSD.
3. Some container depot are not having C&W staff so loading /Unloading .The revalidation is necessary for locking of container is not done due to this locking of container is not so easy reroute.
4. Due to floating bolster on run while functioning of LSD .The plunger of VTA valve is not inserting directly into its housing and bending effects occurs on plunger, resulting damage to the VTA valve.

ROH OF BLC WAGON

A routine overhaul should be given to all wagons as a unit of 5 wagons (car A-2, car B-3) at nominated depot at intervals of 18 months. (Newly built 24 months)

Disconnect 5 car units in to separate individual cars. Each car required to undergo ROH individually. The important points generally to looks into are given below for ready reference. Other details of ROH procedures as per RDSO instructions for ROH on wagons should also be looked into.

1. Remove bogie brake rigging attachments to under frames, break gears & flexible pipes connected to VTA.
2. Remove centre pivot split pin, lock pin and shackle lock etc.
3. Lift the body through lifting pads. Run out the bogie.
4. Clearance of 16 mm between load sensing device sensing tip to stopper are required to be maintained within tolerance of ± 1 mm.
5. Place the under frame on trestles.
6. After removing the side frame key and lifting the bogie remove wheel sets from the bogie. Machine the wheel tread of all the wheels to worn wheel trade profiles.
7. Strip the bogie. After proper cleaning, examine the bogie side frames, bolster and other castings for cracks etc.
8. All the side bearers' liners of the spigot and hole should be replaced. The procedure for attending the repairs of bogie frames, bolster and liners should be as per instructions.
9. Check bolsters springs and side bearer's springs for defective/broken springs. Replace the defective one such that variation in the free height of springs in the same group not to exceed 2mm.
10. Examine center pivot, center pivot pin, center pivot liner.

11. Strip brake gear fittings and examine for wear and damage and serviceability of bogie brake gear levers, replace worn out bushes and pins.
12. Examine the load sensing device on bogie with air brake in respect of item 5.
13. Replace all the worn out brake blocks and repair worn out brake heads.
14. Check the roller bearings for grease leakages and check adapter and replace if found any defect.
15. Reset the load sensing valve on bogie and tack weld the fasteners after setting with the necessary gap (16mm) between tip and stopper as mentioned.
16. Examine under frame brake gear levers; replace worn out bushes and pins.
17. Examine slack adjuster (SAB) and replace or attend to the defects as per RDSO instructions G-92. After fitment set the SAB "A" dimensions 72^{+0}_{-2} mm.
18. Check up hand brake arrangement for repair. Replace the missing or damage parts. Lubricate the gears.
19. Lower the under frame on bogies and provide pivot pin shackle, pin with split pin and connect brake gear pin. Dimensions $60^{+0}_{-0.5}$ mm between center pivot top to side bearer top liner should be maintained.
20. Test air brakes as per G-97 RDSO instructions.
21. Check automatic locks. Replace the defective locks. Check welding failures. Check automatic locks for any damage to lock head and for rotation of head. Change the lock head and rubber spring of locks.
22. Check the distance from side frame top liner to side bearer seat $131^{+0}_{-0.5}$ mm at all four side bearer points under tare conditions of the flat wagons. Keep the distance within tolerance by altering shims provided at the top side bearer locations.
23. Ensure all APD (anti pilferage devices) measures are incorporated after assembly.
24. Paint the bogie, stencil station code and date of ROH.
25. Touch up pain and lettering.
26. Special instructions: The draw bar height and CBC height should not be more than 845 mm and 1105 mm respectively. No packing is given over axle box adapter.

The following items must be attended to during ROH of BLC rakes:

1. Ultrasonic testing of axles.
2. Wheel treads profiling.
3. Side bearer liner renewal for spigot and base.
4. Wide jaw adapter renewal/replacement.
5. Side frame repair/ building up at location of adapter.
6. Brake beam (truss bar) pocket liner replacement/ rotation.

LOADING OF CONTAINER ON BLC WAGON & TROUBLESHOOTING

Container when loaded/unloaded by KALMAR (Special Crane.) pressure on twist lock is not perfectly vertical at JNPT/NSICT. As a result following problems arises:

1. Cases of trolley shifting and sometimes body/trolley and wheels are dismantled.
2. Center pivot pins get bend and ring damaged.
3. EM pads are damaged and broken.
4. Adaptor gets canted.
5. Side bearer springs shifted/ broken.
6. Twist lock head damaged.
7. Due to rough loading excess pressure comes to wheel set bearing and oil film between roller and race may cut which causes grease oozing resulting hot axle.

Remedy: To reduce this problem now at JNPT/NSICT loading is done by EOT.

Examination pattern & Rake formation

1. Generally CC Rakes are formed with 45 wagons (9 units) & 1 Brake Van.
2. CC/Free circuit BPC issued for 6000km/30days in 'A' class Yard.
3. CC/Free circuit BPC issued for 7500km/35days in Special 'A' class Yard.
4. Special attention is to be given during examination on Composite b/block, brake power 100%, EM Pad, T/Lock etc.
5. After every loading/unloading revalidation of BPC is compulsory and then special attention is to be given that loading of container is perfectly set on ATL.
6. Examination must be at base depot.-TKD/GMC/AQ/SBI/DER.
7. 100 man hrs. Is required for examination.
8. Piston stroke of BC is maintained 95 ± 10 mm in empty & 120 ± 10 mm in loaded condition.
9. Periodicity of POH – 6 years & ROH for newly built – 24 months, Subsequent ROH – 18 months

Advantages

1. Due to reduction in CBC coupler chances of train parting almost nil.
2. During acceleration and de-acceleration, impact on loaded consignment is reduced resulted lesser damage of body & trolley.
3. Reduction in maintenance cost.
4. Theft cases are completely eliminated.
5. Transportation time is reduced due to quick loading & unloading of consignments.
6. Carrying capacity is increased as reduction of tare weight.
7. Hence earning of Railways is increased.

Limitations

1. Whenever anyone wagon is marked sick, unit of five wagons has to be detached.
2. Re railing at the time of derailment due to 5 wagon coupled

Up gradation

Up gradation of carrying capacity of BOXN/BCN Wagon from 20.3 ton to 22.9 ton to accommodate CC + 10 ton

Sr.	DESCRIPTION	REMARKS / MODIFICATION
1	Spring arrangement per Nest outer- Inner – Snubber	6 – 4 – 2 (Existing) 7 – 6 – 2 (Modified)
2	Identification	Yellow strip on side panel of size 258 mm. yellow strip on spring & trolley frame below 777 mm from top height. On end panel i.e. E R, E M AND E S .
3	CTRB	AAR approved grease to be used, bearing face plate to be painted with white paint
4	Bolster	Outer jip increased from 234 mm to 241mm. Bolster inclination from centre pivot to side bearer is changed from 15mm to 16mm
5	Side bearer	Spring loaded PU pad type side bearer is used
6	D. V.	D.V. replacement date & station must be stenciled on side panel
7	stamping	Stamping of month & year on CTRB on back plate.

Up gradation of carrying capacity of BOXN/EL Wagon from 22.9 to 25 ton axle capacity

Sr.	DESCRIPTION	REMARKS / MODIFICATION
1.	Selection of wagon	Under framed member and crossed member should be in sound condition and trolley should NLB type .after up gradation trolley is named as NLC
2.	Wheel and axle	Axle wheel seat dia should not be less than 211mm wheel dia. Should not less than 950mm shop issued size is 962 mm.
3.	CTRB	AAR approved grease to be used, bearing face plate to be painted with white paint
4.	CBC and draft pad	High tensile CBC fitted with RF- 361 draft gear is used .
5.	Identification	Olive green paint strip of size 258 mm width should painted on wagon body panel below 777mm from top, red paint marking on spring and trolley H frame stenciling on end panel i.e. E R.
6.	ROH and POH interval	ROH- yearly and POH – 3 YEARLY
7.	Anti corrosion treatment	All anti corrosion measure i.e. 5 nos. per as RDSO pamphlet to be carried out.
8.	Spring arrangement per Nest outer- Inner – Snubber	6 – 4 – 2 (Existing) 7 – 7 – 2 (Modified)
9.	Axle capacity for extreme loading (E I)	25 Tonne
10.	Side bearer	CC pad to be replaced with PU pad

Constructional difference between BOXN and BOXNR

Sr. No	BOX N	BOX NR
	<i>SIDEWALL</i>	
1	<i>Material mild steel</i>	<i>Material Stainless steel (IRS M44)</i>
2	<i>Provided with 6nos.of side stanchions with 8mm thick hat section (on one side)</i>	<i>Provided with 9 nos. of side stanchions with 6 mm thick CRF hat section (on one side)</i>
3.	<i>Side sheet of 5mm thickness</i>	<i>Side sheet of 3.15 mm thickness</i>
4.	<i>Inside height of side wall is 1950 mm from floor level.</i>	<i>Inside height of side wall is 2177 mm from floor level.</i>
5.	<i>Two nos. middle coping provided in side wall</i>	<i>Middle coping are not provided in side walls except corners side panels.</i>
6.	<i>Top coping provided with ISMC-100</i>	<i>Top coping provided with press section of box 100X100X6 mm</i>
7.	<i>Side stanchion riveted with sole bar with two rows rivets</i>	<i>Side stanchion huck bolted with sole bar with in single row.</i>
<i>ENDWALL</i>		
1	<i>Made with mild steel.</i>	<i>Made with Stainless steel (IRS M44)</i>
2.	<i>Each end wall provided with 4 nos. of End stanchions of ISMC-150 mm</i>	<i>End wall provided with two half side stanchions and two middle copings of hat sections 6 mm thickness.</i>
3.	<i>End top coping provided with ISMC 150 channel</i>	<i>End top coping provided with CRF section of box 100X100X6 mm thickness.</i>
4.	<i>End sheet provided with 5 mm thick</i>	<i>End sheet provided with 3.15 mm thick</i>
<i>FLOOR</i>		
	<i>FLOOR Plate provide with mild steel of 6 mm thickness.</i>	<i>FLOOR Plate provide with stain less steel of 4 mm thickness</i>

BOGIE OPEN WAGON ‘BOXNR’



INTRODUCTION TO BOXNR WAGON :

This wagon has been designed for transportation of iron ores, coal etc. BOXNR wagons either fitted with Casnub-22W (without Elastomeric pads) or Casnub-22W (with Elastomeric pads) bogies with a maximum axle load 22.9 t., non-transition center buffer couplers and single-pipe graduated release air brake system.

<u>STANDARD FEATURES OF ‘BOXNR’ WAGON</u>			
S.No.	PARTICULARS		BOXNR
1.	Length over head stock	(mm)	9784
2.	Length over couplers	(mm)	10713
3.	Length inside	(mm)	9784
4.	Width inside/Width Overall	(mm)	2954/3176
5.	Height inside/Height(max.)from R.L.	(mm)	2127/3408
6.	Bogie centers	(mm)	6524
7.	Journal length x dia	(mm)	144x278
8.	Journal centers	(mm)	2260
9.	Wheel dia. on tread (New/Worn)	(mm)	1000/906
10.	Height of C.B.C. from R.L.	(mm)	1105
11.	C.G. from R.L. (empty)	(m)	1.027
12.	C.G. from R.L. (loaded)	(m)	2.033
13.	Floor area	(Sq.M)	28.90
14.	Cubic Capacity	(Cu.M)	61.47
15.	Maximum axle load	(tonne)	22.9
16.	Tare Weight	(tonne)	21.2
17.	Pay load	(tonne)	70.04
18.	Gross load (Pay+Tare)	(tonne)	91.6
19.	Ratio gross load/Tare		4.32
20.	Ratio (Pay load to tare)		3.32
21.	Track Loading density (tonnes/meter)		8.55
22.	No. of wagons per train		58
23.	Brake System		Air Brake
24.	Coupler		C.B.C.
25.	Bearing		R.B.
26.	Maximum Speed	(Loaded) (Empty)	60 kmph 80 kmph

BOGIE L.P.G. TANK WAGON ‘BTPGLN’



INTRODUCTION TO ‘BTPGLN’ WAGON :

This wagon is a variant of BTPGL with air brake and casnub trolley.

STANDARD FEATURES OF ‘BTPGLN’ WAGON			
S.No.	PARTICULARS		BTPGLN
1.	Length over head stocks	(mm)	18000
2.	Length over couplers	(mm)	18929
3.	Length of barrel inside	(mm)	17994
4.	Dia. inside (barrel)	(mm)	2400
5.	Overall Height from R.L.	(mm)	4285
6.	Bogie centers	(mm)	12970
7.	Journal length x dia.	(mm)	RB 144.5 Ø
8.	Journal centers	(mm)	2260
9.	Wheel dia. on tread (New/Worn)	(mm)	1000/860
10.	Height of C.B.C. from R.L.	(mm)	1105
11.	C.G. from R.L. (empty)	(m)	1.6
12.	C.G. from R.L. (loaded)	(m)	2.03
13.	Max. Volumetric carrying capacity(L)		-----
14.	Cubic capacity	(Cu. M)	79.48
15.	Maximum axle load	(tonne)	20.32/19.8
16.	Tare Weight	(tonne)	41.6
17.	Pay load	(tonne)	37.6
18.	Gross load (Pay+Tare)	(tonne)	79.2
19.	Ratio gross load/Tare		1.903
20.	Ratio (Pay load to tare)		0.903
21.	Track Loading density (tonnes/meter)		4.184
22.	No. of wagons per train		31
23.	Brake System		Air brake
24.	Coupler		C.B.C.
25.	Bearing		R.B.
26.	Maximum Speed	(Loaded) (Empty)	75 kmph 80 kmph

ICF BOGIE GOODS BRAKE VAN 'BVZI'



INTRODUCTION TO 'BVZI' WAGON :

This 8-wheeled brake van was designed with ICF bogie to achieve comfort level (Ride Index) equivalent to loco criteria for goods guard and capable of run 100kmph. The brake van is 5 meter longer than BVZC brake van.

<u>STANDARD FEATURES OF 'BVZI' WAGON</u>			
S.No.	PARTICULARS		BVZI
1.	Length over head stocks	(mm)	13540
2.	Length over couplers	(mm)	14469
3.	Length inside	(mm)	-----
4.	Width inside/Width Overall	(mm)	-----/3200
5.	Height inside/Height(max.)from R.L.		2448/3894
6.	Bogie center	(mm)	9026
7.	Journal length x dia	(mm)	
8.	Journal centers	(mm)	
9.	Wheel dia. on tread (New/Worn)	(mm)	915/813
10.	Height of C.B.C. from R.L.	(mm)	1105
11.	C.G. from R.L. (empty)	(m)	0.981
12.	C.G. from R.L. (loaded)	(m)	0.981
13.	Floor area	(Sq.M)	
14.	Cubic Capacity	(Cu.M)	
15.	Maximum axle load	(tonne)	5.875
16.	Tare Weight	(tonne)	23.5
17.	Pay load	(tonne)	-----
18.	Gross load (Pay+Tare)	(tonne)	23.5+....
19.	Ratio gross load/Tare		1.00
20.	Ratio (Pay load to tare)		
21.	Track Loading density (tonnes/meter)		1.624
22.	No. of wagons per train		
23.	Brake System		Air brake
24.	Coupler		C.B.C.
25.	Bearing		R.B.
26.	Maximum Speed	(Loaded) (Empty)	100 kmph 100 kmph

BOGIE COVERED WAGON ‘BCNHL’

<u>STANDARD FEATURES OF ‘BCNHL’ WAGON</u>			
S.No.	PARTICULARS	BCNHL	
1.	Length over head stock	(mm)	10034
2.	Length over couplers	(mm)	10963
3.	Length inside	(mm)	10034
4.	Width inside/Width Overall	(mm)	3345/3450
5.	Height inside/Height(max.)from R.L.		3024/4305
6.	Bogie centers	(mm)	7153
7.	Journal length x dia.	(mm)	144x278
8.	Journal centers	(mm)	2260
9.	Wheel dia. on tread (New/Worn)	(mm)	1000/906
10.	Height of C.B.C. from R.L.	(mm)	1105
11.	C.G. from R.L. (empty)	(m)	1134
12.	C.G. from R.L. (loaded)	(m)	2327
13.	Floor area	(Sq.M)	33.56
14.	Cubic Capacity	(Cu.M)	92.54
15.	Maximum axle load	(tonne)	22.9
16.	Tare Weight	(tonne)	20.8
17.	Pay load	(tonne)	70.8
18.	Gross load (Pay+Tare)	(tonne)	91.6
19.	Ratio gross load/Tare		4.4
20.	Ratio (Pay load to tare)		3.4
21.	Track Loading density (tonnes/meter)		8.35
22.	No. of wagons per train		58
23.	Brake System		Air Brake
24.	Coupler		C.B.C.
25.	Bearing		R.B.
26.	Maximum Speed	(Loaded) (Empty)	65 kmph 65 kmph



INTRODUCTION TO BCNHL WAGON :

This wagon was designed at 22.9t axle load in 2006, for transportation of food grain, fertilizer and bag quantities. The design was made with CRF section and stainless steel materials.

BOGIE OPEN WAGON ‘BOXNHL’



INTRODUCTION TO BOXNHL WAGON :

This bogie open wagon was designed at 22.9t axle load and 250mm longer than BOXNHS wagon. The wagon is manufactured by using stainless steel and cold rolled sections.

<u>STANDARD FEATURES OF ‘BOXNHL’ WAGON</u>			
S.No.	PARTICULARS		BOXNHL
1.	Length over head stock	(mm)	10034
2.	Length over couplers	(mm)	10963
3.	Length inside	(mm)	10034
4.	Width inside/Width Overall	(mm)	3022/3250
5.	Height inside/Height(max.)from R.L.		2028/3301
6.	Bogie centers	(mm)	6690
7.	Journal length × dia.	(mm)	144x 278
8.	Journal centers	(mm)	2260
9.	Wheel dia. on tread (New/Worn)	(mm)	1000/906
10.	Height of C.B.C. from R.L.	(mm)	1105
11.	C.G. from R.L. (empty)	(m)	
12.	C.G. from R.L. (loaded)	(m)	
13.	Floor area	(Sq.M)	30.32
14.	Cubic Capacity	(Cu.M)	61.05
15.	Maximum axle load	(tonne)	22.9
16.	Tare Weight	(tonne)	20.6
17.	Pay load	(tonne)	71.0
18.	Gross load (Pay+Tare)	(tonne)	91.6
19.	Ratio gross load/Tare		4.45
20.	Ratio (Pay load to tare)		3.45
21.	Track Loading density (tonnes/meter)		8.35
22.	No. of wagons per train		58
23.	Brake System		Air Brake
24.	Coupler		C.B.C.
25.	Bearing		R.B.
26.	Maximum Speed	(Loaded) (Empty)	75 kmph 100 kmph

BOGIE COVERED HOPPER WAGON FOR FOOD GRAIN ‘BCBFG’

<u>STANDARD FEATURES OF ‘BCBFG’ WAGON</u>			
S.No.	PARTICULARS		BCBFG
1.	Length over head stocks	(mm)	11861
2.	Length over coupling faces	(mm)	12790
3.	Length inside	(mm)	11770
4.	Width Overall	(mm)	3250
5.	Height inside/Height(max.)from R.L.		---/4260
6.	Bogie centers	(mm)	8661
7.	Journal length x dia.	(mm)	144.5Ø
8.	Journal centers	(mm)	2260
9.	Wheel dia. on tread (New/Worn)	(mm)	1000/906
10.	Height of buffer from R.L.	(mm)	1105
11.	C.G. from R.L. (empty)	(m)	1.130
12.	C.G. from R.L. (loaded)	(m)	2.182
13.	Floor area	(Sq.M)	----
14.	Gross Cubic capacity	(Cu. M)	84.82
15.	Maximum axle load	(tonne)	21.82
16.	Tare Weight	(tonne)	26.44
17.	Pay load	(tonne)	60.84
18.	Gross load (Pay+Tare)	(tonne)	87.28
19.	Ratio gross load/Tare		3.30
20.	Ratio (Pay load to tare)		2.30
21.	Track Loading density (tonnes/meter)		6.82
22.	No. of wagons per train		48
23.	Brake System		Air Brake
24.	Coupler		C.B.C.
25.	Bearing		R.B.
26.	Maximum Speed	(Loaded) (Empty)	65 kmph 65 kmph



INTRODUCTION TO ‘BCBFG’ WAGON :

This wagon has been designed for transportation of food grain in Bulk. This wagon is designed with CASNUB-22HS MOD-I bogie, non-transition CBC, single pipe graduated release air brake system with automatic load sensing device. There are two Nos. gravity discharge gate at bottom for unloading.

FREIGHT TRAIN EXAMINATION

Introduction: *The efficient working of freight stock is closely linked to the standard of yard maintenance. Several factors are responsible for good and quality examination/repairs in the yard. The method of examination is described as under.*

New Pattern of Freight Train Examination: *It is based on the Revised JPO issued by Railway board letter No. 94/M(N)/951/57 Vol-2 dated 25.10.04, letter NO-2005/M(N)/951/13 Dated 07.04.05 and even no dated 25.04.2006 and 5.11.07.*

Following are the main feature of new pattern examination of freights trains.

- 1.** *Word CRT has been deleted as the stock has been phased out.*
- 2.** *The freight train can only be subjected to examine for intensive End to End, Premium End to End and Close Circuit rakes.*
- 3.** *The practice of safe to run examination of freight trains per se may be discontinued.*
- 4.** *En route rolling-in-Examination freight trains may be discontinued. However rolling in examination as part of intensive examination will continue.*
- 5.** *Post loading examination by TXR Staff may be discontinued for all type of stock (except loading of steel consignment). This check is to be carried out by Guard and Driver as per standard proforma issued by Railway Board. The post loading check must be carried out by TXR Staff and securing of steel bundles with lashing chains may be ensured.*
- 6.** *After Tippling the rake will be offered for post Tippling examination, in case less than three rakes are day, the check may be carried out by guard and driver as per standard proforma issued by railway board. In cases 3 or more trains are being tippling, post tippling check will be done by Skelton TXR staff. After tippling the rakes should be subjected post tippling check either by TXR Staff or by GUARD & Driver in case of non provision of TXR Staff in siding.*
- 7.** *It should be ensured that unexamined lead (after unloading before next TXR Point) of freight trains running end to end pattern or invalid BPC in case of premier & cc rakes does not exceed 400 kilometer .*
- 8.** *Since multiple loading and unloading are permitted in CC & Premium Rakes .movement of CC rakes & premium rakes will be monitored through FOIS by Traffic Department with C & W control.*
- 9.** *In case of mechanized loading an unloading (i.e. BOXN wagon examination by TXR will be desirable.*
- 10.** *In case of clearance of stable load instruction contained in Board letter No. 2000/safety (A&R) /19/35/ dated 31.7.01 should be followed.*
- 11.** *The CC rakes shall be offered for PME in empty condition at the cc base depot where the cc base depot where the cc rake was originally formed.*
- 12.** *ROH and POH wagons from CC rake will be marked and detach at base Depot.*
- 13.** *The rake integrity of CC rake as listed in the BPC should be maintained .However up to 4 wagon (10 FWU) may be replaced by good examined wagons in the entire Run between the two PME.(05 BLC or one mark in case of BLC rakes allowed for attended or replaced).*

PREMIUM RAKES: This new type of examination for air brake stock (i.e. BOXN, BOXNHS, BCN, BCNHS, and BOBRN etc) introduced in Indian railway w.e.f. April 06 as per instructions issued by railway board.

The salient features of such types of examination are under:

1. Premium rake will be form out of air brake stock (i.e. BOXN, BOXNHS, BCN, BCNHS & BOBRN etc.) only.
2. Such rakes will be given intensive examination in empty condition at nominated examination yard only.
3. Premium examination point must be upgraded to "A" category yard on top priority.
4. Stipulation to form rake out of Off POH /ROH wagons as in case of CC rakes are not applied in case of premium rakes.
5. Similar types of wagons are taken to form premium rakes, mixed wagons not allowed for such rake.
6. The rakes will be turned out with minimum 95% brake power and BPC will be issued on Green Book only.
7. The validity of BPC to be issued for 12 days including date of issue. However grace period of 3 days is given when rake is loaded condition and on 15 days rake must be unloaded and must be offered for C &W examination.
8. The rake is handed over to Operating Department for multiple loading and unloading within 12 days.
9. After each loading and unloading , the rake must be offered for Guard and Driver check before commencement of journey as per proforma issued by Railway Board and observation will be recorded on the relevant column of the BPC.
10. Movement f premium rakes will be monitored through FOIS by Traffic Department with Mechanical Department.
11. If the rakes stabled in yard more than 24 hours, the rakes must be offered for C&W examination otherwise BPC will be treated as invalid.
12. Man hours are decided as 75 for Premium End to End (PEE) examination.
13. The integrity of rake will be maintained. However 4(10 FWS) wagons may be replaced by examined wagons en route.

Advantage of Premium Rake: Following are the advantage of introducing premium rake examination:

1. Wagon turn round is reduced and loading cycle is increased by 3 times.
2. Rakes are available for maximum to Traffic Department.
3. During the year 2007-08 a profit of 2000 crore is made to Railway by increasing 40 million ton loading on premium rake pattern.
4. Overall average yard detention of Indian Railway is reduced from 15 hrs to 11.15 hrs.
5. Due to introduction of premium rakes the availability of rake for end to end examination over IR decreased from 400 rakes per day to 150 rakes per day due to this over all expenditure on examination reduced.

Disadvantage of Premium Rake: Since the examination of premium rake are being also attended in yard which is yet upgraded to 'A' category, Following are the disadvantages:-

1. Man hours for Examination of premium rakes are not specified and are being examined on end to end pattern.
2. Reject able Items for attending examination and repair of such rake are also not specified.

3. The rake is permitted for multiple loading and unloading on the basis of GDR check. It is experienced in N. C. RAILWAY that GDR check is not effective and derailment of 3 premium rakes during 2006-07 as 2007-08 were occurred due to lack of GDR check which is not safe practice and not safe to rolling stock.
4. Grace period of 3 days is permitted when rakes are in loaded condition (i.e. in rare cases) and must be reached to unloading point within 15 days and on 15th days rake must be offered for C&W examination .

But it is experienced in NC Railway that Traffic Department is taking advantage of grace period, the premium rakes are received after loading on 12th day from the issue of BPC and thus rakes are running on invalid BPC . Such information is being sent to Railway Board in PCDO and operating department is requested to minimize such practice to safe running of rolling stock.

CLOSE CIRCUIT RAKES : CC rake are being formed in term of Railway Board's letter No.94/M(N)/951/57 Vol- II dated 25.10.2004 and letter No. 87/M(N)/951/31 dated 22.08.1989. and time to time instruction issued by Railway Board. Following are the main feature of CC rakes:-

1. CC rakes will run on predefined path and under completion of day / km mention on BPC the rake should be examined at nominated base depot only.
2. Off POH/ROH wagon should be preferred during formation of CC rakes.
3. Examination should be conducted in day light only (morning to sun set).
4. The examination should be conducted on nominated line by CRSE & CFTM of the Railway.
5. All reject able items must be attended during examination of such rake.
6. The air brake wagons of same types of stocks are formed in CC rake .Mixed wagon can not be allowed.
7. For examination of CC rake, 100 man hrs is to be taken.
8. After examination BPC with 100 % brake power is issued on prescribed yellow color certificate.
9. The examination of CC rakes is carried out where the minimum infrastructure facility for examination as standardized by CAMTECH is available.
10. There are 3 types of CC rakes be examined and validity of BPC being issued over Indian railway.
 - 4500 kms/20 days whichever is earlier (examined at "A" ,"B" and "C" Cat. Yard)
 - 6000kms /30 days whichever is earlier (examined in "A" cat. Yard)
 - 7500 kms/35 days whichever is the earliest (being examined at Special "A" cat. Yard)
11. The rakes are handed over to Traffic Department for multiple loading/unloading within the validity of BPC and GDR check.
12. Listed wagons on BPC are allowed to run .En route if detachment or attachment by 4 or more wagons (10FWU) is done without examination by TXR, BPC should be treated as invalid (In case of BLC 5 wagons/ one unit)
13. Movement of CC rakes will be monitored through FOIS by Traffic Department with Mechanical Department.
14. If the rake is instable in yard more than 24 hrs in yard, the rake must be offered C&W examination and if not, BPC should be treated as invalid.
15. The km runs must be endorsed by Driver and Guard on BPC in relevant column.

Further, zonal Railways shall maintain detailed record w.r.t. en route detachments. Brake power and detachment during examination of these rakes and give monthly feed back to board on their performance.

Railway must ensure that infrastructural facilities at all the above points are upgraded to 'A' category.

POST LOADING AND POST TIPLING EXAMINATION:

Vide Rly Bd's letter no. 2005/M(N)/951/13 dated 08.02.2006. Post Loading examination by TXR staff was discontinued. Post loading check by Guard and Driver was introduced. In the para (iv) of Rly Bd's above letter dated 08.02.06. It was stipulated that:

After tipling the rake will be subjected to post tipling examination. In the case less than 3 rakes are being tiplled per day, the check may be carried out by Guard & Driver as per proforma enclosed. In case 3 or more trains are being tiplled, post tipling check will be done by skilled TXR staff.

The same has been reviewed vide Rly Bd's letter No. 98/ M (N)/951/12/pt.1 dated 17.05.07, relevant paras are reproduced as follows:

1. Board has reviewed the subject matter and has decided to revise the instructions contained in para (iv) of aforesaid letter dated 08.02.06 on post tipling checks on freight trains as under:

- a) After tipling rake should be subjected to post – tipling checks either by TXR staff or by Guards and Drivers in case of non provision of TXR staff in the siding.
- b) As local condition may vary from siding to siding, based on recommendations of CME & COM, GMs may decide whether the post tipling check on a particular point will be entrusted to TXR staff or Guard and Driver. While deciding the matter one way or other, the following may be kept in mind:
 - Recovery of necessary charges from the owner of such sidings in case any defects damages are noticed.
 - Post tipling check by Guards and Drivers should be done as per format enclosed with the above mentioned letter of Railway Board.

2. Rules regarding starting of trains from non-TXR points after examination by Guard and Drivers should be strictly enforced.

3. All other provisions of Board's letter no.2005/ M (N)/ 951/13 dated 08.02.2006. It is to be followed.

GDR CHECK: GDR check has been defined as , required to be done only for rakes , which are to be offered to TXR examination, where due after completion of loading and unloading cycle and are required to move another 250-300 km (now 400 km stretch) before hitting the TXR points.

The GDR check by guard and loco pilot should invariably ensure the following:-

1. Adequacy of Air pressure/vacuum pressure in motive power and brake van.
2. Ensure Air pressure / vacuum continuity from loco to last vehicle.
3. Success of brake feel test.
4. Adequacy of brake power by counting operative/ non – operative pistons.
5. Shall ensure by visual examination that there are no loose fitting in the under gear including brake blocks, safety brackets, track area, brake gear pins etc. which may danger the safe running of train. This examination shall be one by walking along the length of the train by loco pilot on one side and by Guard on other side.

6. Guard and loco pilot shall jointly prepare a memo in triplicate indicating the brake power deficiency, if any. They shall append their signature on the same and both of them shall retain a copy of the same and third copy shall be handed over to station master on duty.

7. In case of premium end to end rake the observation by the guard and loco pilot will be recorded under the relevant para of the BPC.

The GDR check should not lead to a false sense of adequacy of brake power in the psyche of the loco pilot. So apart from adequacy of air pressure per vacuum in the locomotive and the last vehicle, the loco pilot should have the confidence on the adequacy of the brake power only after conducting the brake feel test and this aspects of sufficiency of brake power of the train should not be diluted by other visual examination by Guard and Loco Pilot.

Since premium rake s allowed for multiple loading and unloading up to 12 days based on the GDR check, there is an urgent need to bring improvement in quality of GDR checks by imparting suitable training to all goods Drivers and Guards. Some of the areas that need to be specifically covered in such training are as follows:

- i) Significance of post loading and post tipping examination.
- ii) Items to be checked: critical assemblies and components: procedure for checking.
- iii) Type of BPC s: validity of BPC: action to be taken in the case of invalid BPC.
- iv) Type of tipplers / pay loaders: nature of the damage caused to the freight stock during loading per tipping operation.
- v) How to check brake power of air and vacuum brake train? How to check continuity of air pressure/ vacuum in trains?
- vi) How to check condition of couplers, hoses and other under gear components.
- vii) Types of the angle cocks: open / close position of different type of the Angle cocks.
- viii) Empty / load device: principle of working /correct position of empty / load device handle.
- ix) Types of the bogies suspension arrangement, brake gear components, common defect of spring, brake pull load, brake beam, safety bracket, brake block etc. And action to be taken in each defect.
- x) Type of hand brake, procedure of release/application of various types of hand brakes / defect of hand brake.
- xi) Type of doors and their locking in various type of wagons, door opening mechanism of BOBRN wagons.
- xii) Action to be taken for various defect in freight stocks / train.

Post Loading/ Post Tipping Checks by the Guard and the Driver

Items To Be Checked By Guard And Drivers

1.	All CBCs and Air House are properly coupled and locked
2.	All the Angle Cocks are in open condition.
3.	The last Angle Cock is in closed condition
4.	Empty/Loads device handle is in proper position.
5.	There is no loose fitting/hanging parts like spring push-pull rod, Brake Beam, Safety brackets, Brake blocks etc. which may endanger safe running of the train.
6.	Hand Brakes are released.
7.	Doors of wagons are closed and locked / secured.
8.	Check continuity of the air pressure/vacuum before starting.

Proforma for joint check by the Driver and the Guard

1.	Date	:	
2.	Train No. and loco number	:	
3.	From.....	To	
4.	BPC No.; Date & Station of Issue		
5.	Loaded at	Or	Tippled at
6.	Time of Locomotive Attachment	:	
7.	Total Load	:	

(Signature of Driver)
Driver's Name.....

(Signature of Guard)
Guard's Name.....

(This memo should be prepared in three copies, one copy to be kept with Driver , one with the Guard and one will be given by the Guard to the Station Master / Yard Master.)

Examination of Container Trains (In each case of invalid BPC): (Ref: Railway Board's letter No. 2007/M (N)/ 951/67 Dated 19/20. 11.2008);

i) Para 2.9(a)- In empty Condition (container off loaded from wagons): Rake shall be offered at the nearest TXR point for intensive examination, where after examination its BPC will be revalidated for a period of 7 days, with endorsement on BPC by TXR that rake is safe to run up to its base depot. During this period of 7 days, one loading/ unloading shall be permitted in the direction of CC base depot. The revalidation of BPC in above manner is permitted only once and rake shall be back to its CC base depot within 6 days period. Else the rake shall lose its CC character and will become normal end-to-end rake. Re-conversion of such end-to-end rakes to CC shall be permitted only after personal approval of CME & COM of the concerned Zonal Railway.

ii) Para 2.9(b)- In loaded Condition (container loaded on the wagons): Rake shall be offered at the nearest TXR point for safe to run examination and endorsement on BPC by TXR that train is safe to run up to its destination. After unloading of consignment at destination, such potential unsafe rake shall be offered at the nearest TXR point for safe to run examination and endorsement by TXR on BPC that the train is safe to run in unloaded condition up to its CC base depot examination with containers loaded on wagon, shall be done only in case of extreme urgency with prior approval of CME & COM of the concern Zonal Railway.

iii) Para 2.10 of the same Railway Board's letter dated 19/20.11.2008: Container rakes detained for more than 24 hours at TXR point, shall be subjected to safe to run examination and endorsement on BPC by TXR that rake is safe to run for the remaining validity period of BPC.

JPO for Close Circuit Pattern of Examination, Maintenance and Operation of BTPN Wagons. (Ref: EDME/ Frt./Railway Board's Letter No.-2008/M(N)/951/13 CC Rakes dated 05.02.2010)

- i) After successful trial for one month Close Circuit Examination of BTPN wagons shall be taken up at BAAD Depot in Agra Division on trial basis for one year as detailed below:

Sr. No.	Base	No. of Rakes	Circuit	Validity of BPC
1	BAAD/ NCR	30	NCR, NE, NR, WR & WCR	6000 km or 30 days whichever is earlier

- ii) CC examinations will be done only on the examination lines in the Yard having proper facilities of material handling, pucca pathway, welding, lighting etc. the CC examination will be done on the lines where there is no OHE.
- iii) If there is no CC examination line without OHE, CC examination and maintenance of under gear parts of wagons should be carried out in the CC examination Yard and maintenance of barrel portion of the wagon (i.e. barrel mounting and valves) may be done in gantry before loading of the product.
- iv) The existing rakes (ordinary/ non CC/ Premium) shall be thoroughly examined before the rake is declared CC by a supervisor not below the rank of SSE/C&W.
- v) The rake shall be offered for examination in day time only, till the lighting facility is developed.
- vi) Due to acute position of staff at BAAD Sr. DME/ C&W/ AGC may move 8 staff from BAAD sick line to yard for three days a week. On these days , CC examination may be carried out @ one rake per day . This will be done till additional staff is posted at BAAD depot.
- vii) All wagons released from ROH shall be utilized for making CC rakes and Off ROH wagons waiting for formation of CC rakes shall not be considered under ineffective.
- viii) During CC examination, proper attention to the safety fitting ,pipes valves etc shall be given and their maintenance records will be maintained separately.
- ix) Monitoring of CC rakes shall be done through FOIS and also by Sr.DME(C & W) and Sr.DOM of AGC division jointly.
- x) Proper computerized record keeping and documentation at CC base depot BAAD to be ensured to be monitor condition of CC rake.
- xi) The BPC will incorporate the name of CC depot as BAAD depot and will be valid over NCR, NE, WR, NR & WCR with a validity of 6000 kms or 30 days whichever is earlier.
- xii) Rake operating on zonal railways not mention on the BPC will loss their CC character and will be treated as per instruction prevailing for normal end to end rake.
- xiii) All rakes will be returned to BAAD depot, nominated CC base report as per validity of BPC.
- xiv) After each loading /unloading, the rake will be examined by Guard and Driver before commencement of journey and observation will be recorded under relevant column of the BPC.

- xv)** Rake integrity as listed in the BPC should be maintained , however up to 4 wagons may be replaced by duly examined wagons in the entire running between two PMEs.
- xvi)** ROH/POH wagons from CC rake will be marked end detached at base depots only.
- xvii)** In case CC rake is not made available for examination at base depot and offered for examination at some other depot. It will become non CC rake and it will follow normal intensive examination pattern.
- xviii)** Railway shall ensure that other requirements as stipulated by RDSO from time to time & IRCA for carrying and handling of hazardous material in the tank wagon shall be met.
- xix)** Besides special conditions mention here in above , with respect to maintenance and operation of this special CC rakes, all other general instructions issued by railway board from time to time, regularly maintenance of 6000 kms CC rake shall be observed.
- xx)** After completion of 1 year of trail period, division should submit performance report with detail of detachment (with reasons),poor brake power cases and detachment during examination etc. for review.

ITEMS TO BE EXAMINED/ CHECKED DURING PREMIUM/CC RAKE EXAMINATION IN YARD (BRIEF):

Rolling in examination must be done before block of rake to detect Hot Box, flat tyre and other unusual.

1. Complete inspection of running gears, break gears & spring gears including tapping and gauging of wheels. The wheel dia should be within limit .

DISCRIPTION	LIMIT	
	NEW	Condemn
Wheel dia used on BOX/UIC bogie)	1000 mm	860 mm
Wheel dia used on BOXN/CASNUB bogie)	1000 mm	906 mm
Wheel dia used on BLC Wagon	840 mm	780 mm

2. Ensure fitness of all safety fittings, straps & safety loops, safety brackets etc.
3. Replacement of cast iron & composite brake blocks. Brake blocks should be replaced on reaching condemning thickness. After fitment of brake block and key on brake head fitment of split pin should be ensured. The limit of brake block thickness is as under :

DISCRIPTION	LIMIT
Brake block condemning limits	10 mm
Yard leaving thickness of brake block except BOY wagons	20 mm

4. Correct functioning and positioning of empty load device.
5. Checking and proper securing of doors of the wagons.
6. Look for abnormal and/or unequal buffer height/CBC height, wear plates, knuckle, etc to the extent it is possible to detect by visual examination. In case of doubt , the buffer height / CBC height should be measured and maintained as given below :

DISCRIPTION	LIMIT
Buffer height from rail level	10 mm
Yard leaving thickness of brake block except BOY wagons	20 mm

7. The Bogies, complete side frames and bolsters to be visually examined for cracks and missing parts. Bolster springs , Snubbers, spigots, center pivots fastening, roller side bearer in case of CASNUB 22 bogie to be checked for defects , if any. Following nominal clearances may also be observed;

DISCRIPTION	22W/ 22W (R)	22W(M)	22NL/ NLB	22HS
Lateral clearance between Side frame and bolster	18mm	18mm	18mm	25 mm
Lateral clearance between Side Frame and axle box/adopter	25mm	25mm	16mm	
Longitudinal clearance between side frame and axle box/adopter	2mm	10mm	9mm	9mm
Longitudinal clearance between side frame and bolster	6mm	6mm	6mm	6mm
Clearance between anti rotation lug and bolster	4mm	4mm	4mm	4mm

8. Hand brake Wheel should be checked for their proper working in ON & OFF position.
9. Visual examination of under frame members, body, door mechanism, CBC wear or deficiency of parts to be marked and their operation to be checked.
10. Examination of any loose or hanging component.
11. Abnormal behavior of any vehicle indicates if unsafe working condition.
12. The condition of side bearer pads Rubber bonded metal pad (RBMP), E M Pad must be examined and it should be within limit as given below.

DISCRIPTION	Normal dimension	Dimension after permanent set (condemning size)
Elastomeric pad	46mm	42 mm
Side bearer rubber pad	114mm	109 mm

13. A rake of air brake wagons should be tested with rake test rig from one end of rake. Leakages through air house pipe, MU Washers, cut off angle cock to be arrested and air pressure in BP gauge should be maintained as given below:

No. of Wagons	On BP Gauge of rake test rig	On last Wagon
Up to 56 wagons	5.0 kg/cm ²	4.8 kg/cm ²
More than 56 wagons	5.0 kg/cm ²	4.7 kg/cm ²

14. During the testing following parameters are to be checked and maintained as given below:

a) "A" Dimension:

Description	Limit
<i>A' dimension of slack adjuster BOX/UIC bogie Wagon</i>	50^{+2}_{-0} mm
<i>A' dimension of air brake stock fitted with CASNUB bogie except BOBRN wagons</i>	70^{+2}_{-0} mm
<i>A' dimension of BOBRN</i>	27^{+2}_{-0} mm

b) Piston stroke:

Type of wagons	Piston Stroke	
	Empty	Loaded
<i>BOXN, BCN/ BCNA, BRN, BTPGLN</i>	$85 \pm 10 \text{ mm}$	$130 \pm 10 \text{ mm}$
<i>BTPN</i>	$87 \pm 10 \text{ mm}$	$117 \pm 10 \text{ mm}$

15. Brake Power Certificate (B.P.C.): This is a certificate jointly signed by guard, driver and C&W supervisor prepared in triplicate by C&W supervisor after ensuring vehicle attached in train is fit to run and required amount of vacuum/ pressure is maintain in engine and brake van/ last vehicle. It contains train no., engine no., and loads, break up of load, brake power of the train, amount of vacuum/ pressure in engine and brake van and first and last two vehicles number respectively.

BPC to be issued on prescribed colour proforma as under:-

<i>Premium rake</i>	<i>– Green color - 95%</i>
<i>CC rake</i>	<i>– Yellow color - 100%</i>
<i>End to End vacuum rake</i>	<i>– Pink color - 85%</i>
<i>End to End air brake</i>	<i>– Green color - 85 %</i>

BRAKE POWER CERTIFICATE FOR AIR BRAKE (BOXN/BCN) PRIMMUM END TO END RAKE

ISSUED BY: (EXAM Point/Div/Rly.)

_____yard /Railway

INTSTRUCTIONS**A. GUARDS AND DRIVERS**

- Before starting the train , guard and driver should ensure:
 - Continuity of air pressure from first to last vehicle of the train.
 - Validity of BPC, if found invalid inform the section controller immediately and take necessary instructions from carriage control.
 - Driver and Guard should correctly log the dates and 190ilometers earned.

B. STATION STAFF

- They should be vigilant for averting any theft or tempering with this rake. Any incident of theft/ tampering to be reported to section controller / TXR controller and entry of attention done be made in space provided.

- | |
|---|
| 1. THIS CERTIFICATE IS VALID FOR 12 DAYS (WITH A GRACE LIMIT OF 3 DAYS), PROVIDED. |
| 2. More than 4 wagons have not been detached / attached from the rake. |
| 3. The rake is not stopped for more than 24 hours in any train examination yard. |

Note : 3 days additional grace period is given for loaded trains only . Loading of train on or after 12th day will make BPC invalid.

C. TO BE FILLED AT THE ORIGINATING EXAMINATION POINT BY RAKE EXAMINING STAFF

TRAIN No.		LOAD & STOCK	
BPC No.		ENGINE No.	
BPC ISSUED DATE		BPC VALID UP TO	
STATION		DATE OF LOADING	
AIR PRESSURE ON DEP: ON LOCO		kg/cm ² ON BK. VAN	kg/cm ²
TOTAL No. OF CYLINDERS		No. OF OPERATIVE CYLINDERS	
BRAKE POWER %			

LIST OF WAGONS IN ORDER:

S. No.	Wagon No.	S. No.	Wagon No.	S. No.	Wagon No.	S. No.	Wagon No.
1.		16.		31.		46.	
2		17		32		47	

()
DRIVER'S NAME & SIGN.()
GUARD'S NAME & SIGN.()
JE/SE (C&W) NAME & SIGN.

D. DISTANCE TRAVELLED (TO BE FILLED BY DRIVERS)

S.no.	Date	Loco no.	Train no.	Driver's name/hq	From	To	Kms	Cum kms	Driver's sign.

PARTICULARS OF WAGONS DETACHED ENROUTE:

- 1.
- 2.
- 3.

E. ENROUTE PROBLEMS NOTICED & ATTENTION GIVEN:

Date	Loco no.	Dr's name	Dr's h.q.	Problems & action taken		Sign. Of Dr/ Stn. Staff
				Stn.	Nature	

F. BRAKE CONTINUITY/ VALIDATION AT LOADING POINTS.

S. No.	Station	Rly.	Date	Pr. Ready at	Abnormality observed	Sign.

G. GDR CHECK LIST :

S.N	ITEMS TO BE CHECKED BY GUARD AND DRIVER
1.	Rake integrity is not disturbed by more than 10 FWUs. Only intensively examined wagons given fitness by train examining staff may be attached.
2.	All CBC's and Air hose are properly coupled and locked.
3.	All the angle cocks are in open condition.
4.	The last angle cock is in closed condition.
5.	Empty /load device handle is in proper position.
6.	There is no loose fitting/hanging parts like push pull rod, brake beam, safety bracket, brake blocks etc. which may endanger safe running of train.
7.	Hand brakes are released.
8.	Doors of wagons are closed properly and locked/secured.
9.	Any other abnormality noticed and action taken.
10	Guards and drivers shall prepare a memo jointly on a plain sheet triplicate indicating the brake power and deficiency, if any
11.	Continuity of the brake pipe pressure is confirmed through VHF/ Whistle code before starting the train.

BRAKE POWER CERTIFICATE FOR AIR BRAKE (GOODS) CLOSE CIRCUIT RAKES

ISSUED BY: (Intensive Exam. Point/ Divn./Rly.)

NOMINATED CLOSE CIRCUITS

DATE

BPC NO.

INSTRUCTIONS

A. GUARDS AND DRIVERS:

1. Before starting the train, guard and driver should ensure:-
 - (i). Continuity of air pressure from first to last vehicle of the train.
 - (ii). Validity of BPC. If BPC is invalid, inform the control office & take necessary instructions from Carriage Controller/ Sr. DME.
2. Driver and Guard should correctly log the kilometers earned.

B. STATION STAFF:

1. They should be vigilant for averting any theft or tempering with this rake. Any incident of theft/ tempering to be reported to TXR Control and entry made in the space provided.

THIS CERTIFICATE IS VALID FOR 6000 km.

1. Provided the kilometrage has been logged in correctly and continuously, if not BPC will be deemed to be valid for 30+5 days only from the date of issue of BPC.
2. Provided the rake integrity is not changed and only listed wagons are included.
3. Provided the rake is not stabled for more than 24 hours in train examination

C. TO BE FILLED AT THE ORIGINATING EXAMINATION POINT BY RAKE EXAMINING STAFF

Train No.

Engine No.

Load & Stock

Total No. of Bk. Cylinders

No. of Operating Cylinders

Brake Power %

Air pressure on dep: On Loco

kg/cm²

On BK. Van

kg/cm²

Engine on Train

Air pressure ready at

LIST OF WAGONS IN ORDER:

S. No.	Wagon No.	S. No.	Wagon No.	S. No.	Wagon No.	S. No.	Wagon No.
1		16		31		46	
2		17		32		47	

(
DRIVER'S NAME & SIGN.

(
GUARD'S NAME & SIGN.

(
JE/SE (C&W) NAME & SIGN.

D. DISTANCE TRAVELLED (TO BE FILLED BY DRIVERS)

S.no.	Date	Loco no.	Train no.	Driver's name/hq	From	To	Kms	Cum kms	Driver's sign.

PARTICULARS OF WAGONS DETACHED ENROUTE:

- 1.
- 2.
- 3.
- 4.
- 5.

E. BRAKE CONTINUITY/ VALIDATION AT LOADING POINTS (OR THEIR NEAREST EXAMINATION POINTS) BY TRAIN EXAMINING STAFF.

S. No.	Station	Rly.	Date	Loco no.	Pr. Ready at	Abnormality observed	Sign.

F. ENROUTE PROBLEMS NOTICED & ATTENTION GIVEN:

Date	Loco no.	Dr's name	Dr's h.q.	Problems & action taken		Sign. Of Dr/ Stn. Staff
				Stn.	Nature	

INTENSIVE BRAKE POWER CERTIFICATE FOR AIR BRAKE (GOODS) (END TO END RUNNING)

		No.
ISSUED BY : (Intensive exam. Point /divn./Rly.)		
Destination :		
Date		BPC No.
Train No.		Loco No.
Load & Stock		Total No of cylinders
No. of operating cylinders		Brake Power %
Vacuum on departure on loco		On Bk. Van

INSTRUCTIONS**A. GUARDS AND DRIVERS**

Before starting the train guards and drivers should ensure:-

- (i) No driver should move the loaded train from the loading point unless the destination is clearly mentioned of the brake power certificate.
- (ii) Continuity first to last vehicle of the train.
- (iii) If BPC is invalid , inform the control office & take necessary instructions from C&W controller.

THIS CERTIFICATE IS VALID

Provided the destination is mentioned on the BPC of the loaded train

Provided the composition of the rake is not changed by 4 or more wagons

Provided the rake is not stabled for more than 24 hours

B. LIST OF WAGONS IN ORDER

S.no	Wagon No	S.No.	Wagon No.	S.No.	Wagon NO.	S.No.	Wagon NO.
1.		19.		37.		55.	
2.		20.		38.		56.	
3.		21.		39.		57.	

C. ENROUTE PROBLEMS NOTICED AT STATION GIVEN

Date	Loco No	Drivers Name/HQ	Problems and action taken	Sign. Of Driver

D. NOTES :

1. The incoming driver shall handover the brake power certificate to relieving driver . If he is leaving the train without relief . It shall be deposited with nominated authority who will give it to the outgoing driver.
2. The outgoing driver and guard will satisfy themselves from the listed wagon numbers that the brake power certificate pertains to their train.

()	()	()
DRIVER'S NAME & SIGN.	GUARD'S NAME & SIGN.	JE/SE (C&W) NAME & SIGN.

Details of intensive examination:

1. Rolling in examination including axle box feeling
2. Intensive examination and repairs:

Intensive examination and repairs:

- A Inspection and repairs of running gear fittings.
- B Inspection and repairs of brake gear and spring gears.
- C Inspection and repairs of draw and buffing gear.
- D Checking and making good the deficiency of safety fittings, safety brackets, safety loops etc.
- E Replacement of brake blocks.
- F Correct functioning and positioning of Empty Load Box Device.
- G Checking and proper securing of doors of covered wagons.
- H Checking of CBC and knuckle.
- I Meticulous check of BC, DV, AR.CR and other pipe joints.
- J After brakes are released, the wheel profile should be examined.
- K Where a reject able defect cannot be attended on the train in the yard, wagon shall be damaged labeled for attention in the sick line.

Difference between Safe to run and intensive examination:

S.N.	Safe to run / Revalidation	Intensive
1	➤ Validity of BPC is given up to next TXR examination point and it is endorsed on the same BPC, No fresh BPC is issued.	➤ The rake is examined in empty condition intensively and fresh BPC is issued as per required Brake power % as 85% or 95% or 100 %in case of EE or PEE or CC respectively.
2	➤ Only safety "S" marks defects are examined and attended to.	➤ All reject able, chargeable and "S" marks defects are examined and repairs if required.
3	➤ This type of examination is being carried out for air brake stock only.	➤ All type of stock offered by traffic given intensive examination.
4	➤ C&W staff with a least facilities may examine the load for STR.	➤ All minimum infrastructural facilities as per CAMTECH are required to examine and repairs the fitting during examination.
5	➤ It may be carried out at any traffic line.	➤ It is done only on nominated yard as well as selected line where infrastructural facilities are available.
6	➤ 10 man hours are required to carry out safe to run.	➤ The man hours required as 40 for Vacuum brake stock EE, 56 for air brake EE and 100 man hours for CC rake, 75 per PEE.

Difference between CC and premium rake examination:

S.N.	CC rake	Premium rake
1	➤ CC rake can only examine at base depot (A,B or C cat. Yard).	➤ Base depot is not nominated and can only be examined in "A" cat. Yard and nominated yard.
2	➤ 100 % brake power is ensured and BPC is issued on yellow color certificate.	➤ 95% brake power is ensured and BPC is issued on green color certificate.
3	➤ It is run on predefined (closed circuit) path only.	➤ It may be run over any route of Indian railway.
4	➤ 100 man hours is specified to attend CC examination of standard rake (58 BOXN+ 1 BVZC)	➤ Man hours are not decided. (Proposed 75 man hours)
5	➤ CC rake is preferred to form out of off POH/ROH wagons	➤ Stipulation of off POH/ROH wagons is not applied to premium rakes.
6	➤ Validity of BPC is 4500 km/20 days, 6000kms/30 days or 7500 km/35 days	➤ Validity of BPC is 12 days and a 3 days grace period is allowed for the rake in loaded.(kilometer is not decided)
7	➤ Items (reject table & chargeable) are specified to examine, repair & maintenance.	➤ Items (reject able & chargeable) are not specified for examination and its repair.(Only being examined on EE pattern)

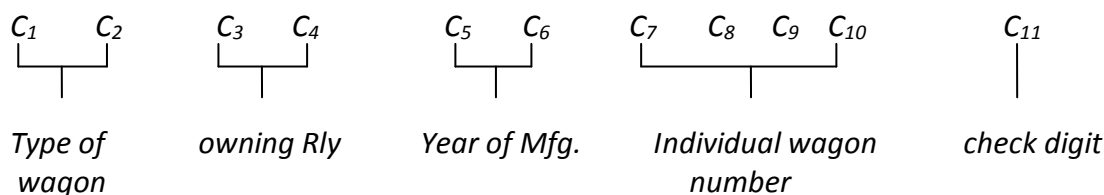
STEPS OF INTENSIVE EXAMINATION:

- a- Rolling in examination including axle box feeling.
- b- Intensive examination of originating train including repairs, detachment of damaged/sick wagons, brake testing etc.
- c- Issue of Intensive Brake Power Certificate after ensuring brake continuity of the formed load.

NEW WAGON NUMBERING SYSTEM

The new wagon numbering system is being done as per railway board's instruction issued vide letter vide letter Number. 2000/M (N)/60/2/wagon census dated 4th July 2003.

As per new scheme, the wagon number shall consist of 11 digits. First two digits will indicate types of wagon, next two digits will indicate owning railway, next two digits will indicate year of manufacture, and next four digits will indicate individual wagon number and the last digit will be a check digit. Brief is as under:



Check digit is calculated as under :

1. Add all the character in the even number (S_1)= $C_2+C_4+C_6+C_8+C_{10}$
2. Multiply S_1 by 3 = $3 S_1$
3. Add all the character in the odd number (S_2)= $C_1+C_3+C_5+C_7+C_9$
(Except check digit)
4. Add $3S_1 + S_2$ = S_4
5. Round this total up to next multiple of 10.

Check digit is the number required to be added to roundup to the next multiple of 10.
If the total in S_4 is already a multiple of 10, then the check digit will be Zero.

For example let a Wagon has its number as 12030345679

$$S_1 = 2+3+3+5+7=20$$

$$3S_1 = 20 \times 3 = 60$$

$$S_2 = 1+0+0+4+6 = 11$$

$$\text{Now } 3S_1 + S_2 \text{ i.e. } S_4 = 60+11$$

$$\text{Or } = 71$$

The next multiple of 10 will 80, to made round figure 80, more 9 is required to be added ,
hence 9 will be check digit.

The type of wagon and individual railway has unique code issued by railway board.

ACCIDENT

Accident is an occurrence in the course of working of Railway which does or may affect the safety of the Railway, its engine, rolling stock, permanent way and works, fixed installations, passengers or servant or which affect the safety of others or which does or may cause delay to train or loss to the Railway.

Classification of Accident:-

Accidents are classified into 16 parts which are following:

1. *A' class : Collision : Collisions are of 3 types –*

- *Head on collision.*
- *Side on collision.*
- *End on collision.*

Collision involving a train carrying passenger, resulting in loss of human life & damage to railway property exceeding Rs. 25 lakhs.

2. *'B' class: Fire In Train*

Fire in a train carrying passenger resulting loss of human life & damage to railway property exceeding Rs. 25 lakhs.

3. *'C' class: Train running in to road traffic at Level crossing.*

The road surface meeting with railway track on same level is known as 'Level Crossing'. The most causes of accident turns on account of an unmanned level crossing resulting into loss of human life.

4. *'D' class: Derailment*

Derailment is of two types – 1) Sudden, 2) Gradual

Derailment of trains carrying passengers resulting in loss of human life.

5. *'E' class : Other train Accident*

Train running over any obstruction (land slide, flood, & earthquake) fixed structure other than resulting in loss of human life or damage in railway property.

6. *'F' class : Averted Collision*

Averted collision between trains at least one of which carrying passengers & obstruction.

7. *'G' class : Breach of Block Rules*

Train carrying passengers, entering a block section without any authority or entering a wrong line a station.

8. *'H' class : Train Passing signal at Danger.*

Train carrying passengers running past a stop signal at danger without proper authority.

9. *'J' class : Failure of Engine & Rolling Stock Failure*

Failure in rolling stock such as failure of tyres, wheels, axles. Etc. on passenger carrying train.

10. *'K' class : Failure of Permanent Way*

Buckling of track, weld failure, rail fracture.

11. 'L' class : Failure of Electrical Equipment
Snapping or any damage to OHE wire requiring switching of OHE for more than 3 minutes.
12. 'M' class : Failure of S & T
Failure of block instrument, interlocking, signal, station communication for more than 15 minutes.
13. 'N' class : Train Wrecking
Attempted wrecking of or sabotage to train carrying passengers.
14. 'P' class : Casualties
Person falling out of a running train resulting loss of human life or grievous hurt.
15. 'Q' class : Other Incidents
Murder or suicide in a train, robbery, attempted robbery or attempted theft in railway premises, & blockage to train services due to agitation.
16. 'R' class : Miscellaneous
 - i) Train running over cattle
 - ii) Floods, breaches & landslides etc., were resulting interruption of an important through line of communication more than the threshold value.
 - Most serious class 'A'
 - Least serious class 'R'
 - I & O is not included

Role of Supervisor at accident site: As per accidental manual following duties would be performed by C & W supervisors at accident site.

- Protect the site of accident to restrict further movement on that line or yard as the case may be.
- Ensure that the necessary message regarding the details of the accident and casualties have been relayed to the SS/SM, control and civil authorities.
- Make first-aid arrangements for the injured persons, if any.
- Examine and make a note of all evidence which may prove useful in ascertaining the cause of accident.
- Try to save life and alleviate sufferings. Ensure everything possible is done. Provide security to the injured and help to all other passengers.
- To relay the prima-facie cause of the accident with the expected time of restoration.
- Seize and freeze all records as laid down in accident manual.
- Arrange to take photographs from different angles to assist in reconstructing the scene of the accident.
- Co-ordinate with train crew, station staff and control office for providing relief to the injured and restoration of traffic.
- Record the statement of the concerned staff available at site for ascertaining the cause of the accident.
- Take written evidences of as many witnesses as possible in case passenger train is involved. Their names/addresses should be recorded (witnesses should not be from Railways).
- Ensure relaying the progress report to control after every one hour.

- Obtain clearance from civil/police before starting restoration in case of sabotage/doubt of sabotage.
- When a senior officer arrives at the spot, relevant records should be handed over to him and he must be briefed about the situation.
- If the accident is within station limits, note down the condition of points, fixed signals, position of levers, position of block instrument and SM control.
- Scrutinize as early as possible the train signal register, reply books, PN book and other relevant records pertaining to the train movement.
- Make a rough sketch showing the position of derailed vehicles, marks on sleepers etc.
- To give fit certificate to the accident involve train up to the next examination point towards direction of movement.

Following operating features must be checked while investigating into a derailment:

- Speed of the train just before the accident.
- Uneven load/ shifted load – load in all the vehicles must be checked to get an idea of loading and lashing/securing of loads.
- Application of brakes.
- Brake power of the train and location of vehicles without brake power.
- Whether all hand brakes are in released condition.
- Sudden reversal of points.
- S&T failure reported before the accident – how and when was it set right?
- The position of loose components wherever found.
- Whether the engineering staff was on work.

Wheel marks, points of mount, flange travel mark on the rail, point of drop and other damages to sleepers, rails and other fitting

DERAILMENT

Definition: -Derailment means off loading of wheel/wheels causing detention to rolling stock/P. Way.

Classification: Derailment classified in 'D' class accident. It is further classified as –

- D-1- Derailment of a train carrying passenger resulting loss of human life/Hurt or damage to railway property of the value exceeding Rs.25,00,000/- or interruption of communication for at least 24 hrs.
- D-2- Derailment of a train not carrying passenger resulting loss of human life/Hurt or damage to railway property of the value exceeding Rs.25,00,000/- or interruption of communication for at least 24 hrs.
- D-3- Derailment of a train not falling under D-1
- D-4 - Derailment of a train not falling under D-2
(The loss of the property is more than the threshold value.)
- D-5 - Derailment of a train not carrying passenger, not falling either D-2 or D-4
- D-6 Other derailments occurring in shunting, marshalling in yard, loco yard and siding but not involving a train.

Categories: There are two broad categories of derailment

- i) Sudden derailment – Instant dismounting of wheel from rail.
- ii) Gradual Derailment – Gradual climbing of flange on the rail.

Sudden derailment: When derailing forces are high on a wheel it may suddenly jump off from the rail table and the rolling stock derails. In this case no flange mounting marks are available on the rail table. However the wheel drop marks can be seen on ballast or sleepers.

The possible causes for sudden derailment are –

- (i) Sudden shifting of load
- (ii) Improper loaded vehicle

Gradual Derailment: The forces acting on the rail & wheel contact are the weight of the vehicle transmitted by the wheel on the rail which keeps the vehicle on the track

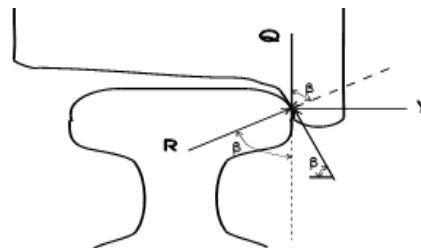
When the wheels of the vehicle roll parallel to the rail there is no angularity, when there is positive angularity the conditions creates the chance of derailment.

Derailment Mechanism: Nadal has derived the formulae as follows –

$$Y = R \sin \theta - \mu R \cos \theta$$

$$Q = R \cos \theta + \mu R \sin \theta$$

$$\text{So, } Y/Q \leq \tan \theta - \mu / 1 + \mu \tan \theta$$



FORCES AT RAIL-WHEEL CONTACT

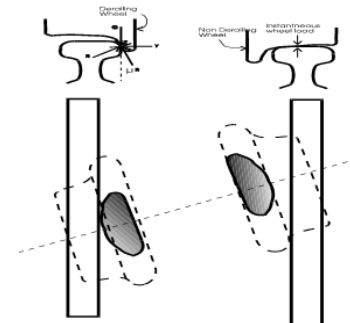
Cause of Derailment: The cause of derailment can be largely classified into following two major categories –

- (i) Equipment failure
- (ii) Human failure

Apart from the above cattle run over, sudden falling of boulders, trees etc. on the track, sinking of track also may be the cause of derailment of rolling stock which do not require thorough investigation.

Derailment occurred due to one or more of the following factors:

- (a) Operational factors
- (b) Track
- (c) Rolling stock
- (d) S&T
- (e) Others



Operational factors:-

1. **Speed** – The exceed speed of train creates more lateral forces on the flange there by creating the chances of derailment.
2. **Loading** – Excess loading may leads to derailment of a vehicle as the wheel may float off due to excess loading.
3. **Wrong marshalling** – Empty stock if marshaled in between two loaded wagon may cause the derailment
4. **Operating** –
 - I. Improper setting of point during shunting operation
 - II. Undetected obstruction between toe of switch and stock rail
 - III. Improper train operation i.e. sudden application of brake causing bunching and off loading of wheel

Track Investigation:

- (1) If point of mount is known.-the track length should be taken for measurement: - 45 m. In rear side of the point of mount & 45 m in front side of the point of mount.

(2) If point of mount is unknown/ disputed. - 90 m. In rear side of the point of mount & 45 m in front side of the point of mount.

Point of mount :- The first wheel flange climb marks found on rail table is known as point of mount.

If there is more than one point of marks found on rail table

The rear most 1st point of mount mark will be treated as actual point of mount. No any point of mount is found in case of sudden derailment.

Point of drop:- The wheel flange drop marks shown on rail table is known as point of drop.

In case of gradual derailment, the point of mount & point of drop both are found on rail table .but in case of sudden derailment only point of drop is found.

Track parameters which to be recorded

(1) **Track gauge**:- The distance between two running rails is known as track gauge. The track gauge should be measured 13mm below from the top of the rail.

Track gauge should be :-

- I. On straight track - 1676 ± 6 mm.
- II. Up to 5 degree curve - 1676_{-6}^{+15} mm.
- III. Above 5 degree curve - 1676_{-0}^{+20} mm.

(2) **Cross level** :- The variation in top level of the both rails on same point on straight track is known as cross level..cross level is always measured on left rail in reference to right rail as per direction of the train.

Permitted cross level is - 13 mm/station.

(3) **Twist**: The variation in two continuous cross levels is known as twist.

Or

The algebraic variation in two continuous cross levels is known as twist. It is always measured in mm/m.

For example. ---- if cross level on 1st station is +5 mm and on station 2nd is - 5 mm,

Then twist will be:-

$$\text{Twist} = \frac{(+5) - (-5)}{3} \text{ or } \frac{(+5) + (5)}{3} \text{ or } \frac{10}{3} = 3.3 \text{ mm/m.}$$

The allowed twist:-

- On straight and circular curve - 2.8 mm /m.
- On transition curve - 1.0 mm/ m.
- A class track (160 kmph) - 1.37 to 1.41mm/m
- B class track (130 kmph) - 1.41 to 1.78 mm/m
- C class track (100 kmph) - 1.78 to 2.41 mm/m
- D class track (Main Line) - 1.78 to 2.41 mm/m
- E class track (Branch line) - 2.41 to 2.78 mm/m

(4) **Cant or super elevation**: To maintain the centre of gravity or to neutralize the effect of centrifugal force, the outer rail on curve is lifted in respect to inner rail, is known as cant.

Or

The lifting of outer rail in reference to inner rail on curve on same point is known as cant or super elevation.

Cant is always measured on outer rail in mm. The max Cant allowed in Indian Rly B.G. Is 165mm.

For A & B class track - 165 mm.

For C, D & E class track - 140 mm

Variation in cant is possible due to :- Cant deficiency or Cant excess.

Max. Cant deficiency permitted - 75mm (gen), 100mm (in spl. Case).

Max. Cant excess permitted - 75 mm .

Causes of variation in cant

(1) Due to poor maintenance of track by Engg. Deptt. (excess or low cant)

(2) Due to poor engineman ship by the driver. (Excess or low speed)

The equilibrium cant

$$E = \frac{GV^2}{127R}$$

Where G = 1676+74 mm (gauge + width of the rail)

V= speed in kmph

R = radius in metre

(5) VERSINE

The perpendicular distance joining the centre of chord to the midpoint of arc is known as versine. Versine is always measured on outer rail in mm.

If the versine is measured on 11.8 mtrs chord basis, the allowed variation in versine is only 10mm. Or if the versine is measured on 20 mtrs chord basis, the allowed variation in versine is only 29 mm. The versine measured on 11.8 mtrs chord in cm, is tells directly the degree of curve.

$$\begin{array}{llll} \text{Versine} & V & = & \frac{125 C^2}{R} \\ \text{Where} & C & = & \text{length of chord in metre} \\ & R & = & \text{radius in metre.} \end{array}$$

RAIL DEFECTS

(6) HOGGING:- bending of rail ends at rail joints is known as hogged rail. Allowed max. 2mm

(7) BATTERING:- wear of rail ends at rail joints is known as battered rail. Allowed max. 2mm

(8) RAIL WEAR

(i) Vertical wear

For B.G. (a) 60 kg rail--- - 13mm

(b) 52---,, ---,,----- 8mm

1 90 r ----- 5 mm

(ii) Lateral wear

A&B track ---- 8mm

C&D --,,----- -----10 mm

(iii) Angular wear. – Permitted max. 25 degree

(9) UNEVENNESS:- To be measured on 3.6 m Gen Isolated Chords.

Limit --- On long term basis ---6mm – 10mm

On short term basis ---10 mm- 7mm

(10) STRAIGHT NESS:- To be measured on 7.2 m chords.

Limit --- On straight track ---5mm–10mm

On curved track ---5 mm –7mm

(11) CREEP:- Shifting of rail in longitudinal direction due to poor fastening, or due to emergency rake braking or due to skidding of engine wheel.

(12) BUCKLING:- Increase of rail length due to effect of temperature.

(13) **CORROSION**

(14) **BURNING**

(15) **KINK**

(16) **BALLAST**: To provide Cushing & to absorb vibration, ballast is provided.

Types of ballast:-

(1) CUSHION BALLAST :- depth of ballast below the sleeper.

For A cat. Track ---300mm
B&C --,, --- 250 mm
D -- ,, --- 200 mm
E ----,, ----- 150 mm

(2) CRIB BALLAST :- Between two sleepers in the level of sleeper height.

(3) SHOULDER BALLAST:- Heap of the ballast at the end of the sleepers , to minimize the effect of lateral thrust.

Generally it should not be less than 150 mm.

SIZE & SHAPE OF BALLAST :- In 50mm size with at least 30% sharp edges

(17) **SLEEPER**

ROLE Of SLEEPER :- To hold the rails, distributes the load equally and to maintain the track gauge on prescribed limit.

TYPES Of SLEEPER :- Wooden sleeper, CST-9 , Iron sleeper & concrete sleepers. Now a days max. concrete sleepers are being utilized .

SLEEPER DENSITY :- Number of sleepers per km is known as sleeper density.

(1) If track density is more than 10 GMT:-

For A,B,C&D route ----- M+7.

For E ,, ----- M+4

(2) If track density is less than 10 GMT:-

For A,B,C route ----- M+7.

For D & E ,, ----- M+4

Where M is the master length of rail i.e. 13 m

Means no. of sleepers required in 13 m length is----- $13 + 7 = 20$

for. ---,, ---,, in 1 m ,,----- $20/13 = 1.540$.

so that ,,----- in 1000 m ,, ----- $1.54 \times 1000 = 1540$.

(18) **CONDITION OF FASTENERS, & FORMATION: -**

During investigation, the condition of fasteners, & formation of track also to be observed for any type of abnormality.

(19) **ROLLING STOCK**

During investigation the following parameters of rolling stock to be taken into account

Wheel

wheel gauge – Wheel gauge to be measured at 4 locations and average OF them will be considered (1600 +2/-1mm.)

Thin flange – A thin flange increase the lateral play between wheel set and track which increases lateral oscillation (y/Q) / Angularity of wheel (Cond.-16 mm for goods,22 mm for coaching stock).

Sharp flange – Due to sharp flange wheel set can take two roads of slightly gaping point. (Cond. Limit- 5mm.)

Deep flange – Deep flange of the wheel may hit the fish plate of rail causing derailment (Cond. Limit – 35 mm.)

Hollow tyre – It increases the conicity of the wheel which reduces the critical speed of the rolling stock and increase the lateral force (y)

Flat tyre – It can damage the rail due to successive impact and cause high stresses leading to rail fracture (Cond.- 50mm for coaching stock & 60mm. For goods stock)

Root radius too small – It increases the coefficient of friction between rail & wheel flange which increases the frictional force causing derailment (Cond. Limit – 13mm)

Buffer

- I. Height of buffer from rail level should be within 1105 mm. to 1030 mm.
- II. Buffer projection to be within 635 mm to 584 mm.
- III. Condition of buffer / CBC to be noticed for any crack/worn.

Wagon body – If the wagon is in loaded, the condition of consignment to be observed for over loading / uneven loading.

S&T –

- I. Improper setting of point during shunting operation
- II. Undetected obstruction between toe of switch and stock rail
- III. Approaching signal may be indirect cause of bunching and off loading of wheel

Others :

Other derailments occurring in shunting, marshalling yard, loco yard and siding

Notable points:-

- (i) Before investigation first of all 1st derailed vehicle and wheel to be decided and point of mount & Point of drop to be marked.
- (ii) Track measurement to be taken on loaded condition.
- (iii) There is no need of measurement in case of train wrecking and sabotage.

In any case of derailment, observations to be made jointly by Sr. subordinates of different departments like CWI, SLI TI, PWI and representative of S&T department. A rough sketch of site to be prepared after that they will prepare a joint note mentioning the followings

1. Subject
2. Brief history
3. Observations
4. Conclusion
5. Responsibility

The joint note will be duly signed by all the nominated subordinates. If any point mentioned in the joint note not agreed by anybody he will submit dissent note by mentioning the logical points and copy of the same will be given to all of the signatories. After getting the dissent note others will submit their rejoinder.

PURPOSE OF ENQUIRY & TYPES OF ENQUIRY

Purpose of an Enquiry

- To ascertain cause of an accident.
- To formulate proposals for preventing a reoccurrence.
- To ascertain if any inherent defect in the system of working or in physical appliances such as, track, rolling stock & other working apparatus.
- To ascertain negligence or avoidable delay in relief & restoration.

Need of an Enquiry

- Provision of section 113 of railway act 89.
- Any accident attended with loss of any human life, or with grievous hurt, as defined in the IPC (45 of 1860), or with such serious injuries to property as may be prescribed.

- Any collision between trains of which one is a train carrying passengers.
- The derailment of any trains carrying passengers, are of any part of such train.
- Any accident of a description usually attended with loss of human life or with such grievous hurt as aforesaid or with serious injury to property.
- System working suspected defective.
- Railway staff responsible-prima facie or on police report.
- Cause of accident is not clear.
- On advice of CRS.
- Railway administrations decision.

Types of Enquiry

1. **Commission of Enquiry**
 - Appointed by central government under commission of enquiry act 1952.
 - Commission has powers of a civil court.
 - Its findings are not binding on govt.
 - Such type of enquiry is very rare.
2. **Commissioner of Railway Safety Enquiry**
 - It is an independent agency, constituted under provisions of Section 113 of The Railway Act 1989.
 - Such enquiry is obligatory in cases of-
 - 1) Loss of human life as defined in IPC.
 - 2) Serious damage to railway property.
 - 3) Accident at unmanned level crossing only if loss of life/ grievous injury of passenger involves.
 - 4) Any other accident-at his discretion.
 - CRS can summon and enforce attendance of any person it can examine and receive evidence under oath. It can requisition any public record or copies thereof from.
 - CRS ENQUIRY is not obligatory if passenger travelling in the train is not affected and no passengers are hurt at level crossing accident.
 - CRS enquiry withheld in case of constitution of commission of enquiry by Central Govt.
 - Final report copy is sent to Railway Board, Railway Administration concerned, and Director IB-Ministry of Home. Affairs in case of sabotage.
3. **Magisterial / Judicial Enquiry**
 - DM or any other magistrate appointed by state govt. may make an enquiry.
 - No enquiry if commission of enquiry or any other authority nominated by central govt.
 - Magistrate may order judicial enquiry for trial of any person criminal offence.
4. **Joint Enquiry**
 - It can be at SAG, JAG, Sr. scale, Jr. Scale or Sr. subordinate level.
 - Enquiry committee has representatives from traffic, civil, mechanical, electrical, S&T etc.
 - Committee is nominated by DRM/GM depending upon the level.
 - Joint enquiry may be dispensed with if:
 - a) CRS enquiry is ordered.
 - b) Any other agency is appointed by central govt.
 - c) There is no reasonable doubt about cause of accident.
 - d) Any one of the departments has accepted the responsibility.

In case of c & d above, departmental enquiry is to be held

Proceedings of joint enquiry

- Conduct enquiry proceedings carefully.
- President of enquiry to warn all the witness against the untruthful all false statements.
- Certificate signed by the president to be forwarded with proceedings.
- Simple language in logical order.
- End of statement to be certified as “Read “accepted and correct or” Read”, explained, and accepted as correct.

Documents to be accompanied with proceedings

- Title page and particulars of accidents.
- History of accident.
- Description of the site if the accident.
- Sketch of the site of the accident.
- Findings , and reasons for conclusions
 - Clear, brief, and to the point
 - indicate cause of accident, rules violated,
 - Staff held responsible
- Note of dissident, if any.
- Reading of track, engine, wagons/ vehicles.
- Recommendation of the enquiry committee.
- Statement of witness with a list, and analysis of evidence.
- Plan of accident site and or other drawing in case of serious accidents.
- List of passenger, railway employee killed or injured.
- List of damages to loco, R/stock, track, OHE,S&T with estimated cost
- Relevant extract from train register log book, station diary, guard rough journal.
- Date of start and finalization of enquiry.
- Photograph of the site of serious accident.
- Note on serious or important discrepancies.
- Bio-data of staff held responsible.
- Relief and restoration.
- All the documents to be signed.

5. **Departmental Enquiry:** The departmental enquiries shall be ordered by DRM. The proceedings of departmental enquiry shall be drawn up in the same form as those prescribed for joint enquiries and the rules in regard to the conduct of joint enquiries shall apply to the conduct of departmental enquiries, in so far as they are applicable.

As per railway board latest instructions, following guidelines should be followed for accident enquiries by railways:-

- i. All serious accident shall be enquired into by the Commissioner of Railway Safety.
- ii. In case CRS or CCRS is not in position to inquire into serious accident cases the inquiry shall be done by JA grade inquiry committee with DRM as the accepting, authority subject to the review by CSO.
- iii. All the cases of collision falling under A1 to A4 categories shall be inquired into by committee SAG officers with GM as the accepting authority unless same is being inquired by CRS.

- iv. All other consequential train accidents except Unmanned Level Crossing Accidents shall be inquired into by a committee of JA grade officers and in its absence by branch officers. DRM shall be accepting authorities for these inquiries subject to the review by CSO.
- v. Consequential Unmanned Level Crossing accidents and all other train accidents shall be inquired into by a committee of Senior Scale or Junior Scale Officers as decided by respective DRM with DRM as the accepting authority.
- vi. All yard accidents shall be inquired into by a committee of senior supervisors with Sr. DSO/DSO as accepting authority.
- vii. All cases of indicative accidents shall be inquired into by a committee of Senior or Junior Scale Officers with DRM as the accepting authority.
- viii. GM or DRM can have the inquiry conducted by a committee of higher level of officers then the above mentioned levels depending upon the seriousness of the accidents.
- ix. In accident cases wherein the Inquiry Committee determines responsibility on the State Foreign Railway, the inquiry report should be put up to the PHOD of the concerned department of the railway on which the accident took place through CSO after which such inquiry report shall be accepted by the AGM (instead of DRM). Finalization of Inter Railway DAR cases arising out of such inquiry reports followed up by the PHOD of the concerned department of Railway on which the accident took place. If suitable response is not received from respondent Railway at GM's level, then the case should be referred to Railway Board.
- x. All case of equipment failures shall be inquired into by senior Supervisors/Supervisors of respective departments.
- xi. All inquiries will be ordered by the concerned DRM except for enquiries into collisions as per item (iii) as above wherein GM will order the inquiries.

Schedule of procedure of completion of accident enquiry at zonal railway level

Following target dates have been given by Board of prompt finalization of enquiry:-

Ordering inquiry by DRM/GM	D+1
Commencement of Enquiry	D+3
Submission of Preliminary report	D+7
Acceptance of inquiry report by GM/DRM	D+10
Finalization of inquiry report by CSO/AGM	D+15
Submission of inquiry reports	D+20
Finalization of accident cases	D+90
*D-Date of accident	

Time limits prescribed above are the maximum period of time. Railway should make efforts to finalize the enquiry report and D & AR action as early as possible but not beyond the prescribed time limits.

Preservation of Records	
Accidents of class A & B	- 5 Years
Accidents of class C	- 3 Years

ART AND ARME

Introduction: Accidents and Disasters are unpredicted and unavoidable. Indian Railway has its own mechanism to deal with rescue and restoration work during accidents and disaster is ART & ARME.

Accidents: Any unusual which does or may affect the safety and transportation of Railway in form of damaging rolling stock, permanent structure and serious injuries / casualties to the Railway passengers / Railway staff is known as Rail accidents.

Any accident involving more than 100 injuries should be termed as disaster.

Disasters: Disasters in Railway context is defined as a major accident leading to serious casualties and long duration of interruption of trains.

Disasters caused by Human and equipment failures. i.e. Collisions, Derailments, Fire, Natural calamities, Sabotages.

ART- ACCIDENT RELIEF TRAIN

At present there are 174 ARME and 184 ARTs stationed at strategic locations over Indian Railway system

CLASSIFICATION OF ART.

- A class
- B class
- C class
- Other types of ART
- ✓ SPART (Self propelled ART)
- ✓ Road ART (Modified road vehicle)

Composition of A class ART

1. 140 ton Crane with Match truck.
2. BCNHS (For Engineering and C & W Materials)
3. BRNHS (For Rails & Wheels)
4. RT (Staff car)
5. RT (Officers car)
6. RT (Packing van)
7. RT (Equipment van)
8. RT (HRE & Power van)
9. RT (Crane tool van)

Composition of B class ART

1. BCN (For Engineering and C & W Materials)
2. BRN (For Rails & Wheels)
3. RT (Staff car)
4. RT (Packing van & OHE van)
5. RT (Equipment van)
6. RT (HRE & Power van)
7. RT (Tool van)

Composition of C class ART

1. RT (Brake down van)

ARME- Accident Relief Medical Van

Classification of ARME

1. **Scale 1 ARME**: Scale Equipments are stored in special medical vans stabled in separate sidings. Medical Van (RH), have operation theatre & medical equipments. Auxiliary Medical van (RHV) have all rescue equipments i.e. HRD (cutting equipments), S & T and Electrical equipments.
2. **Scale 2 ARME**: Equipments stored in special rooms on platforms.

Authorities to order ART & ARME:- CME, CMPE, Sr.DME, DME

Codes for accident alarm siren / hooter

- Two hooters each of 45 seconds duration with 5 sec. interval indicate ART required at home stations.
- Three hooters each of 45 sec duration with 5 sec. interval indicate ART required at out station.
- Four hooters each of 45 seconds duration with 5 sec. interval indicate ART& ARME required at home station.
- Five hooters each of 45 seconds duration with 5 sec. interval indicate ART & ARME required at out stations.
- One long hooter of 90 seconds duration indicate cancellation of ART & ARME.

TURN OUT TIME

- Turn out time of ART in day 30 minutes.
- Turn out time of ART at night 45 minutes
- Turn out time of ARME with double exit 15 minutes.
- Turn out time of ARME with single exit 20 minutes.

List of tools & equipments on class A & B ART

1. Generator & electrical equipments
2. Illumination
3. Oxy cutting equipments
4. Compressor pneumatic tools
5. Hydraulic Re railing equipments
6. Jacks
7. Wire ropes, slings & shackles
8. Chains, slings & wooden packing
9. Other mechanical equipments
10. Fitters tools
11. Blacksmith tool
12. Measuring instruments
13. Oil & grease
14. Locomotive spare parts
15. C & W spare parts
16. Engineering tools
17. Fire fighting equipments
18. Medical equipments
19. S&T equipments
20. Books & Manuals
21. Records
22. Camera & Video Recorder

SPEED

- Existing speed of ART - 75 kmph*
- Proposed speed of ART - 100 kmp*
- Existing speed of ARME -100 kmph*
- Proposed speed of ARME -140 kmph*

140 T CRANE

- *140 ton crane is available with A class ART*
- *There are two types of 140 T crane in IR*
 - 1. Jessop 2. Gotwal*
- *All cranes are self propelled at a speed of 30 kmph.*

Books & Manuals Reqd. in ART

- 1. Transportation manual*
- 2. Accident manual*
- 3. G & S R*
- 4. IRCA part 3 & part 4*
- 5. Telegraph code book*
- 6. First aid manual*
- 7. Safety first instruction book*
- 8. Rules for working of crane*
- 9. Working time table.*

Sequence of Movement Of ART & ARME To The Accident Site

- *ARME*
- *ARME from the other end*
- *ART*
- *ART from the other end*
- *Additional brake down special*
- *Special train carrying GM & other officers*
- *Unaffected portion of accident involved train if possible*
- *Empty coaching special for clearing passengers.*

Duties of Mechanical Deptt.

- 1. Ensure marshalling of ART according to site requirement before moving to accident site.*
- 2. Ensure OHE power supply switched off before commissioning rescue/restoration work.*
- 3. Toppling those coaches whose searches has been completed.*
- 4. In case of suspected sabotage, ensure minimum interference to clues.*
- 5. Extracting injured passengers and dead bodies from coaches.*
- 6. Take assistance from other dept.*
- 7. Maximum number of coaches to be tackled simultaneously.*
- 8. Road crane to be arranged to assist 140T crane.*
- 9. Examine unaffected and re railed rolling stock and certify their fitness for further movements.*

IRCA- INDIAN RAILWAY CONFERENCE ASSOCIATION

1. HEAD QUARTER- DRM Building New Delhi
2. WORKING: Under Rly Board(Member Traffic)
3. ADMINISTRATIVE CONTROL-GM/Northern Rly
4. HEAD: General Secretary (Traffic /Commercial Deptt.).

AIM:

- To upgrade the maintenance quality of rolling stock (carriage & wagon).
- Rate fixing and preparation of passenger ticket and goods charges.
- To dissolve the operational dispute between two interchanging point.
- Pre & final examination during POH & ROH to be carried out by neutral TXR (IRCA) and final fit memo to be issued to the concerning officer of the w/shop or sick line.
- Repair cost during POH & ROH & Other repair of carriage & wagon to be assess & repair cost sent to railway board.
- Suggestion and rake assessment of amount paid by commercial department. On account of railway claim to be service time to time and proposal sent to railway.

WORK:

- Mechanical deptt. : Final fitness of off POH / NPOH, off ROH & other wagon lying in sick line for miscellaneous defects to be issued by IRCA men i.e. NTXR.
- Operating deptt. To dissolve the operational dispute between two railways, preparation of time table etc.
- Commercial deptt: Rate fixing & preparation of passenger ticket & goods stock.
- Accounts: Accountant assessment of the expenditure to be checked by IRCA.

Mechanical Deptt. Concern: IRCA gives out the rules for the standards condemning sizes of various components used on rolling stock. They also give the guidelines for the maintenance of rolling stock in workshop and in open line. The rule books used for the carriage & wagon issued for the carriage & wagon branch of mechanical deptt. Are: part III for wagon stock & part IV for coaching stock .

IRCA part III & part IV contain 4 chapters

Chapter	Details
Chapter I	Definitions
Chapter II	Workshop repair practice
Chapter III	Maintenance practice in open line
Chapter IV	Rejection Rules

Rejectable defect in coaching stock

As per IRCA. Rule book part iv, chapter iv rejections are given from **Rule No. 4.0-4.25.2:-**

- (4.1) Coaching stock must not be allowed with any chargeable, reject able and any other defects from work shop.
- (4.2) Maintenance depots shall ensure the maintenance procedure givens in chapter II and III and No any reject able defects is allowed.
- (4.3) Notwithstanding any provisions in the rules, coaching stock must not be allowed to run if in the opinion of a Train Examiner, it is in such a condition as if may cause an accident.
- (4.4) Coaching stock must not be allowed with any defects having 'S' marks for Guidance of staff concern.

Example:- Axle Box Defects;

- Axle box lug/wing broken.
- 'S' - Hot box;**
- Over due oiling
- Due repacking.
- 'S' - Axle guard defects;**
- Axle guard bridle bar crack, deficient , broken.
- Hydraulic dash pots broken
- **Body defects;** door bent
- Alarm chain damaged/deficient.
- **Brake gear defects;**
- Brake gear fitting deficient.
- Brake gear (Air brake system)-Any defects in brake system;
- 'S' - Any buffer dead.**
- **Draft gear + Coupling;**
- 'S' - Draw bar, hook draft hook broken**
- Draft gear + buffing in EMU;
- Infringement of O.D.C
- **Trolley frame defects;**
- 'S' - Bolster spring plank broken.**
- Spring gear;
- Shifted more than 13mm
- **Tyre defects**
- 'S' - loose tyre**
- Wheel defects**
- 'S' - thin tyre, sharp flange etc**
- Under frame defects**
- crack or bent
- 'S' - Under slung tank suspension bracket broken**
- 'S' - Buffer Height Variation more than 64 mm on same end**
- 'S' - Axle Pulley Loose.**
- 'S' - Brake Block Deficient. Or excessive worn**
- 'S' - Foot Board, Hand Hold deficient**
- 'S' - Any Buffer Dead**
- 'S' - Wheel Shifted on Axle**

MARSHALLING

A marshalling yard is a place where goods trains originate & terminate. Goods wagons, loaded & empty, arriving from different directions are shorted out & classified according to a plan so as to enable fresh trains to be formed with optimum load for the locomotive & running of goods wagons in full loads or large blocks through to destination or to the farthest possible common point. The need of marshalling yards, of course, arises from the fact that on the railways the unit of movement, which is the train, is much larger than the unit of booking which typically is the wagon & therefore, facilities for consolidating the wagons together into train as well as for braking up trains & sorting outs into individual wagons become essential.

Marshalling yards are thus a very vital part of the goods operation scheme and, in fact, the predominant proportion of total wagon time is spent not on trains but in marshalling yards. That is why the marshalling yards are often referred to as 'necessary evil' which, if not properly worked can easily become a 'graveyard' for wagons. The yards are normally located at the following types of places:

1. Junction routes.
2. Junction of gauges.
3. Major ports.
4. Major industrial complexes.
5. Collieries.

Marshalling yards may be hump yards or flat yards. In India on the M.G. only flat yards are permitted, & on the B.G. also by far the most numerous are the flat yards. However, all the most important large yards on the B.G. are the hump yards. A hump yard is one wherein an artificial gradient, shaped like a camel's humps, is introduced on the shunting neck so that wagons pushed up to the crest of the hump by a shunting engine can free run down on the other side, & can be send into separate lines or sorting sidings by a operation of the relevant points as the wagons are moving. This enables a hump yard to handle or sort out a vastly increased number of wagons. Thus under Indian conditions, where as a flat yard with a single shunting neck can handle about 400 to 500 wagons a day, a hump yard can handle more than 1500 to 1600 wagons per day. However, the speed picked up by a wagon while rolling down the hump into the sorting sidings necessitates special arrangement for braking of the wagons to prevent damage because of excessive impact. This is the reason why hump yards are not permitted on M.G., where the existing wagon designed is unable to take heavy impact. The main function of marshalling yards may be summarized under:

1. To pass through trains after changing engine or crew, TXR examination, detached sick wagons, if any, & adjust the load of the train, if necessary, according to the maximum permissible on the next engine run.
2. To sort out the terminating trains & local loads originating in the yards.
3. To form originating through trains for farthest point possible or farthest destination possible in accordance with marshalling orders laid down by the H.Q. office.
4. To form shunting & van goods trains for different sections served by the yards including suburban sections, if any.
5. To hold back trains & wagons till they are able to be got out.
6. To move train engine & shunting engine pilot between yards & loco shed.
7. To place & withdrawn wagons from various point in the local area such as goods sheds, sick lines, transshipments sheds & departmental sidings, industrial sidings, etc.

A large hump marshalling yard consisting of reception yard and classification yard and dispatch yard and also facilities for bypassing through trains. A loco shed, sick line, transshipment shed and grid yard for formation of shunting trains are also indicated. A yard of this type where in wagons enter at one end and progress forward in one direction from stage to stage until departure is called a uniflow yard. The layout indicates that trains from both, up and down directions can be dealt with in this yard. That is why it may be called a 'single yard'. It is also possible to have separate yard for receiving UP trains and DOWN trains as is the case, for example, at Mughalsarai, such layout can be justified only for the very largest amount of traffic.

Organization and working- A large yard may handle up to 3000 and more wagons per day. It is, therefore, increasingly becoming the practice to place the bigger yard under the direct supervision of a junior or senior scale officer who is called 'Area Officer'. Although the officer directly controls only the operating staff, he is also given the responsibility of coordinating the functions of various departments working in the yard-commercial, carriage and wagon, loco, signal and telecommunication, electrical, security and civil engineering-who are all contributing directly or indirectly to the functioning of the yard. An idea of the magnitude of operation and the number of men involved may be obtained from the Mughalsarai yard, where the total staff strength of all department including both Northern and Eastern railway, is over 10,000.

APPENDIX – I

Revised Scheme for marking Return Date to Work Shop (Freight Stock)				
Sr.No	Residual Life (Years) 'R'	Maintenance Schedule to be given	Month of dispatch from Workshop	Return date
1	More than 4 years	POH	a	a+4 years
2	3 or $3 < R < 4$ or 4	POH	a	a+R (i.e end of codal life)
3	$R < 3$	ROH	a	a+24 months
<i>The above table covers BOXN wagons</i>				
1	More than 4.5 years	POH	a	a+4.5 years
2	3 or $3 < R < 4.5$ or 4.5	POH	a	a+R (i.e end of codal life)
3	$R < 3$	ROH	a	a+24 months
<i>The above table covers BRN wagons</i>				
1	More than 4.5 years	POH	a	a+4.5 years
2	3 or $3 < R < 4.5$ or 4.5	POH	a	a+R (i.e end of codal life)
3	$R < 3$	ROH	a	a+18 months
<i>The above table covers BOX, BCX, BRH, BOXK, BOXKH, BFK, BFKI & BTPH wagons</i>				
1	More than 6 years	POH	a	a+6 years
2	4 or $4 < R < 6$ or 6	POH	a	a+R (i.e end of codal life)
3	$R < 3$	ROH	a	a+24 months
<i>The above table covers BCN, BCNA, BTPN, BOBR, BOBRN & BLC A / B wagons</i>				
1	More than 4 years	POH	a	a+4 years
2	$R < 4$ or 4	ROH	a	a+24 months
<i>The above table covers BTPGL, BTPGLN, BTAL & BTALN wagons</i>				
1	More than 3 years	POH	a	a+3 years
2	$R < 3$ or 3	ROH	a	a+24 months
<i>The above table covers BOY wagons</i>				
1	More than 3 years	POH	a	a+3 years
2	$R < 3$ or 3	ROH	a	a+18 months
<i>The above table covers BOBX & BOBS wagons</i>				
<i>The wagons having POH periodically less than 3 years should be undertaken for ROH in case the residual life is less than one POH cycle and return date may be marked as 'a' + ROH periodicity is the month of dispatch from workshop.</i>				

APPENDIX – II
TRANSPORTATION CODES FOR COACHES

S.No	Transportation code	Details
1	W	VESTIBULE
2	G	SELF-GENERATING
3	S	SECOND CLASS
4	F	FIRST CLASS
5	L	LUGGAGE VAN
6	R	GUARD BRAKE VAN
7	Y	LADIES COMPARTMENT
8	J	ICE COMPARTMENT
9	Q	ATTENDENT
10	D	VENDORS COMPARTMENT
11	P	POSTAL VAN
12	U	KITCHEN
13	CB	PANTRY CAR
14	CD	DINING CAR
15	CN	SLEEPER CLASS THREE TIER
16	CW	SLEEPER CLASS TWO TIER
17	CZ	CHAIR CAR
18	CT	TOURIST CAR
19	AC	AIR CONDITIONED
20	FC	FIRST CLASS WITH COUPE
21	FF	UPPER CLASS
22	SC	SECOND CLASS WITH COUPE
23	GS	SECOND CLASS WITH SELF GENERATING EQUIPMENT
24	LL	LUGGAGE VAN WITH LAVATORY
25	JJ	REFRIGERATOR COMPARTMENT
26	M	MILITARY CAR ORDINARY
27	MA	MILITARY CAR AMBULANCE
28	ML	MILITARY CAR KITCHEN
29	MF	MILITARY FIRST CLASS
30	ART	ACCIDENT & TOOL VAN OR RELIEF VAN
31	ARME	AUXILIARY RELIEF TRAIN WITH MEDICAL EQUIPMENT
33	CTS	TOURIST CAR FOR 2 ND CLASS PASSENGERS
34	CZACEN	AIR CONDITIONED CHAIR CAR WITH END ON GENERATION
35	ERR	FOUR / SIX WHEELER
36	EN	END ON GENERATION
37	MN	MID ON GENERATION
38	ERU	FOUR / SIX WHEELER SELF PROPELLED TOWER VAN
39	FCS	FIRST CLASS COUPE AND SECOND CLASS
40	FSCN	FIRST CUM II CLASS 3-TIER SLEEPER
41	GS	SECOND CLASS FITTED WITH SELF GENERATING EQUIPMENT
42	LR	LUGGAGE WITH BRAKE VAN

43	NMG	NEW MODIFIED GOODS
44	OHE	OVER HEAD EQUIPEMNT INSPECTION CAR
45	PPS	FULL BOGIE POSTAL VAN
46	RA	INSPECTION CARRIAGE (ADMINISTRATIVE)
47	RAAC	AIR CONDITIONED INSPECTION CAR
48	RD	INSPECTION CARRIAGE (SUBORDINATE)
49	RE	INSTRUCTION VAN (MOBILE TRAINING CAR)
50	RH	MEDICAL VAN
51	RHV	AUXILIARY MEDICAL VAN
52	RK	DYNAMOMETER CAR
53	RN	GENERATING VAN
54	RR	TRAIN CREW AND REST VAN
55	RS	STORES VAN
56	RT	ACCIDENT AND TOOL VAN OR RELIEF VAN
57	RZ	TRACK RECORDING CAR
58	SLR	SECOND CLASS LUGGAGE AND BRAKE VAN
59	SMN	POWER CAR WITH MID ON GENERATION
60	VP	PARCEL VAN
61	VPC	PARCEL VAN CONVERTED
62	VPU	MOTOR CUM PARCEL VAN
63	VPH	HIGH CAPACITY PARCEL VAN
65	WCD	VESTIBULED DINING CAR
66	WCRAC	VESTIBULED AIR CONDITONED TWIN CAR
67	WCTAC	VESTIBULED AIR CONDITIONED TOURIST CAR
68	WFAC	VESTIBULED AIR CONDITIONED FIRST CLASS
69	WFC	VESTIBULED FIRST CLASS
70	WGACCN	VESTIBULED SELF GENERATING AIR CONDITIONED THREE TIER
71	WGACCW	VESTIBULED SELF GENERATING AIR CONDITIONED TWO TIER
72	WACCW	VESTIBULED AIR CONDITIONED TWO TIER
73	WGFAC	VESTIBULED SELF GENERATING AIR CONDITIONED FIRST CLASS
74	WGFACCW	VESTIBULED SELF GENERATING FIRST CUM AC 2-TIER
75	WGSCN	VESTIBULED SELF GENERATING SECOND CLASS THREE TIER
76	WGSCNLR	VESTIBULED SECOND CLASS THREE TIER WITH LUGGAGE & BRAKE VAN
77	WGSCZ	VESTIBULED SELF GENERATING SECOND CLASS CHAIR CAR
78	WGSCZAC	VESTIBULED SELF GENERATING SECOND CLASS CHAIR CAR AIR CONDITIONED
79	WGSD	VESTIBULED SELF GENERATING SECOND CLASS DOUBLE DECKER
80	WLRRM	VESTIBULED LUGGAGE CUM BRAKE VAN WITH DIESEL GENERATING MOTOR CAR
81	WSCZAC	VESTIBULED SECOND CLASS AIR CONDITIONED CHAIR CAR
82	WSLRN	VESTIBULED SECOND CLASS, BRAKE CUM LUGGAGE AND POWER CAR
83	FIAT	FABRICA ITALINA De AUTOMOBILE TORINO

APPENDIX – III
TRANSPORTATION CODES FOR FREIGHT STOCK

Wagons are designated by codes viz. Transportation code and Mechanical code. Transportation codes are allotted to wagons on the basis of their general classification whereas Mechanical codes are allotted on the basis of design features. Transportation codes and Mechanical codes used for various Wagons are as under.

FOR FOUR WHEELER WAGONS

Sr. No	Description	Transportation code	Mechanical code
1	Open wagon (coal or general)	K	O
2	Open wagon (military)	KM	OM
3	Covered wagon (general)	C	CR
4	Covered wagon with light wt. Tr. Type CBC	CRT	CRT
5	Covered wagon, Horse or Cattle	CA	CMR
6	Covered wagon (Explosive)	X	CE
7	Brake van (ordinary)	V	BVG
8	Covered BK van with CBC and Transition type coupling on both ends	BVGT	BVGT
9	Tank wagon, Kerosene	TK	TORX/TO
10	Tank wagon, Vegetable oil	TV	TORX/TO
11	Tank wagon, Lube oil	TZ	TORX/TO
12	Tank wagon, Petrol/Benzol/hexane	TP/TB/TH	TRP
13	Tank wagon, Heavy Oil	TL	TOH
14	Tank wagon, liquefied petroleum gas	TG	TPGL/TPGIN
15	Flats for containers	FK	FK

FOR EIGHT WHEELER

Sl. No.	Description	Transportation code	Mechanical code
1	Open	K	O
2	Covered	C	C
3	Low sided	L	M
4	High sided	C	H
5	Explosives	X	E
6	Horse/Cattle	A	MR
7	Military	M	M
8	Tank	T	T
9	Brake Van	V	BVG
10	Bogie Stock	B (prefixed)	B (prefixed)
11	Hopper	B	B
12	Well	W	W

13	Flat	F	R
14	Container	FK	FK
15	Types of coupling	R,C,T (suffix)	R,C,T (suffix)
16	Types of discharge Centre Side Centre & Side Rapid (Bottom)	C S X R	C S X R
17	Brake system (Air)	N (suffix)	N (suffix)
18	Sliding Roof	S (suffix)	S (suffix)
19	Metre Gauge	M (First suffix)	M (First suffix)
20	Narrow Gauge	N (First Suffix)	N (First Suffix)

Sl. No.	Description	Transp./Mech. code
1	Open box wagons with Screw coupling	BOXR
2	Open box wagon with Transition couplers	BOXT
3	Open box wagon with CBC at both ends and also with CBC at one end and transition coupler at other end.	BOXC
4	Open box Wagon with swing cum flap doors and sliding roof.	BOXES
5	Bogie open ore wagon with St. CBC (Open Wagon Gondola)	BOI
6	Bogie open ore wagon with St. CBC (Casnub Bogie)	BOY
7	Bogie open Box wagon with pneumatic brake	BOXN
8	Bogie open wagon, hopper	BOB
9	Bogie open wagon, hopper with centre discharge	BOBC
10	Bogie open wagon, hopper with side discharge	BOBS
11	Bogie open wagon, hopper with center and side discharge	BOBX
12	Bogie open bottom rapid discharge hopper wagon	BOBR
13	Bogie Open Special Wagon with Transition Couplings	BOXT
14	Bogie Brake Van Air Brake Equipped with ICF Trolley	BVZI
15	Bogie Tank Petroleum Equipped with Air Brake System	BTPN
16	Bogie Tank Ammonia Liquefied Tank Wagon with Air Brake System	BTALN
17	Bogie Well Wagon	BWT